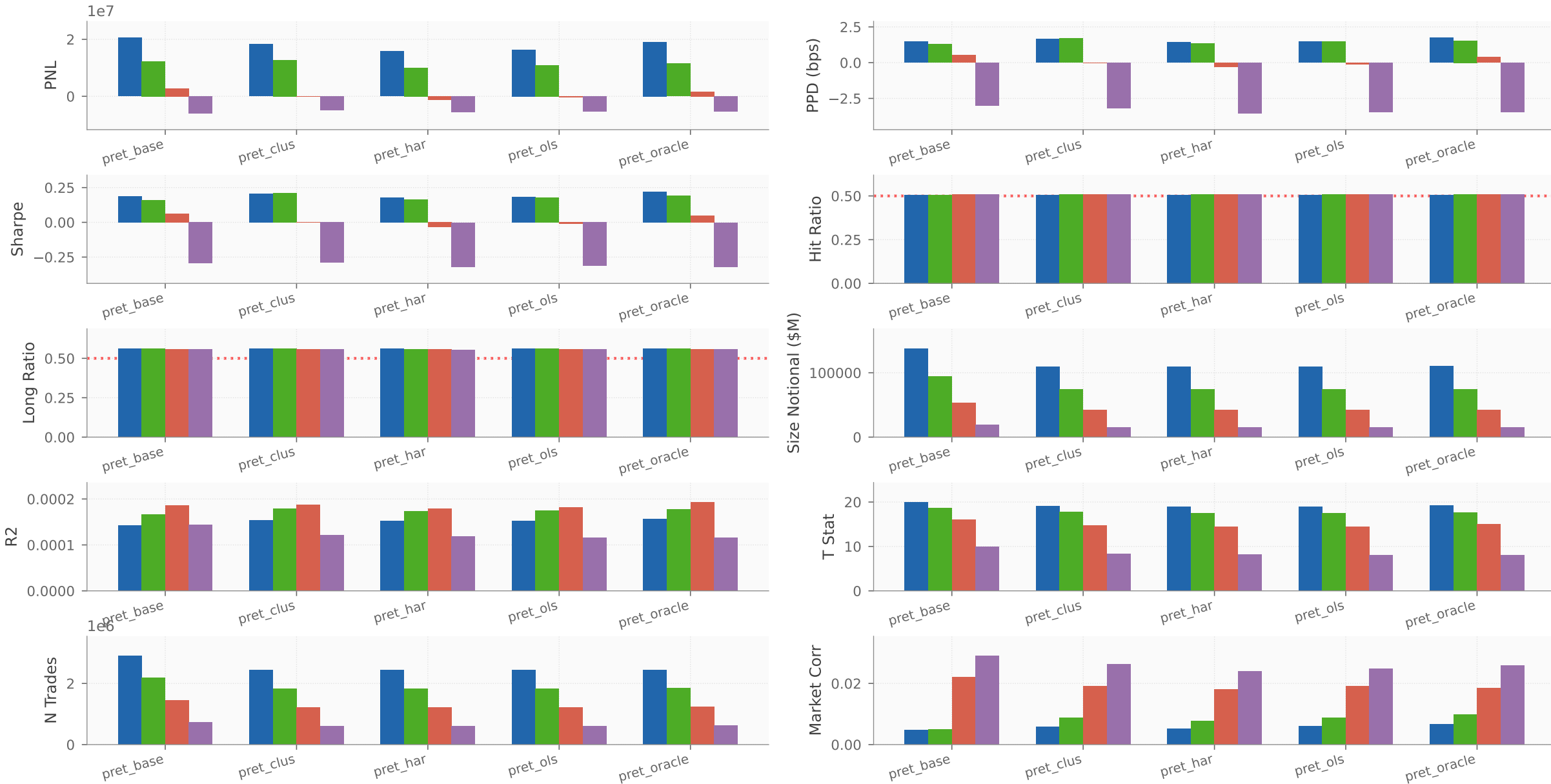


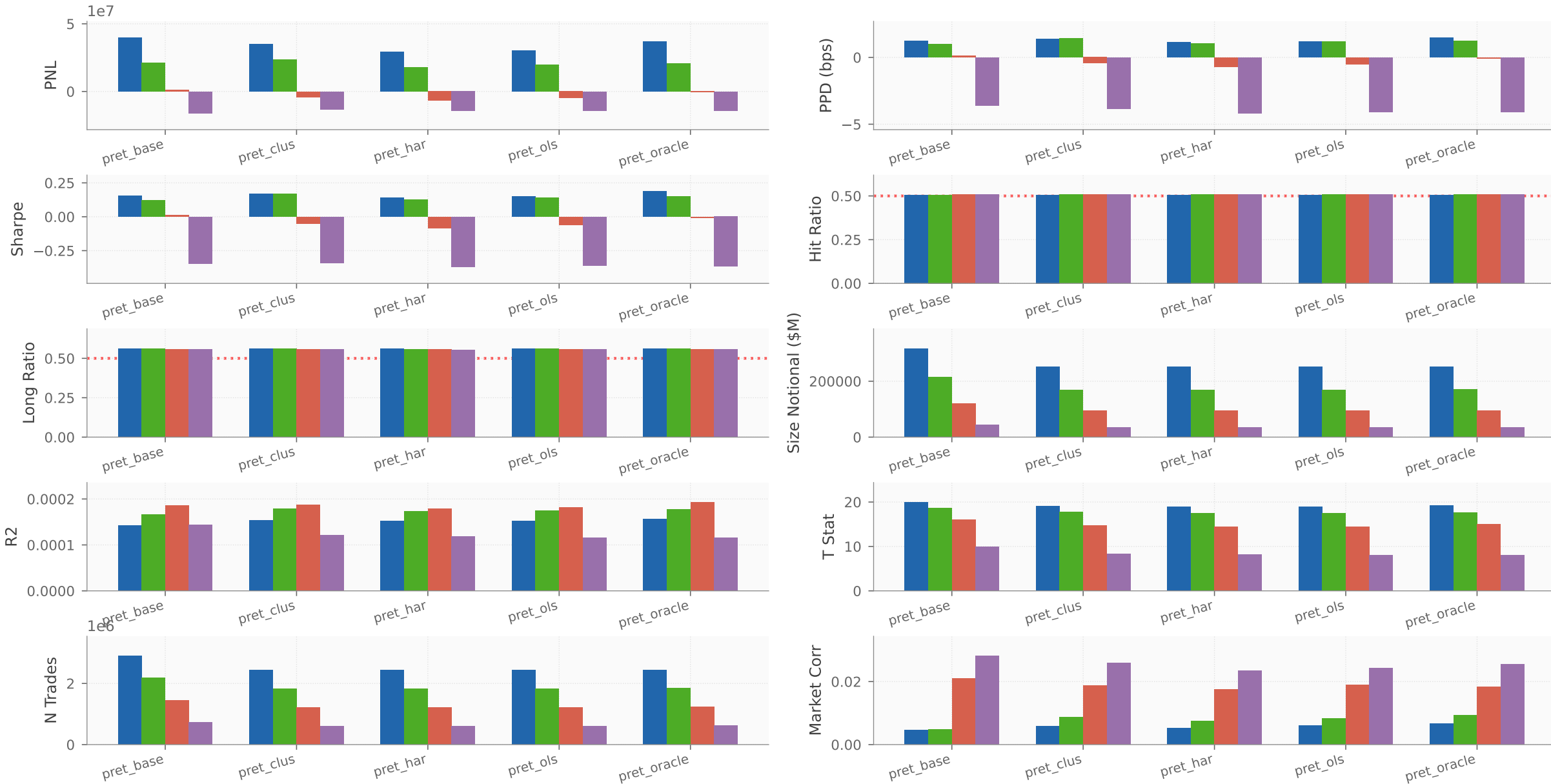
Summary Metrics · target: fret_OP5CL_MR · bet_size_col: bs_m025_c100k

grouped by signal



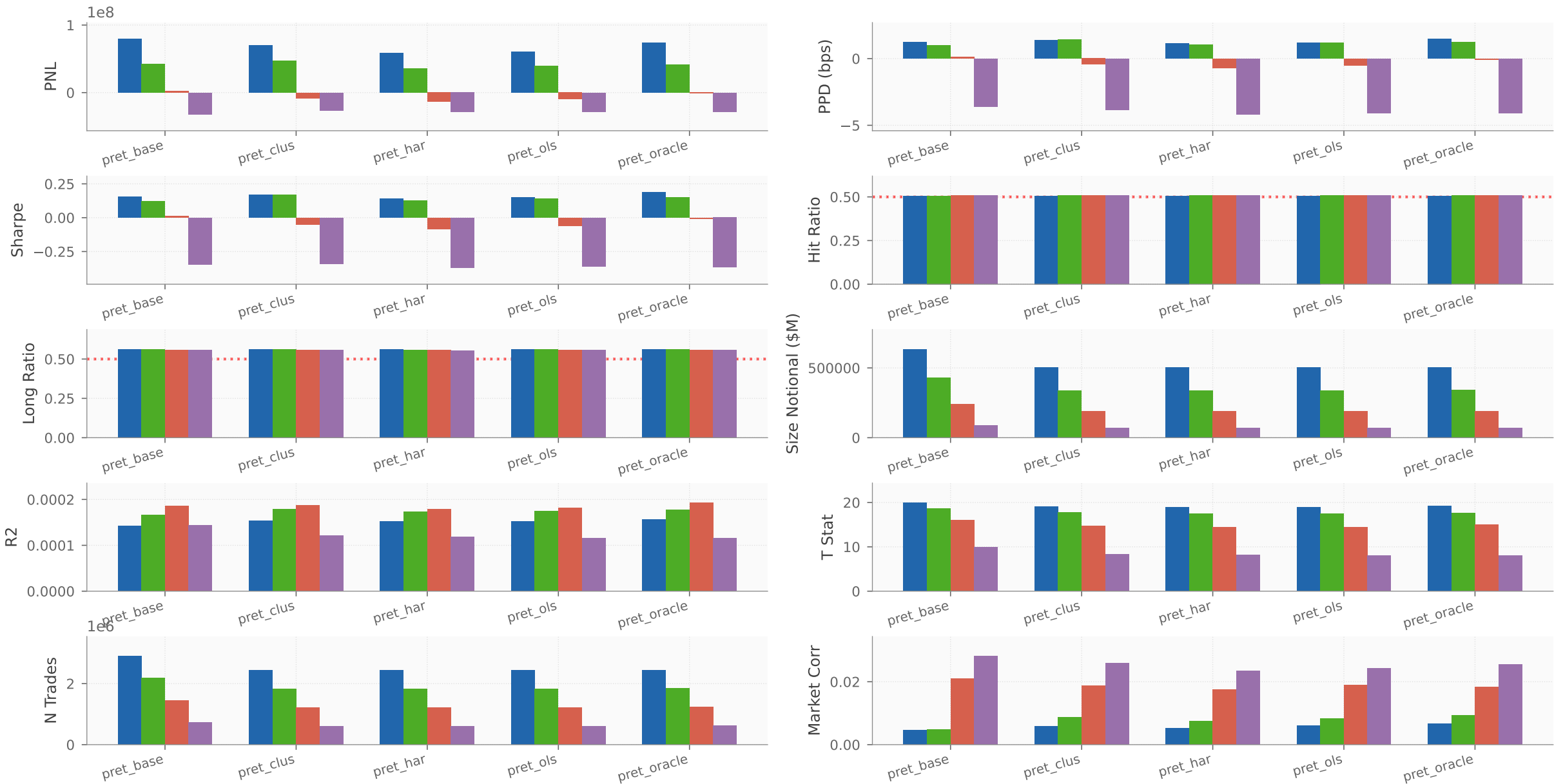
Summary Metrics · target: fret_OP5CL_MR · bet_size_col: bs_m05_c250k

grouped by signal



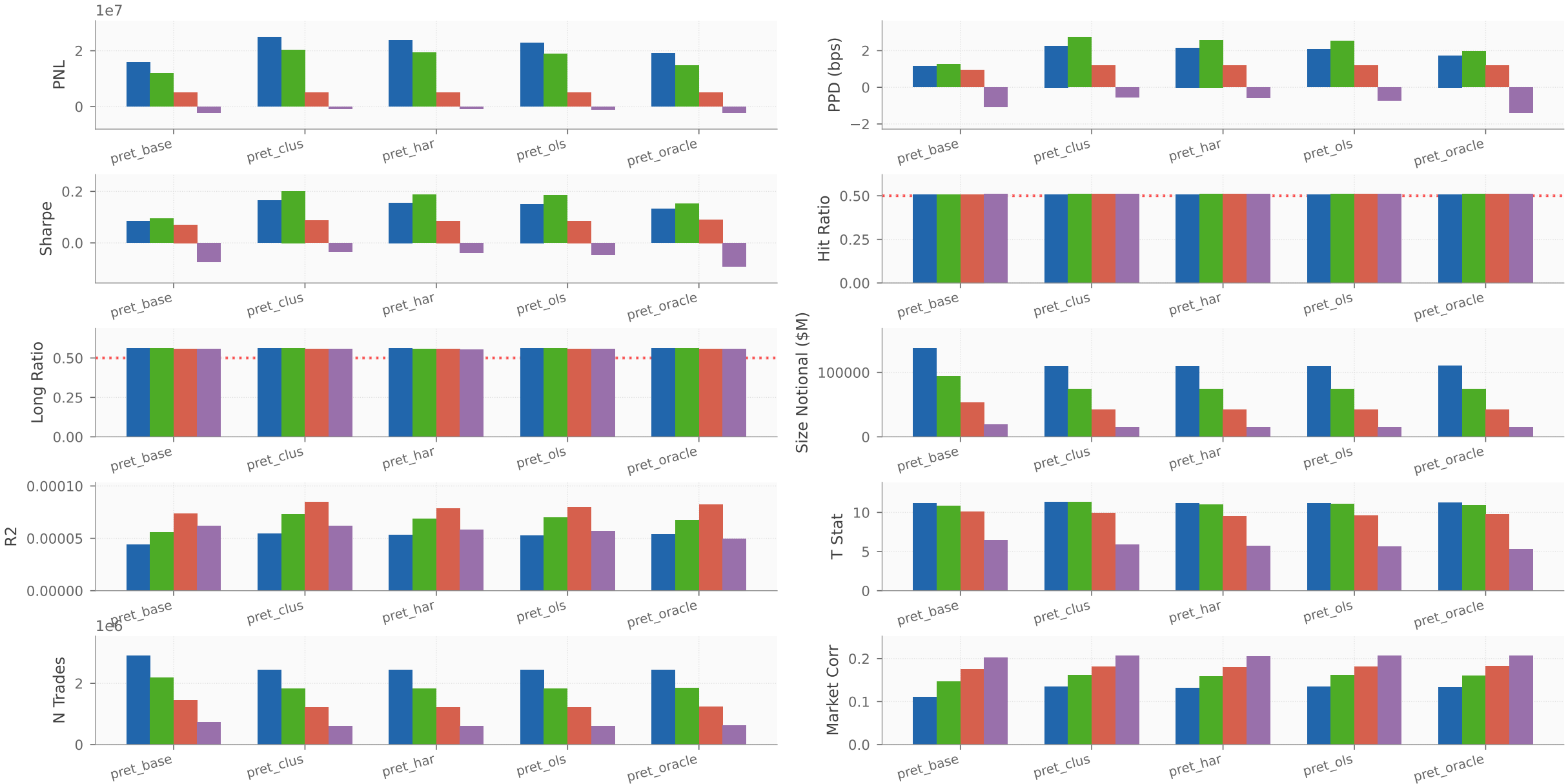
Summary Metrics · target: fret_OP5CL_MR · bet_size_col: bs_m1_c500k

grouped by signal



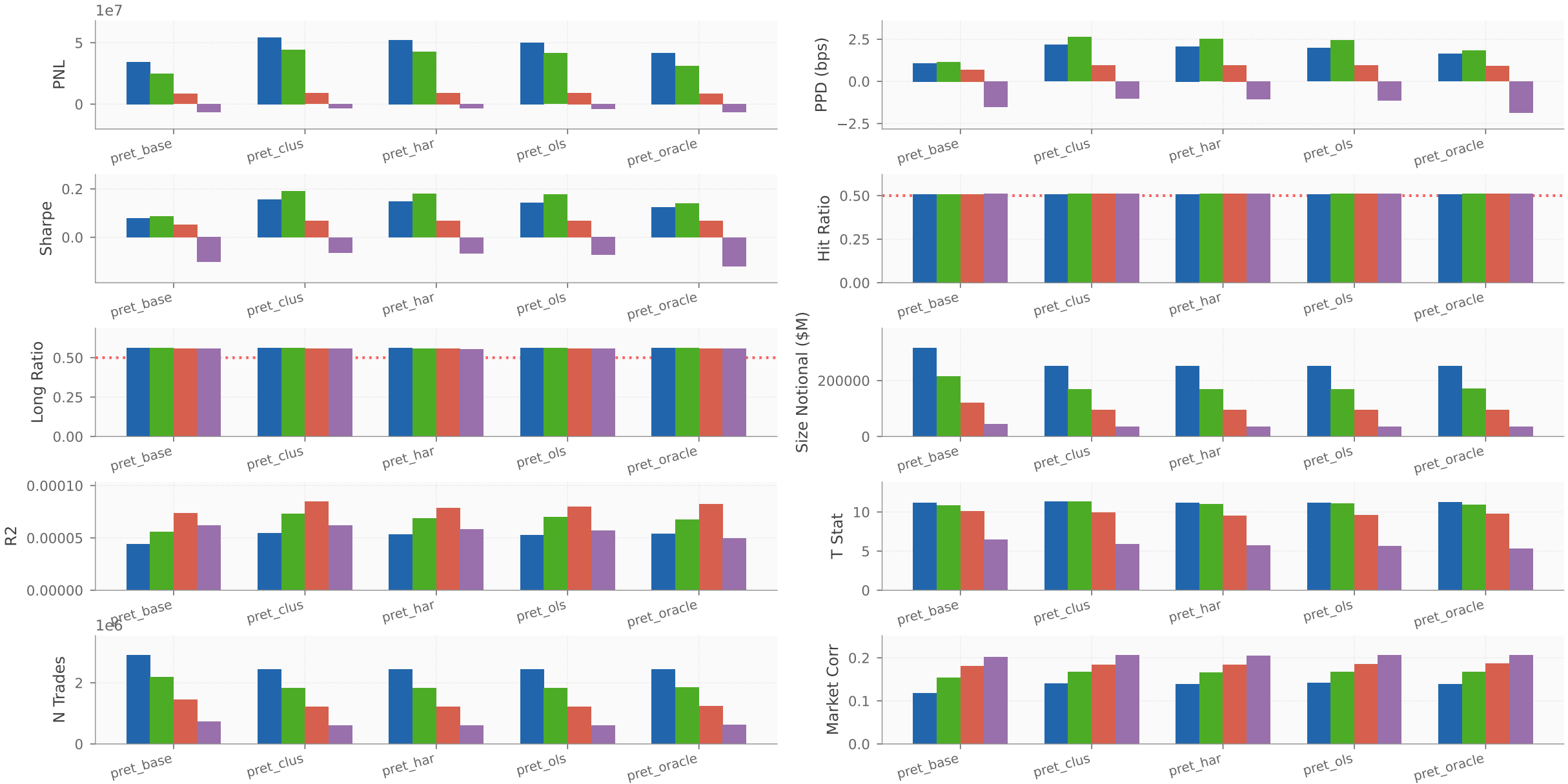
Summary Metrics · target: fret_OP5CL_RR · bet_size_col: bs_m025_c100k

grouped by signal



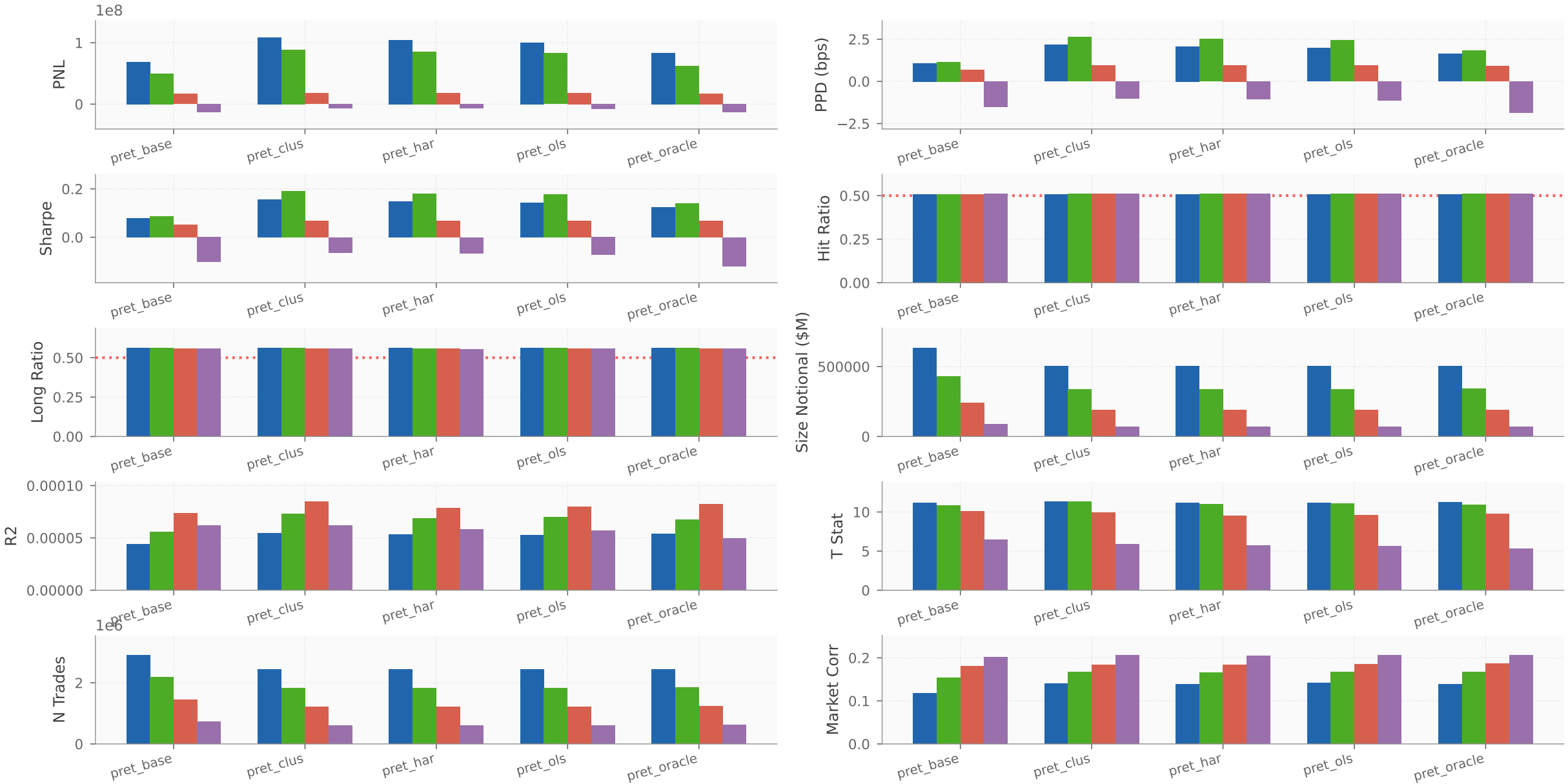
Summary Metrics · target: **fret_OP5CL_RR** · bet_size_col: **bs_m05_c250k**

grouped by *signal*



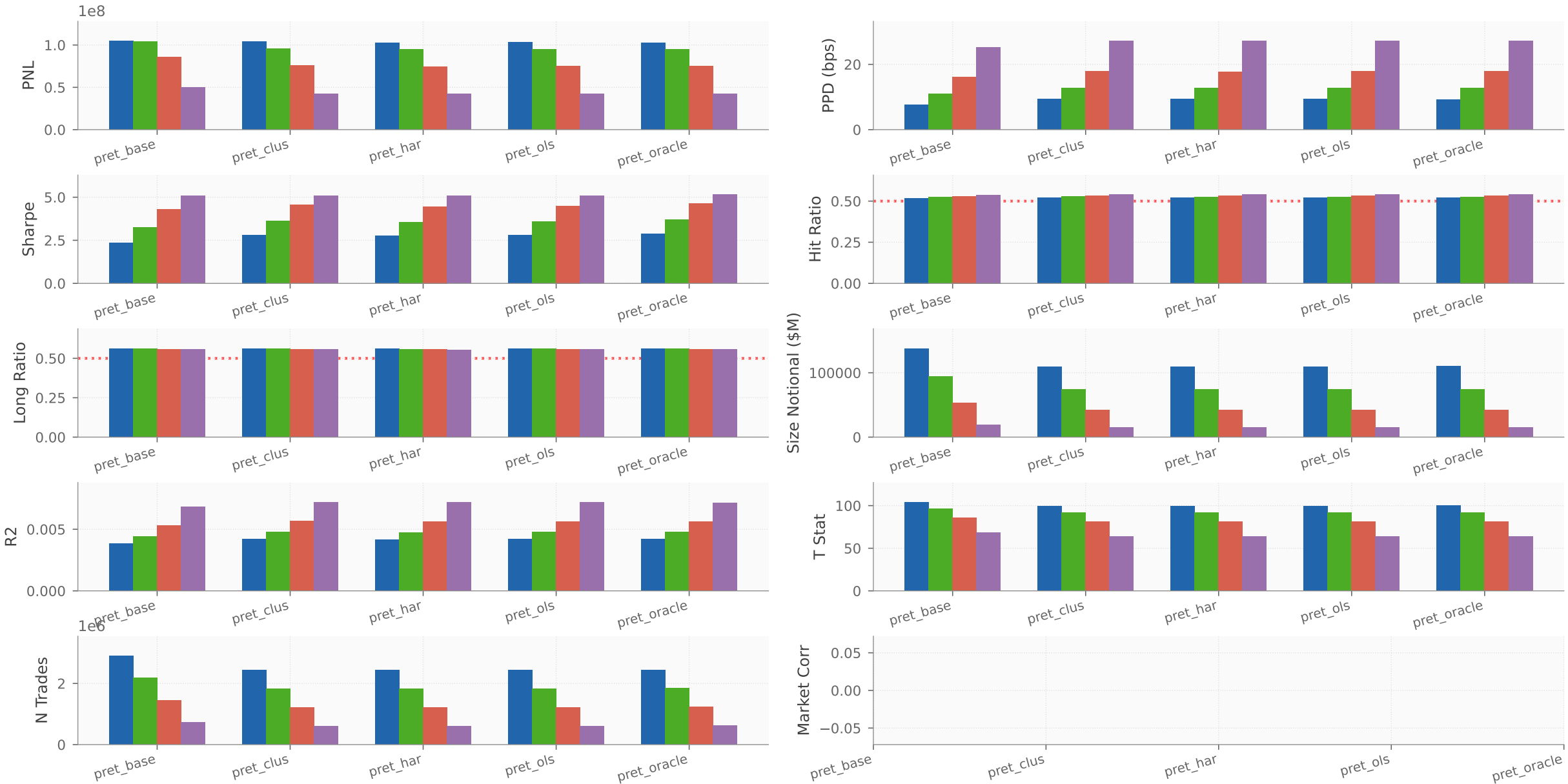
Summary Metrics · target: fret_OP5CL_RR · bet_size_col: bs_m1_c500k

grouped by signal



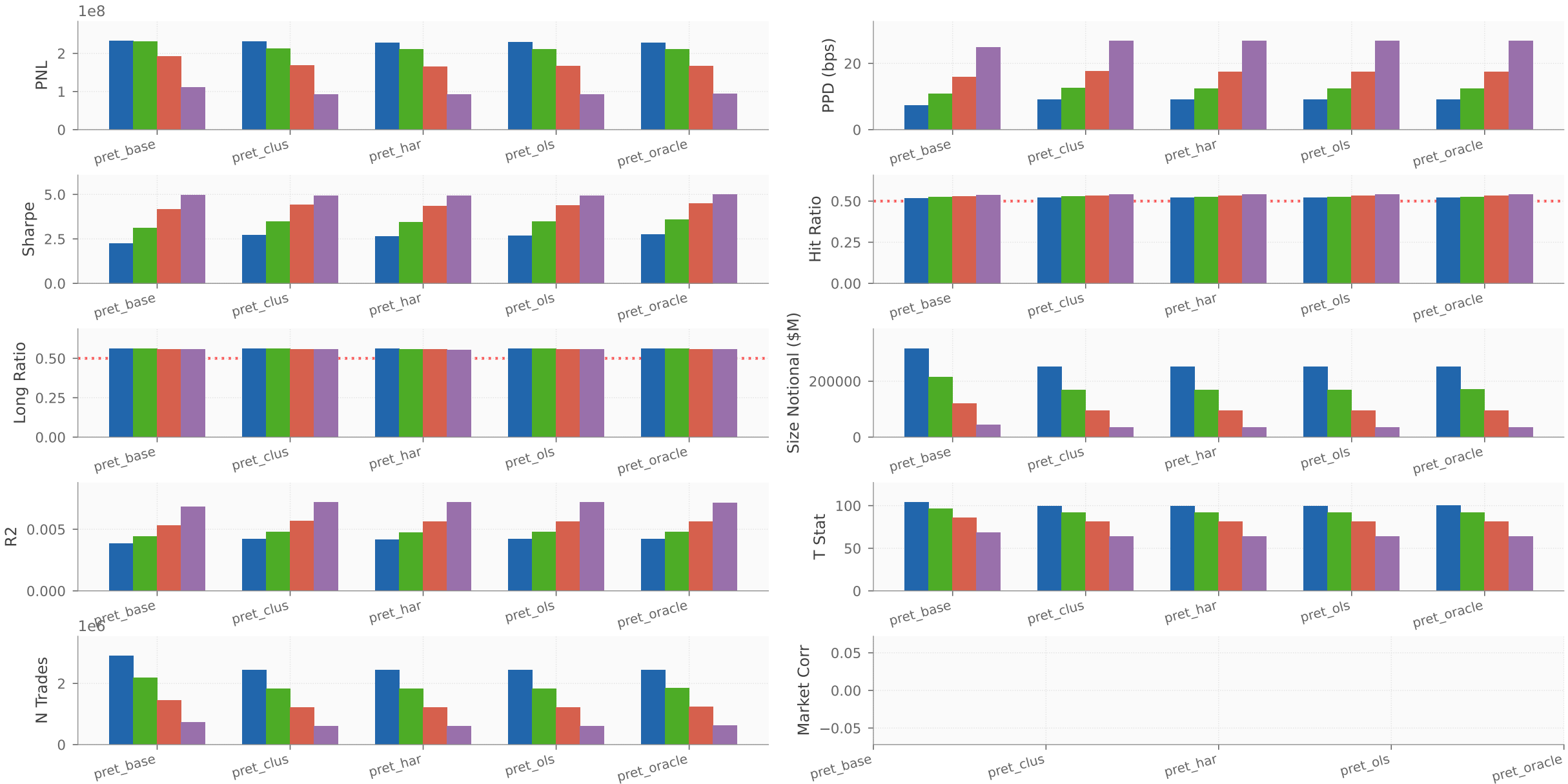
Summary Metrics · target: fret_OPCL_MR · bet_size_col: bs_m025_c100k

grouped by signal



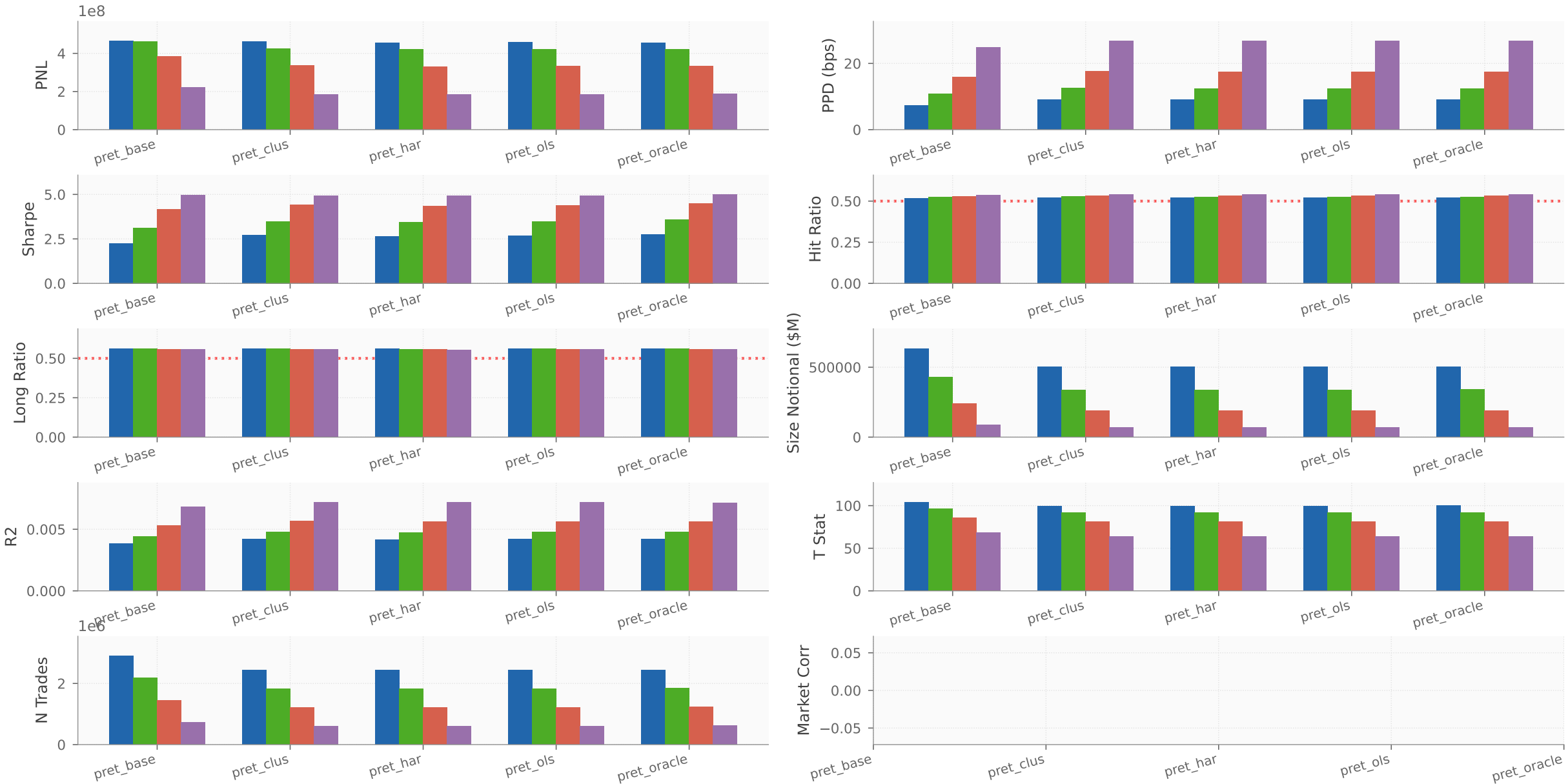
Summary Metrics · target: fret_OPCL_MR · bet_size_col: bs_m05_c250k

grouped by signal



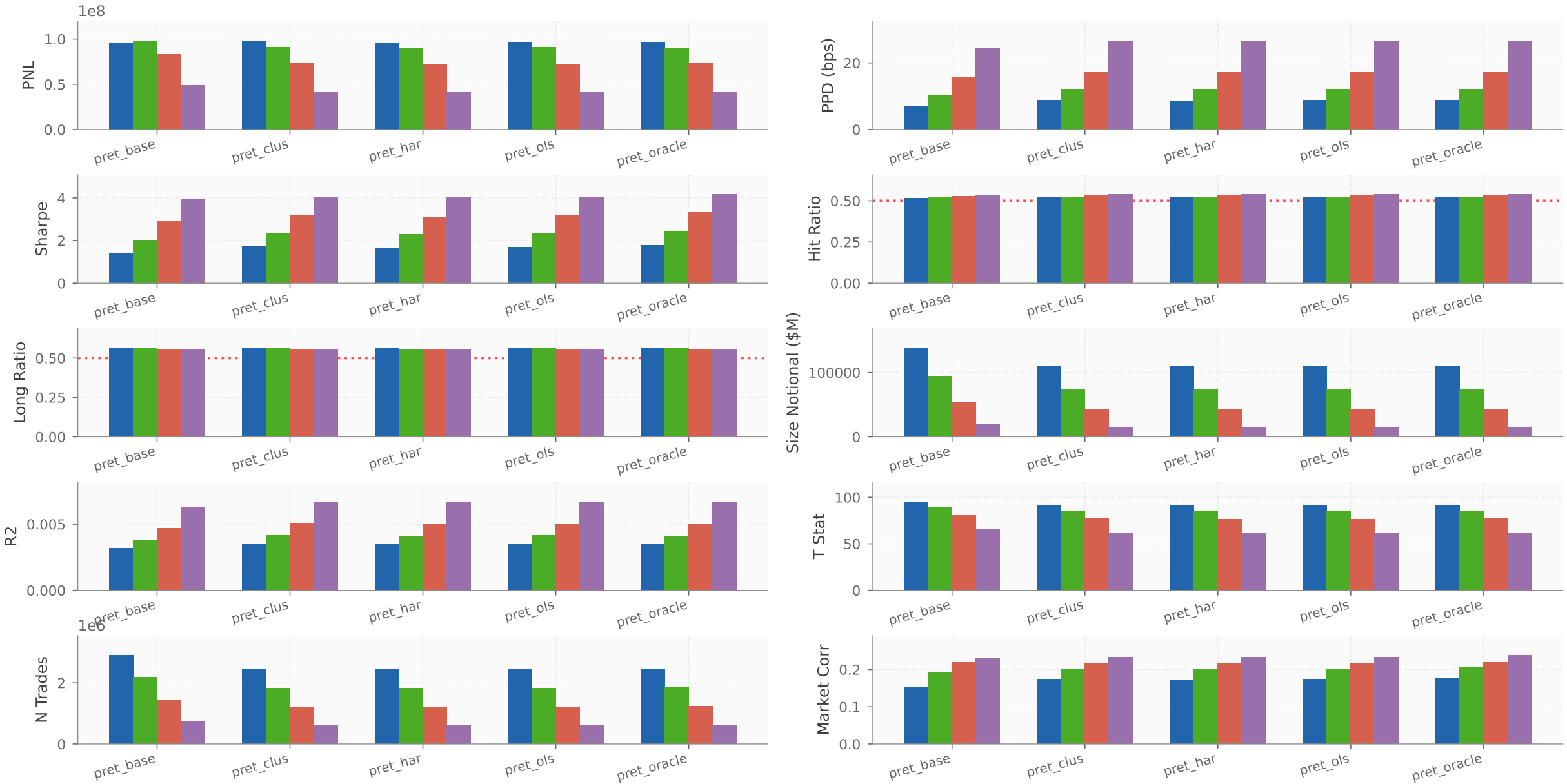
Summary Metrics · target: fret_OPCL_MR · bet_size_col: bs_m1_c500k

grouped by signal



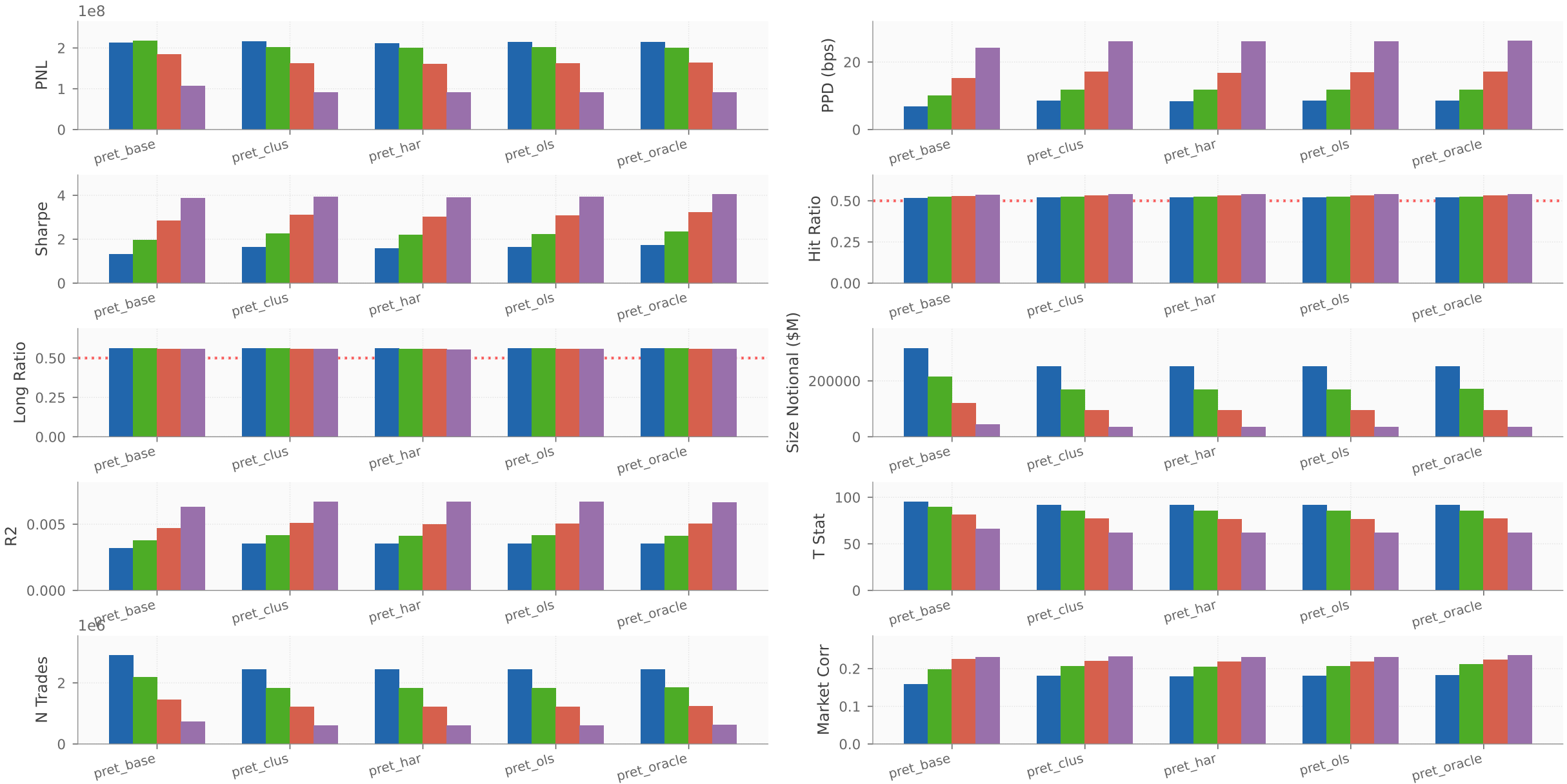
Summary Metrics · target: fret_OPCL_RR · bet_size_col: bs_m025_c100k

grouped by signal



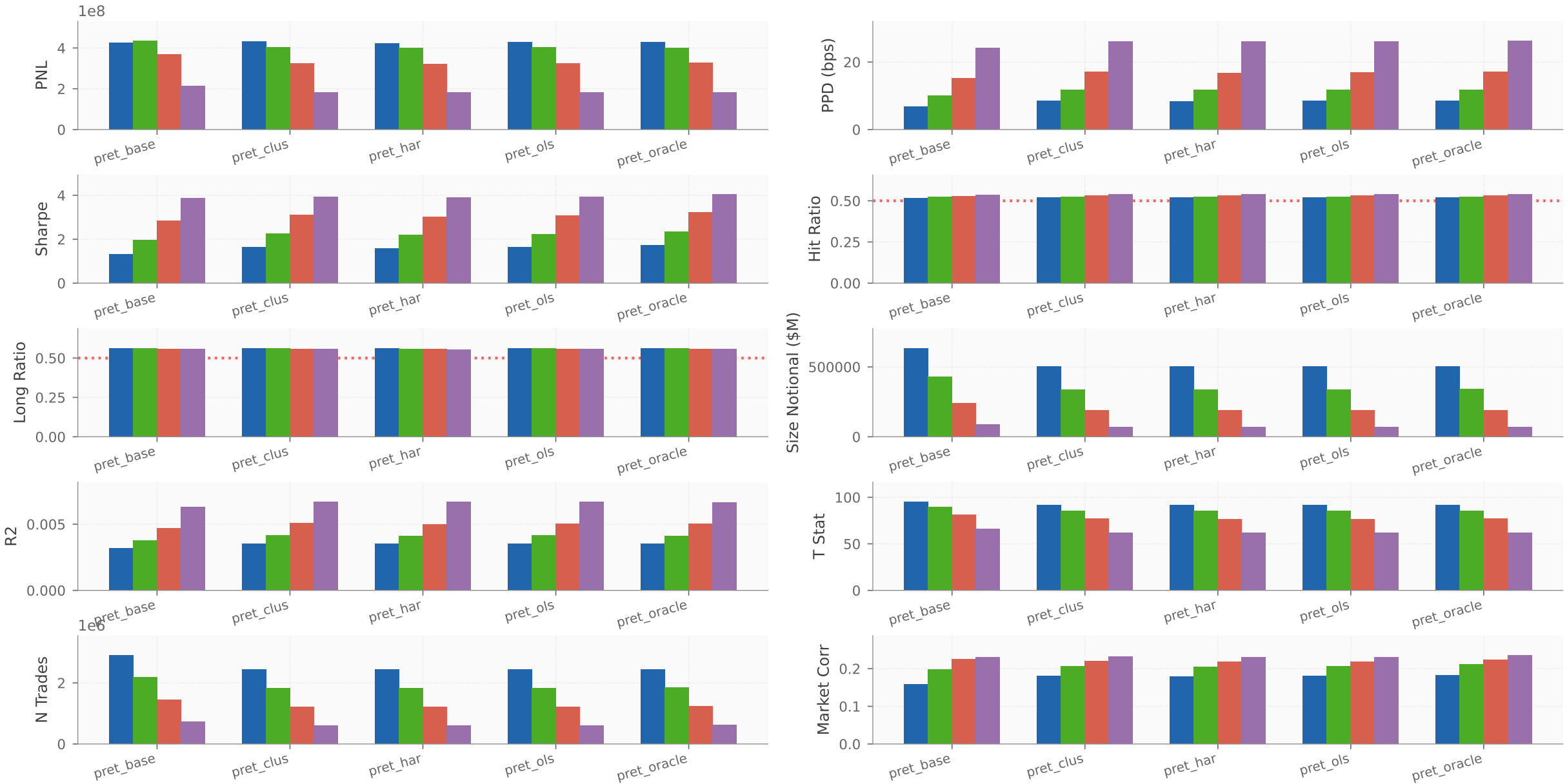
Summary Metrics · target: fret_OPCL_RR · bet_size_col: bs_m05_c250k

grouped by signal

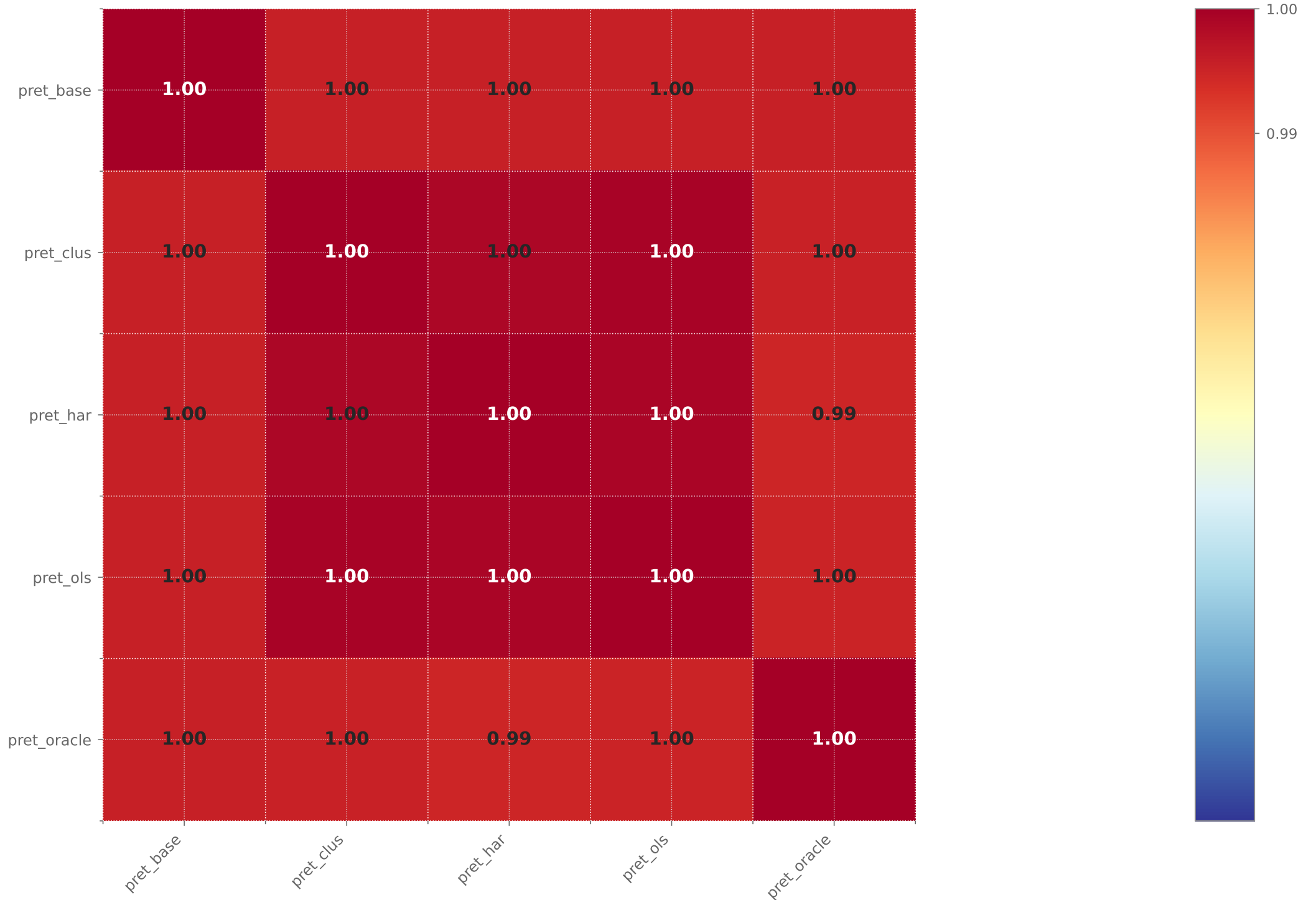


Summary Metrics · target: fret_OPCL_RR · bet_size_col: bs_m1_c500k

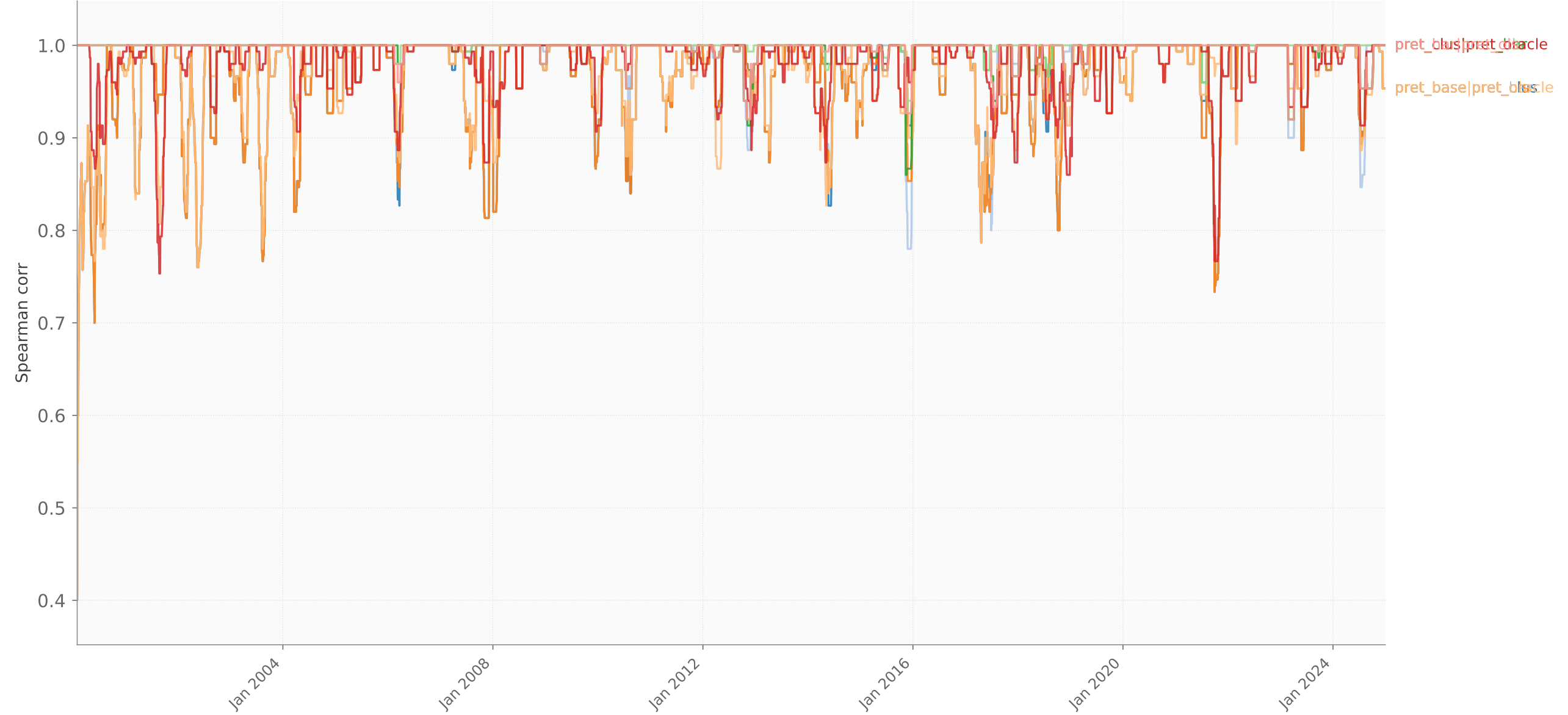
grouped by signal



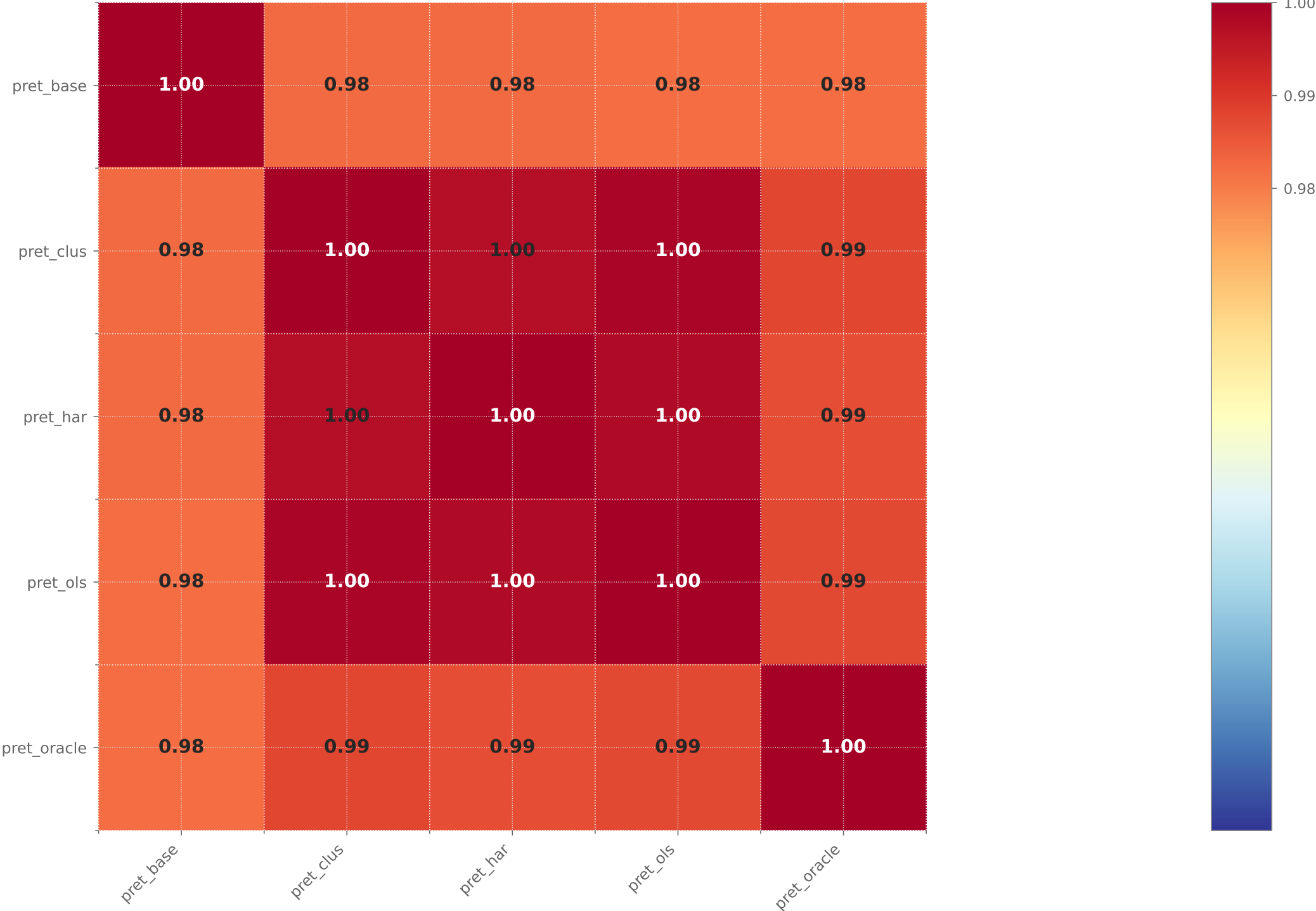
Heatmap 1 — Alpha Time-Series Corr (Spearman, alpha_sum) (avg 6262 days)



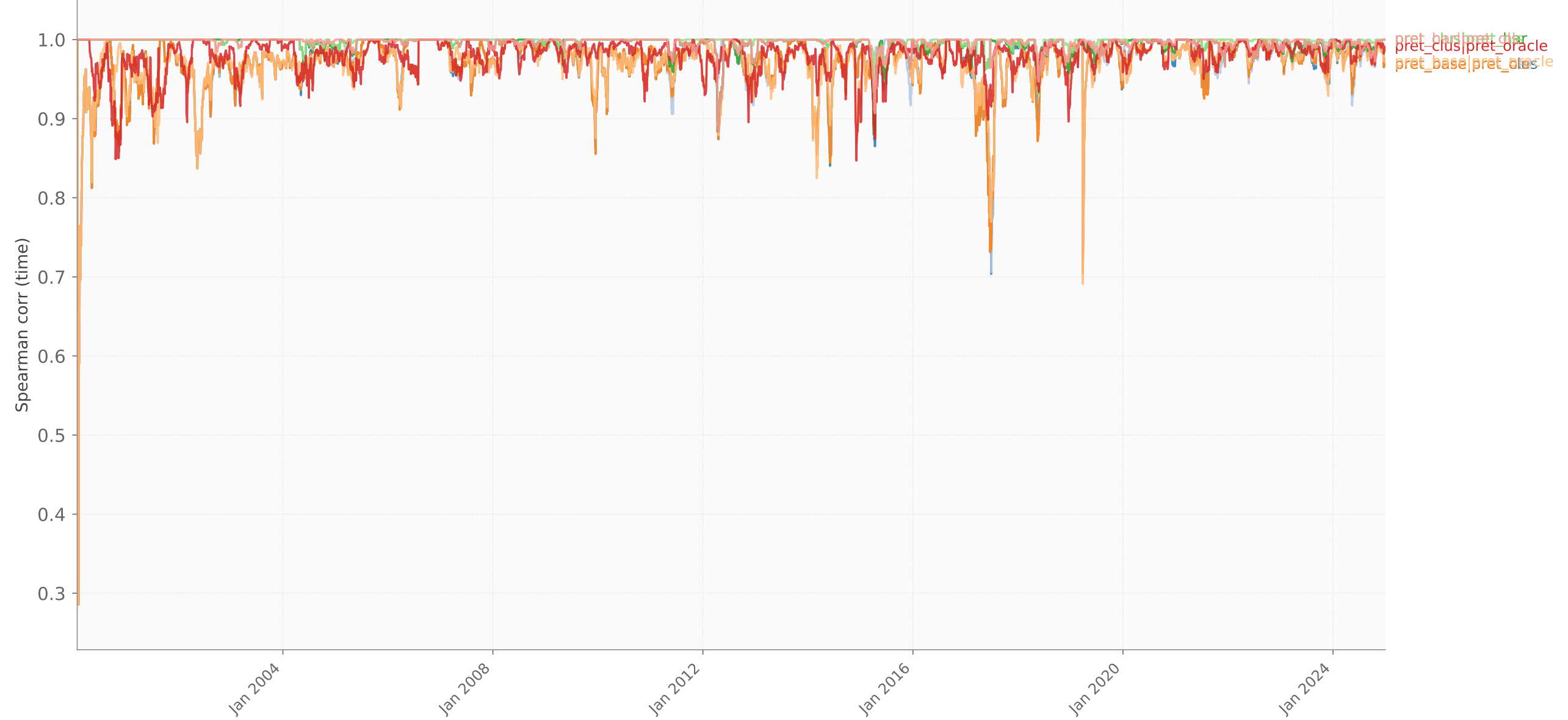
[H1] Alpha vs Alpha — smoothed 30D



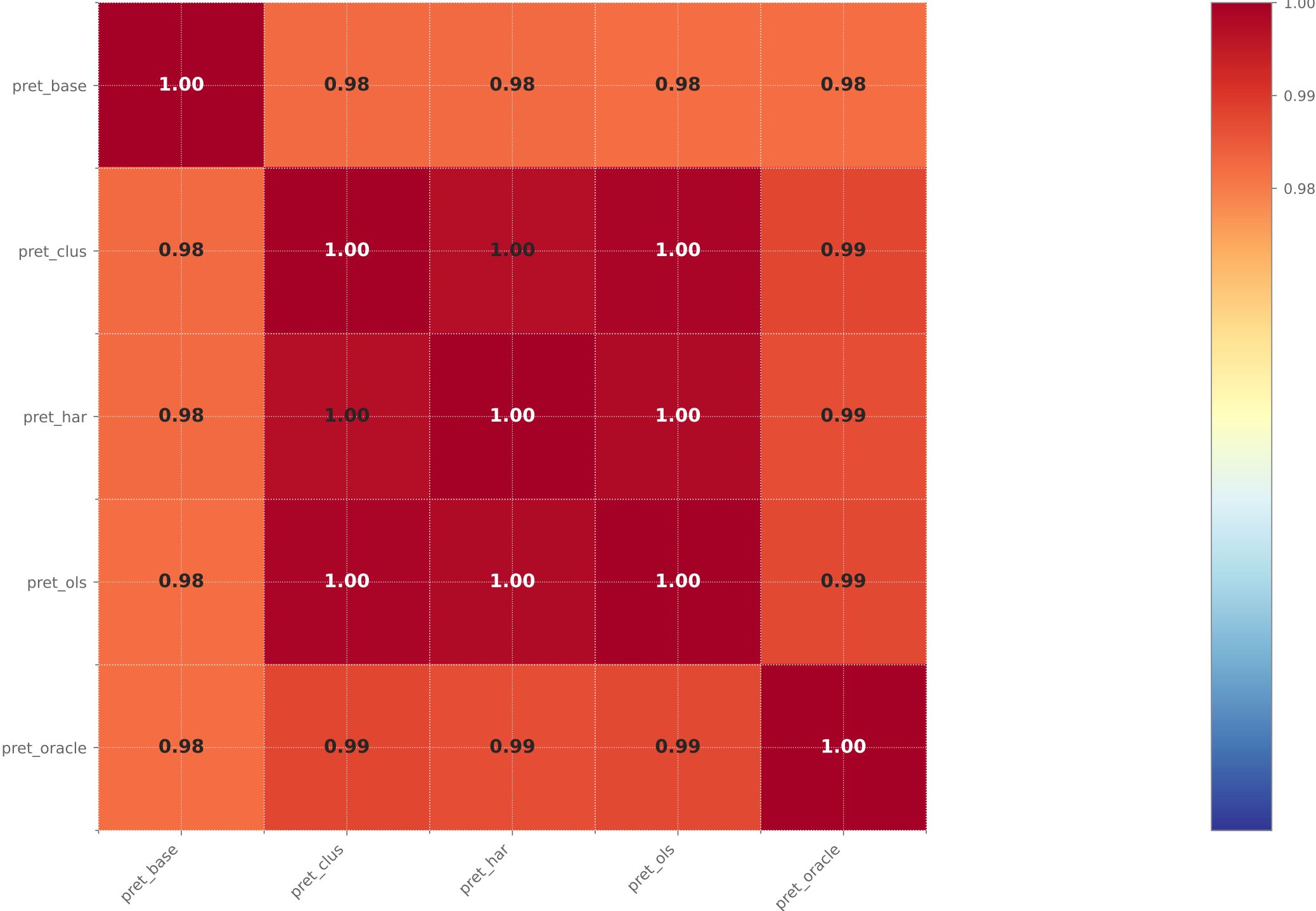
Heatmap 2 — Time-series Spearman of daily PnL (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_100] (avg 6262 days)



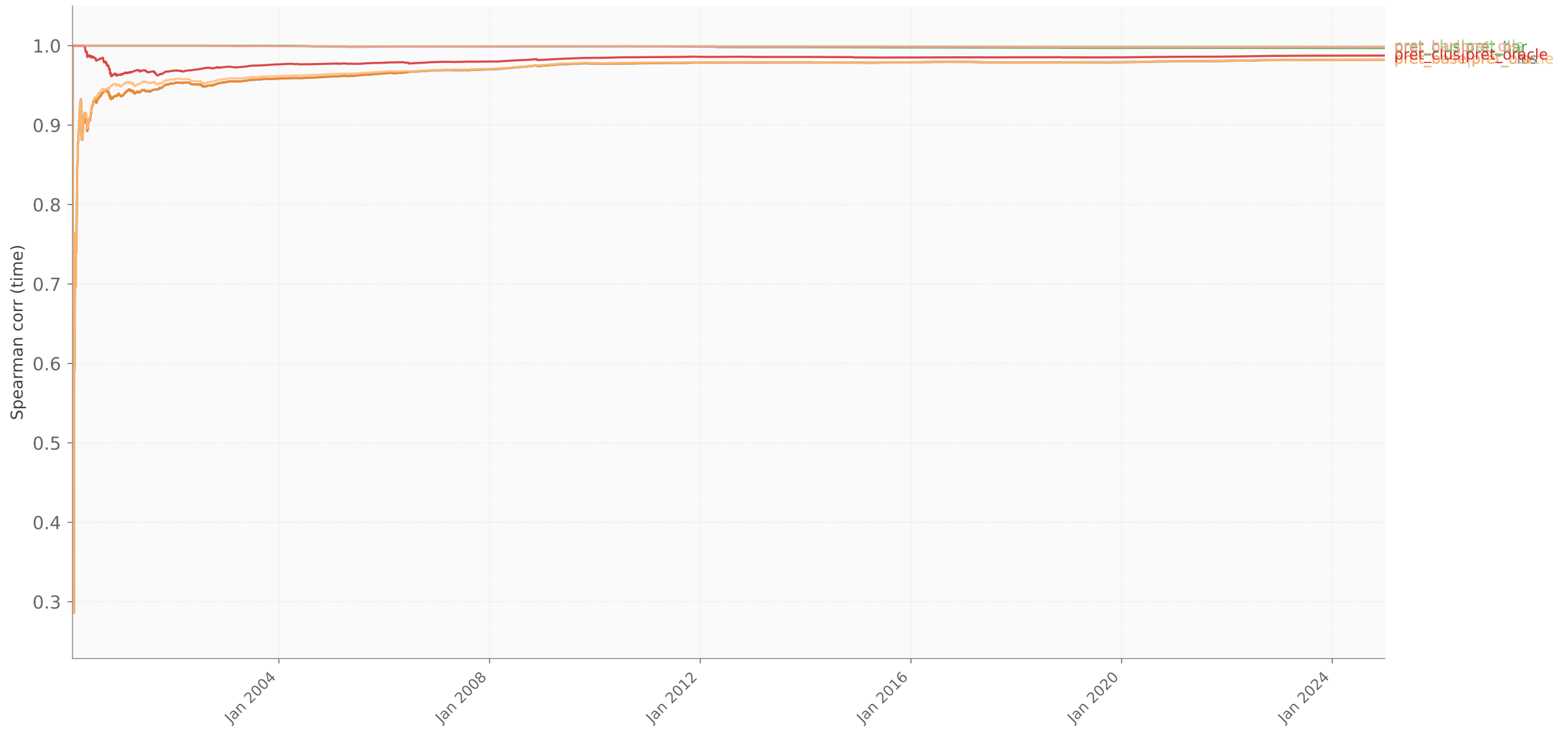
[H2|qr_100] Alpha vs Alpha — Time corr (PnL, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Rolling Spearman 30D



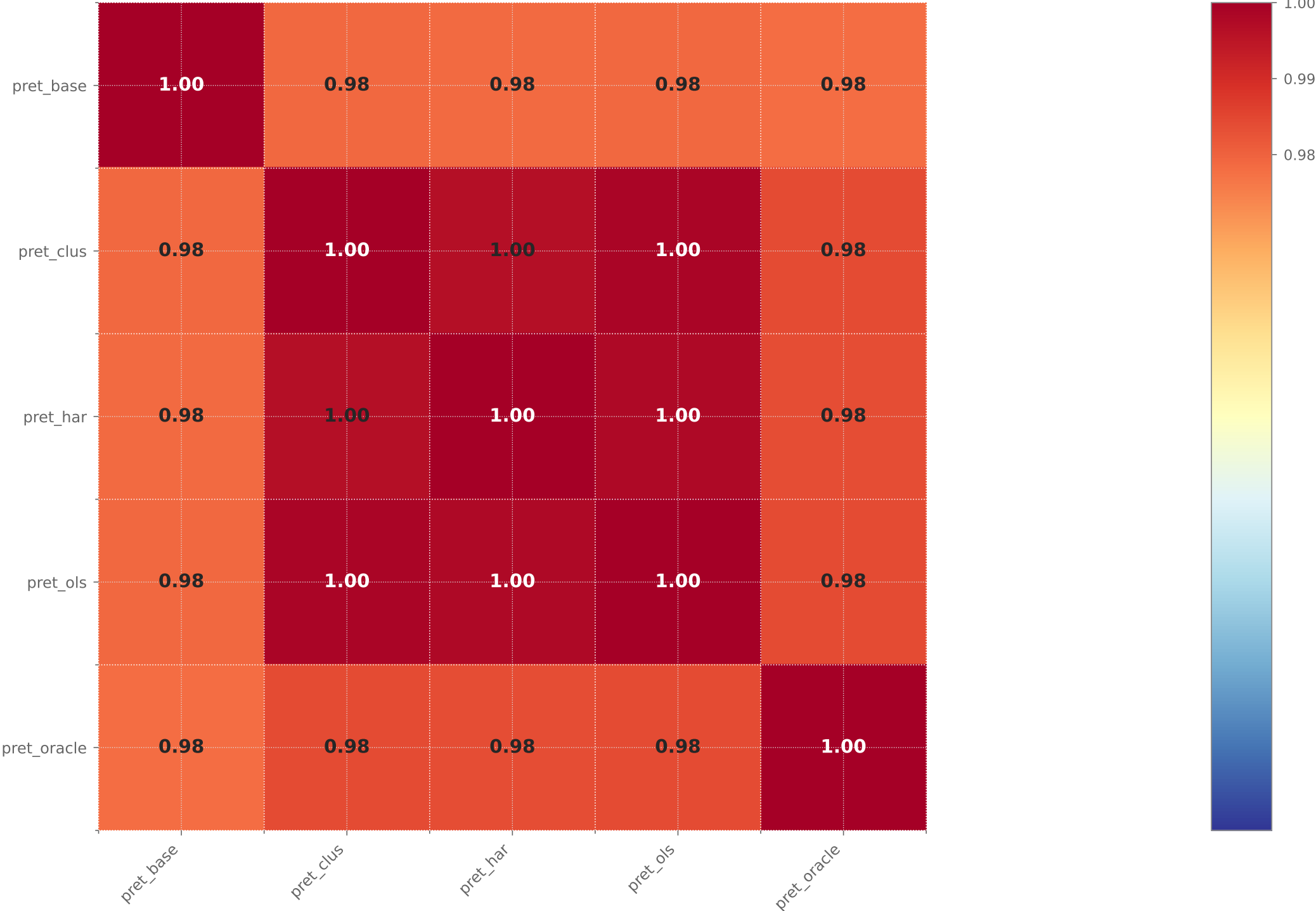
Heatmap 3 — Time-series Spearman of daily PnL vectors (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_100] (days=6262)



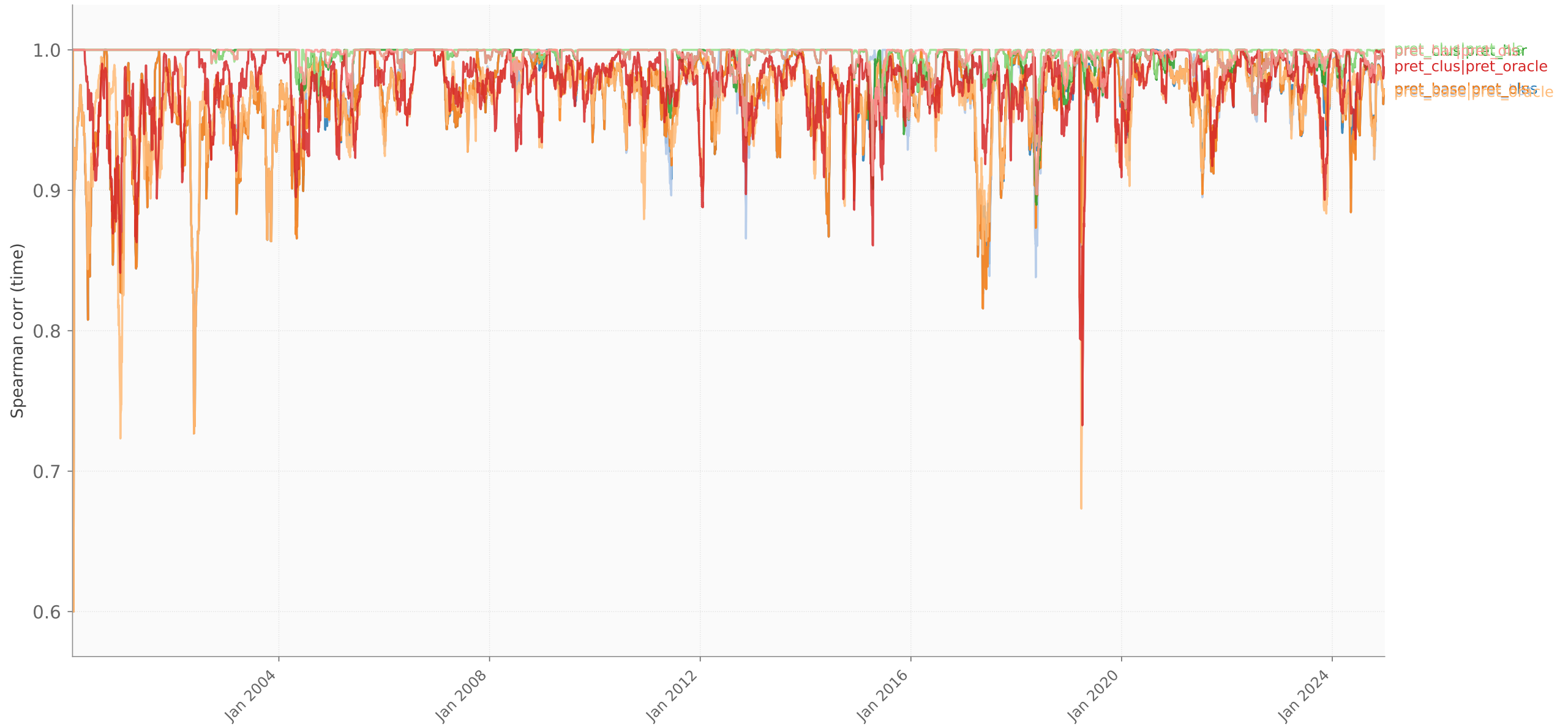
[H3|qr_100] Alpha vs Alpha — Time corr (PnL vectors, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Expanding Spearman



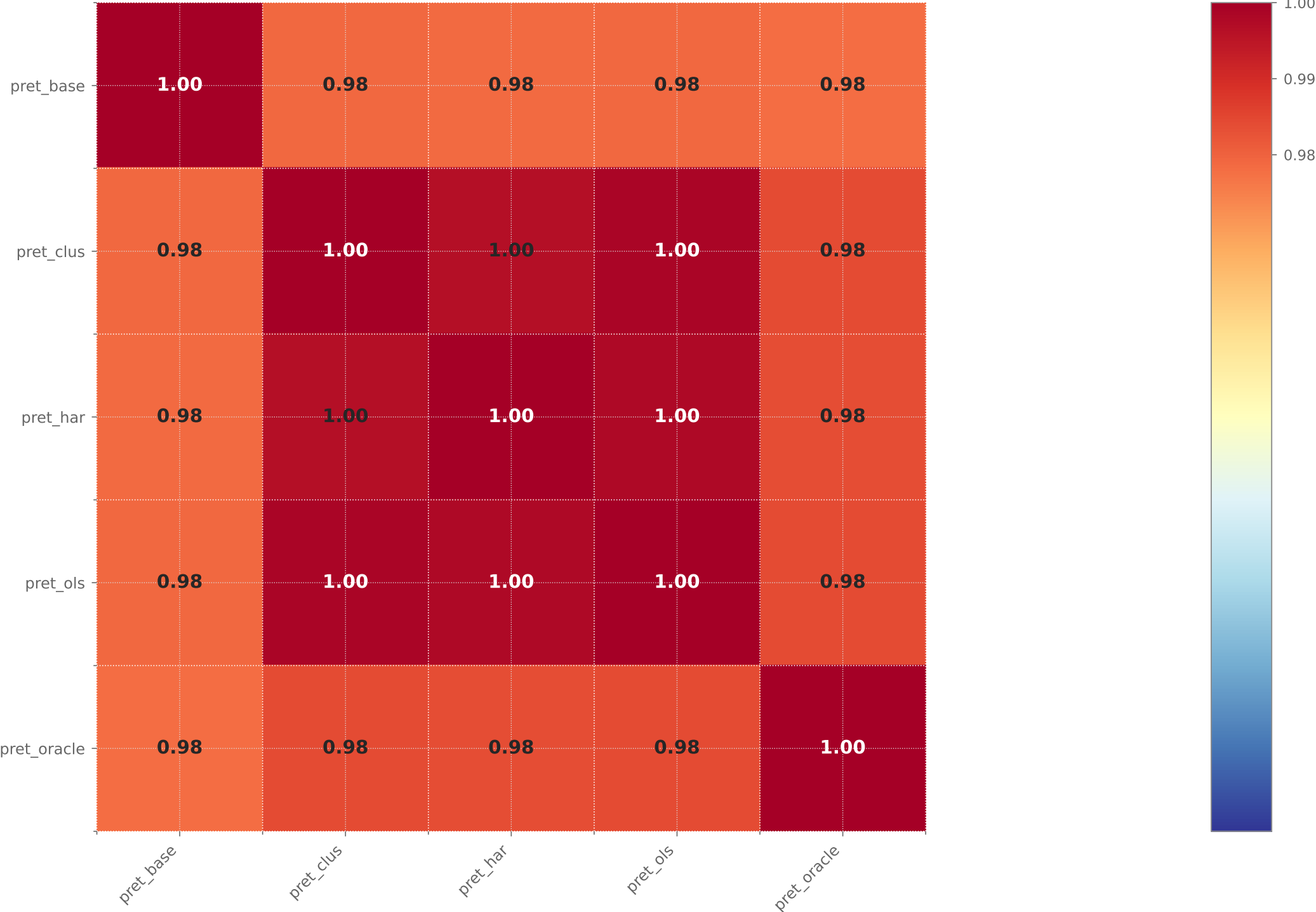
Heatmap 2 — Time-series Spearman of daily PnL (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_75] (avg 6262 days)



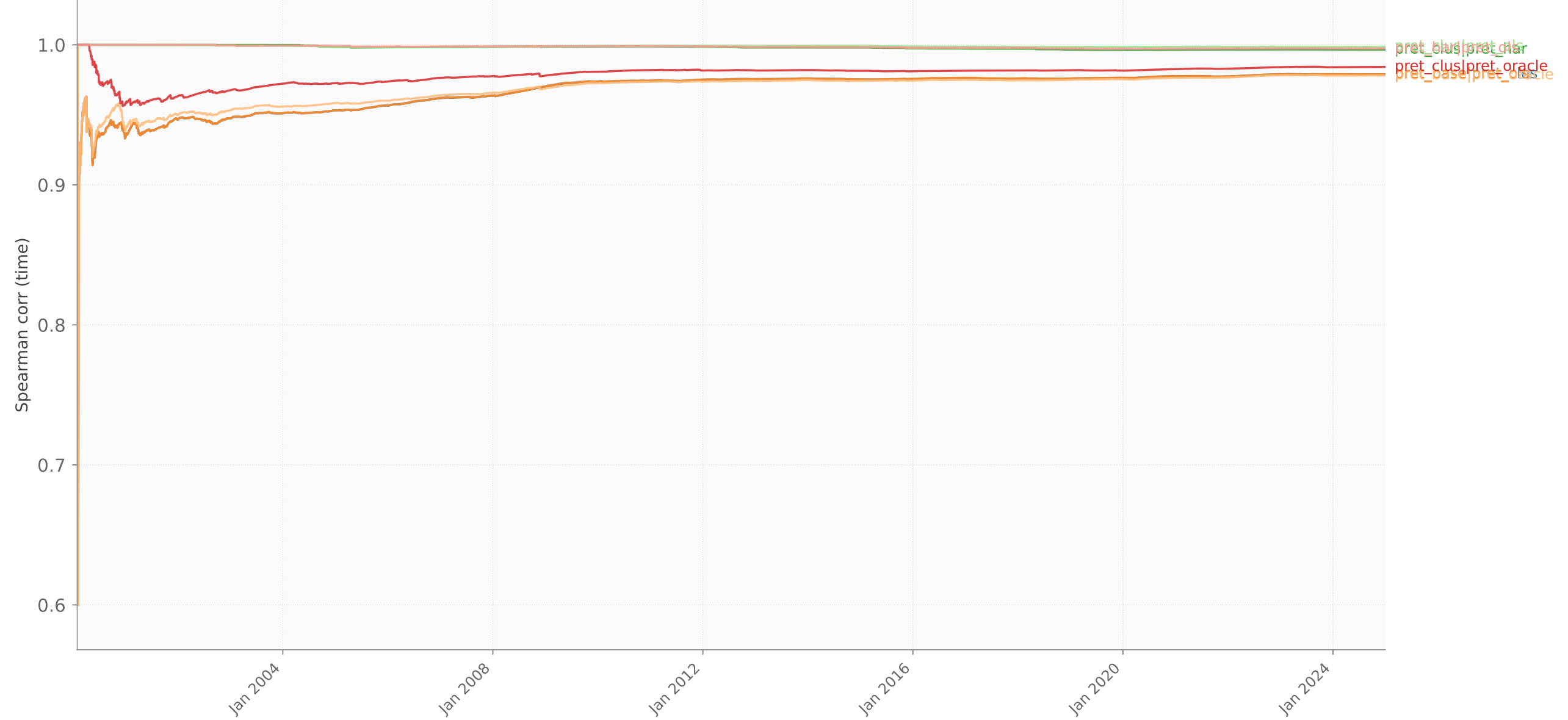
[H2|qr_75] Alpha vs Alpha — Time corr (PnL, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Rolling Spearman 30D



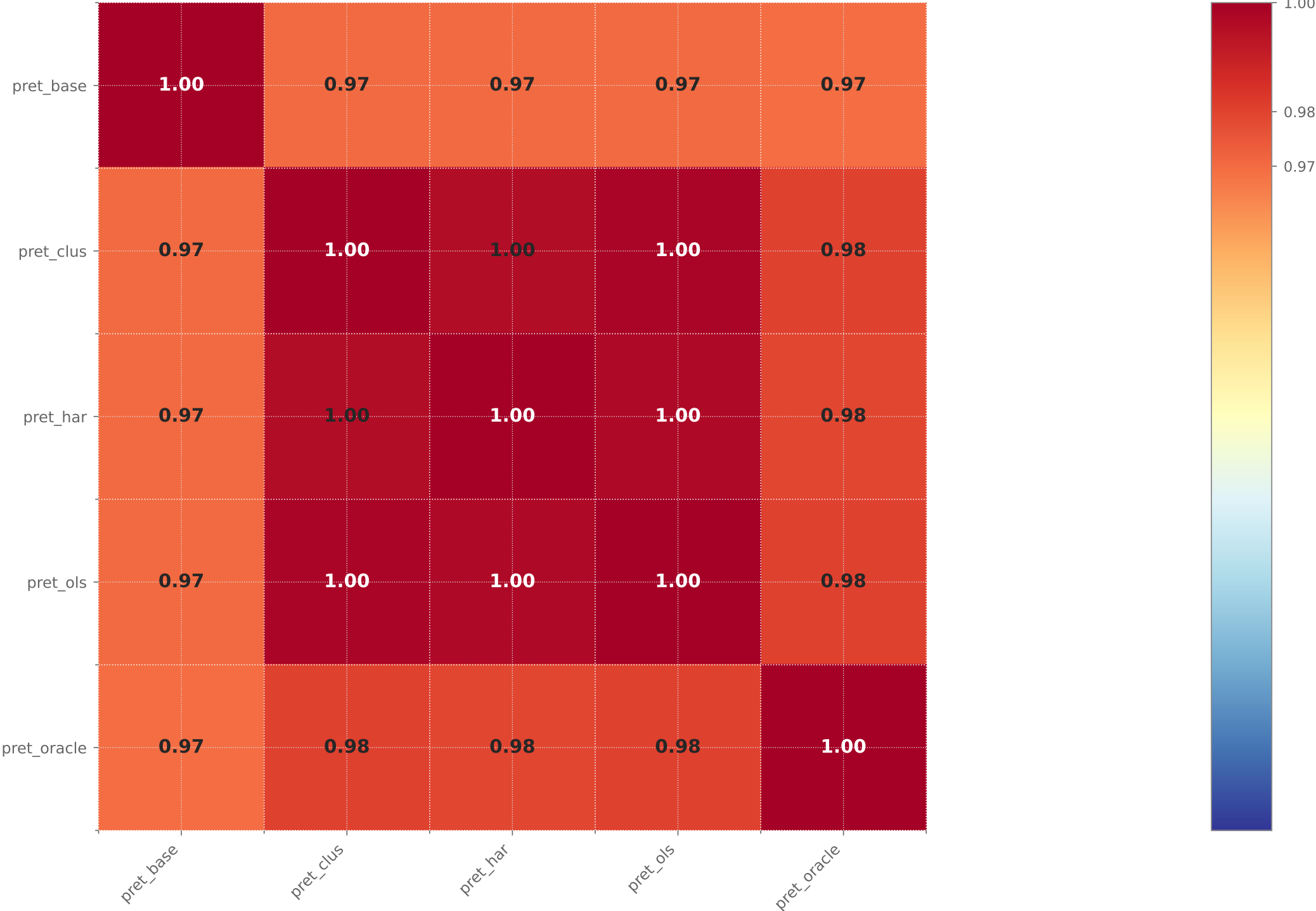
Heatmap 3 — Time-series Spearman of daily PnL vectors (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_75] (days=6262)



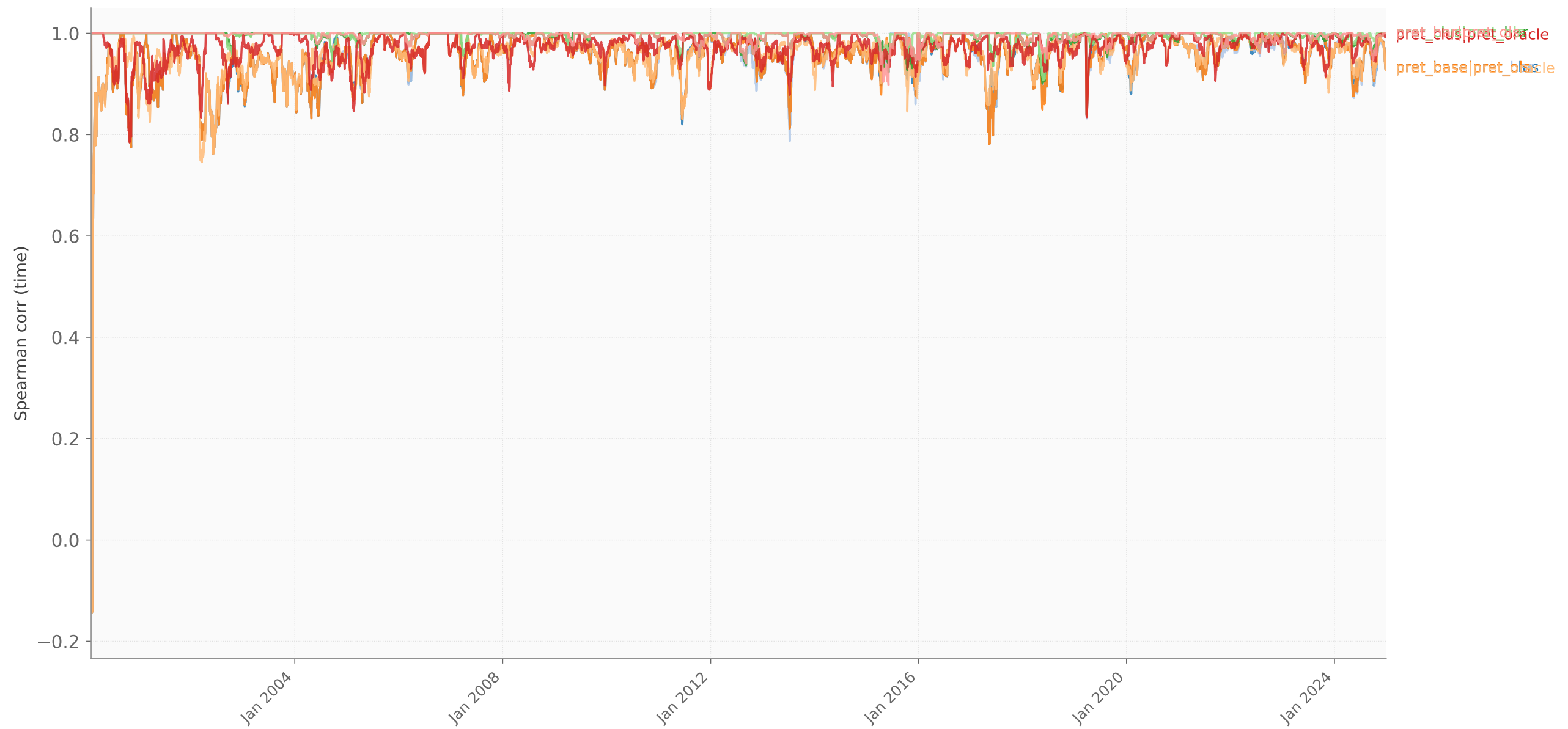
[H3|qr_75] Alpha vs Alpha — Time corr (PnL vectors, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Expanding Spearman



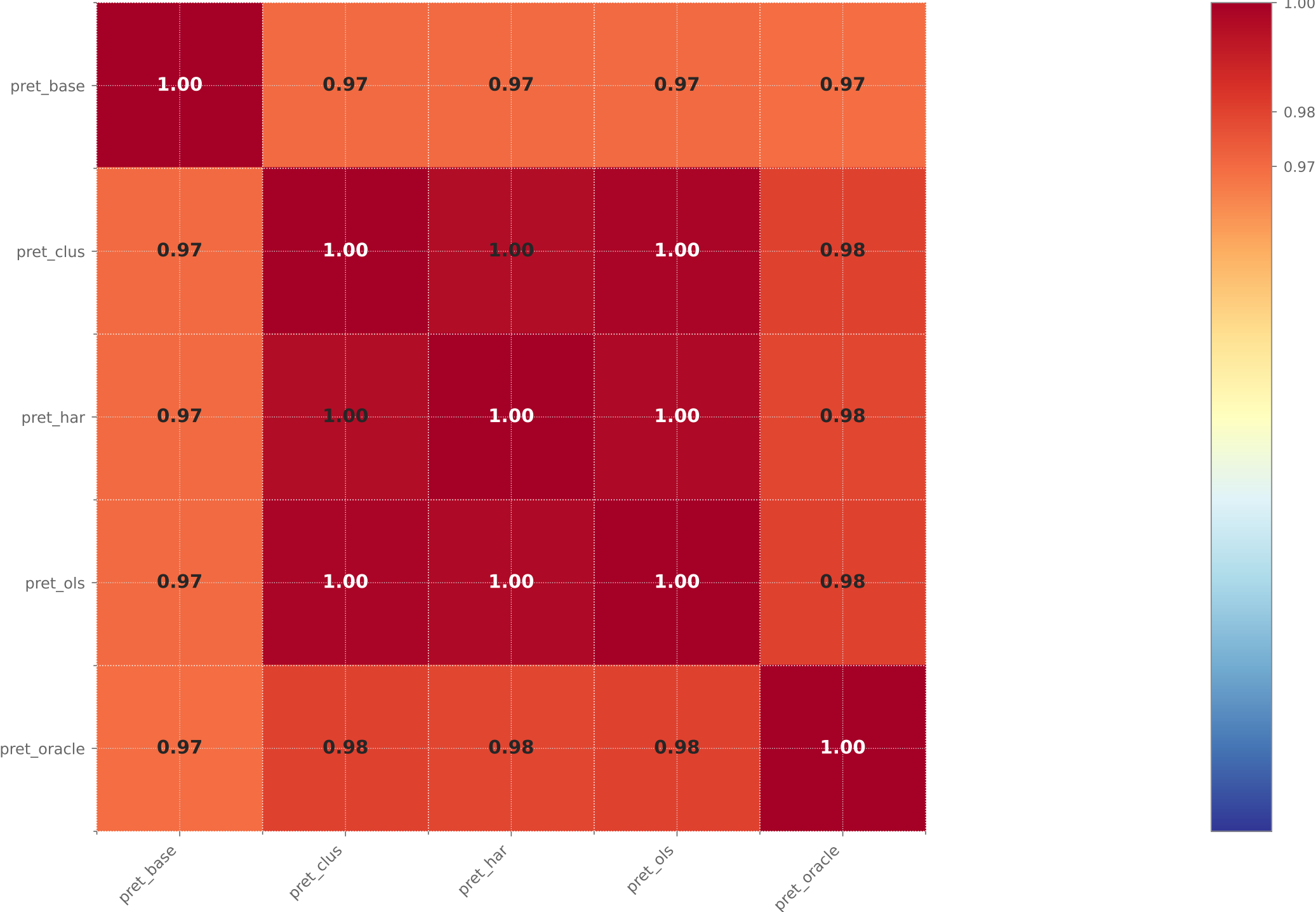
Heatmap 2 — Time-series Spearman of daily PnL (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_50] (avg 6262 days)



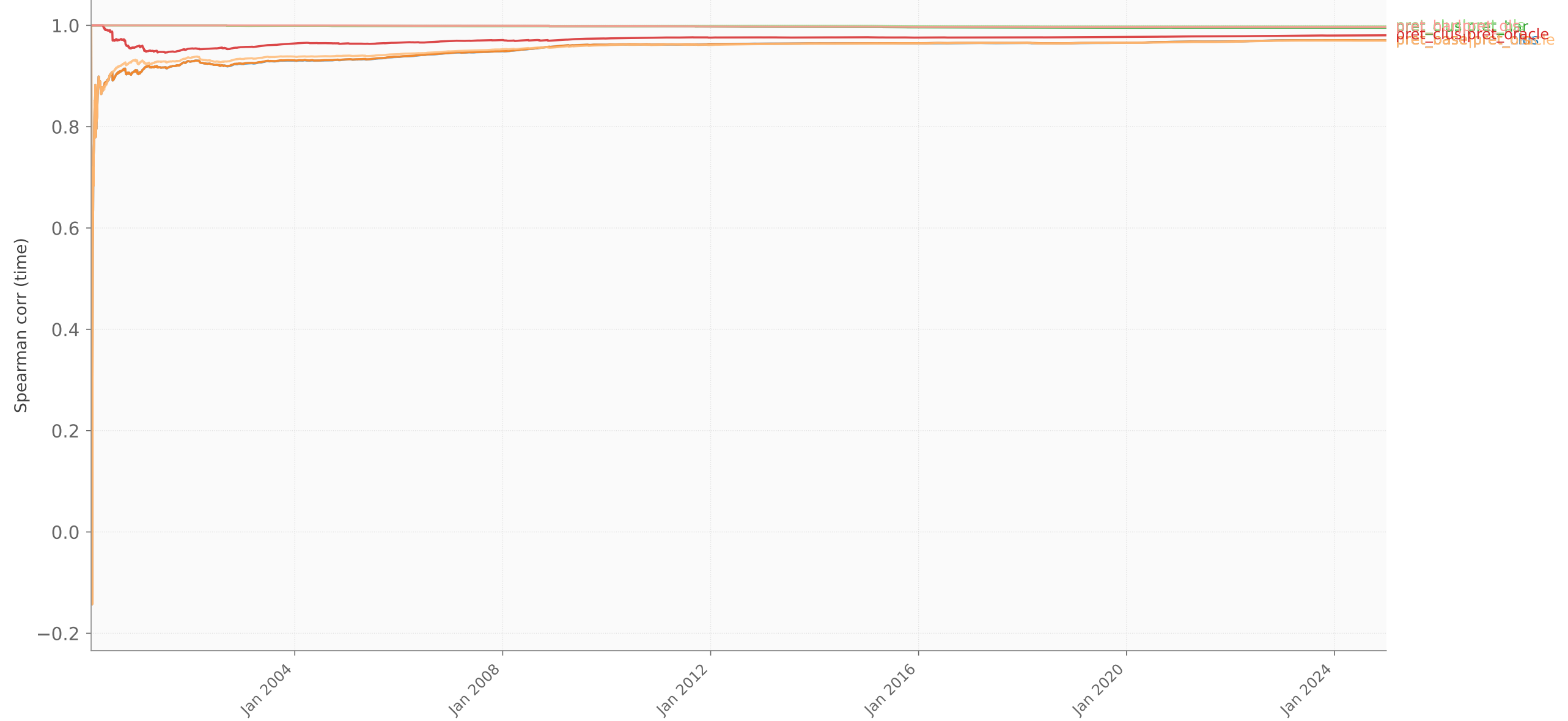
[H2|qr_50] Alpha vs Alpha — Time corr (PnL, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Rolling Spearman 30D



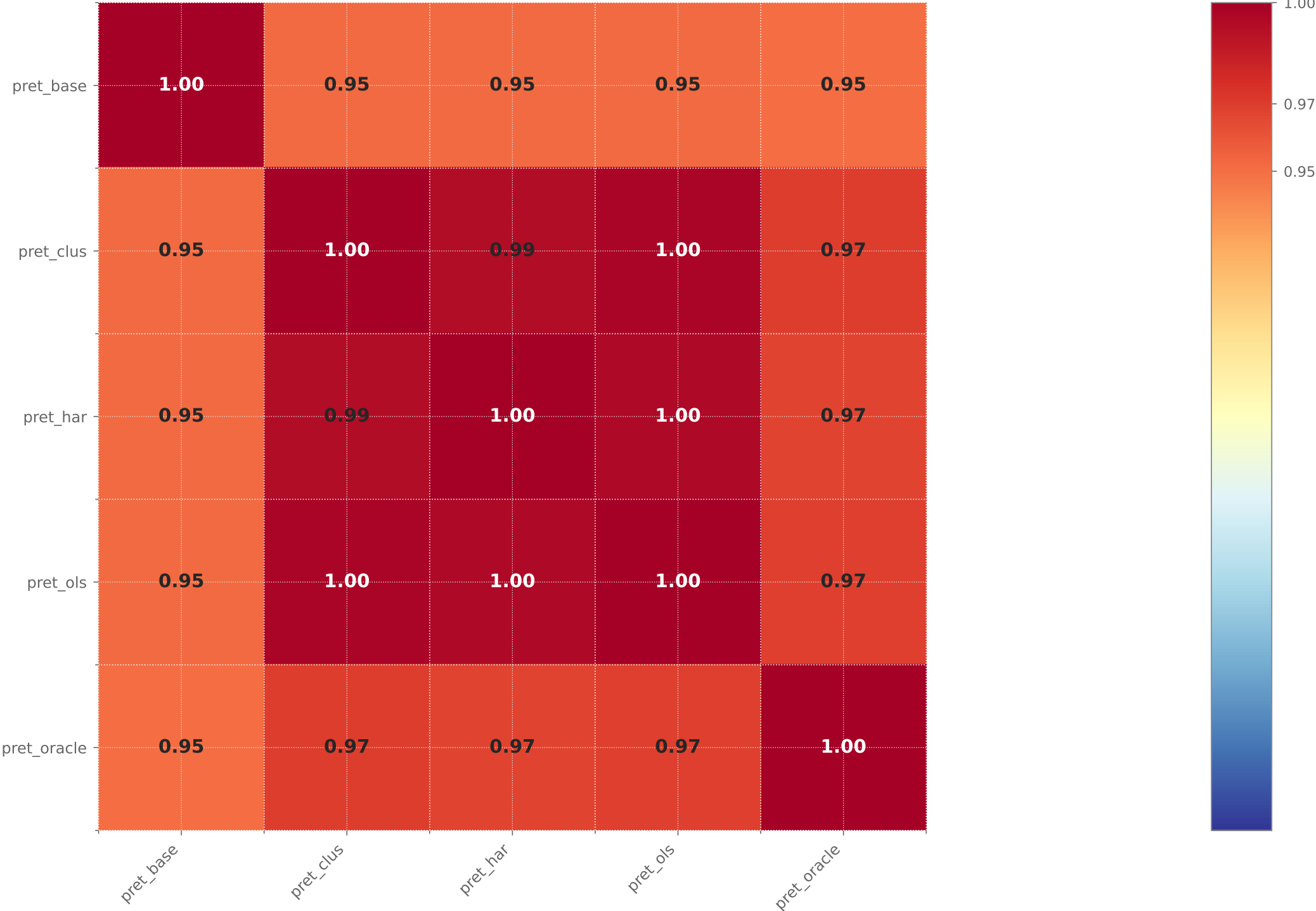
Heatmap 3 — Time-series Spearman of daily PnL vectors (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_50] (days=6262)



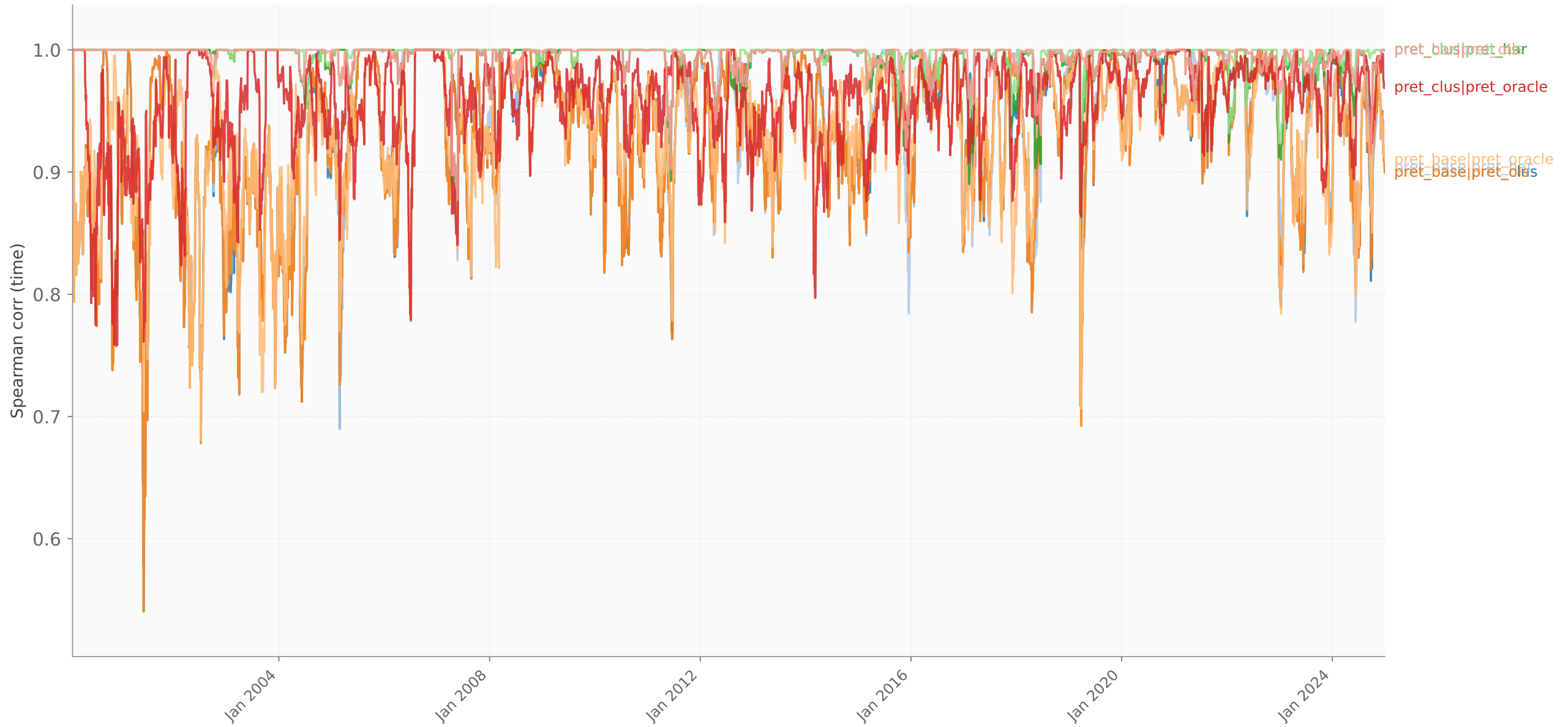
[H3|qr_50] Alpha vs Alpha — Time corr (PnL vectors, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Expanding Spearman



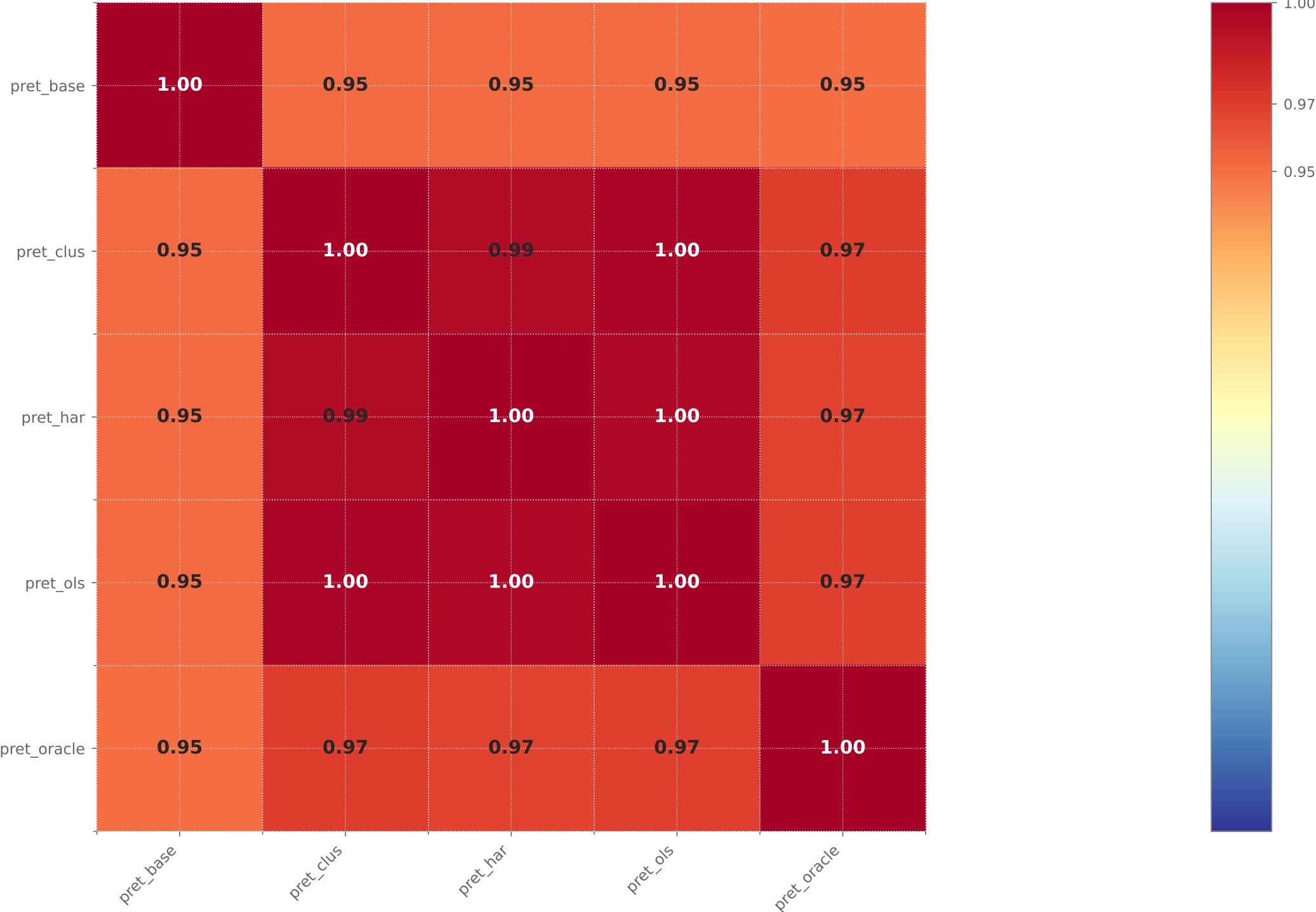
Heatmap 2 — Time-series Spearman of daily PnL (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_25] (avg 6262 days)



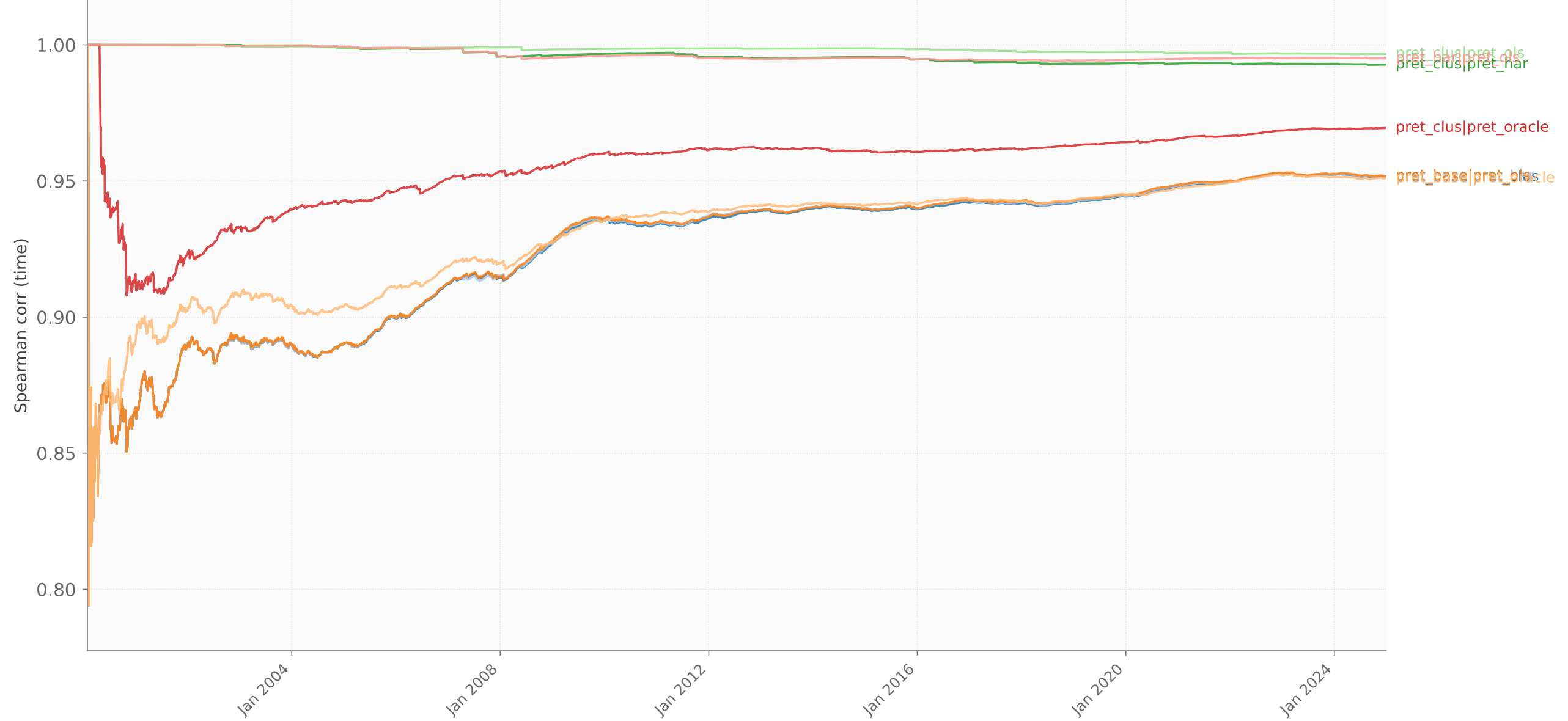
[H2|qr_25] Alpha vs Alpha — Time corr (PnL, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Rolling Spearman 30D



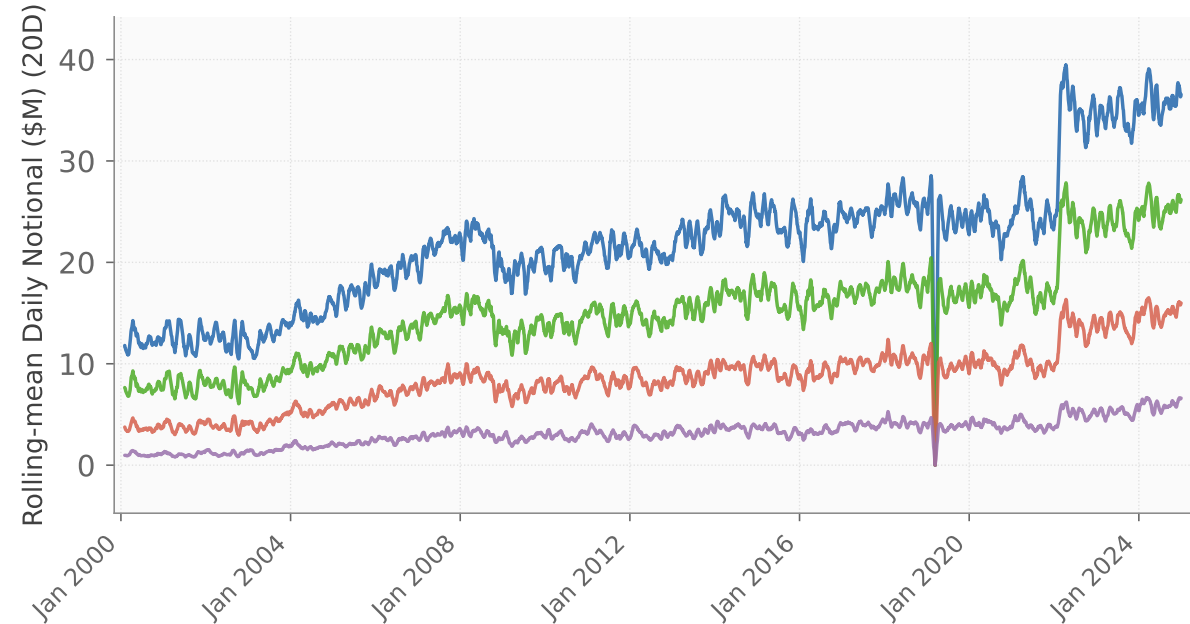
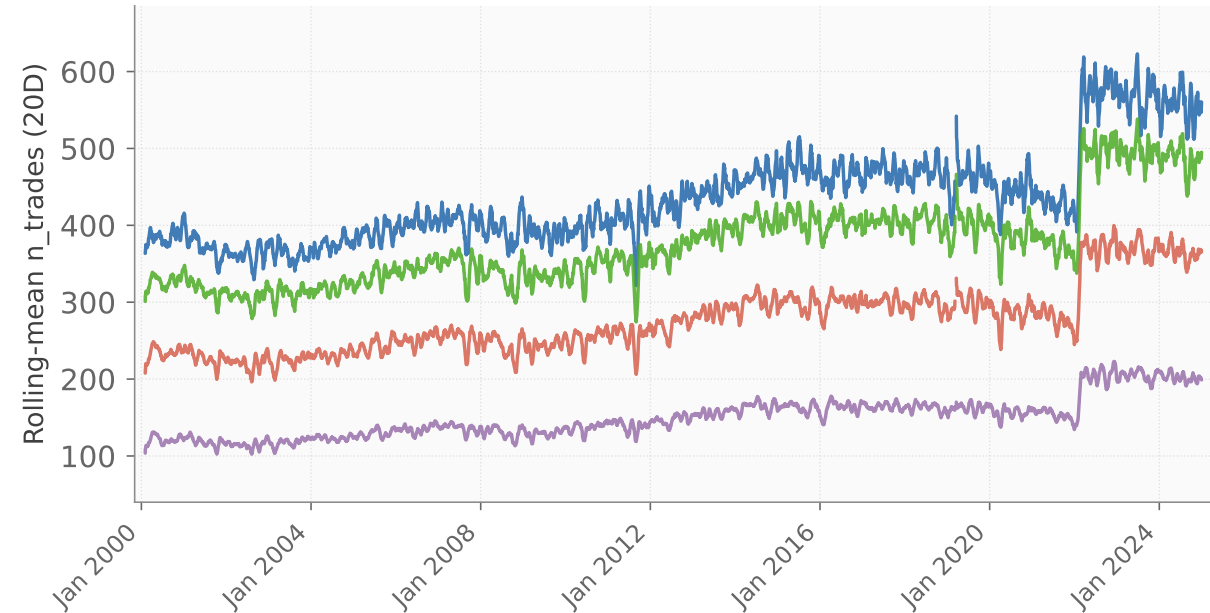
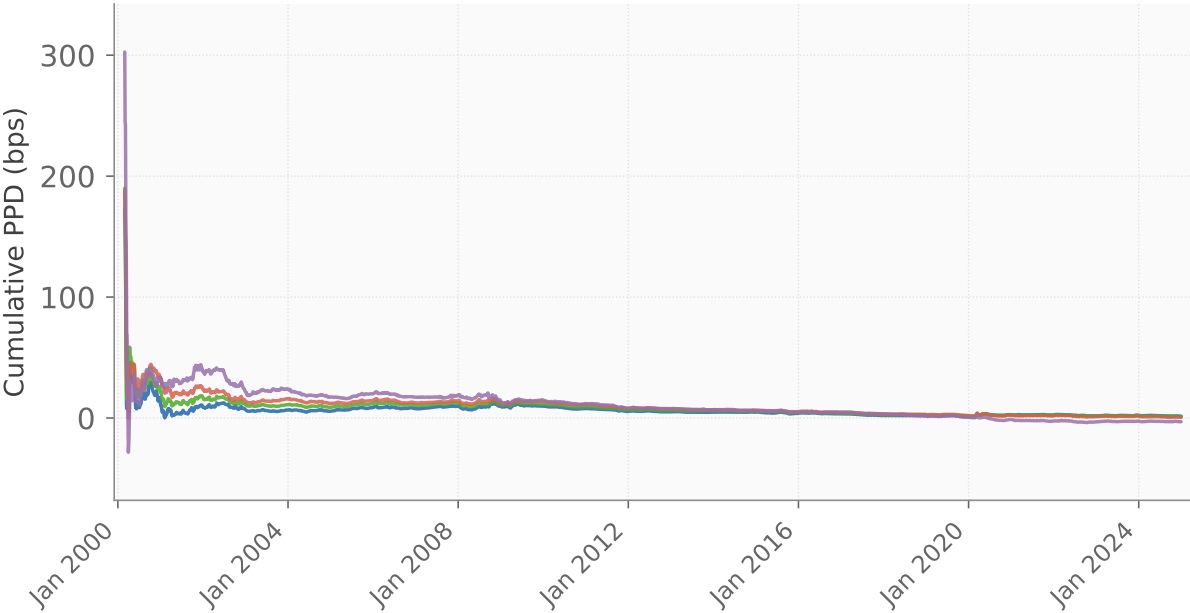
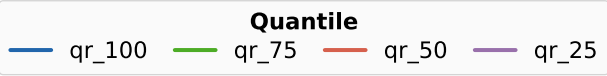
Heatmap 3 — Time-series Spearman of daily PnL vectors (targets=fret_OPCL_RR, bets=bs_m025_c100k) [qr_25] (days=6262)



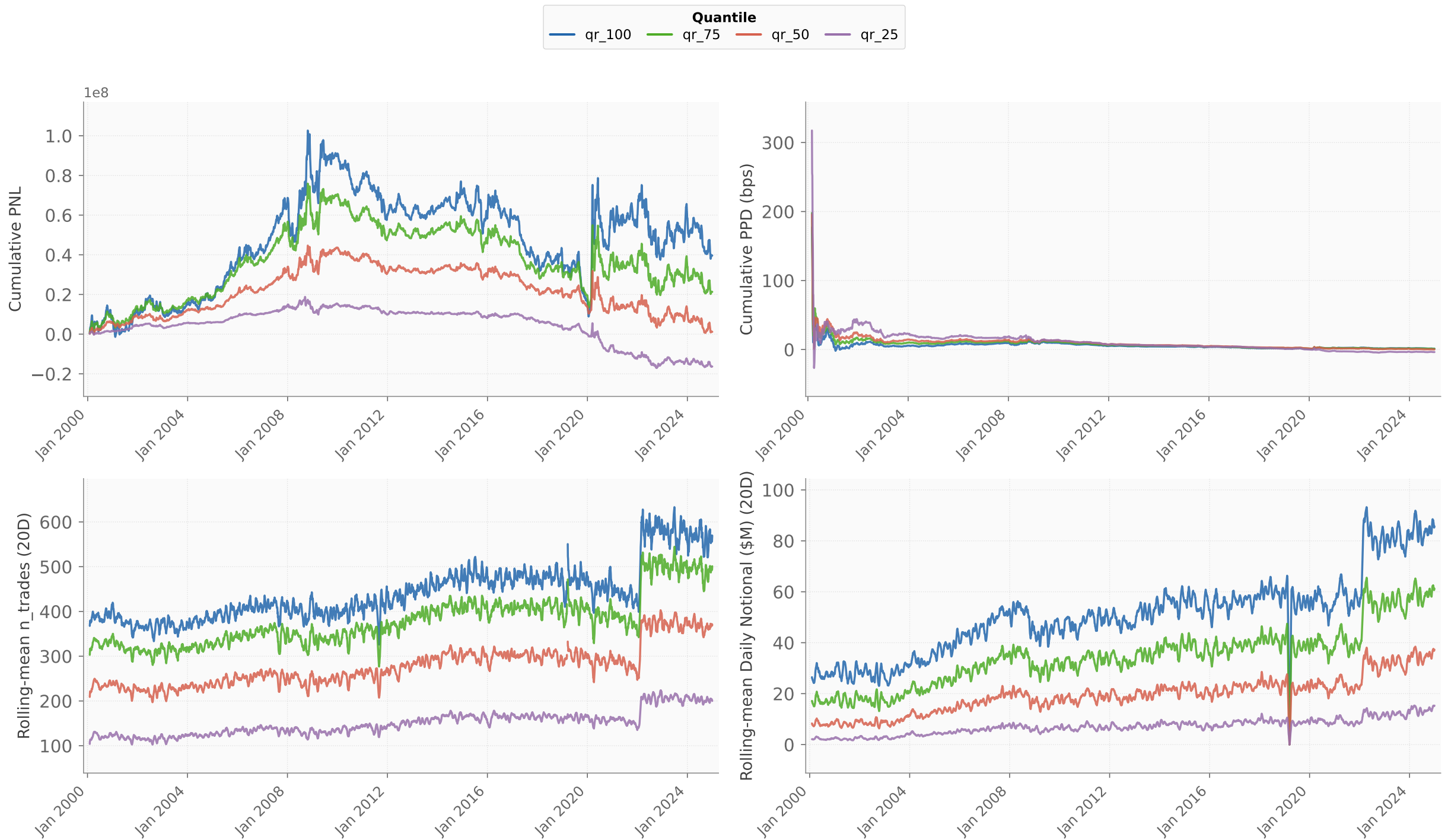
[H3|qr_25] Alpha vs Alpha — Time corr (PnL vectors, targets=fret_OPCL_RR, bets=bs_m025_c100k) — Expanding Spearman



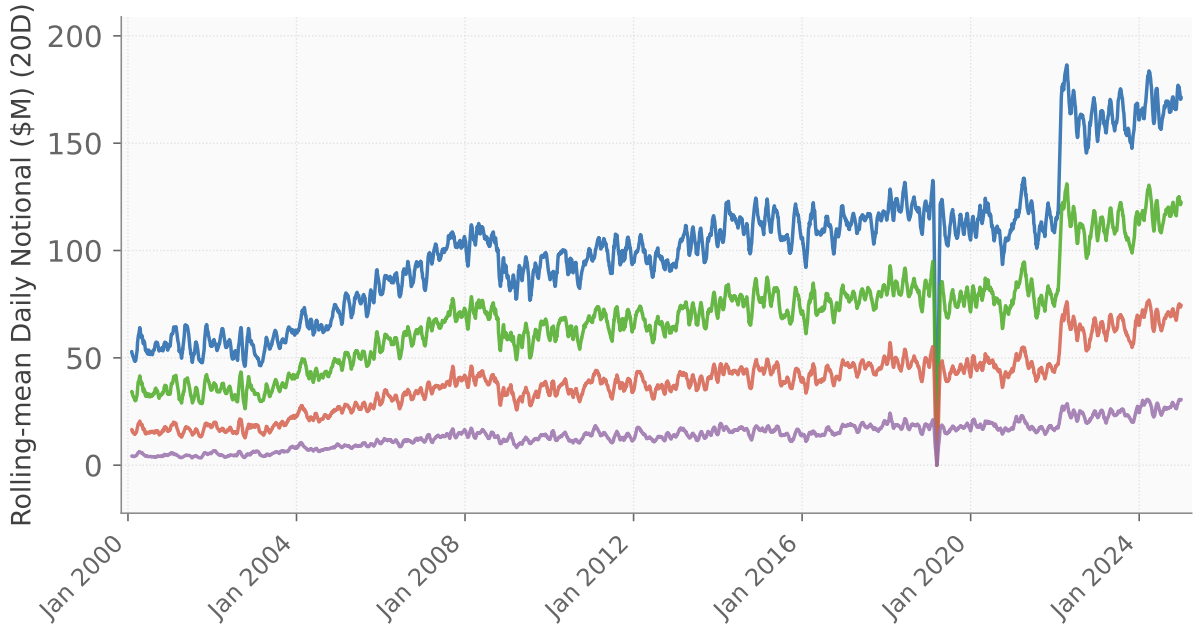
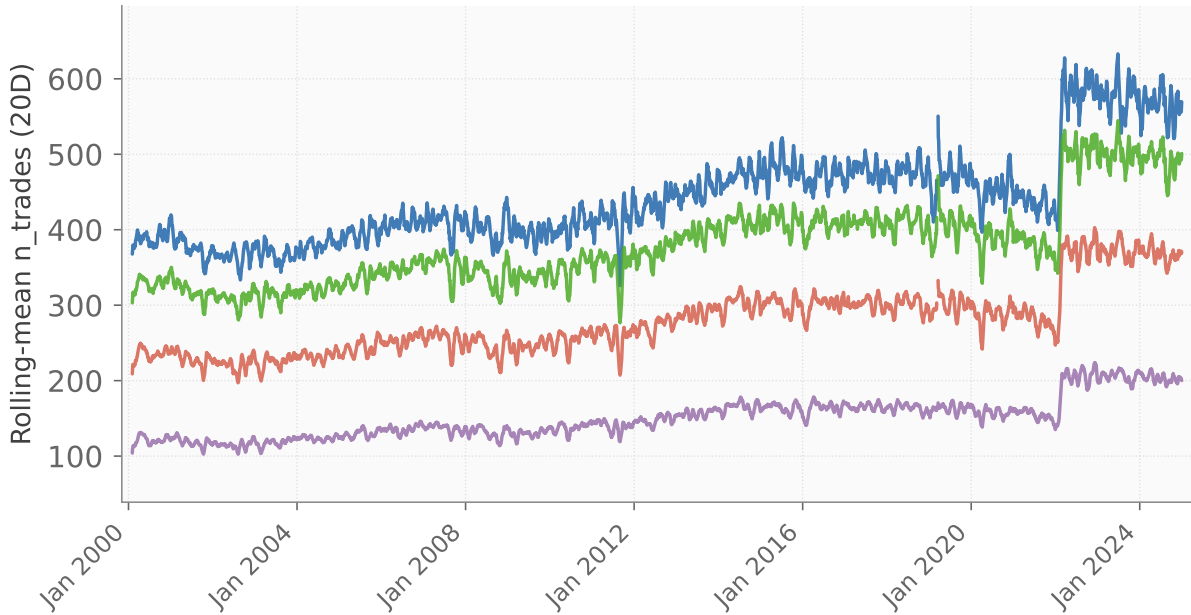
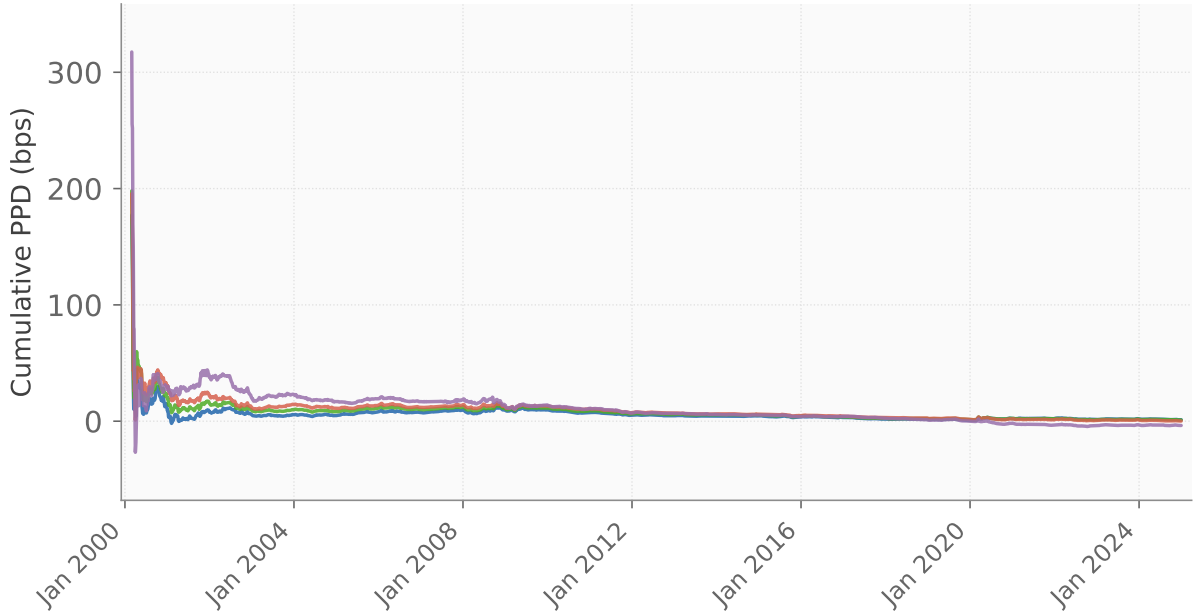
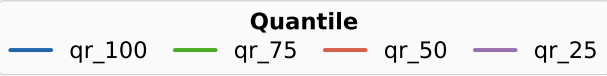
fret_OP5CL_MR | pret_base | bs_m025_c100k — Temporal metrics



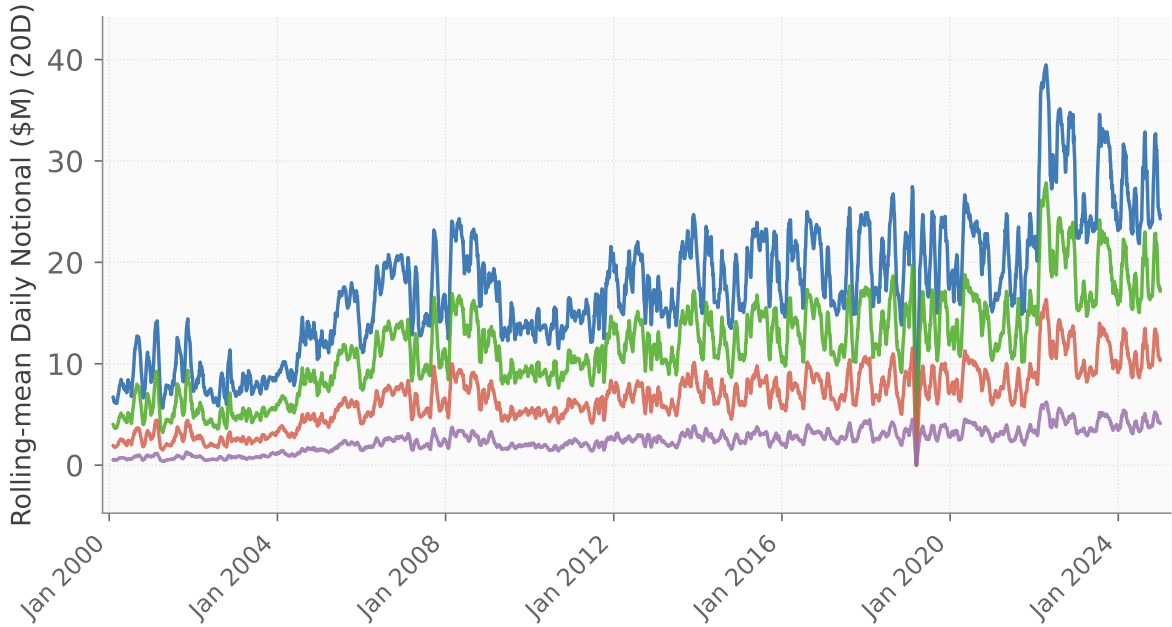
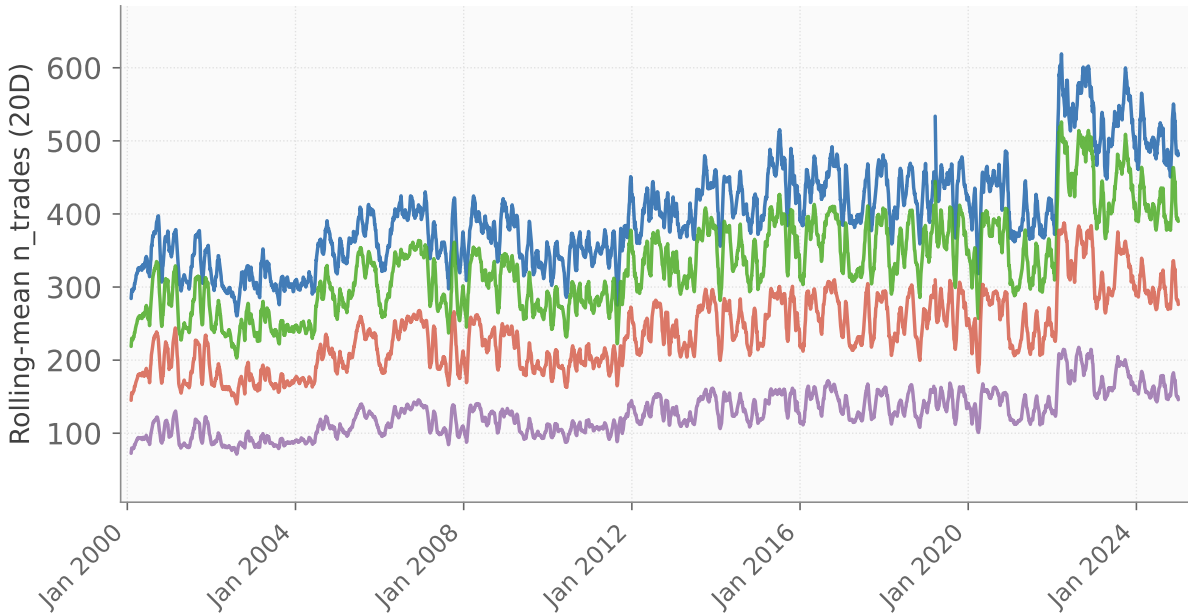
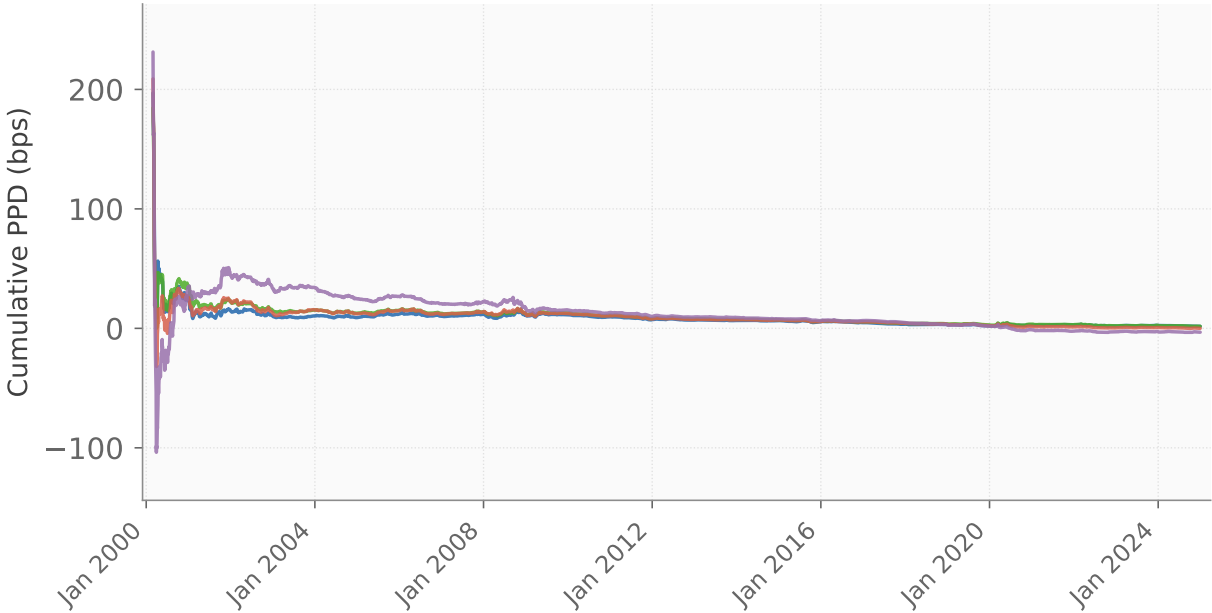
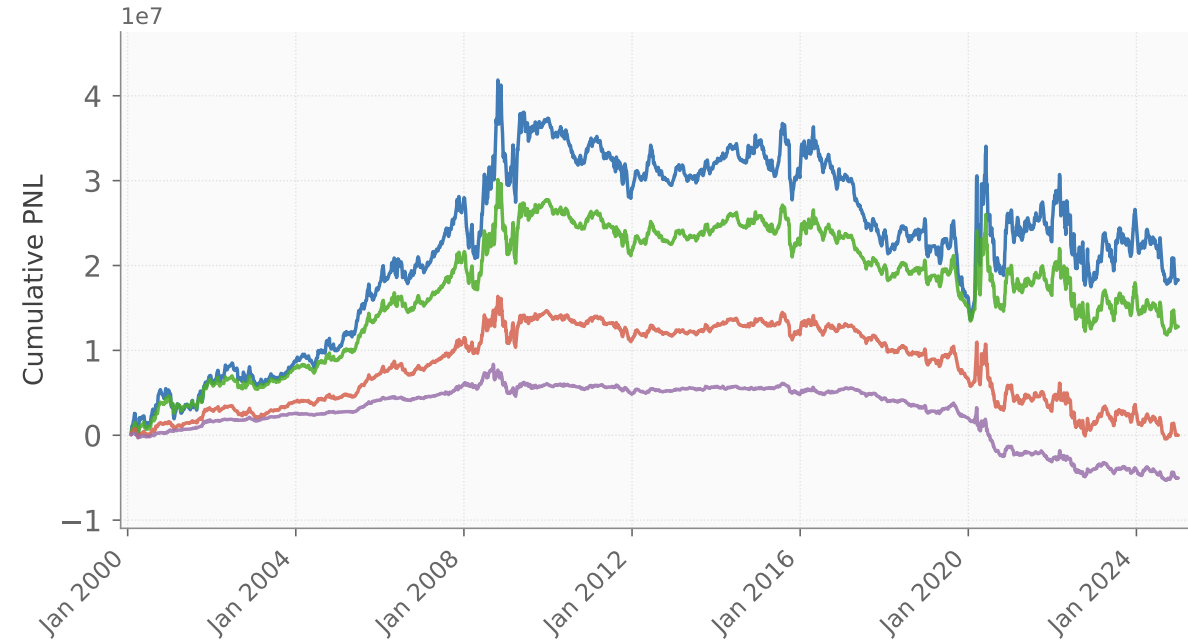
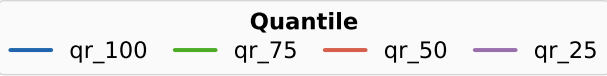
fret_OP5CL_MR | pret_base | bs_m05_c250k — Temporal metrics



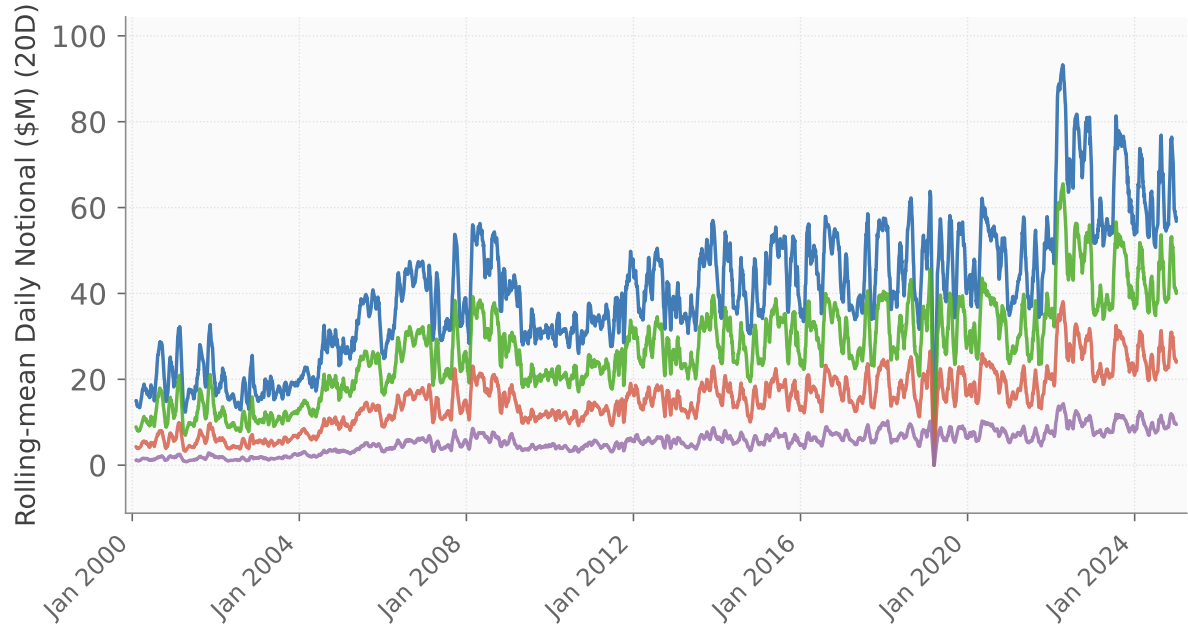
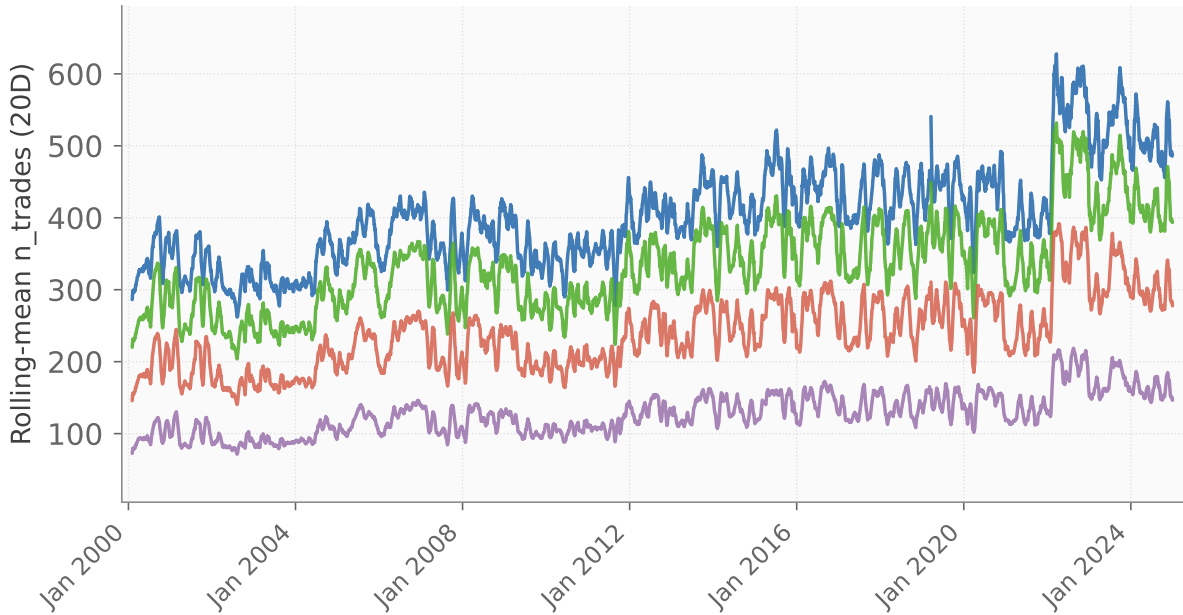
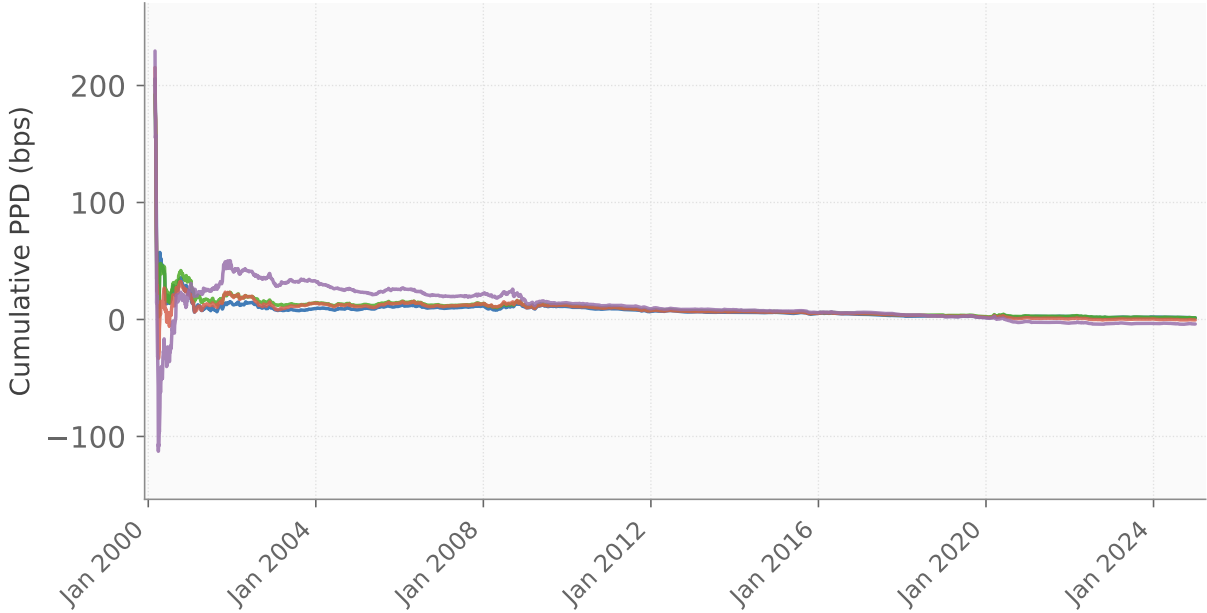
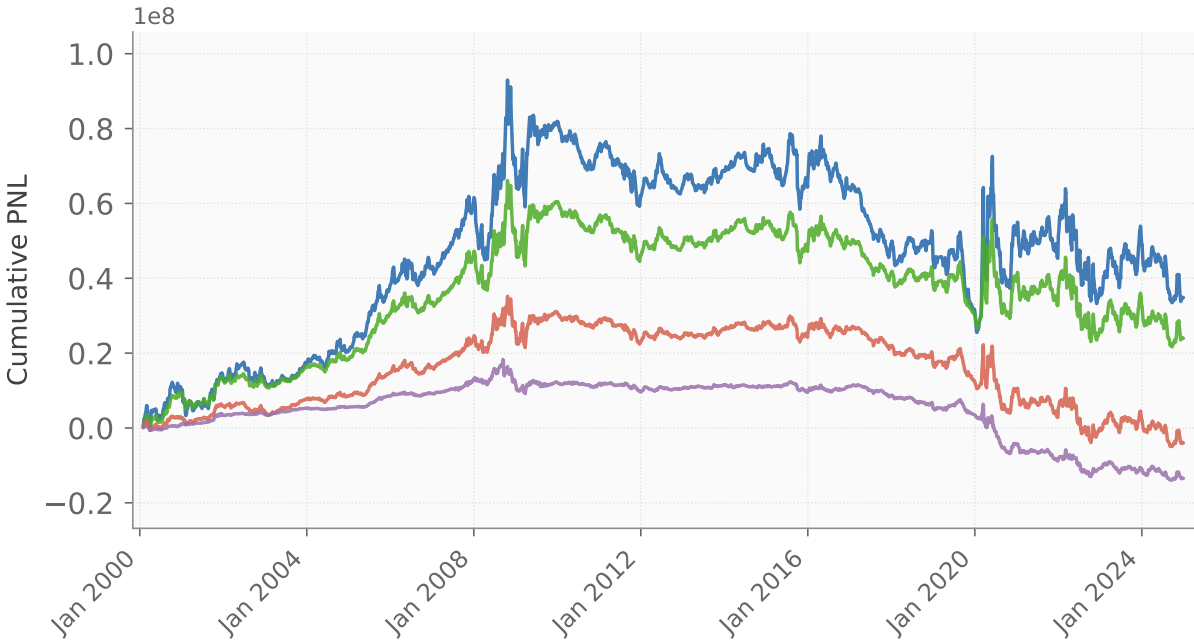
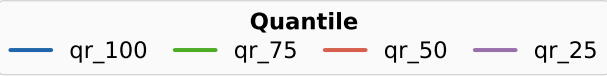
fret_OP5CL_MR | pret_base | bs_m1_c500k — Temporal metrics



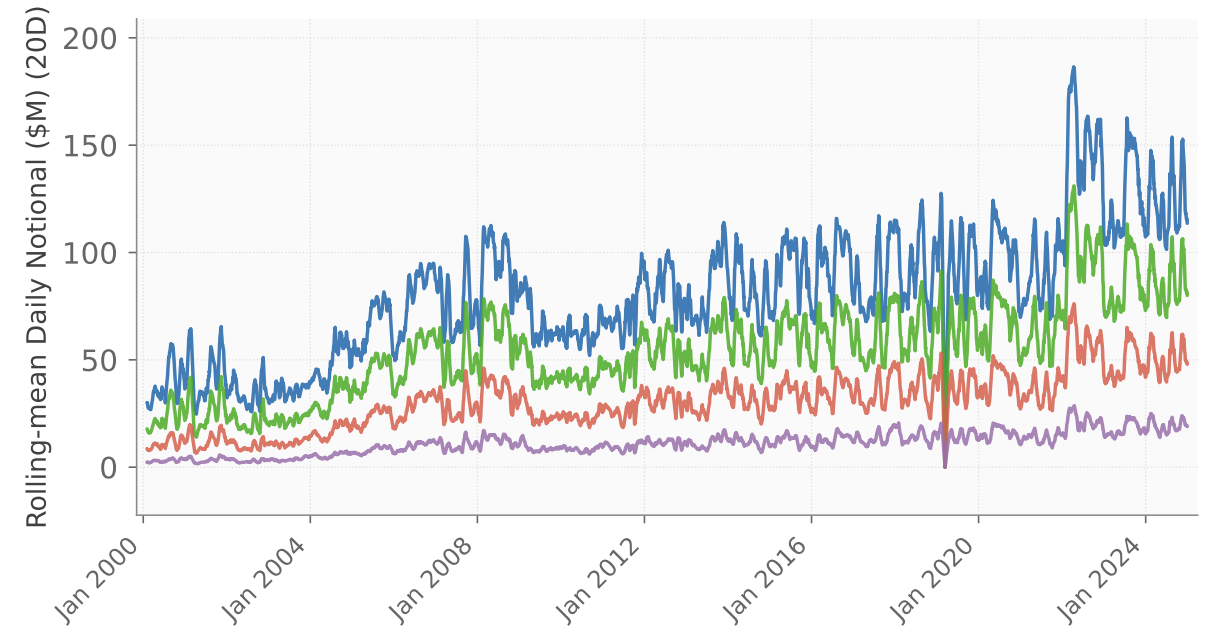
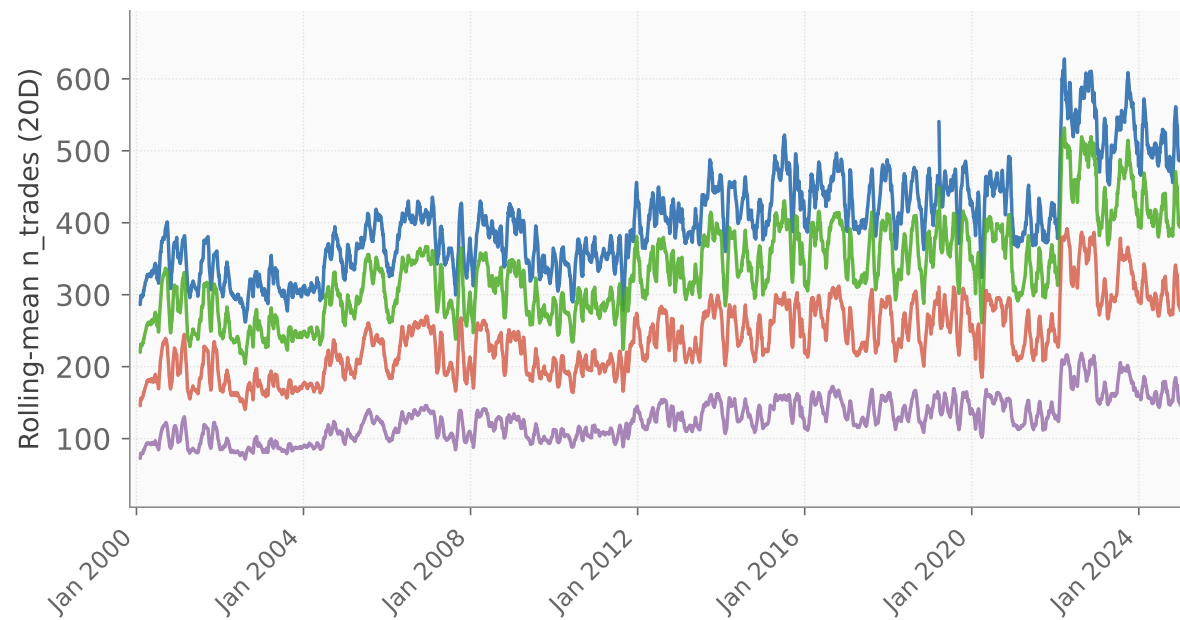
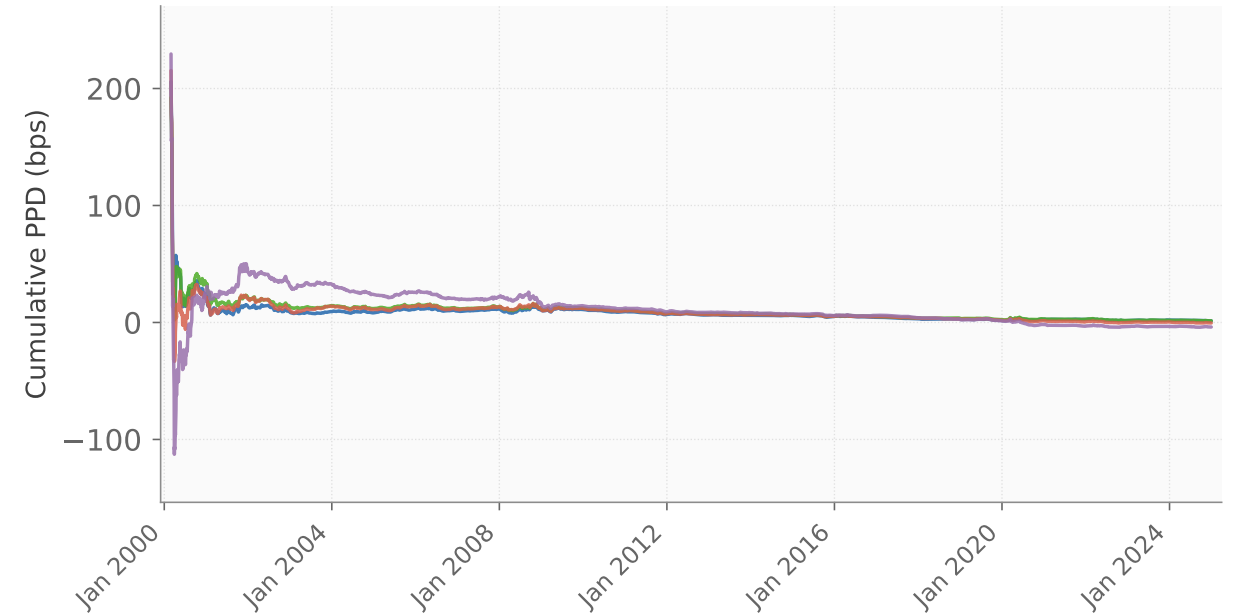
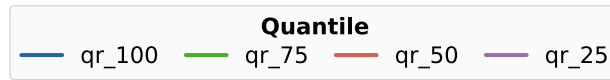
fret_OP5CL_MR | pret_clus | bs_m025_c100k — Temporal metrics



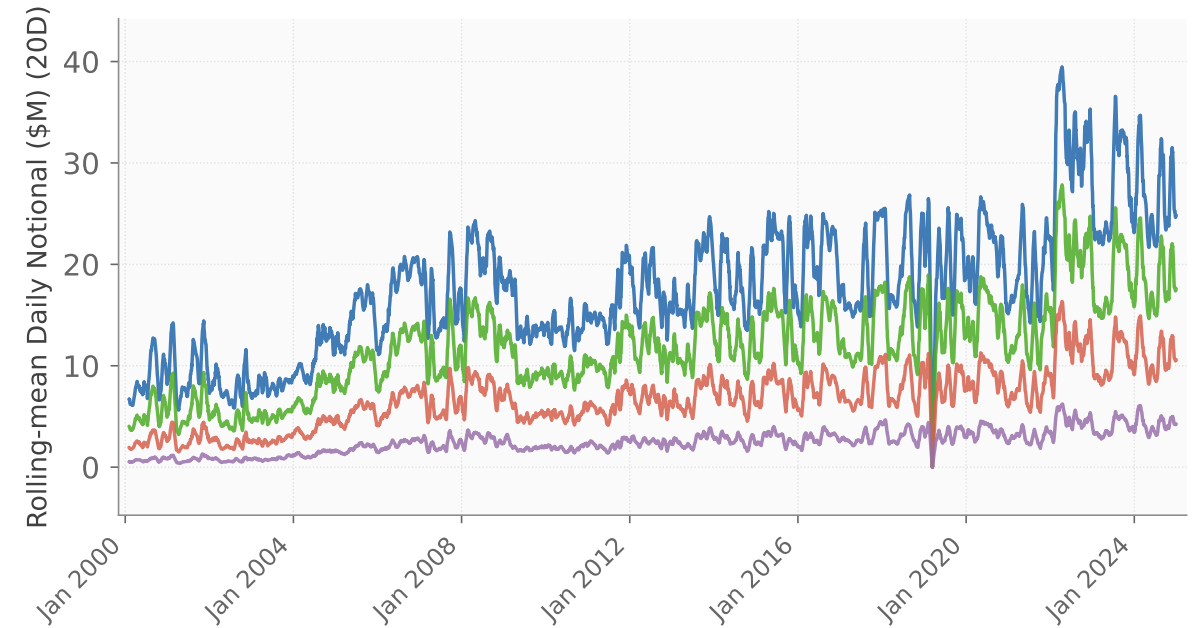
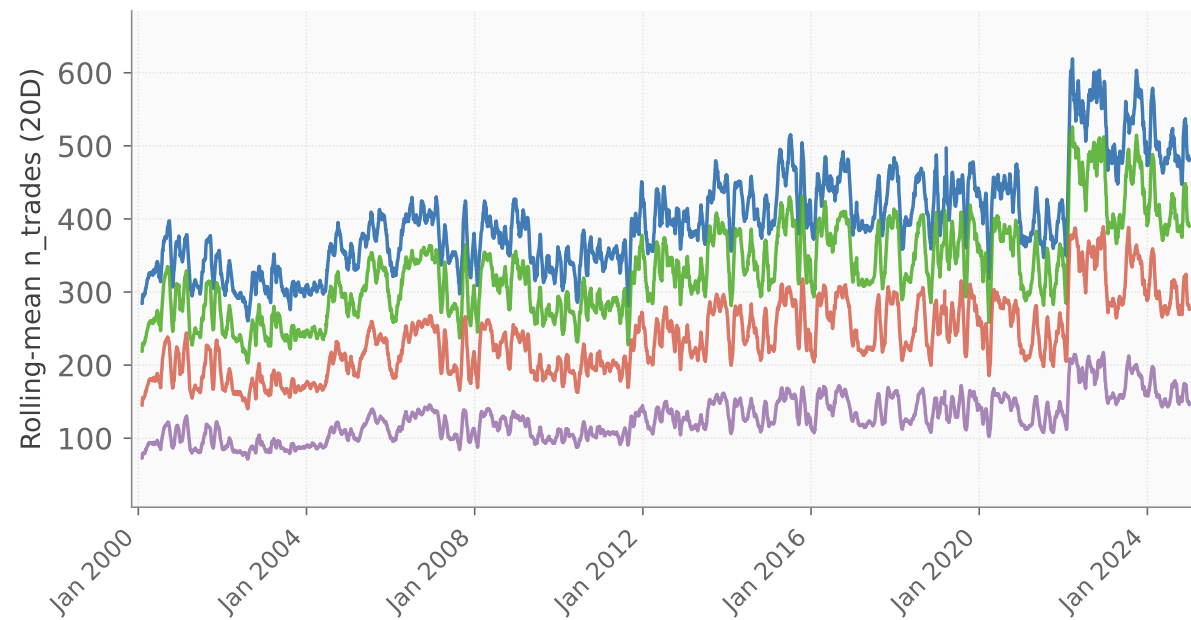
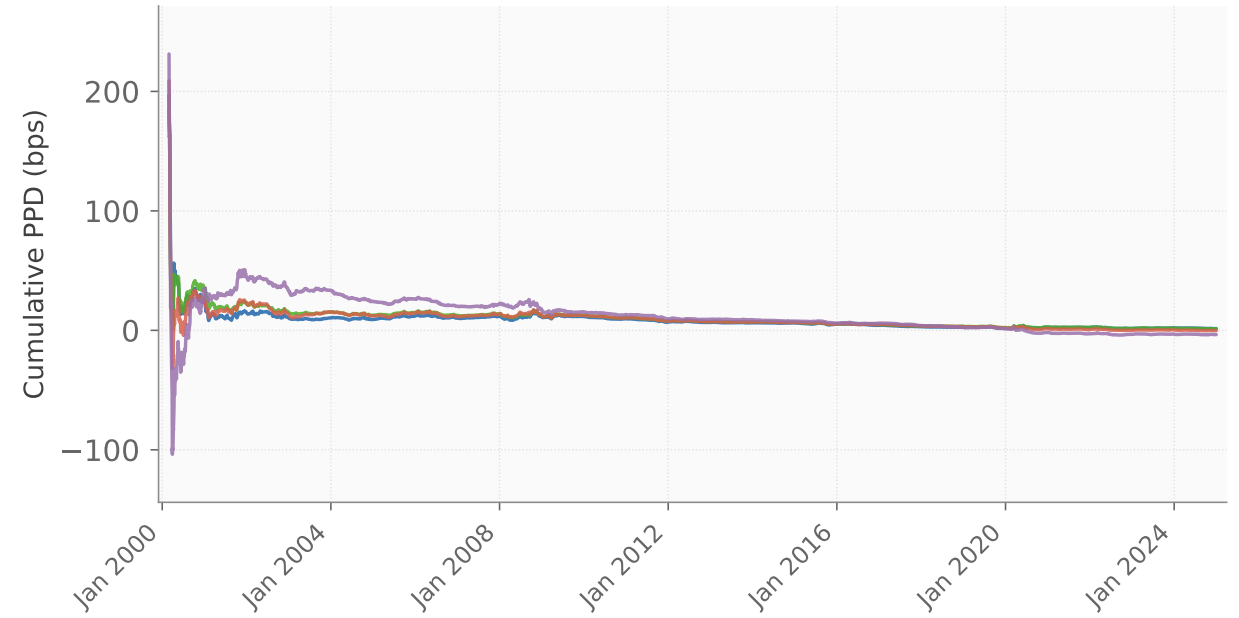
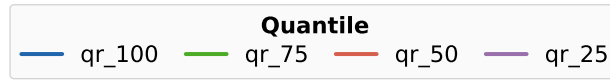
fret_OP5CL_MR | pret_clus | bs_m05_c250k — Temporal metrics



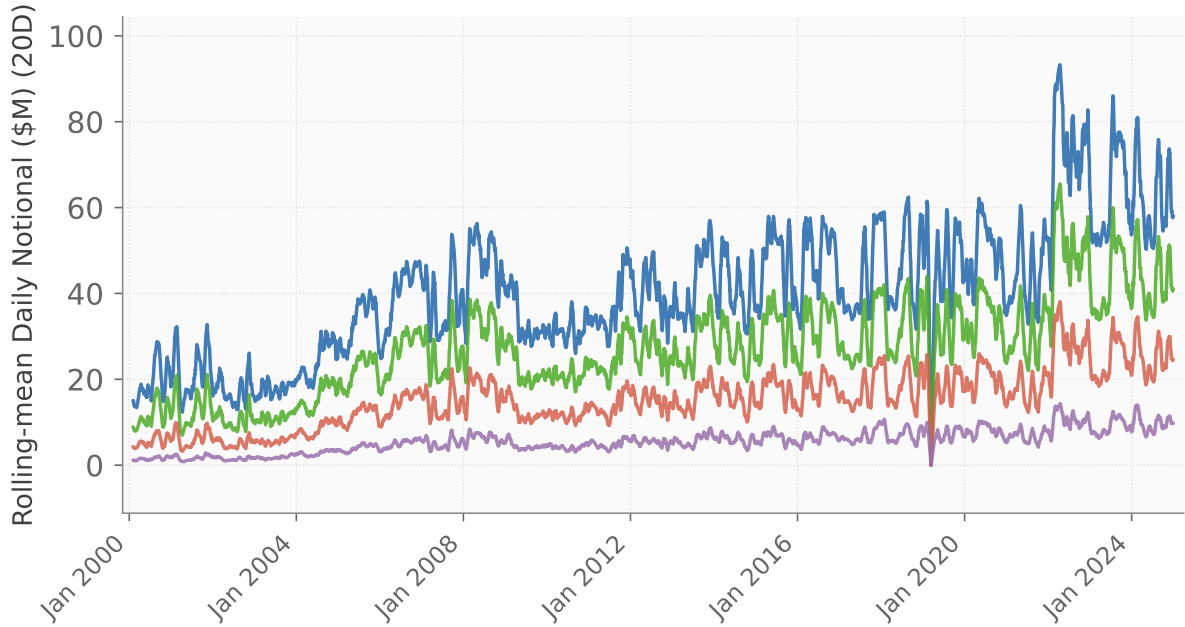
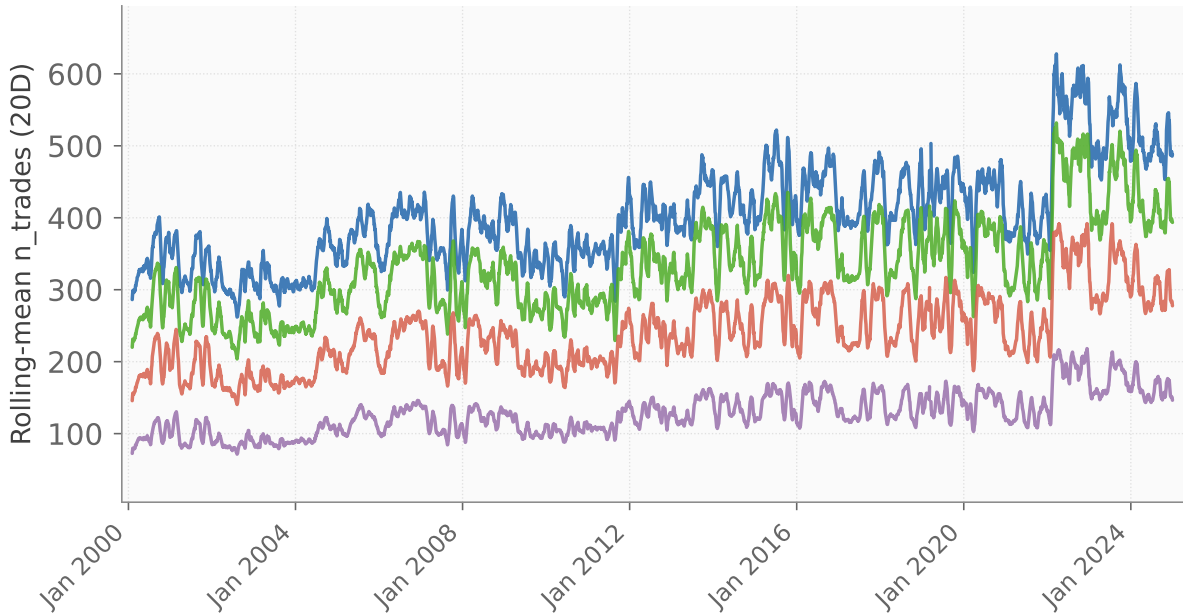
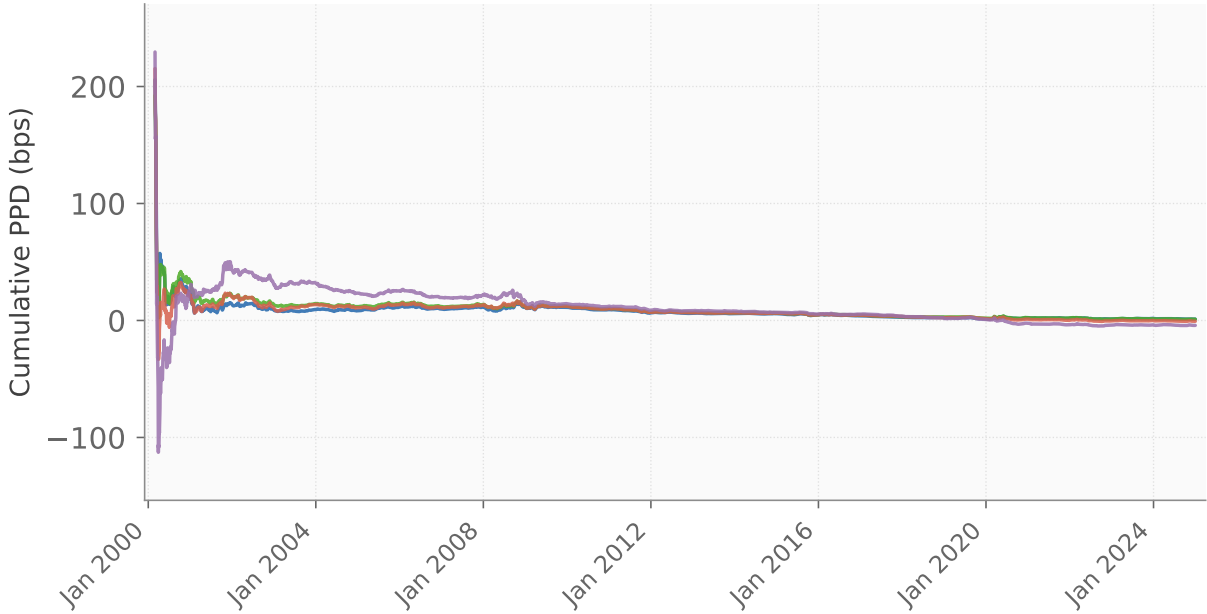
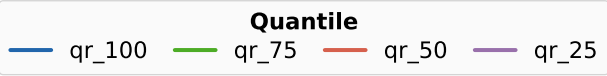
fret_OP5CL_MR | pret_clus | bs_m1_c500k — Temporal metrics



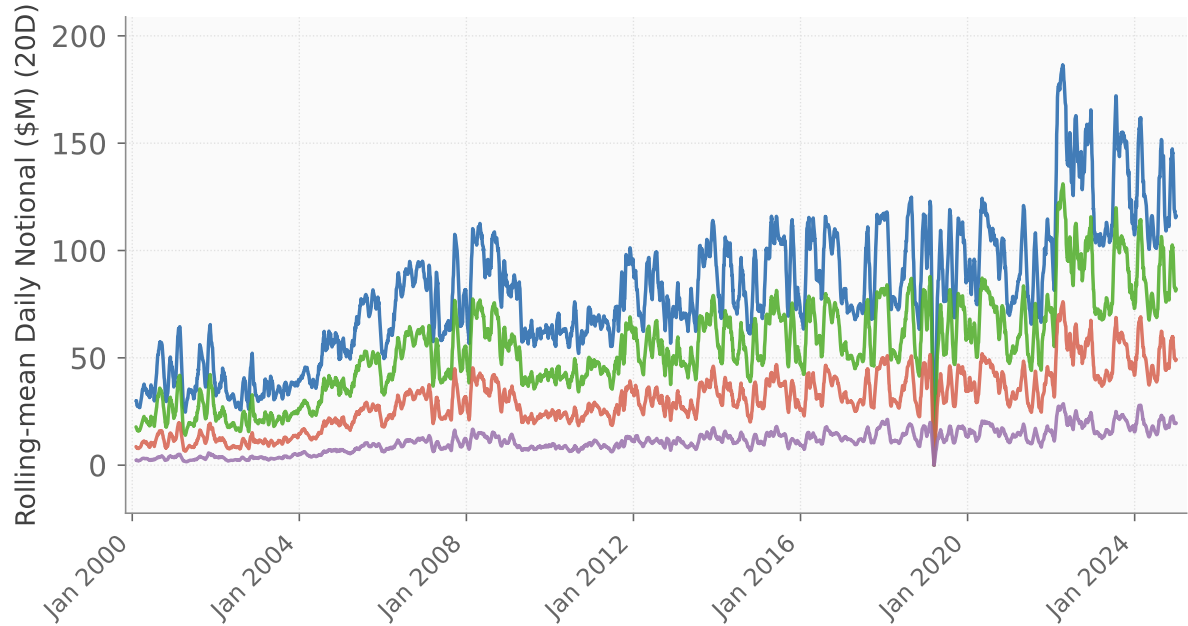
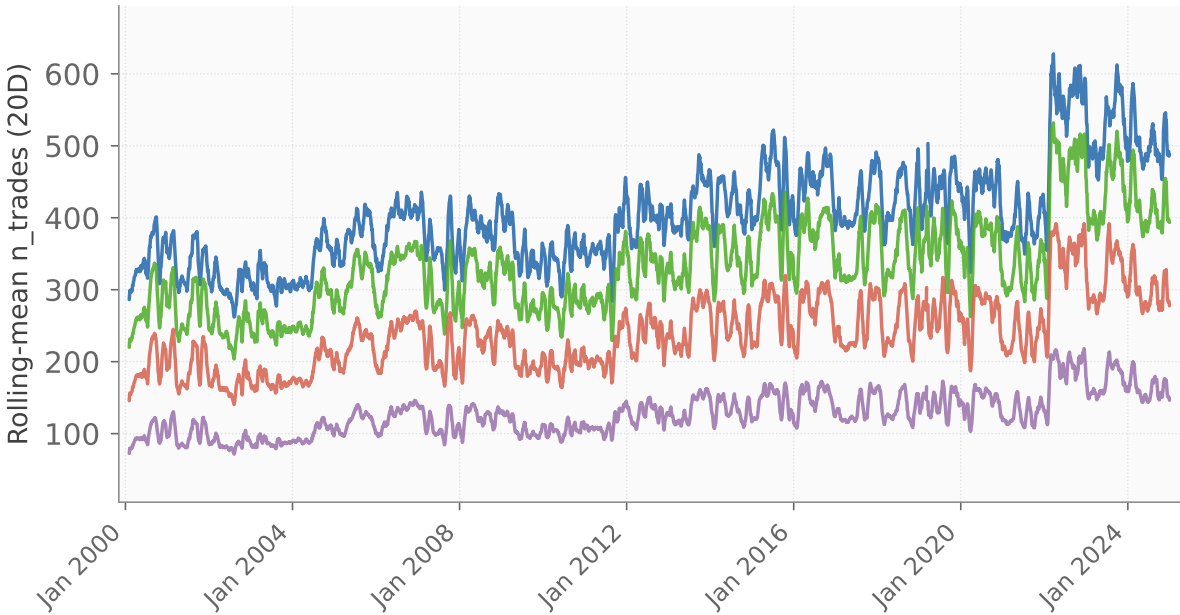
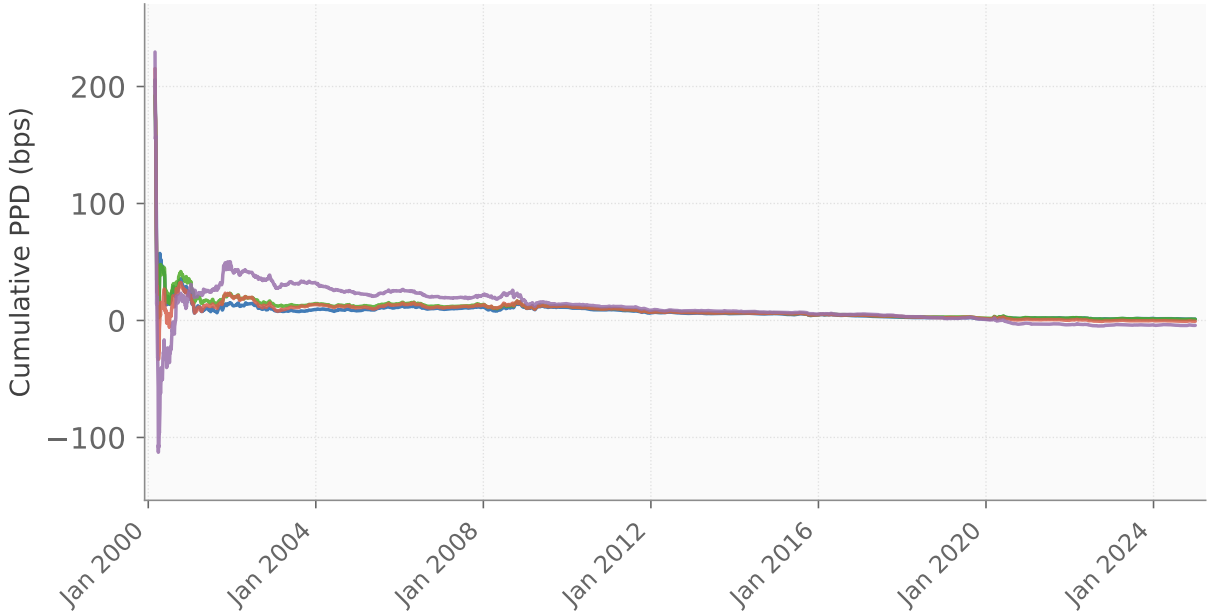
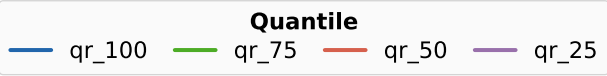
fret_OP5CL_MR | pret_har | bs_m025_c100k — Temporal metrics



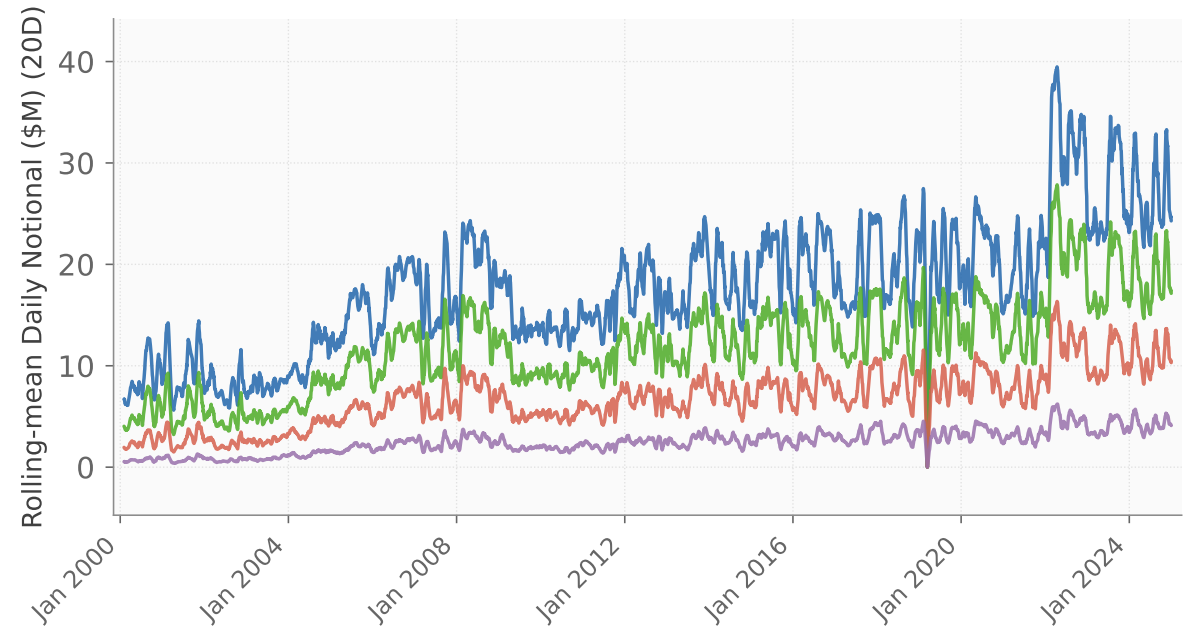
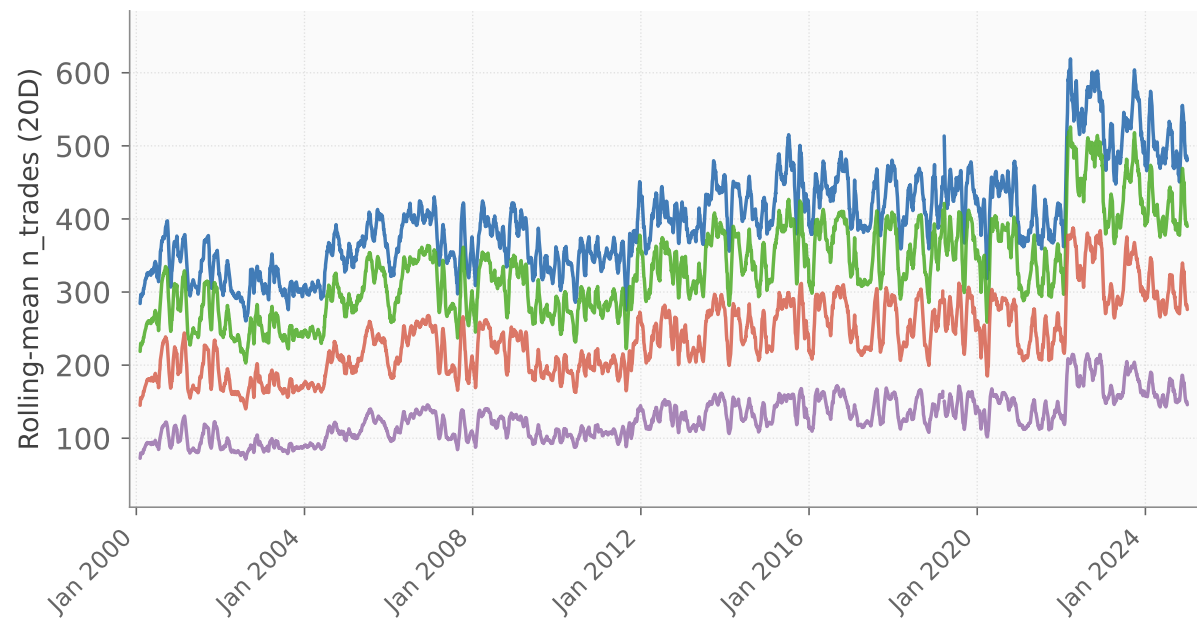
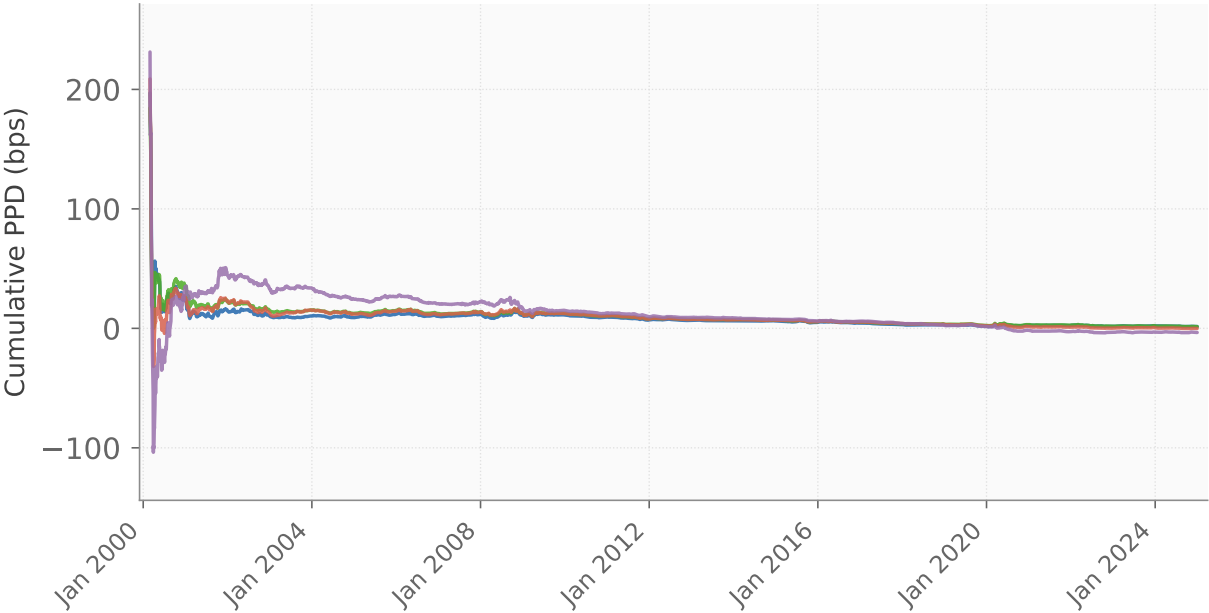
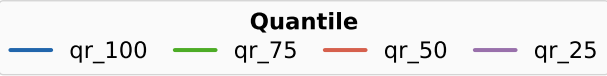
fret_OP5CL_MR | pret_har | bs_m05_c250k — Temporal metrics



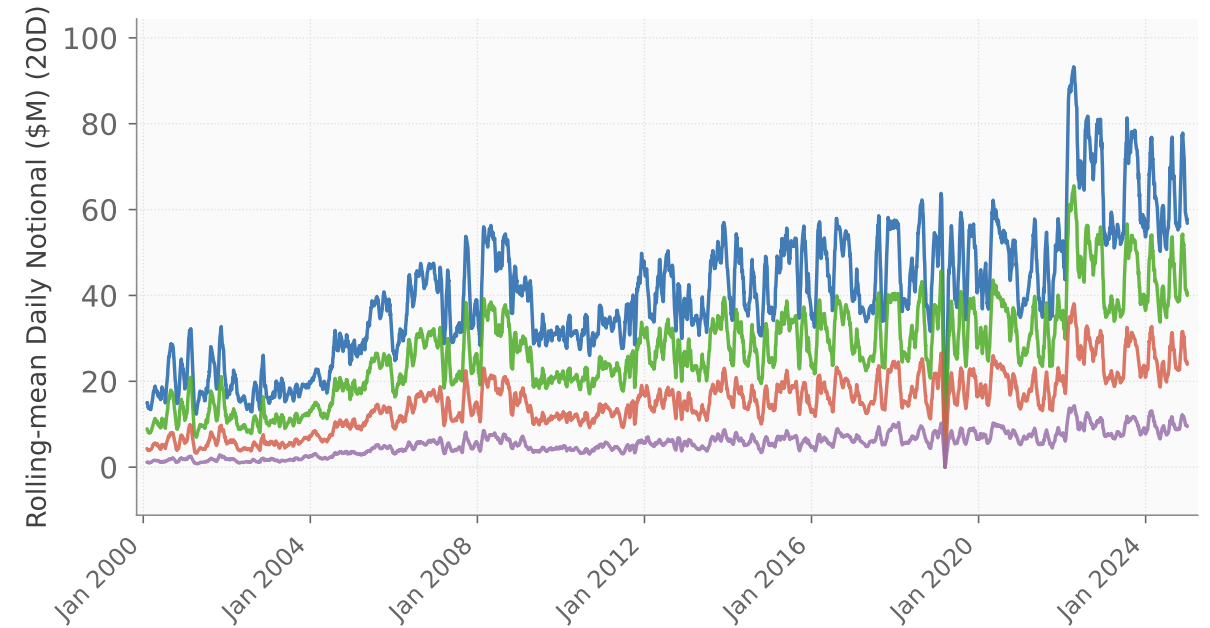
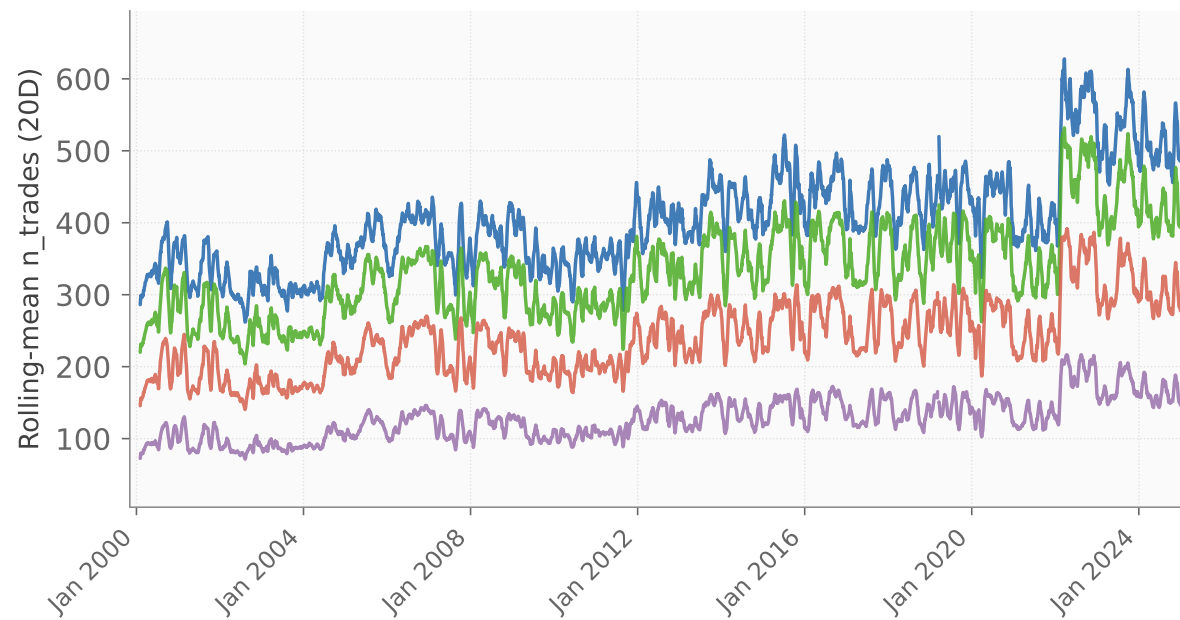
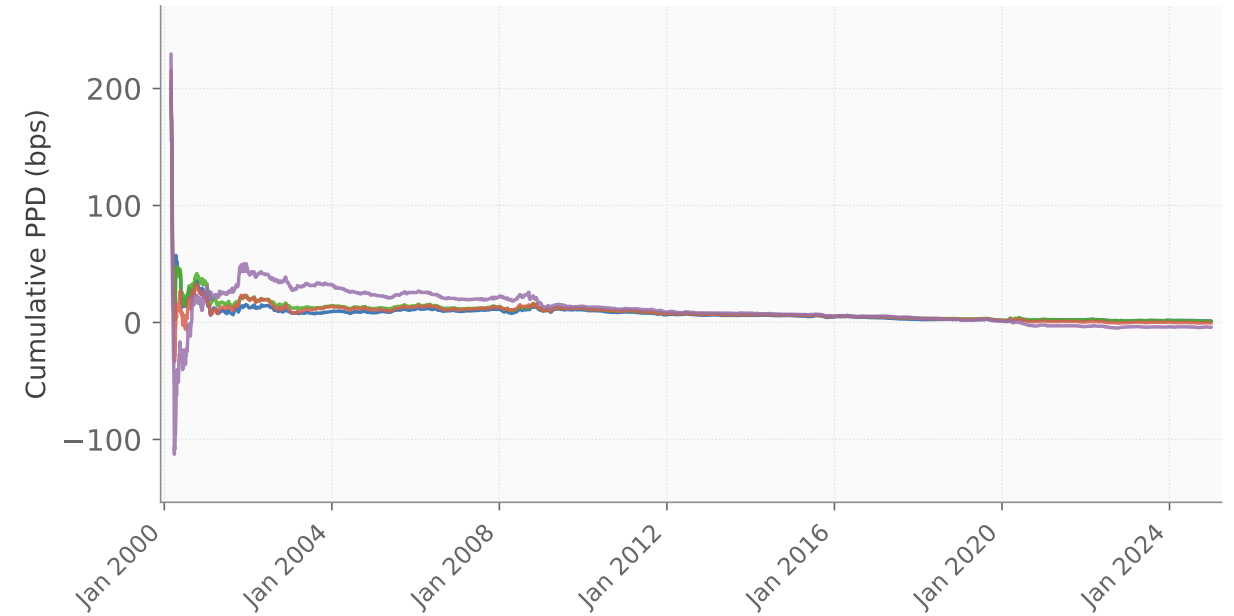
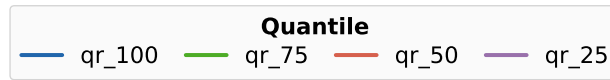
fret_OP5CL_MR | pret_har | bs_m1_c500k — Temporal metrics



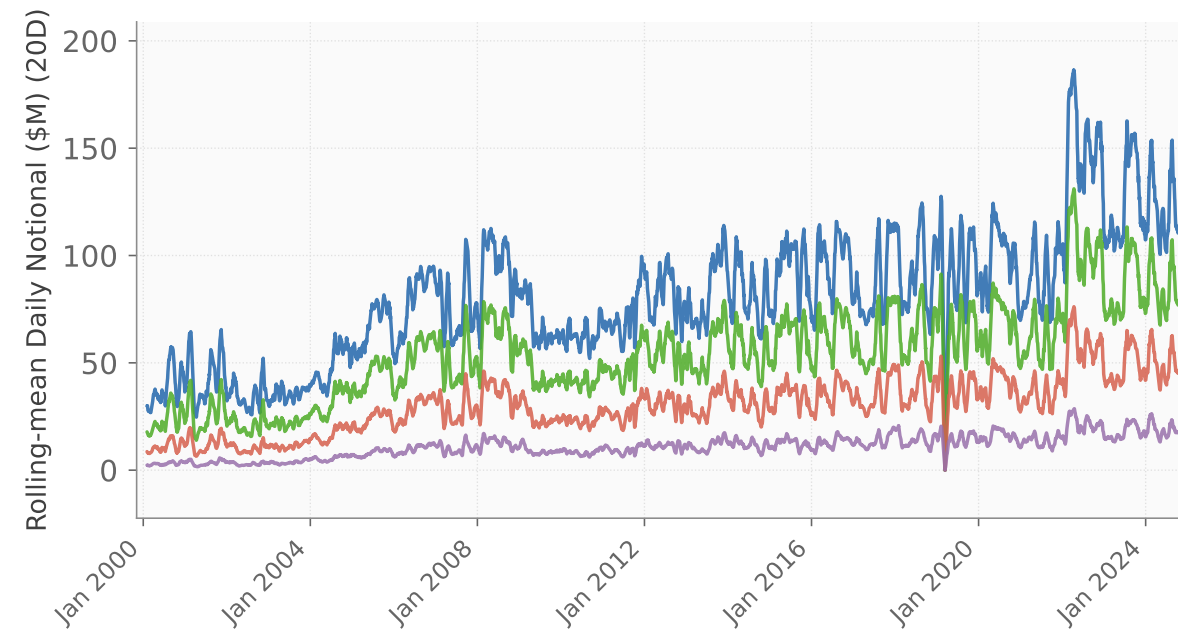
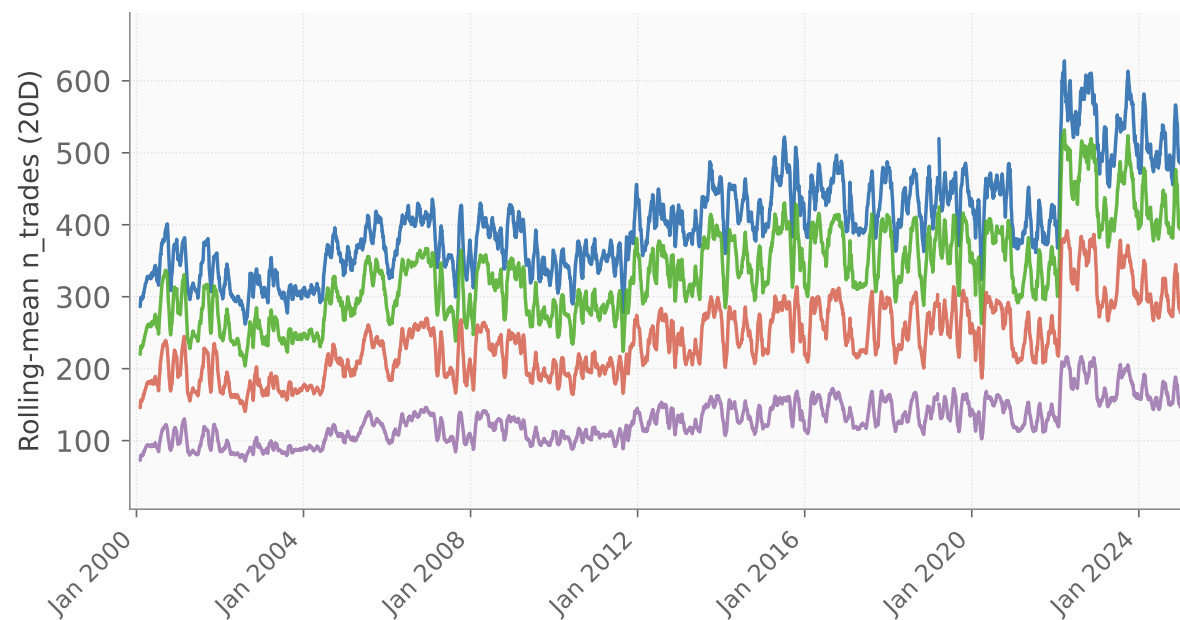
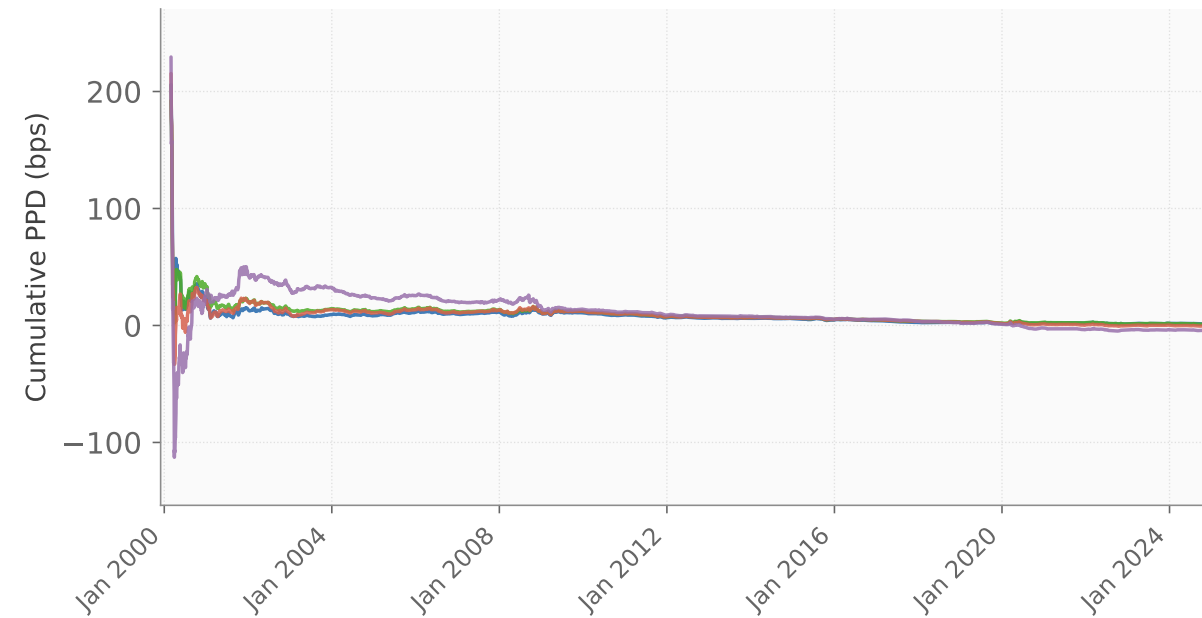
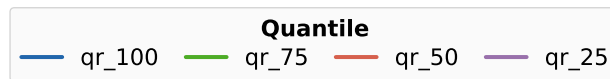
fret_OP5CL_MR | pret_ols | bs_m025_c100k — Temporal metrics



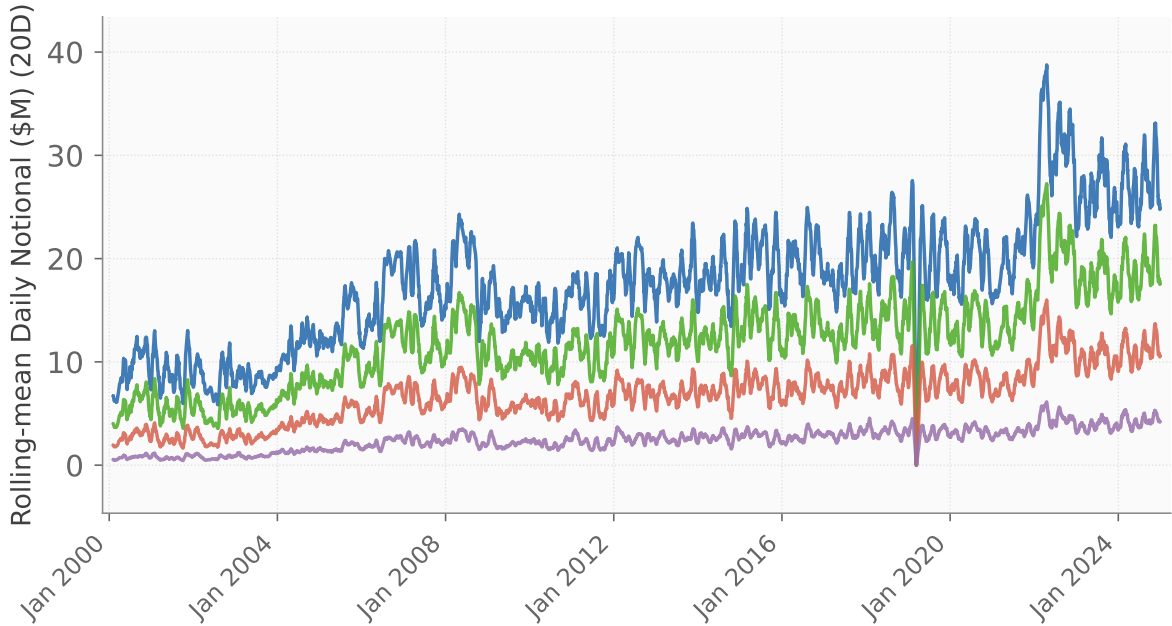
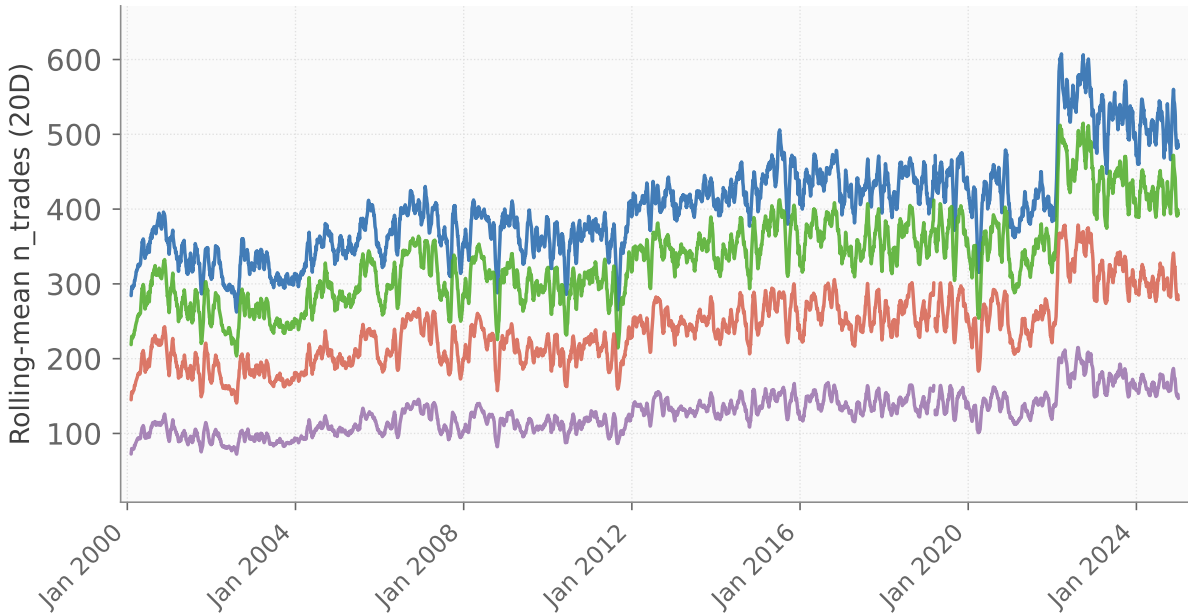
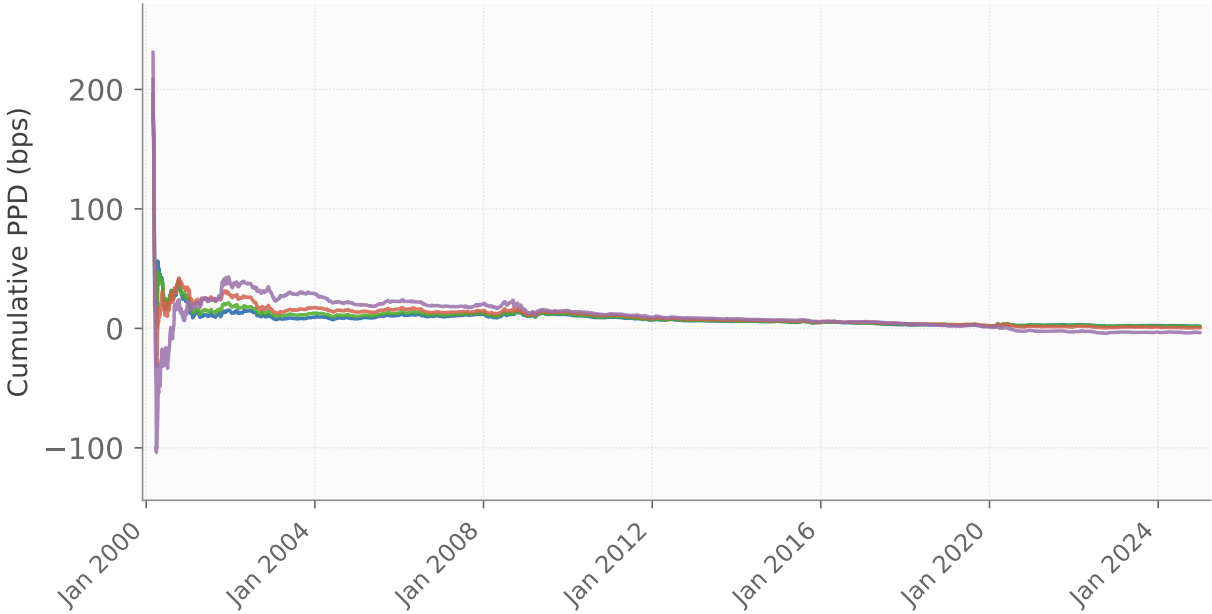
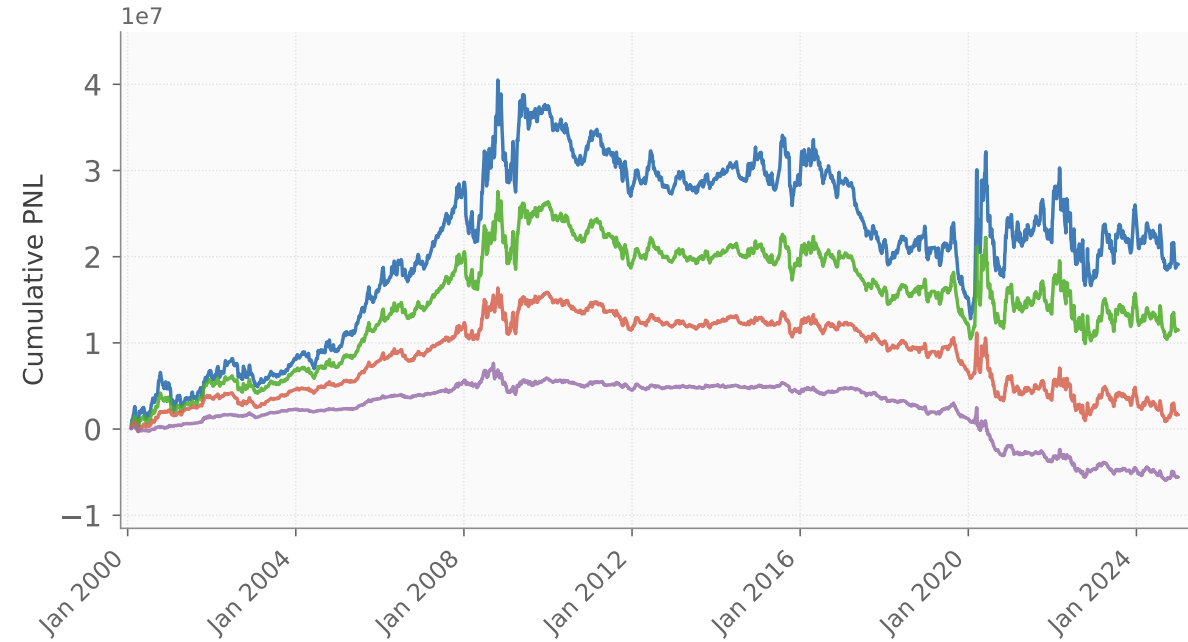
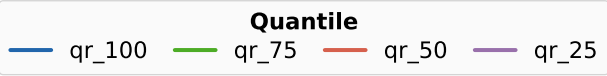
fret_OP5CL_MR | pret_ols | bs_m05_c250k — Temporal metrics



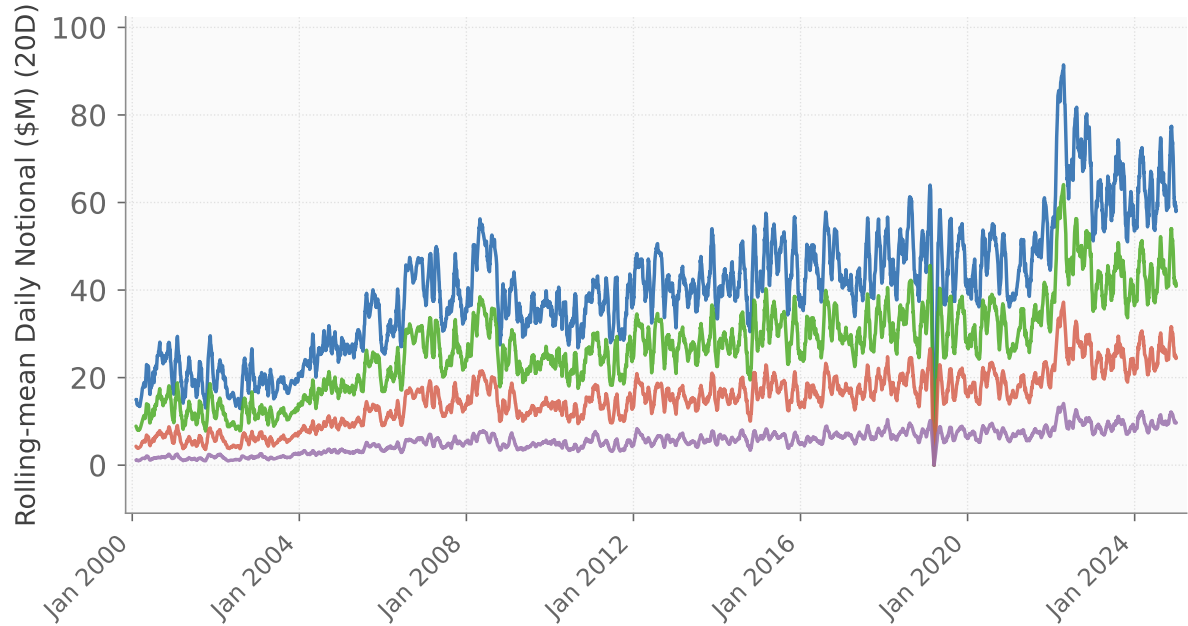
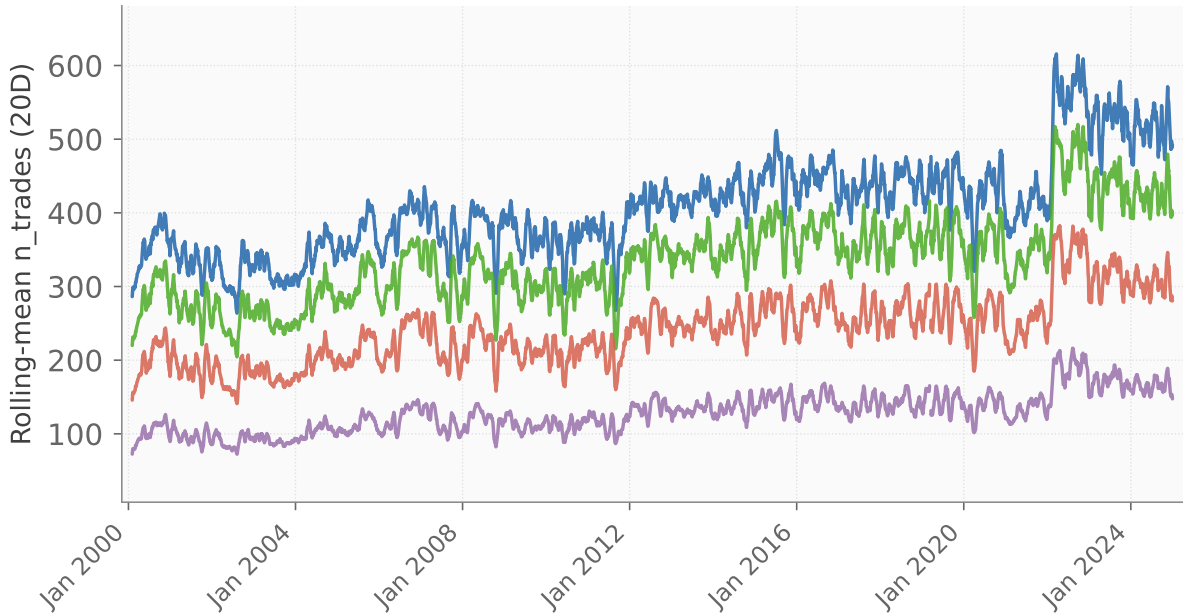
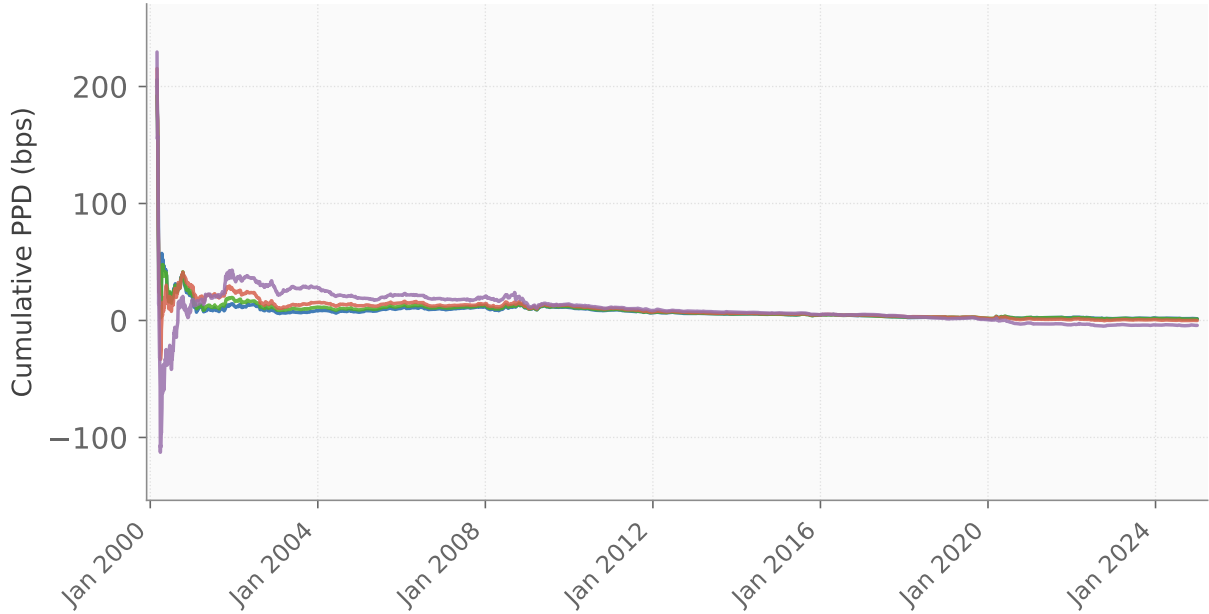
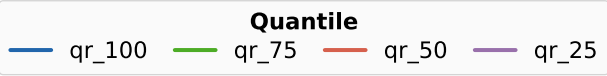
fret_OP5CL_MR | pret_ols | bs_m1_c500k — Temporal metrics



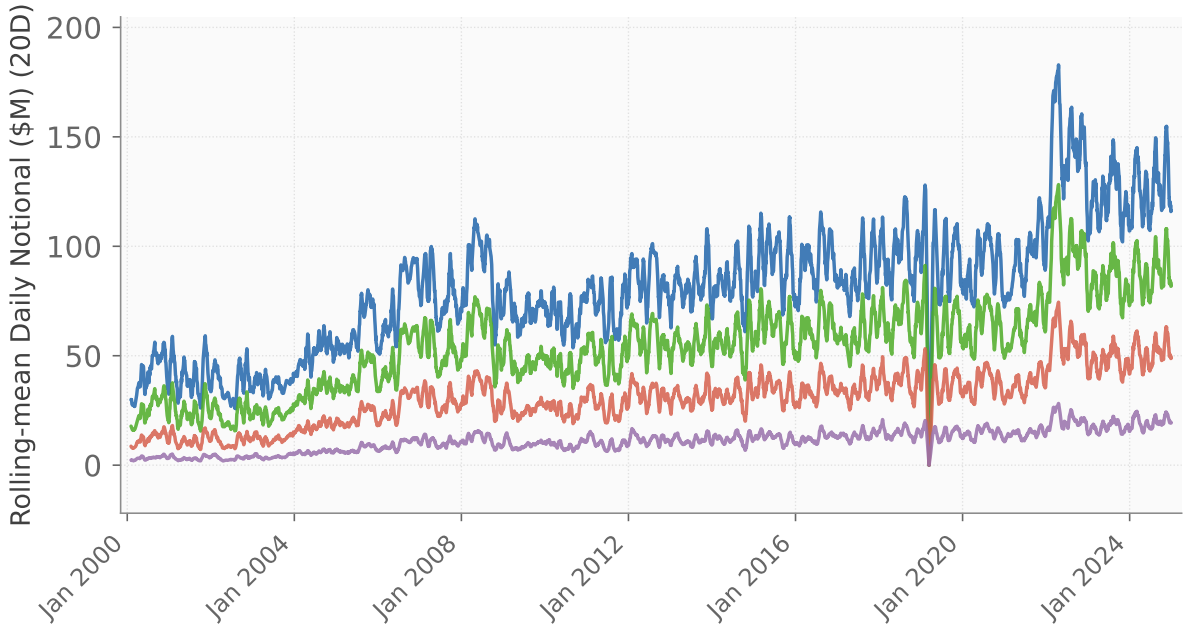
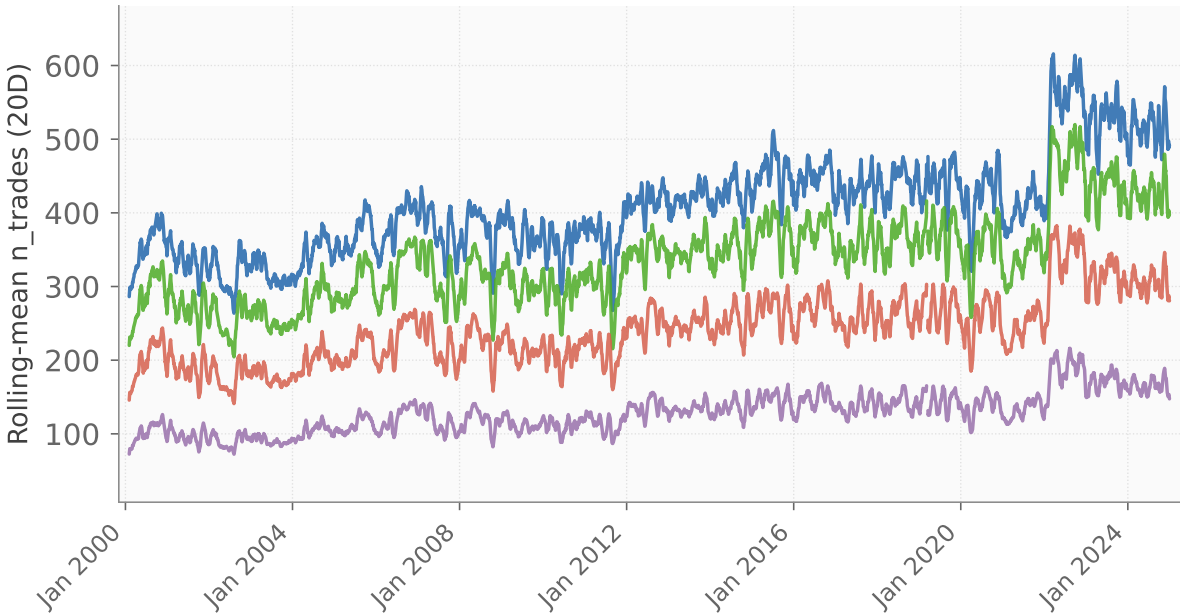
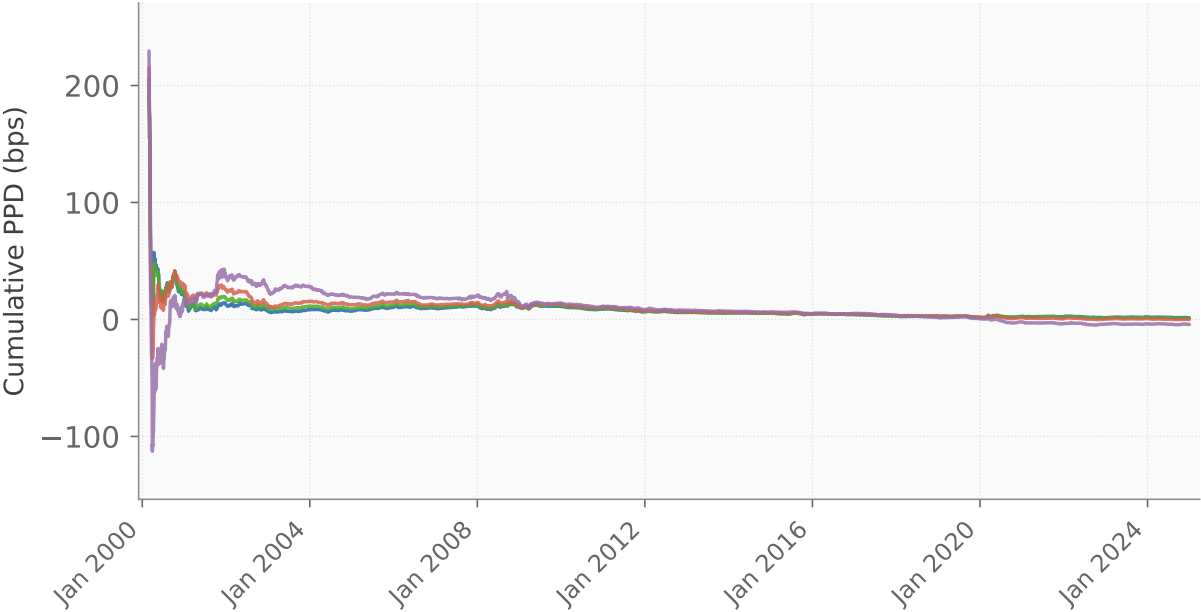
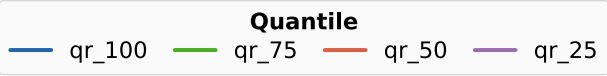
fret_OP5CL_MR | pret_oracle | bs_m025_c100k — Temporal metrics



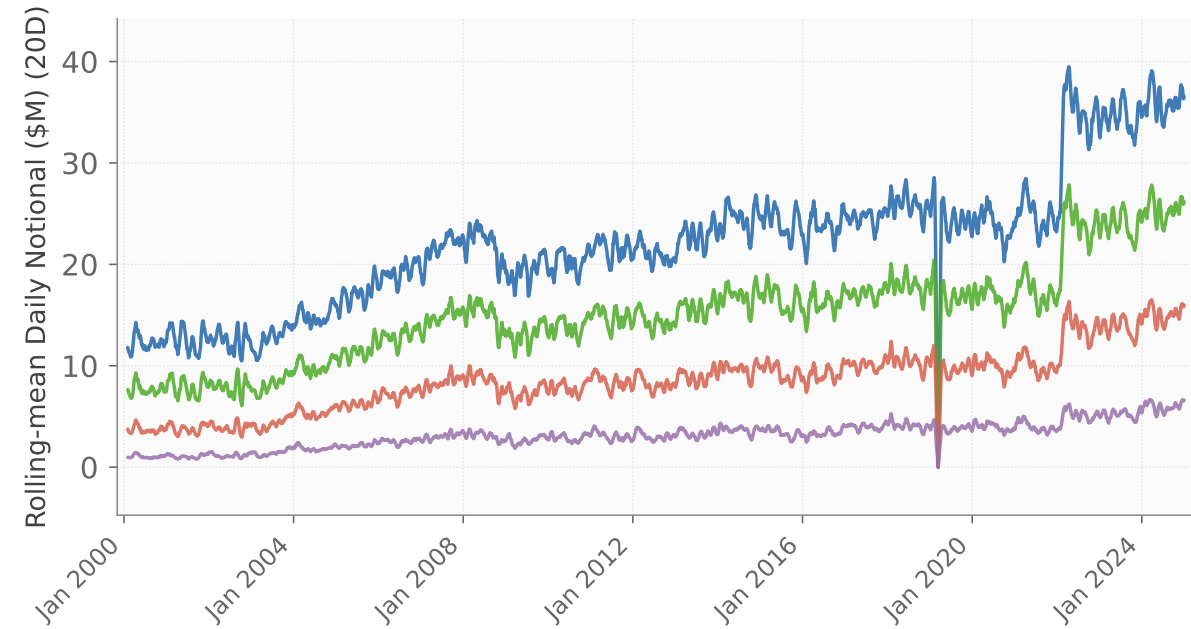
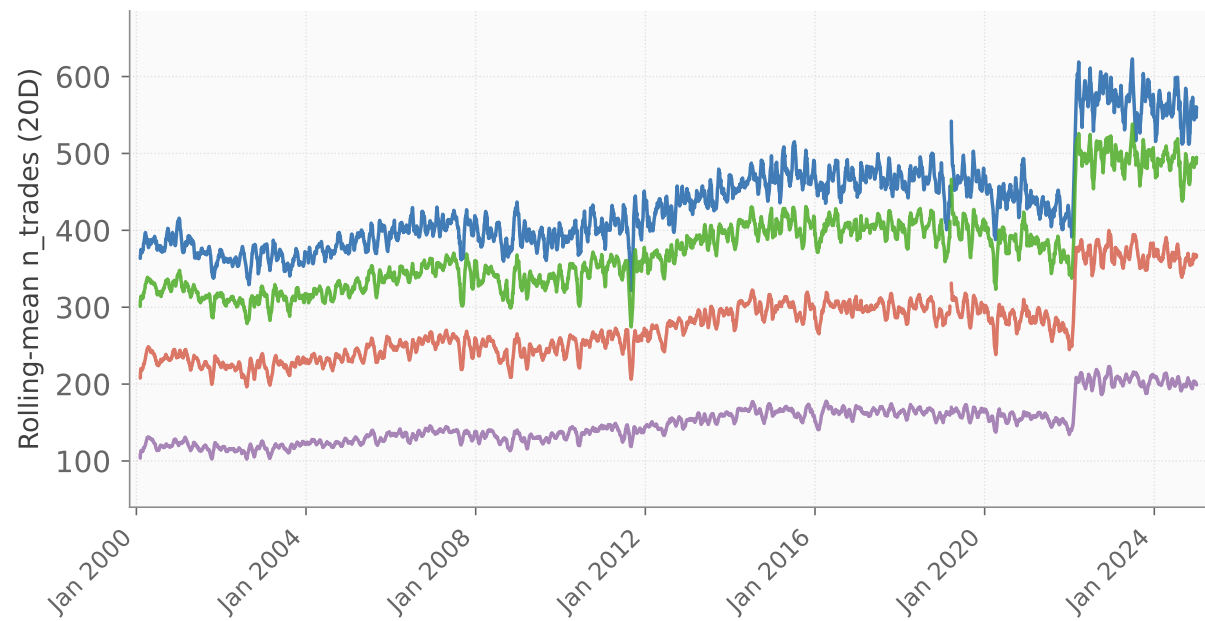
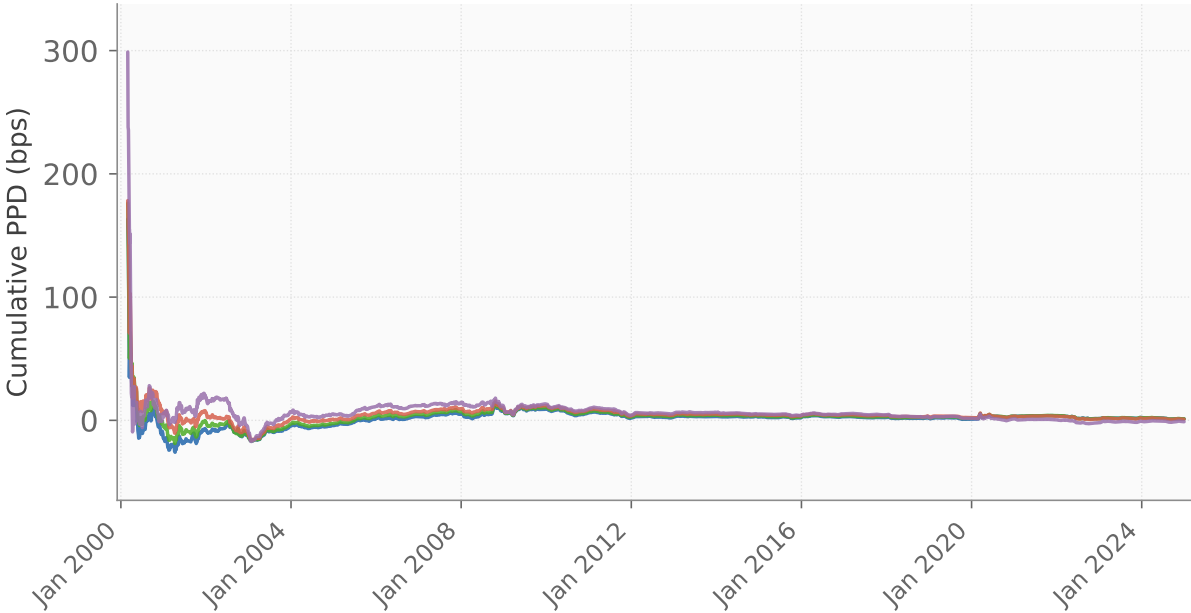
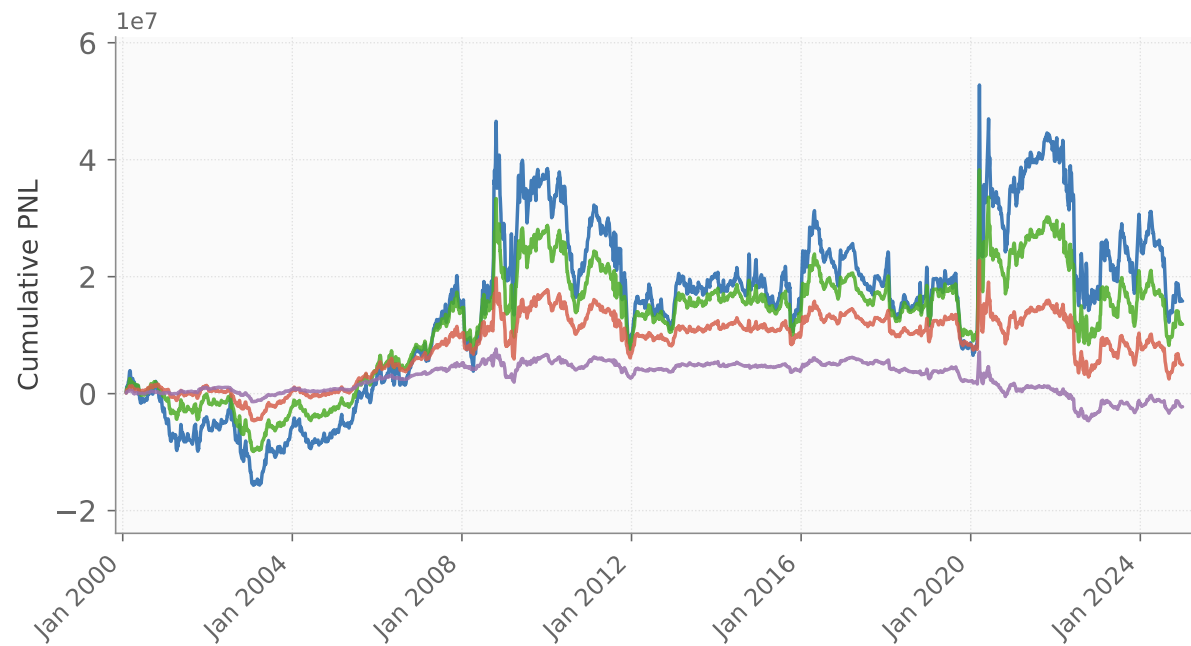
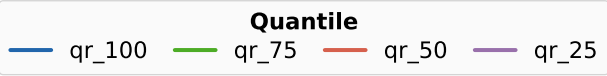
fret_OP5CL_MR | pret_oracle | bs_m05_c250k — Temporal metrics



fret_OP5CL_MR | pret_oracle | bs_m1_c500k — Temporal metrics



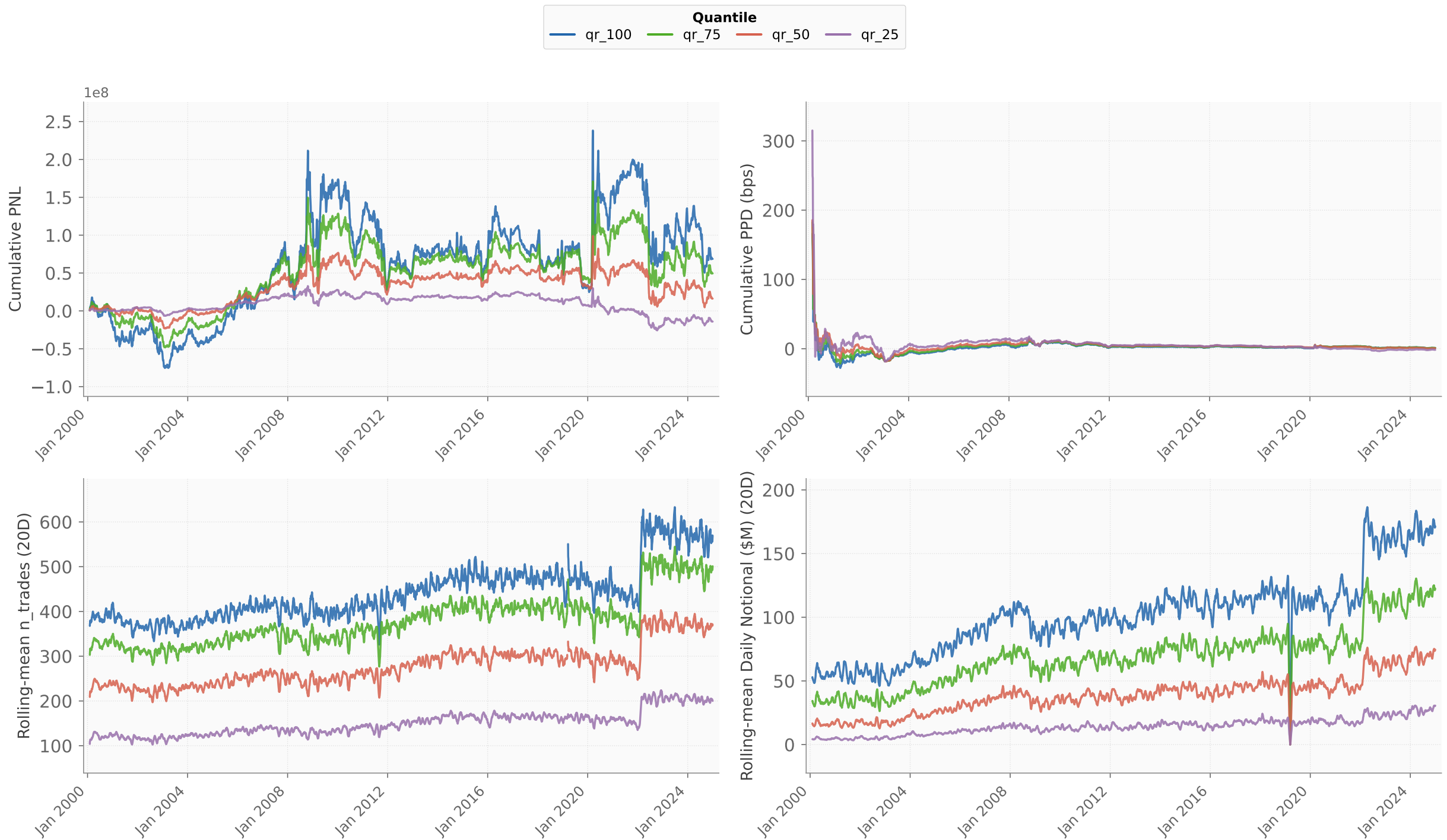
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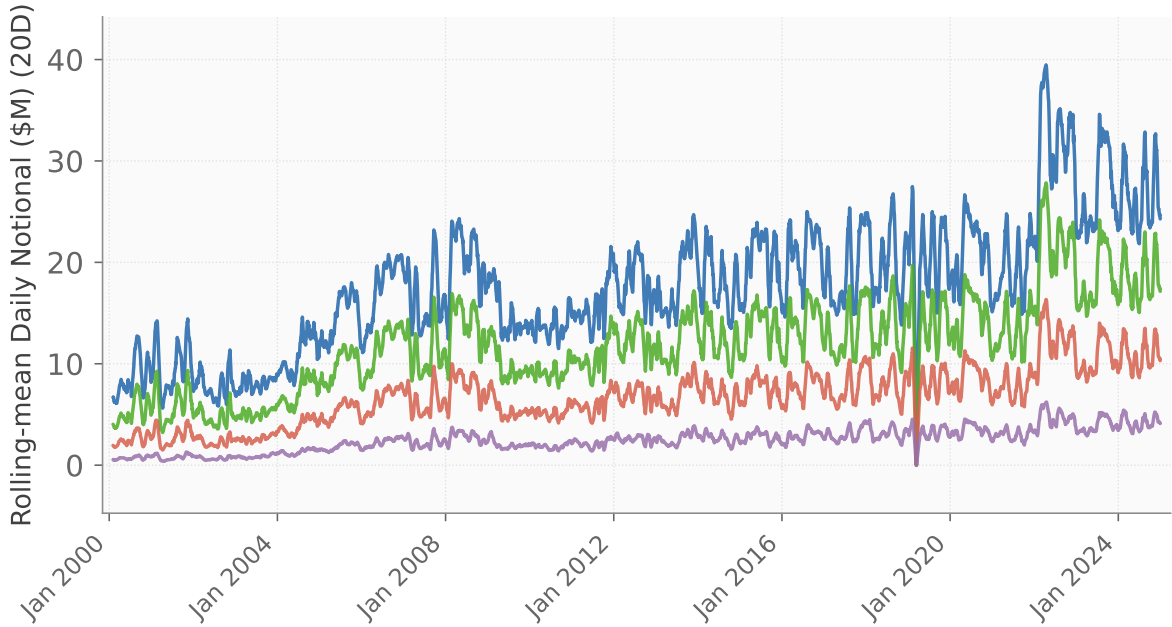
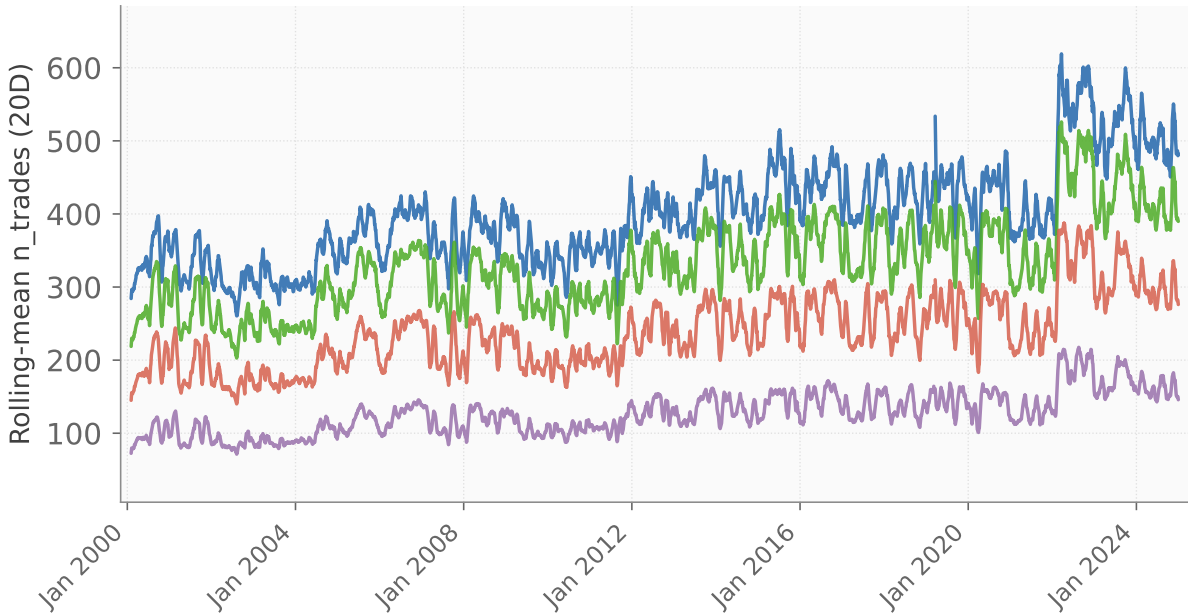
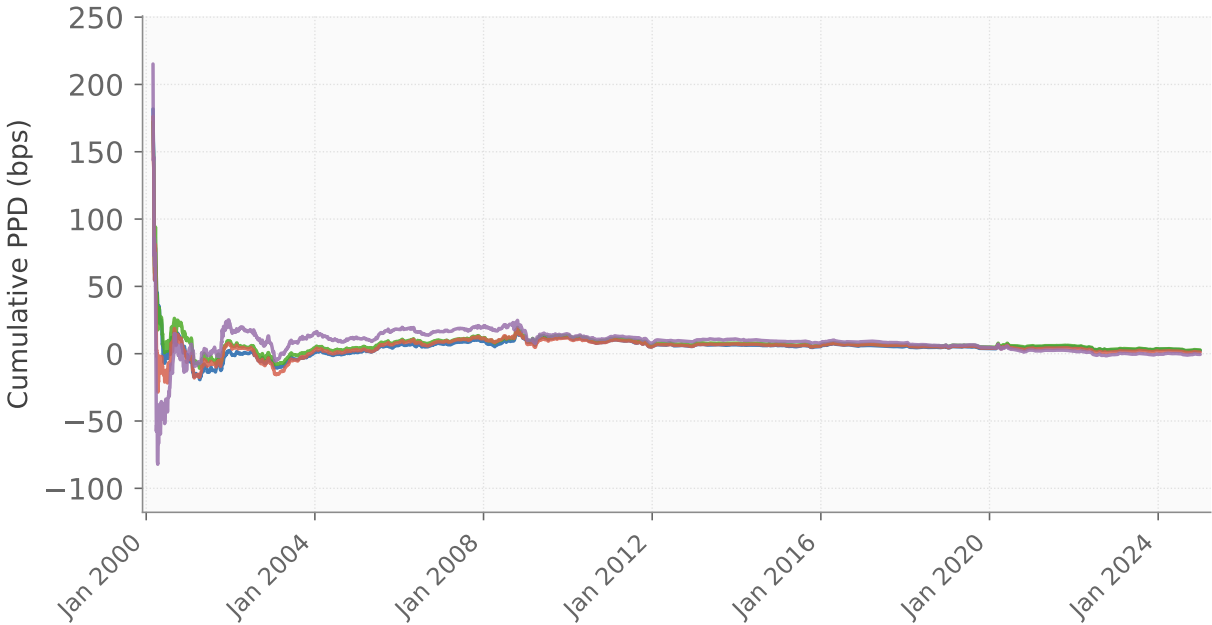
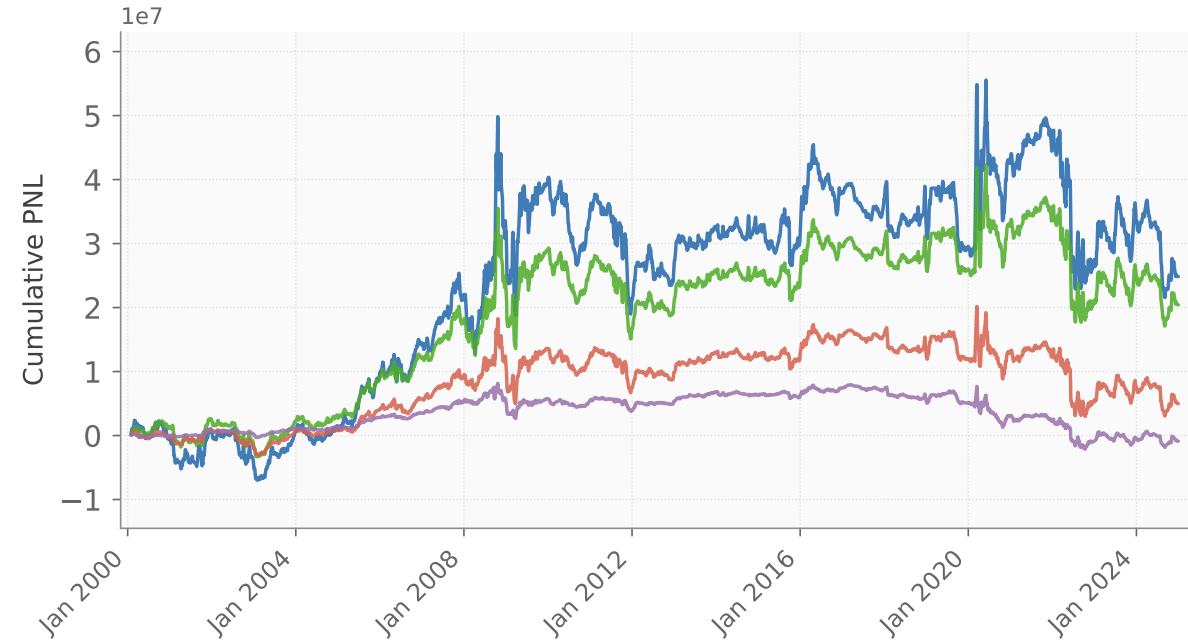
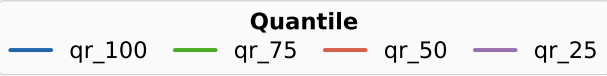
fret_OP5CL_RR | pret_base | bs_m05_c250k — Temporal metrics



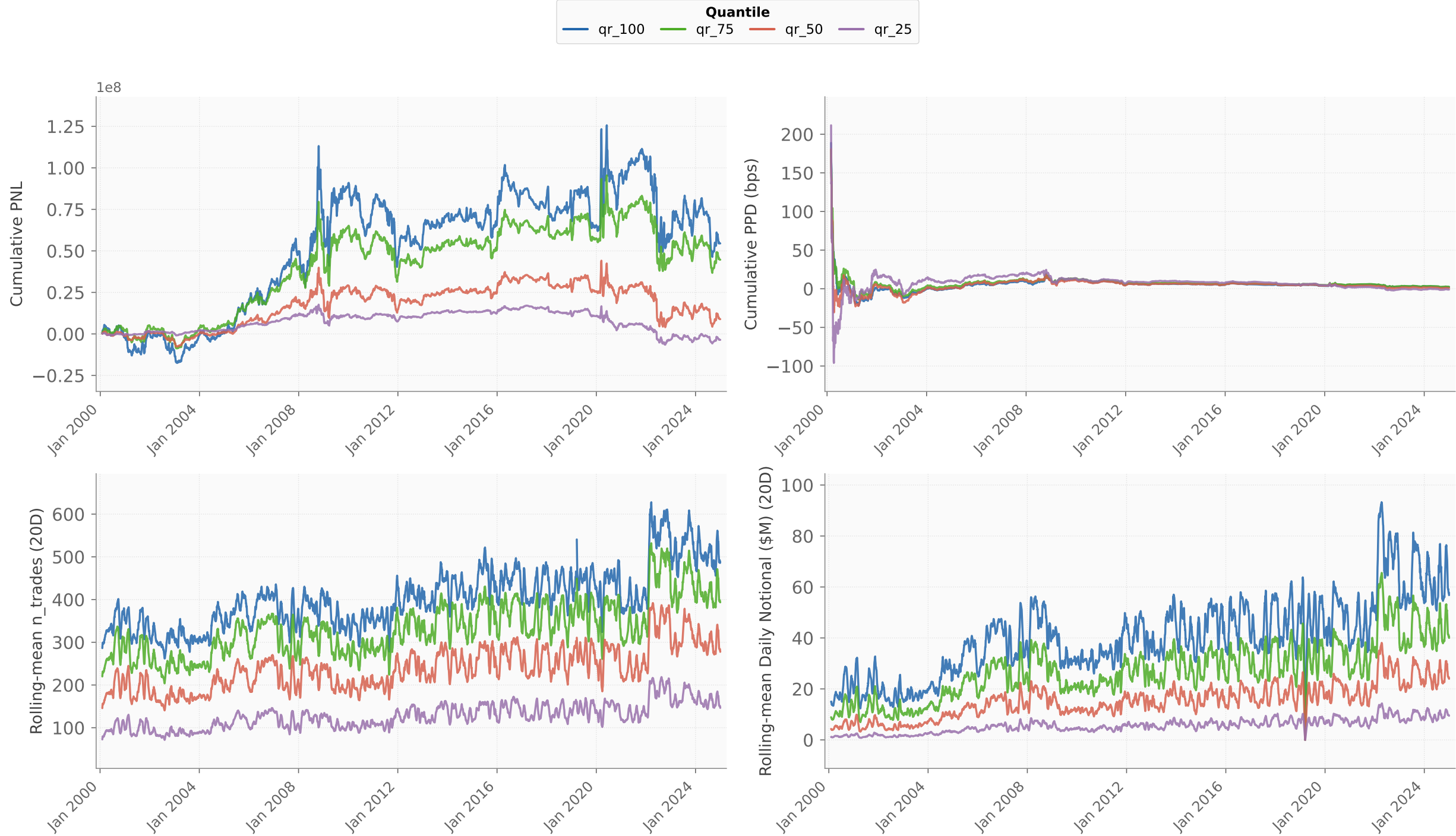
fret_OP5CL_RR | pret_base | bs_m1_c500k — Temporal metrics



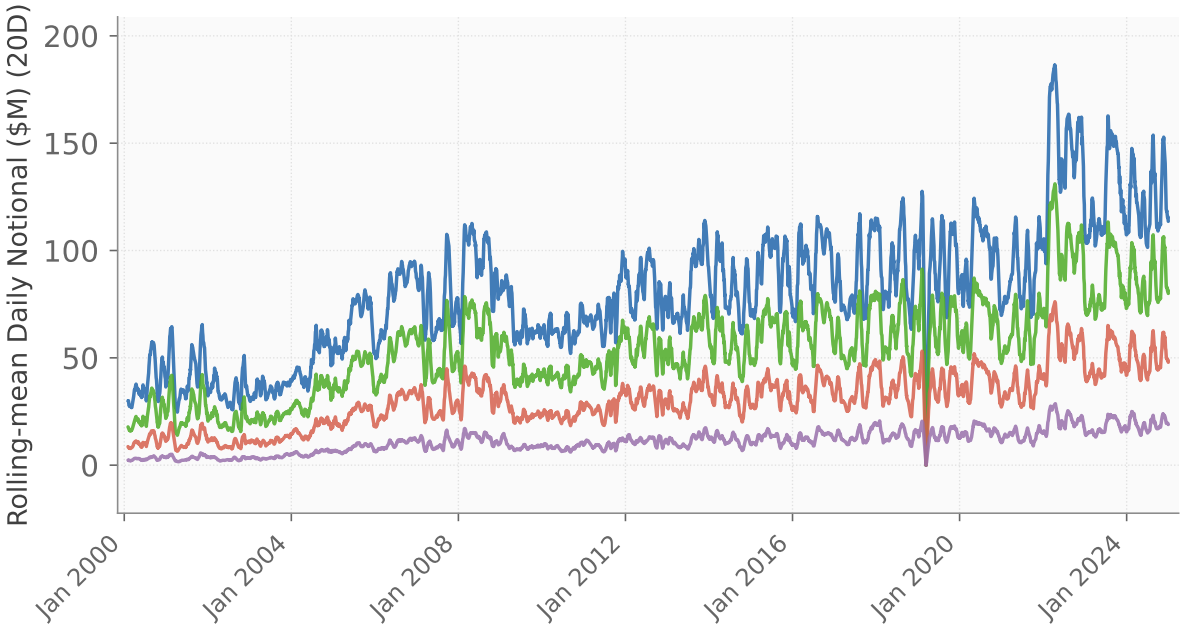
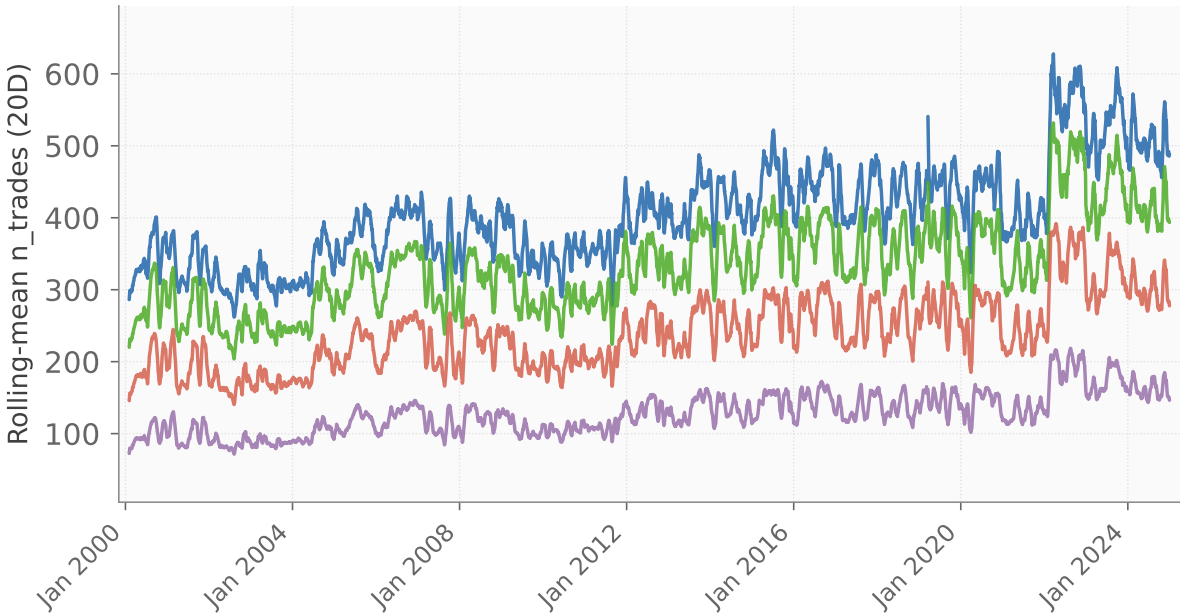
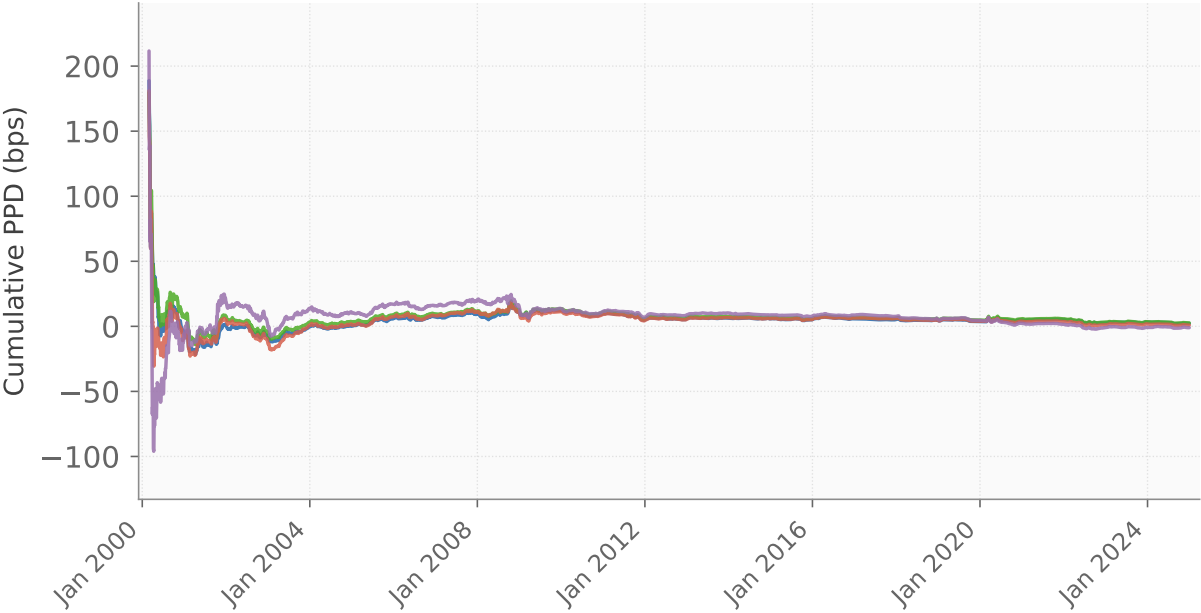
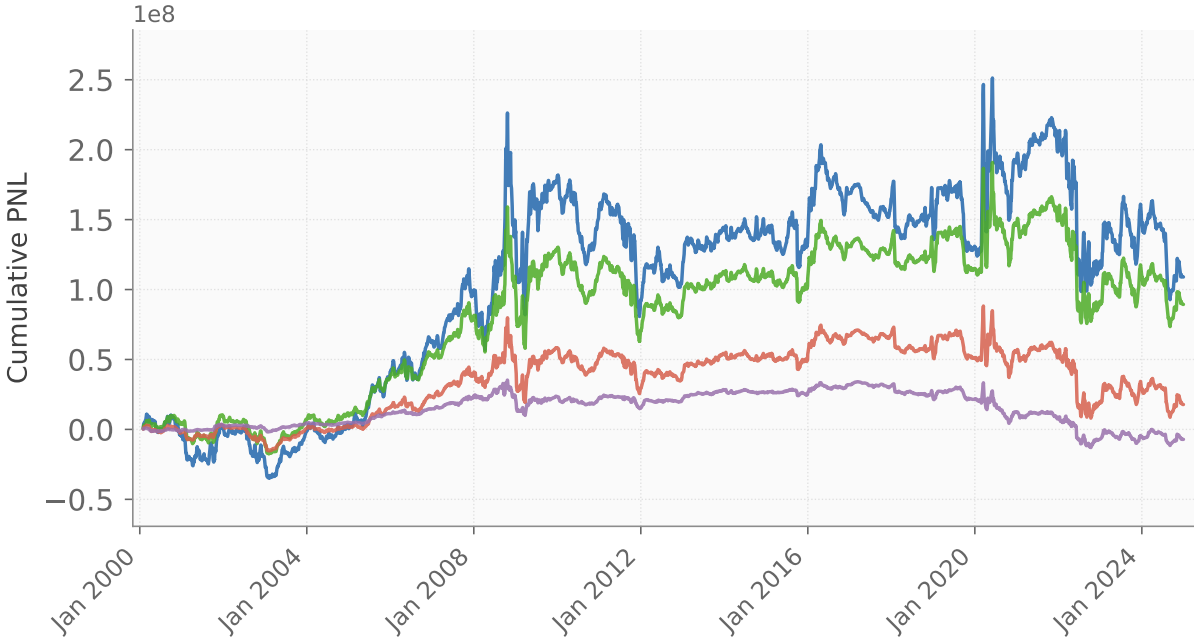
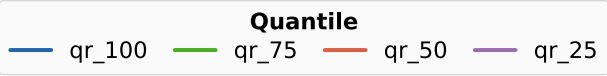
fret_OP5CL_RR | pret_clus | bs_m025_c100k — Temporal metrics



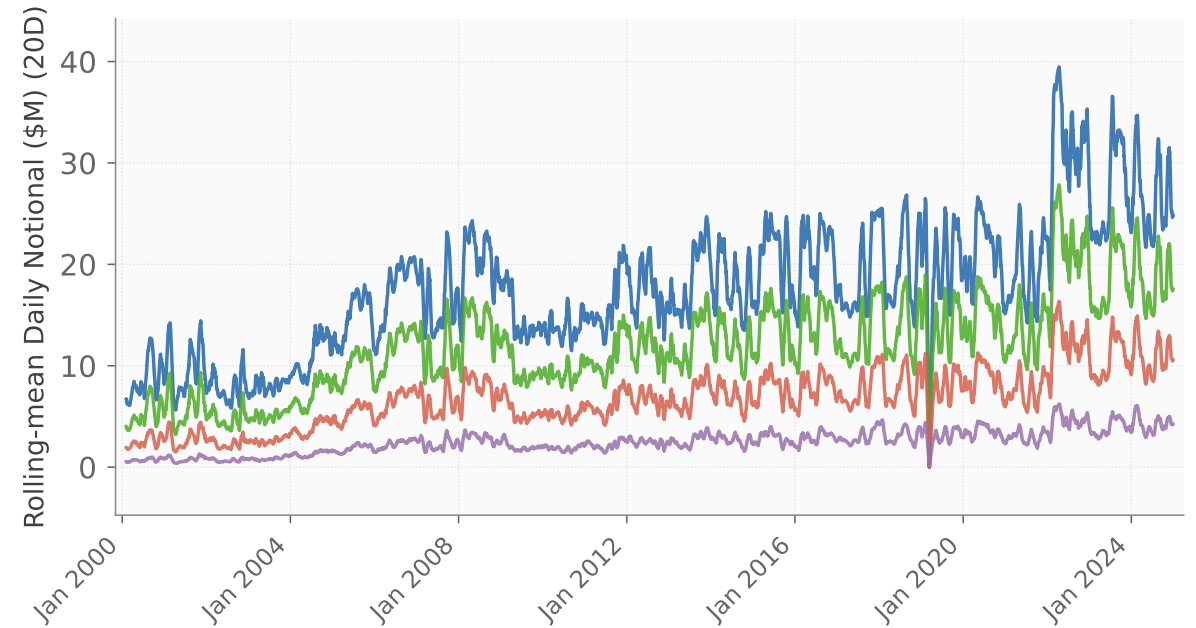
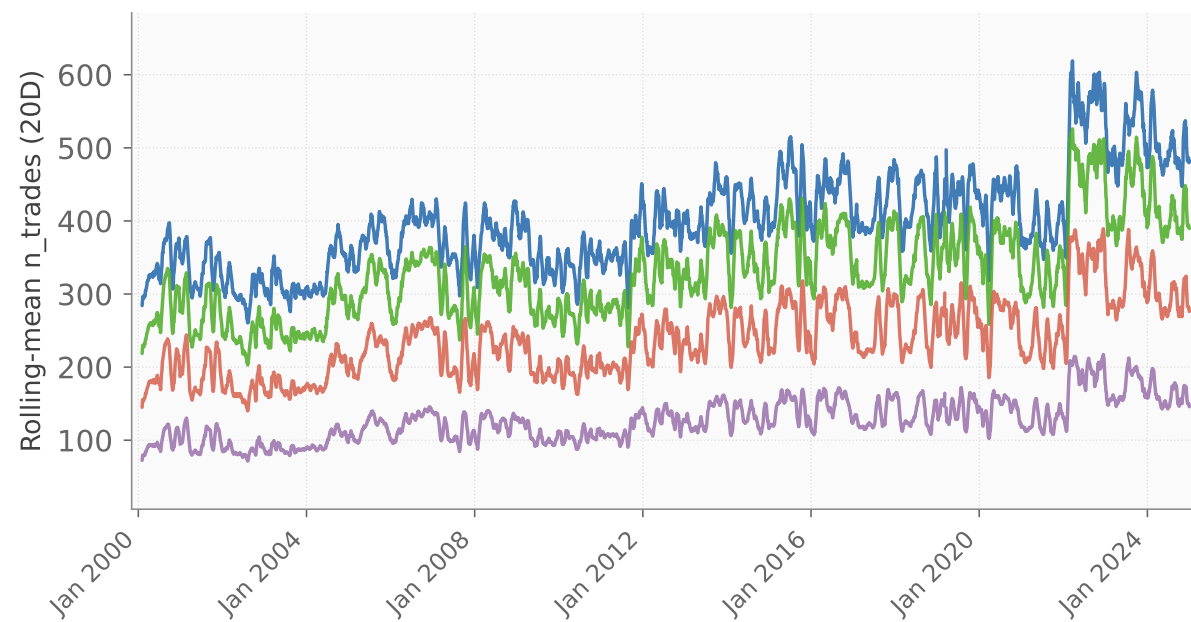
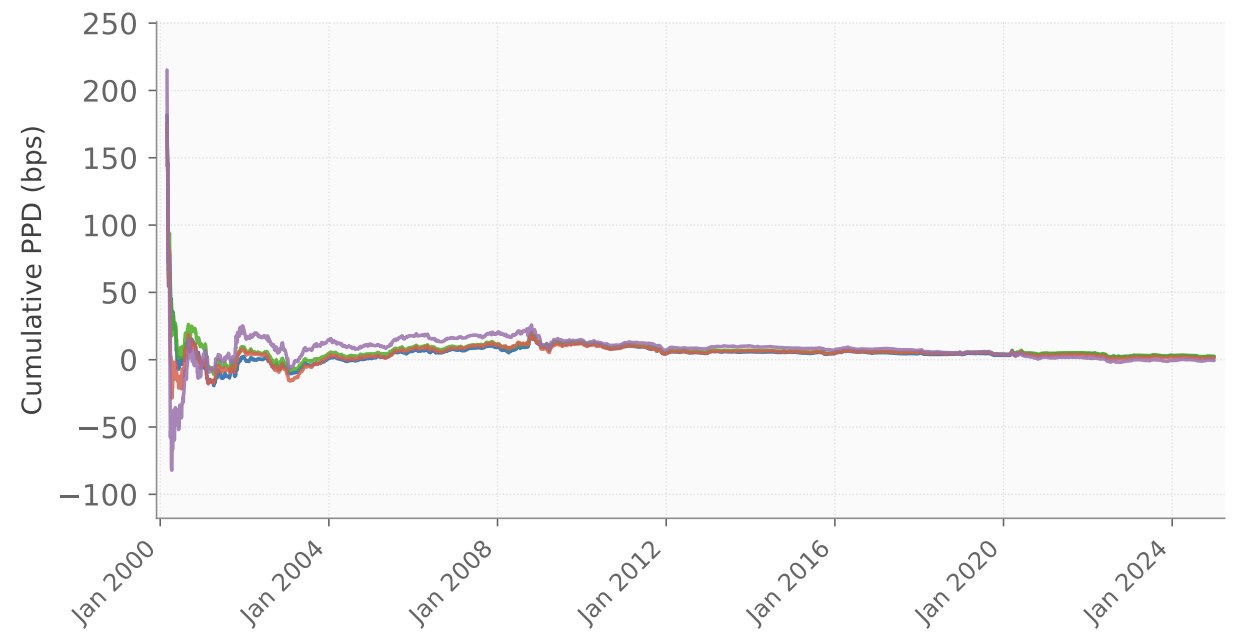
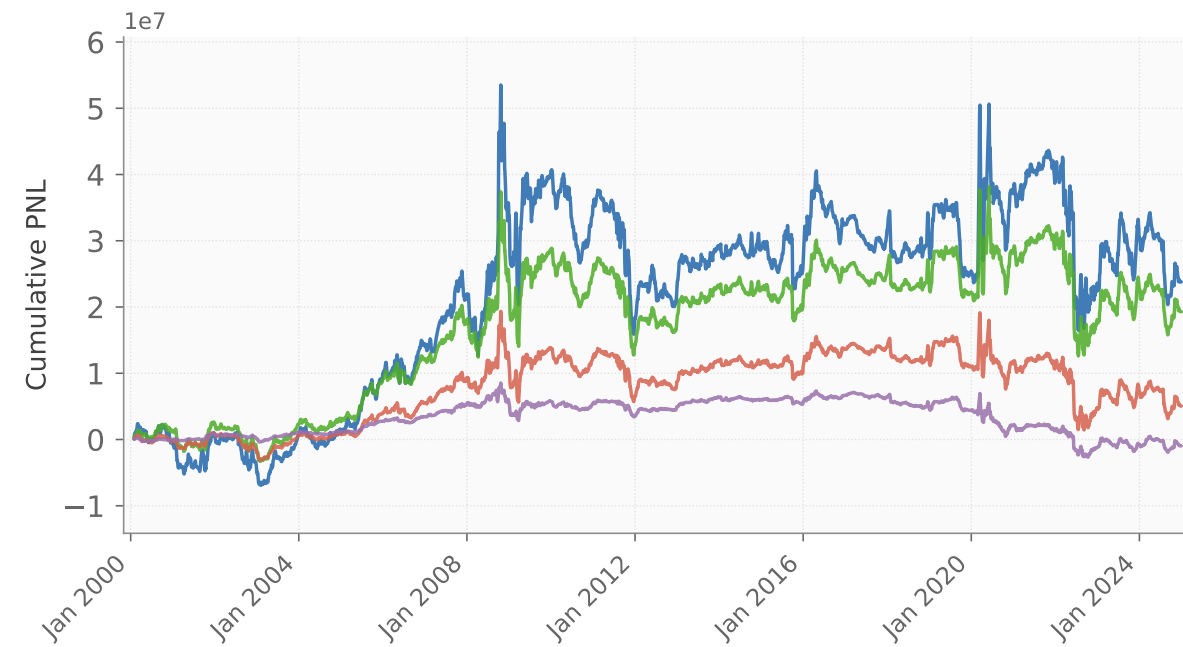
fret_OP5CL_RR | pret_clus | bs_m05_c250k — Temporal metrics



fret_OP5CL_RR | pret_clus | bs_m1_c500k — Temporal metrics



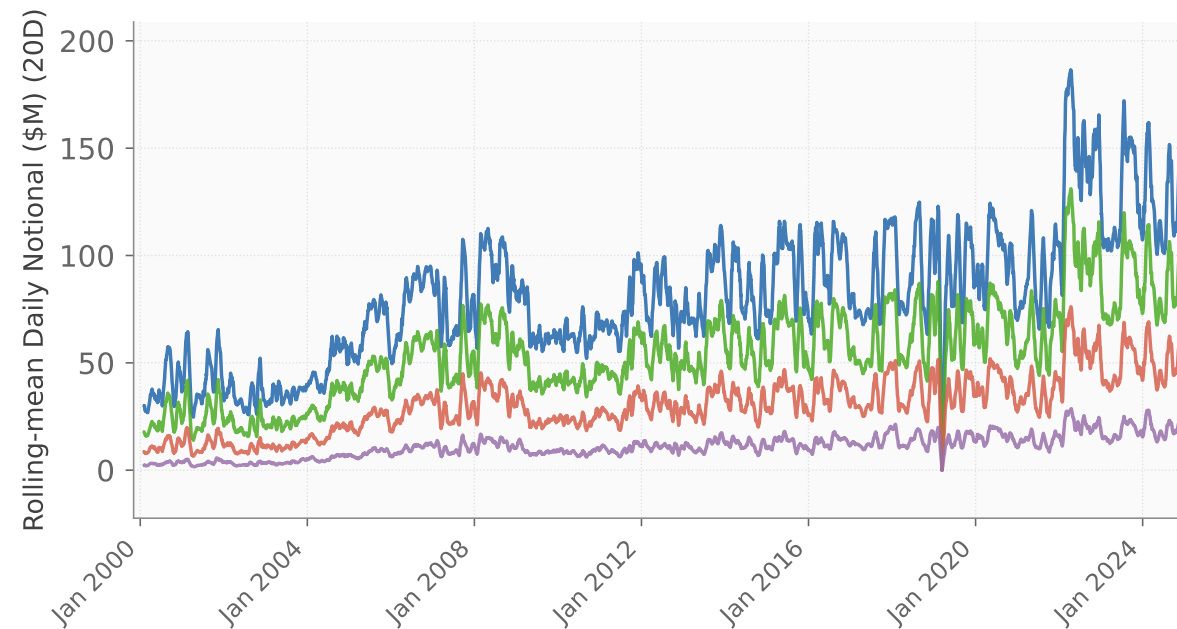
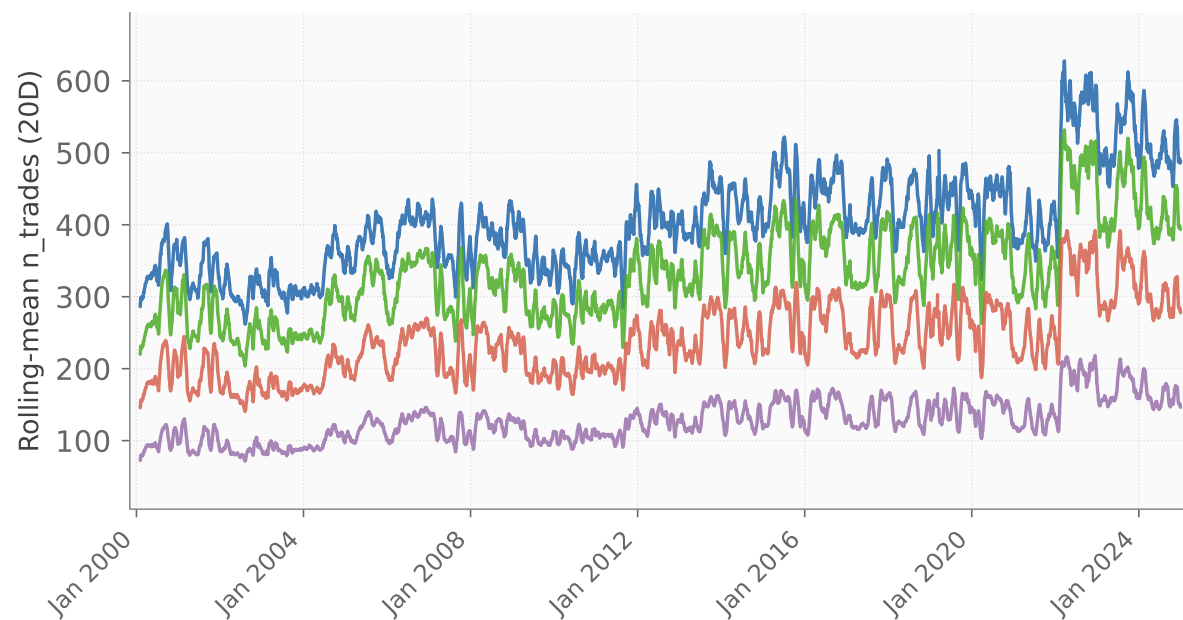
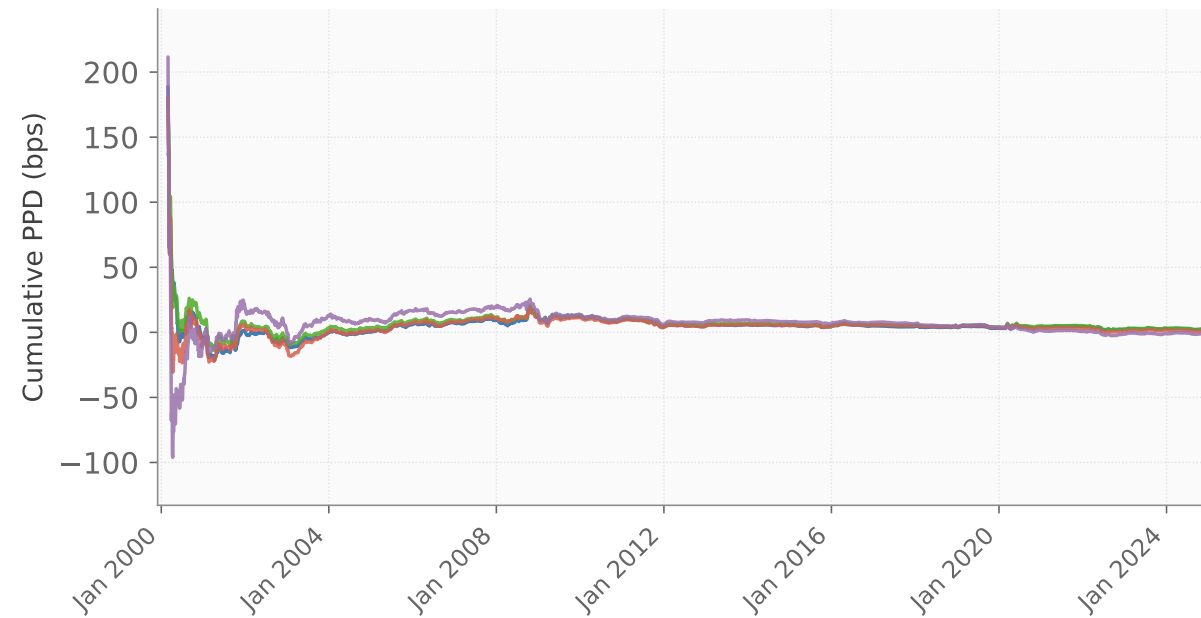
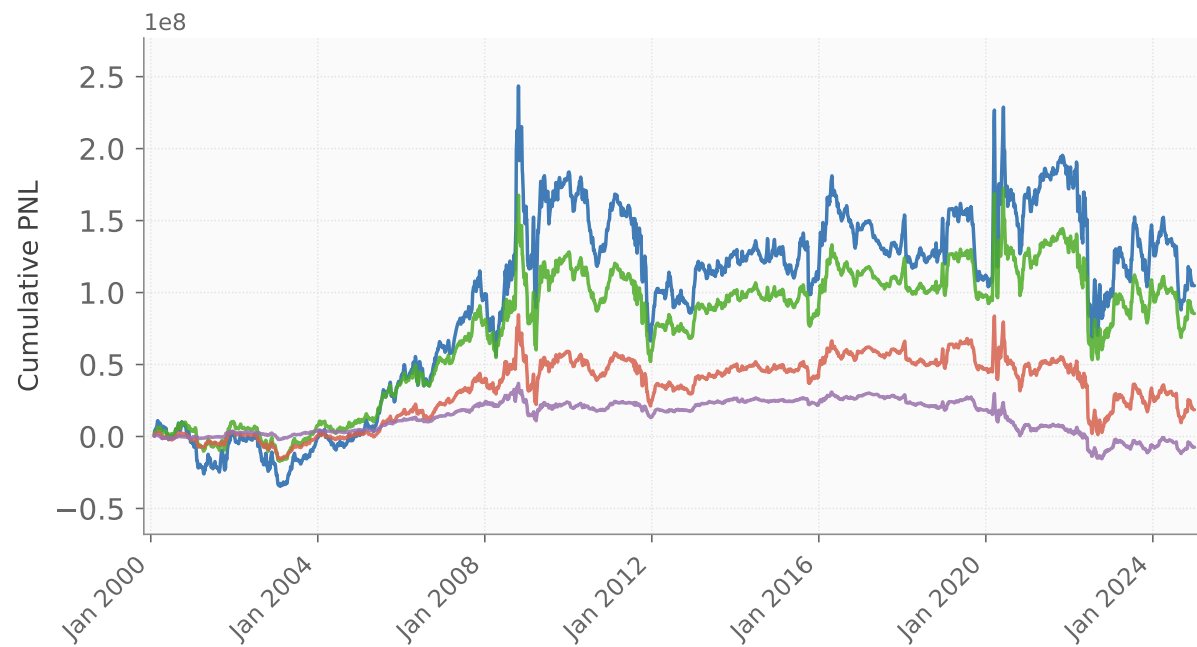
fret_OP5CL_RR | pret_har | bs_m025_c100k — Temporal metrics



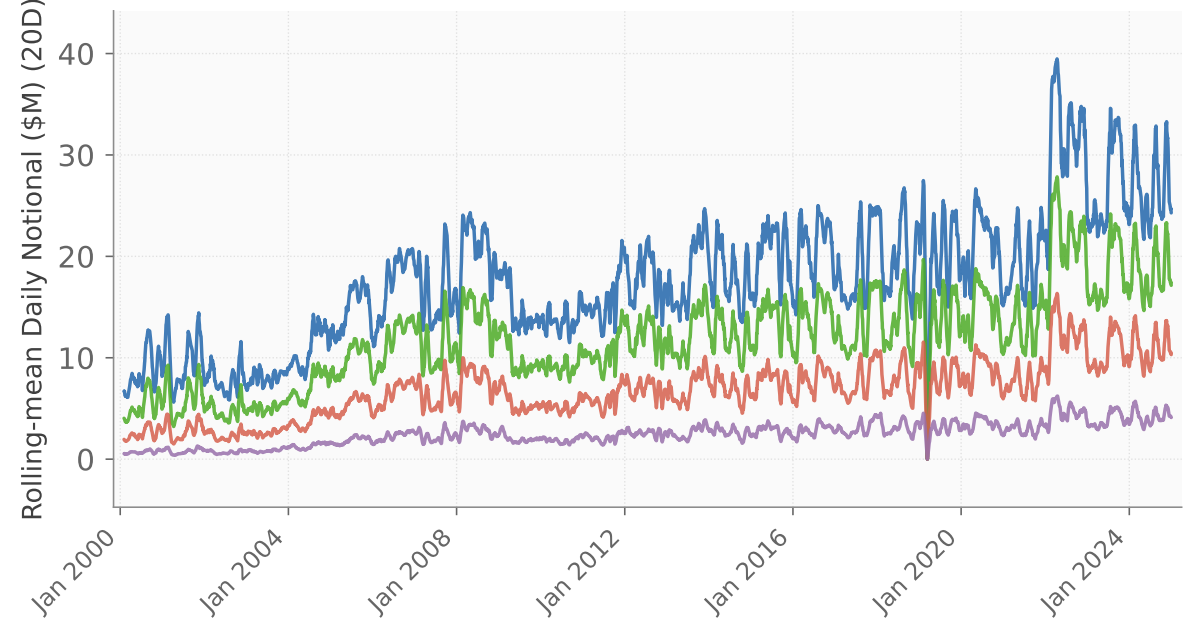
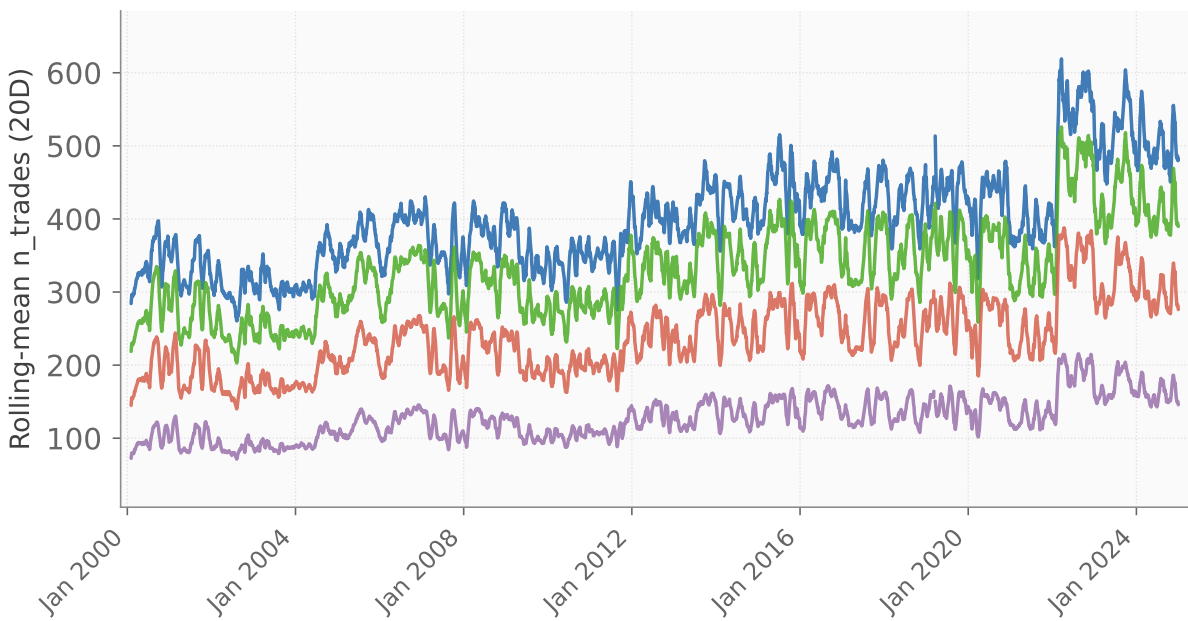
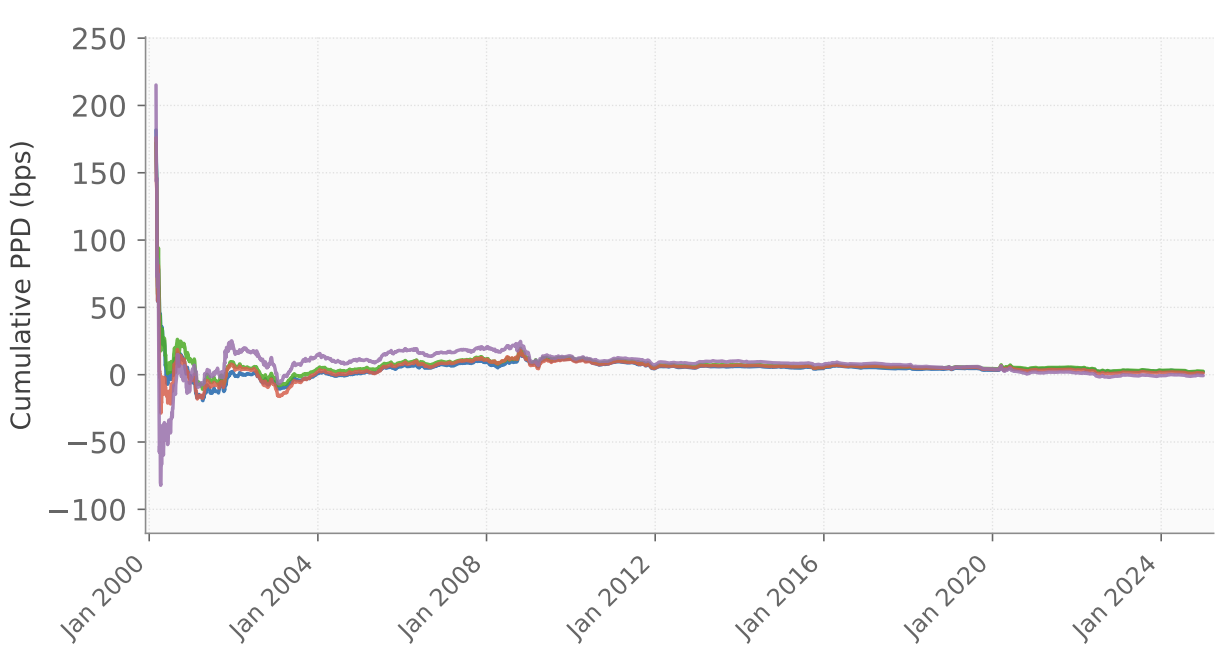
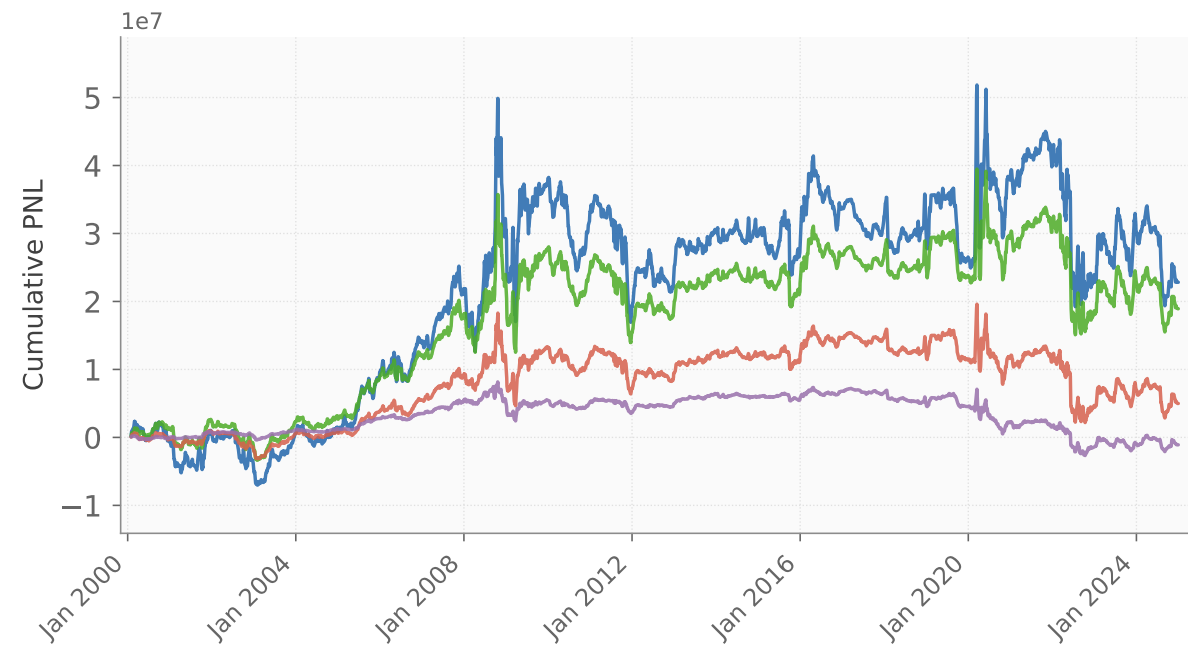
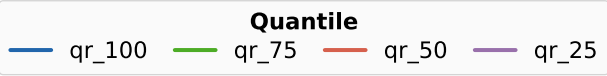
fret_OP5CL_RR | pret_har | bs_m05_c250k — Temporal metrics



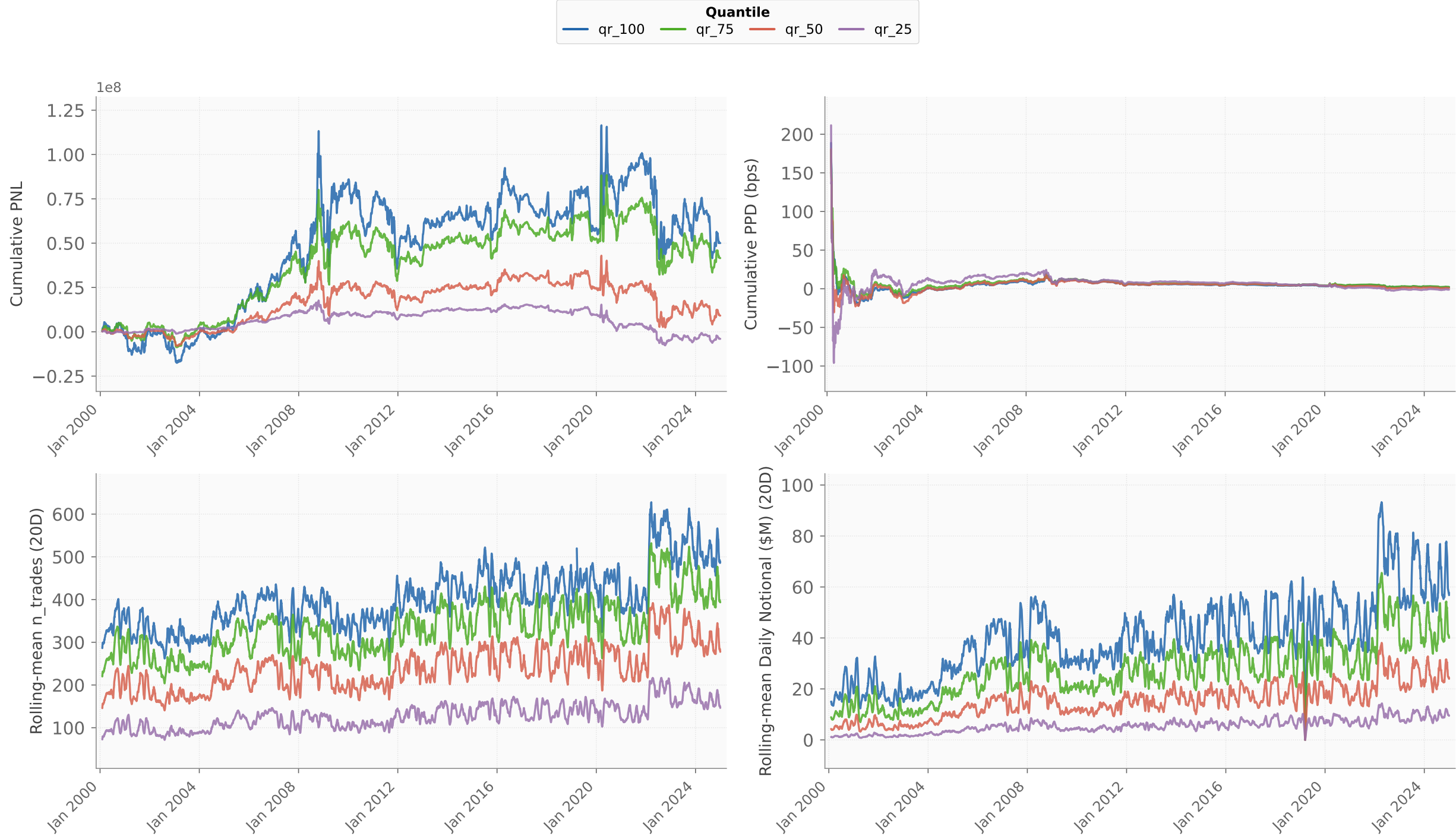
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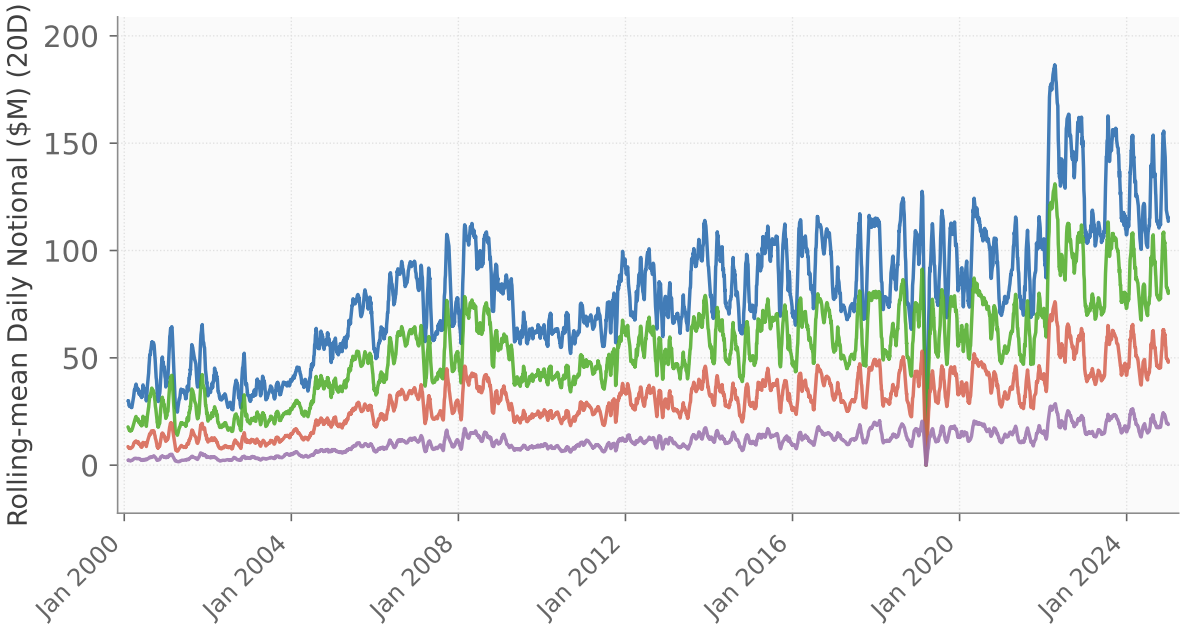
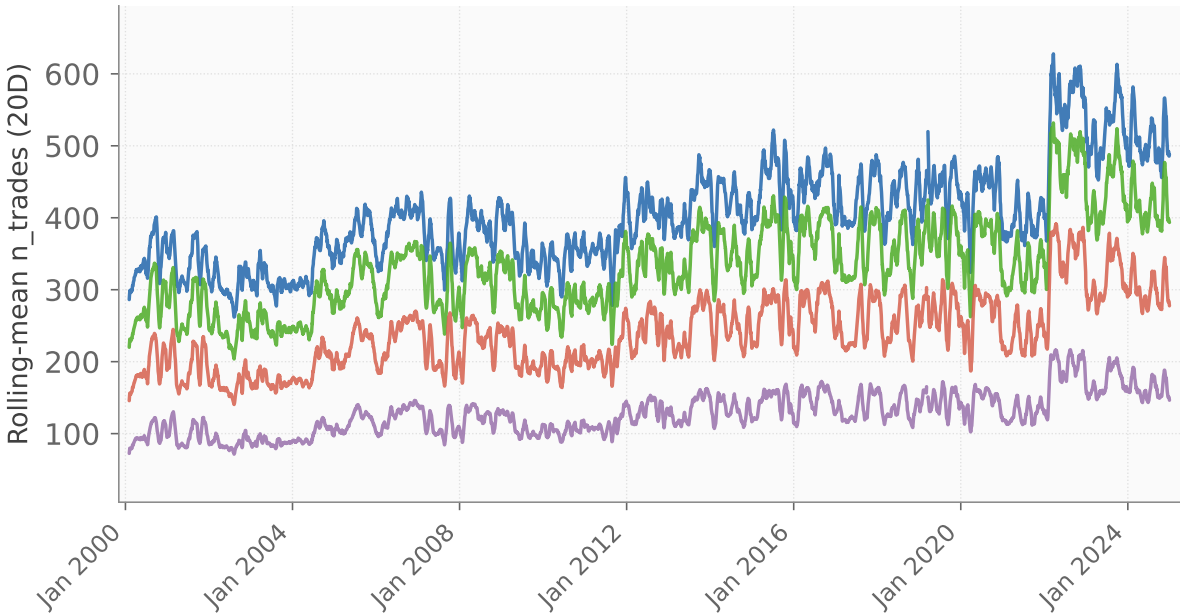
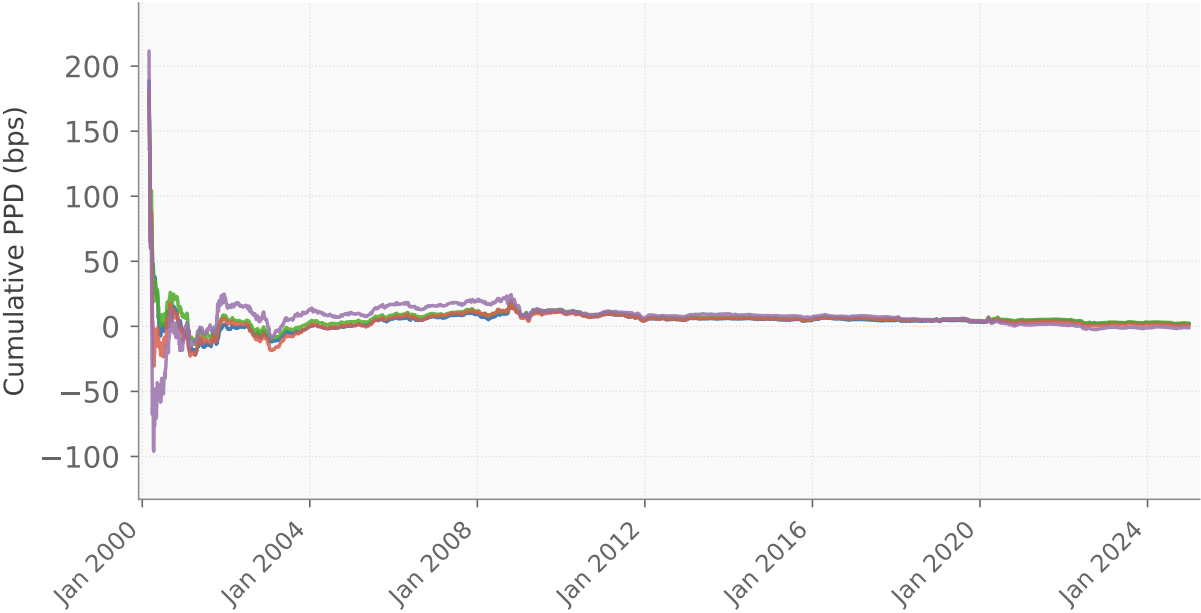
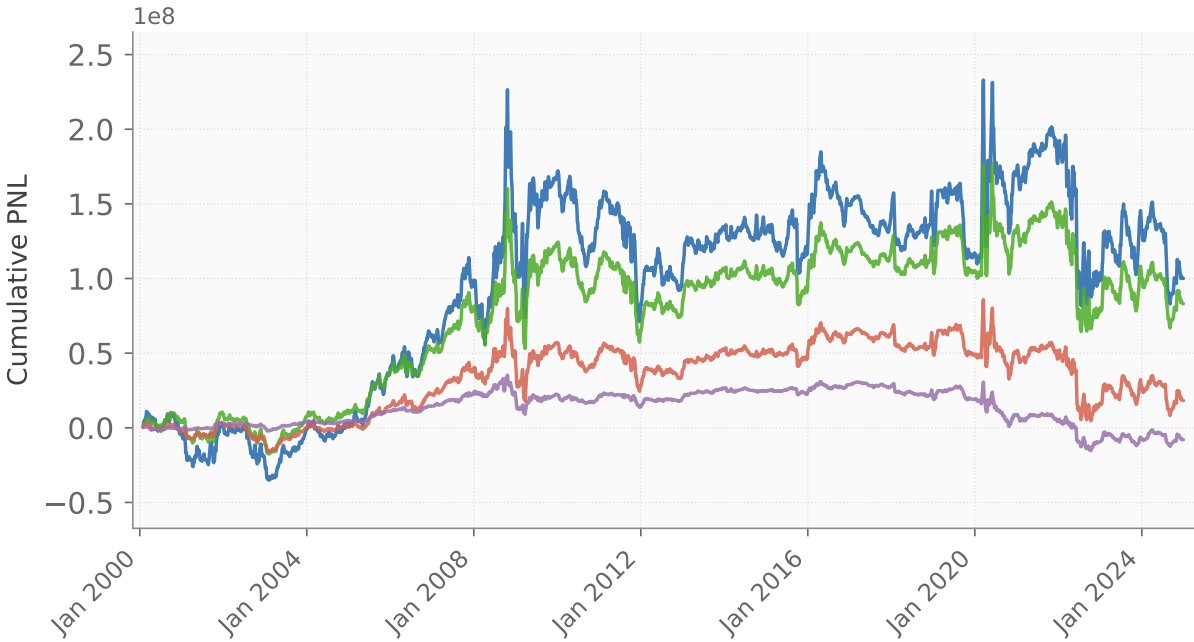
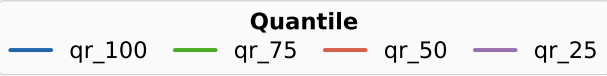
fret_OP5CL_RR | pret_ols | bs_m025_c100k — Temporal metrics



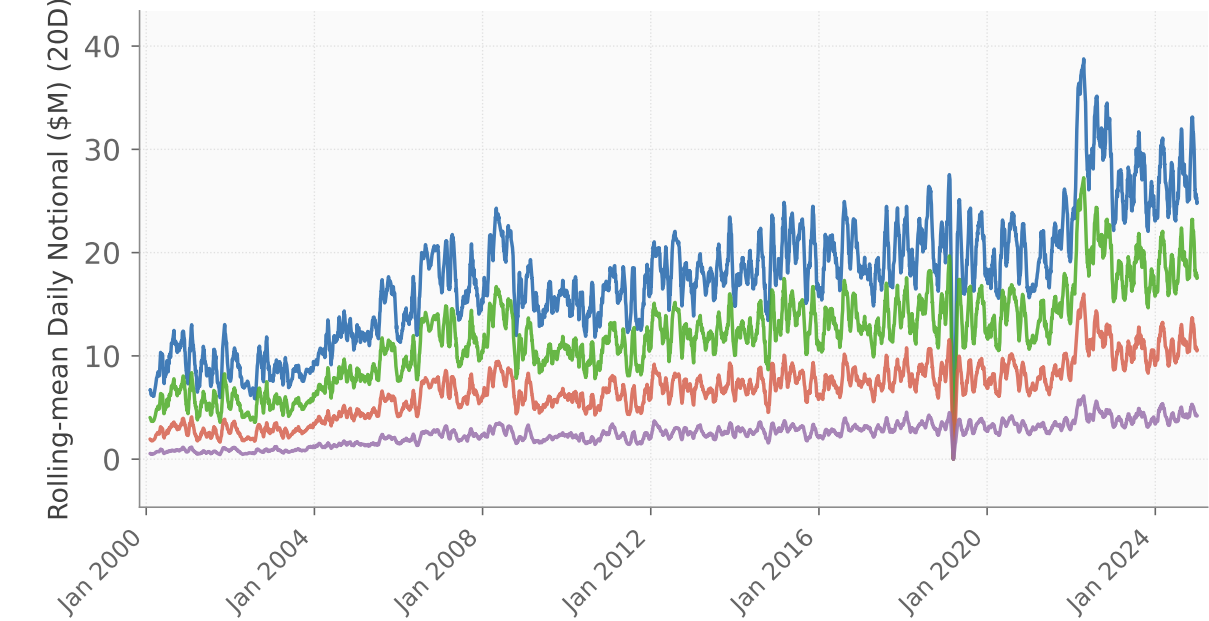
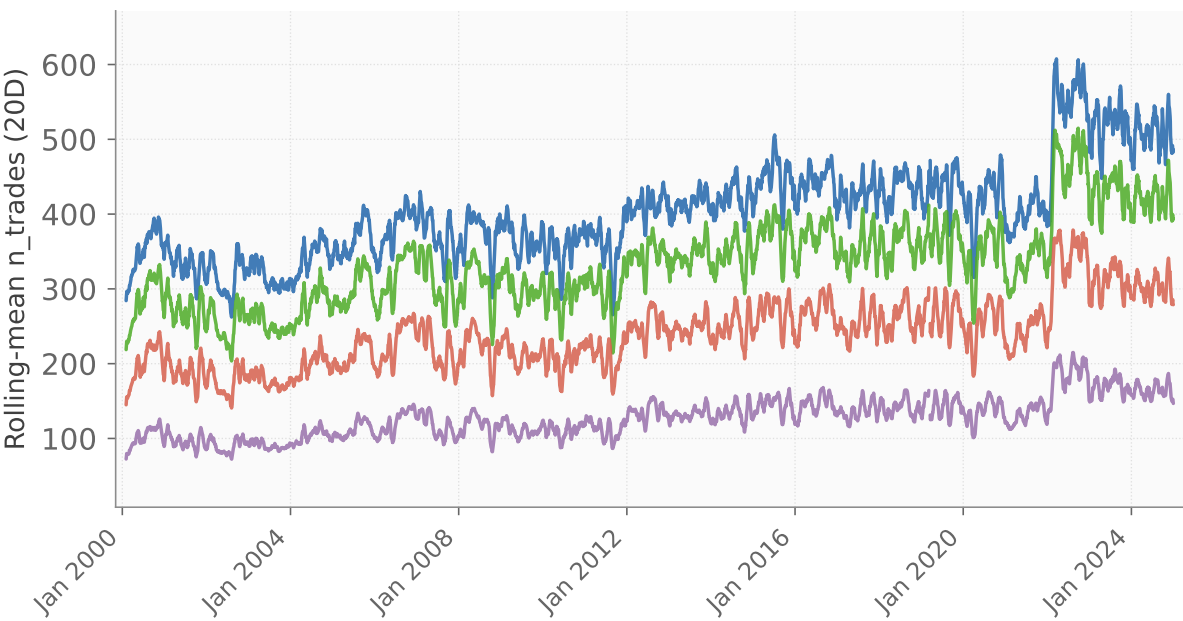
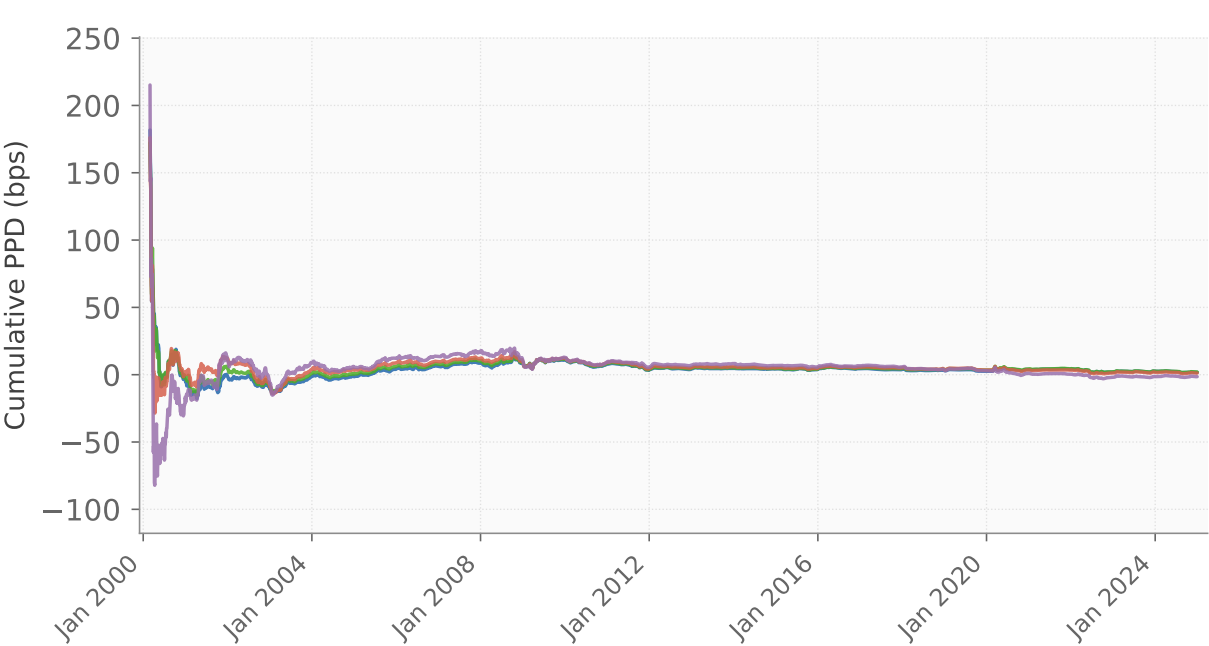
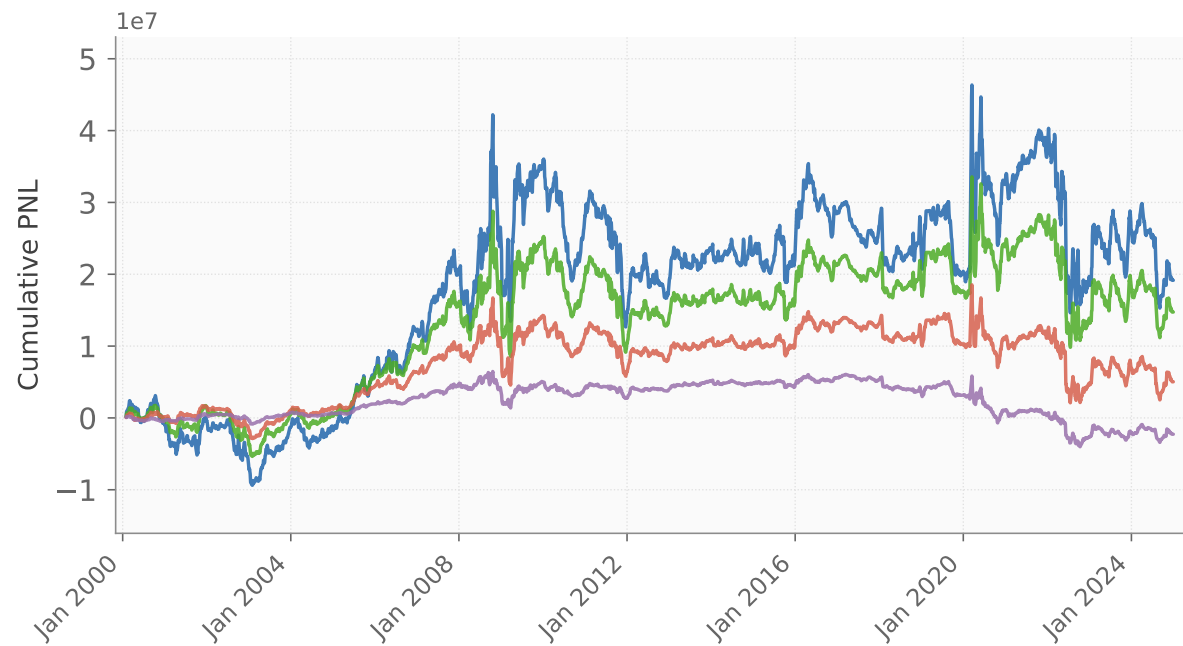
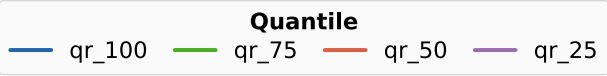
fret_OP5CL_RR | pret_ols | bs_m05_c250k — Temporal metrics



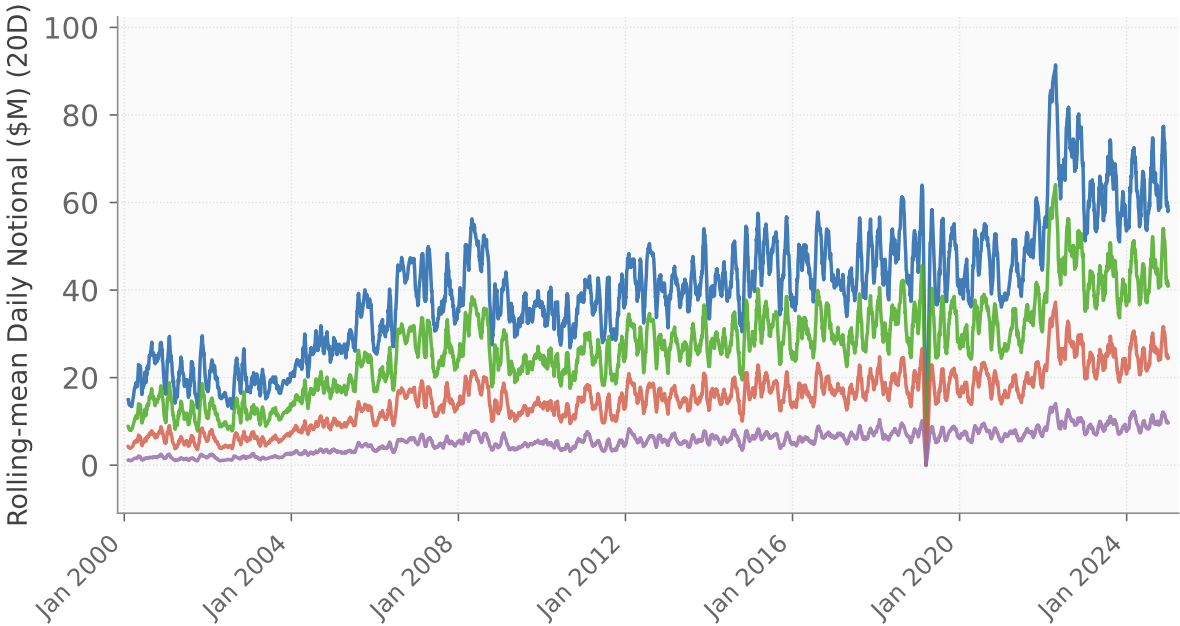
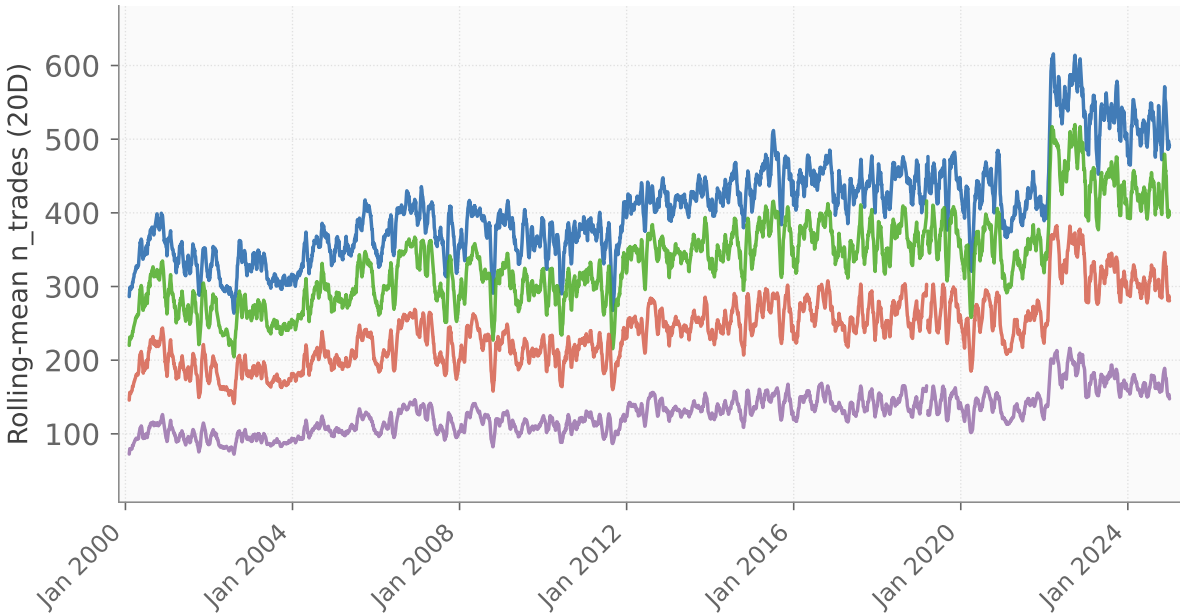
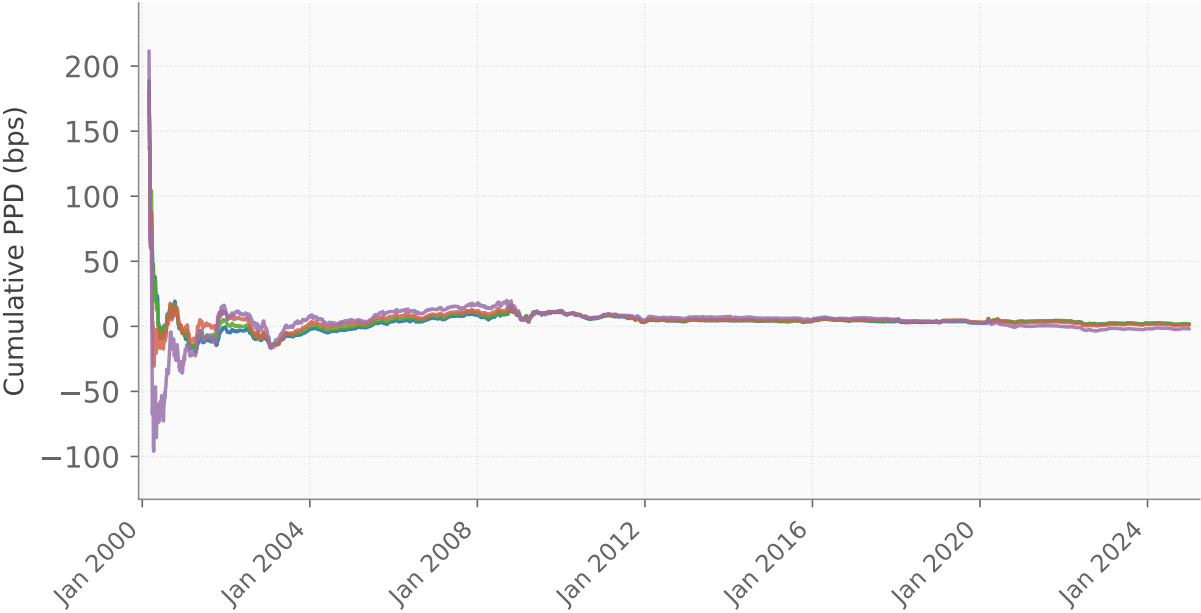
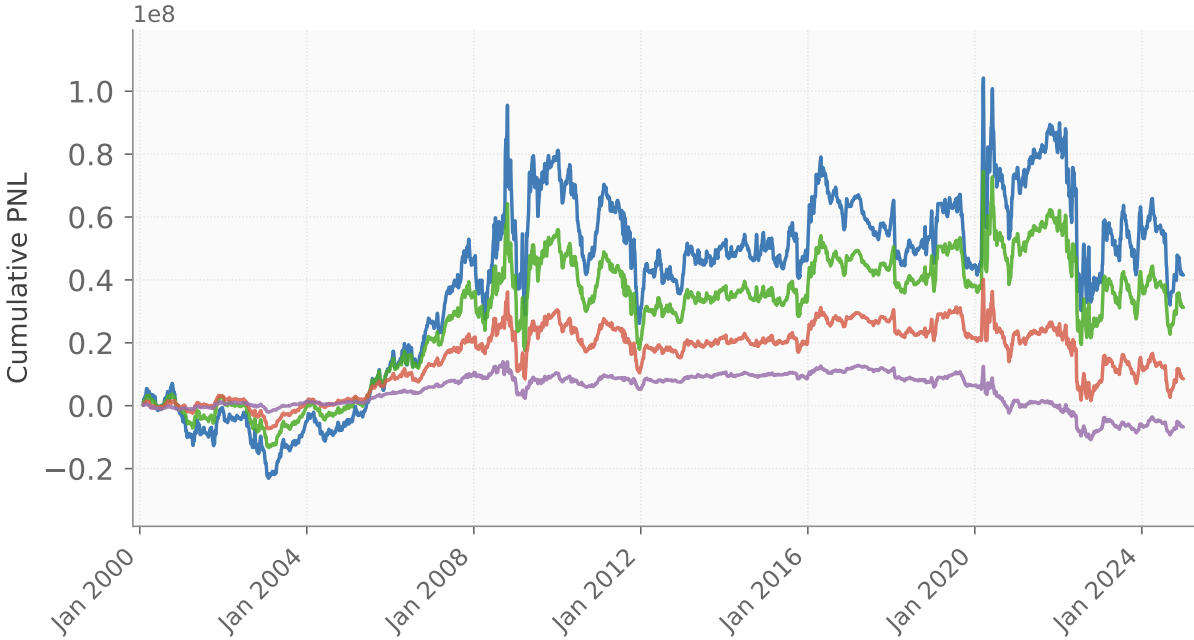
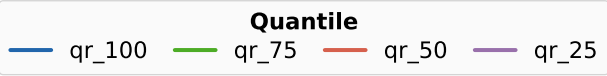
fret_OP5CL_RR | pret_ols | bs_m1_c500k — Temporal metrics



fret_OP5CL_RR | pret_oracle | bs_m025_c100k — Temporal metrics



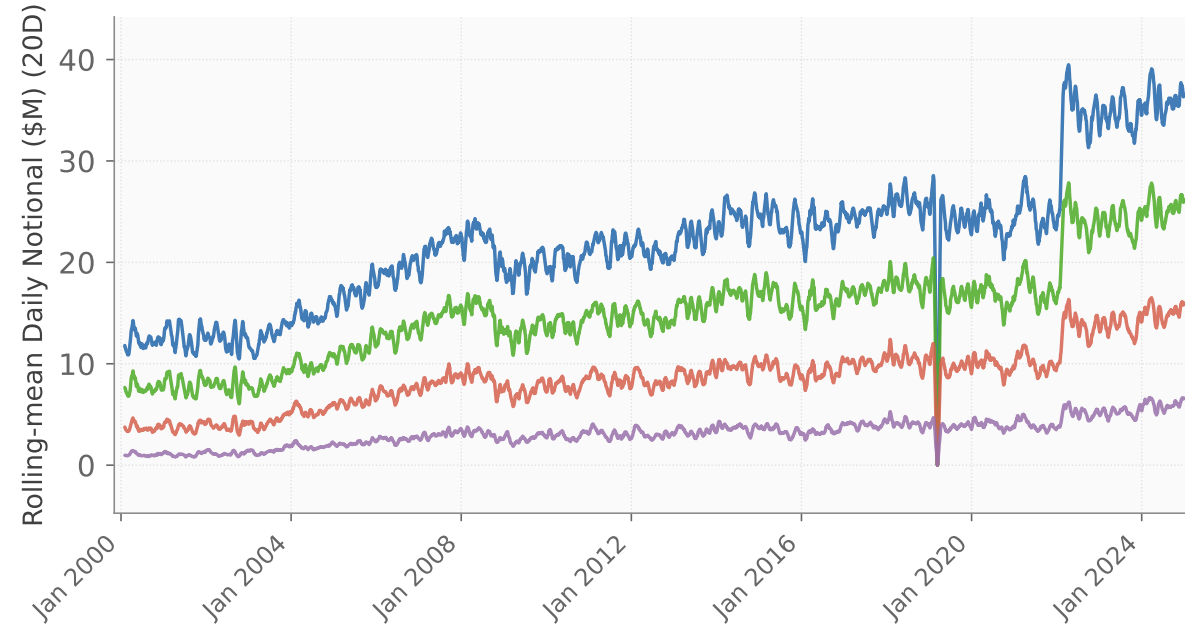
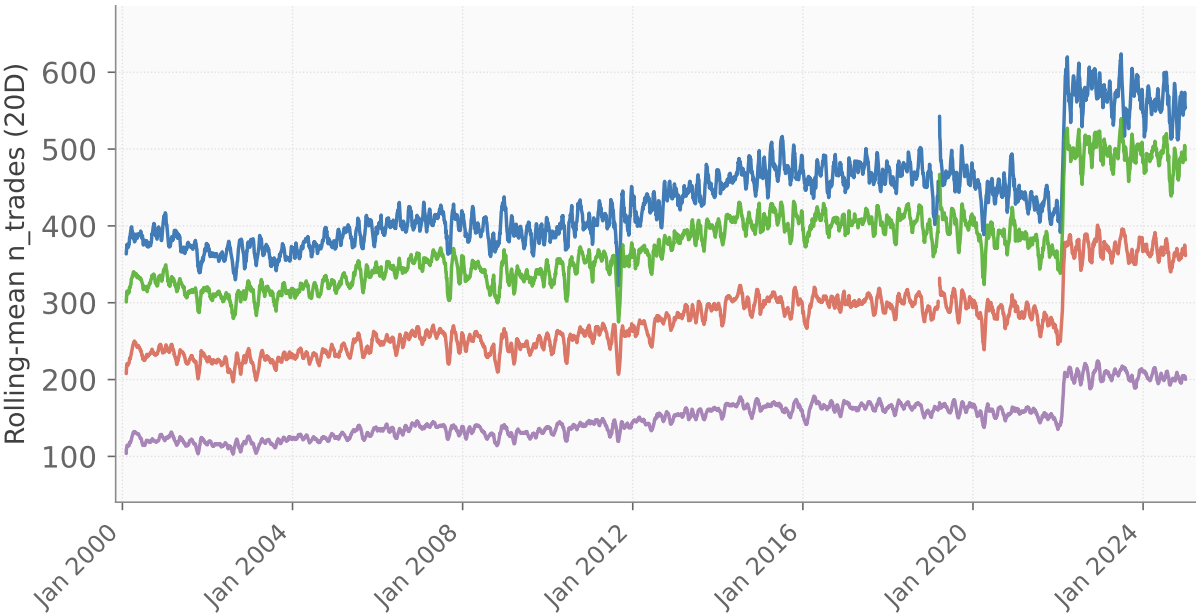
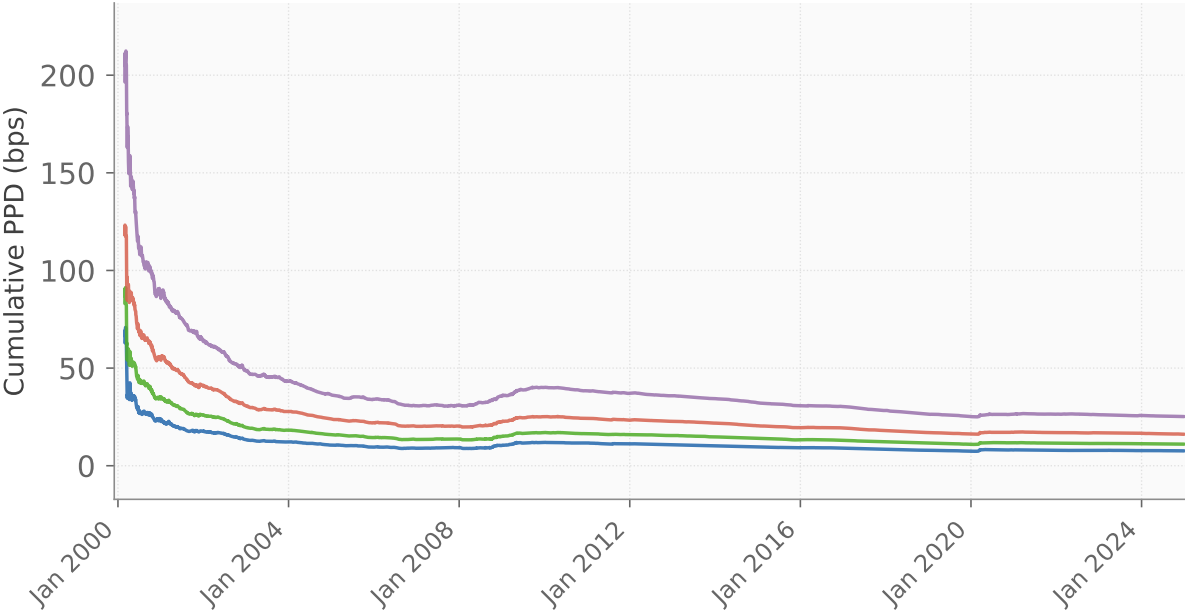
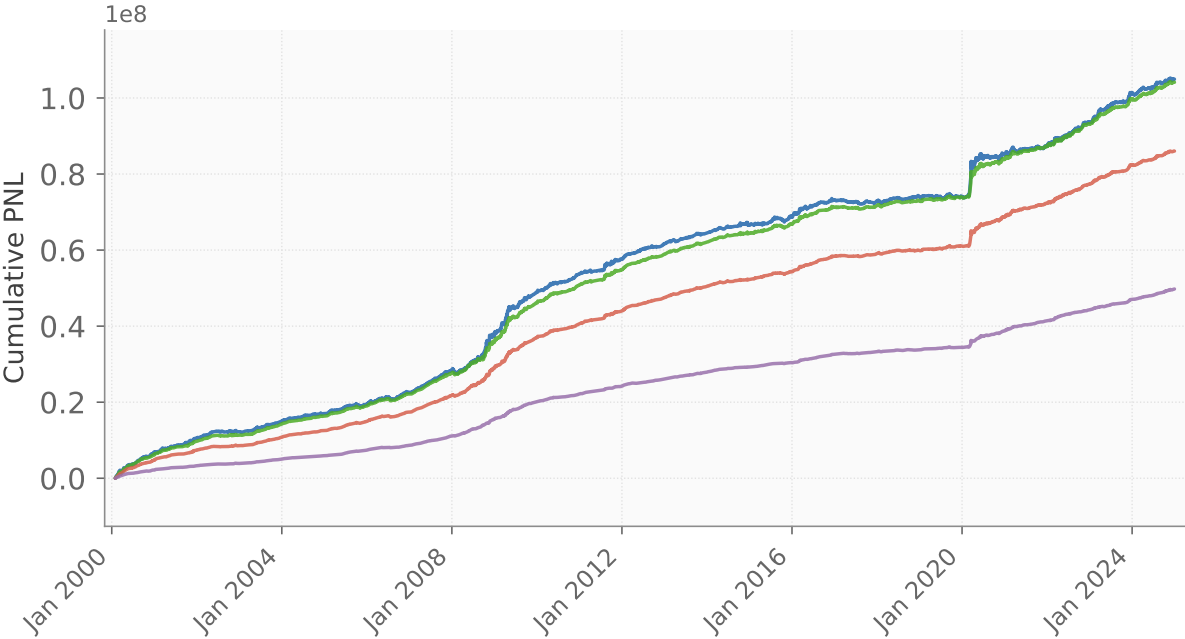
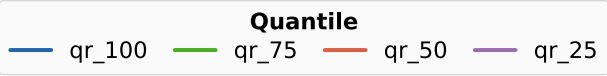
fret_OP5CL_RR | pret_oracle | bs_m05_c250k — Temporal metrics



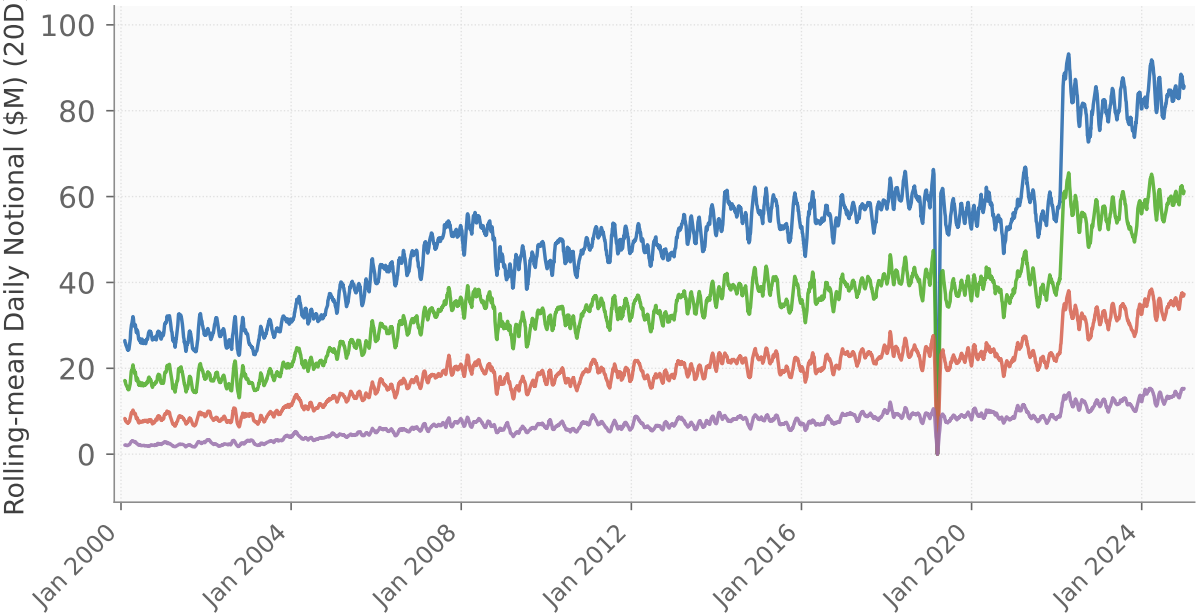
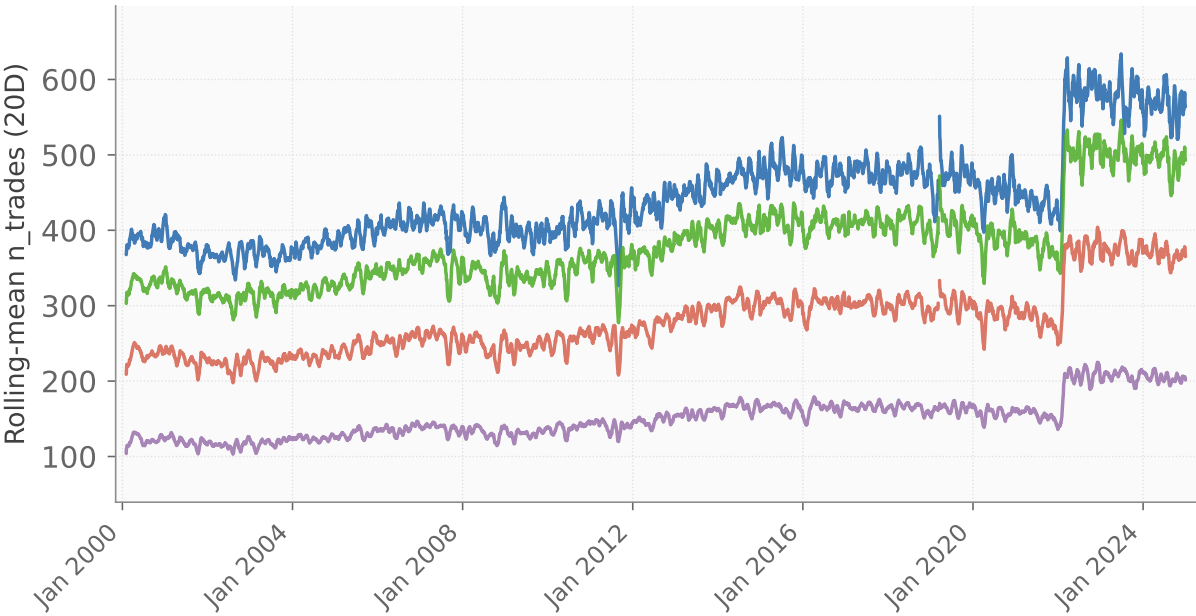
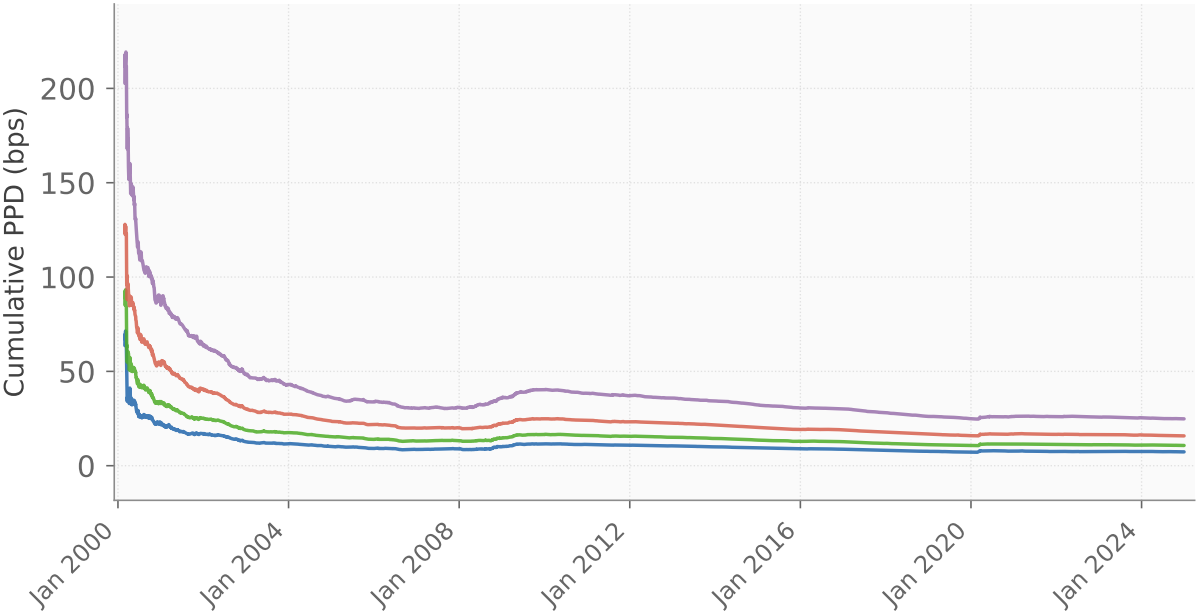
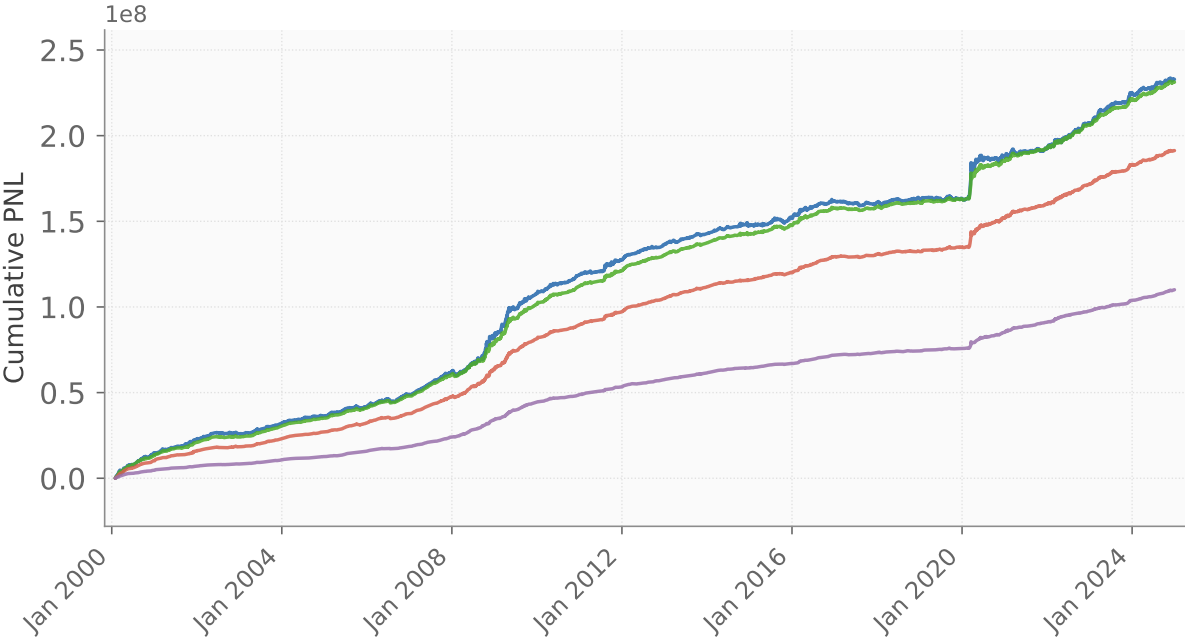
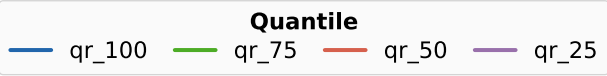
fret_OP5CL_RR | pret_oracle | bs_m1_c500k — Temporal metrics



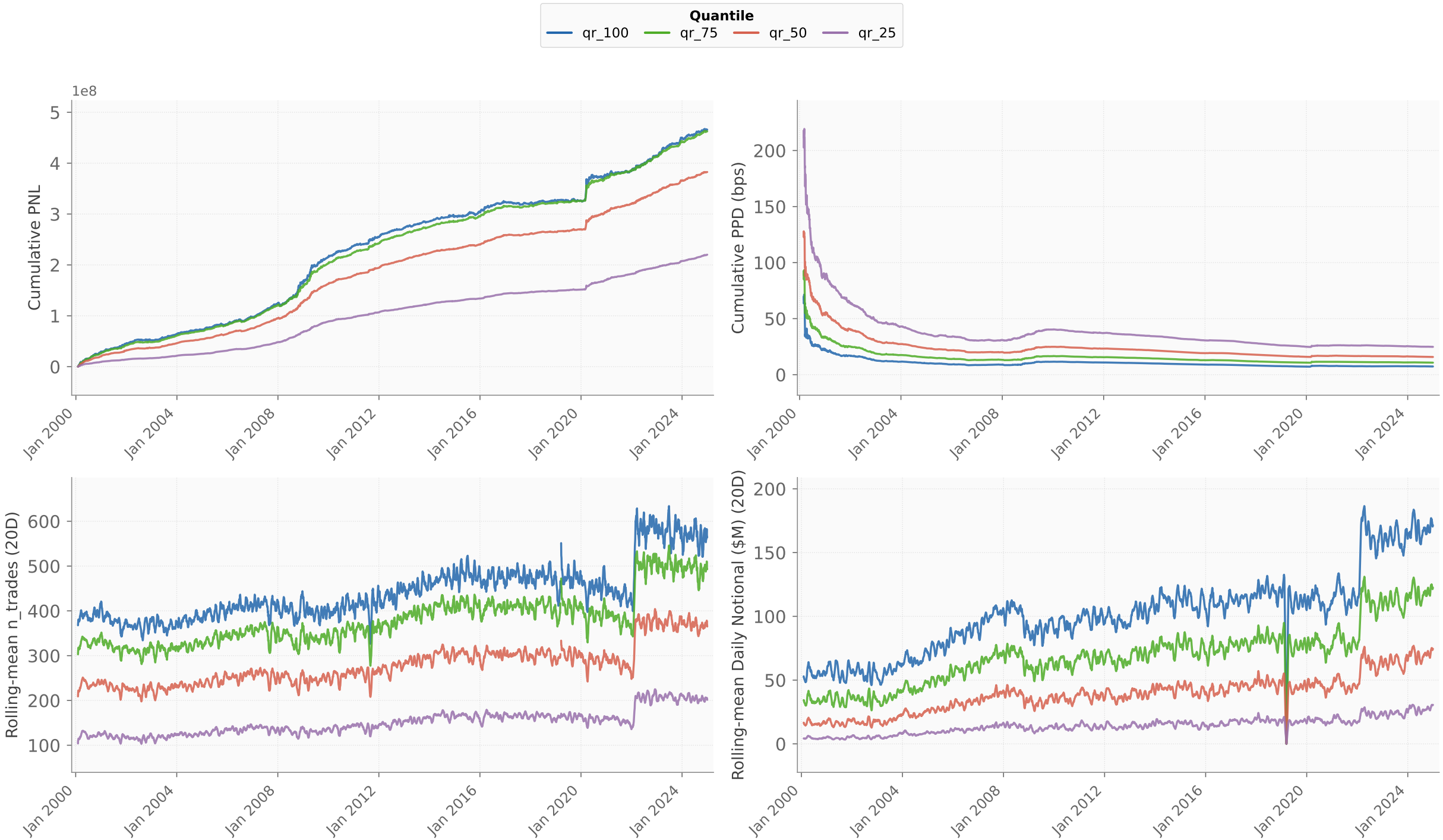
fret_OPCL_MR | pret_base | bs_m025_c100k — Temporal metrics



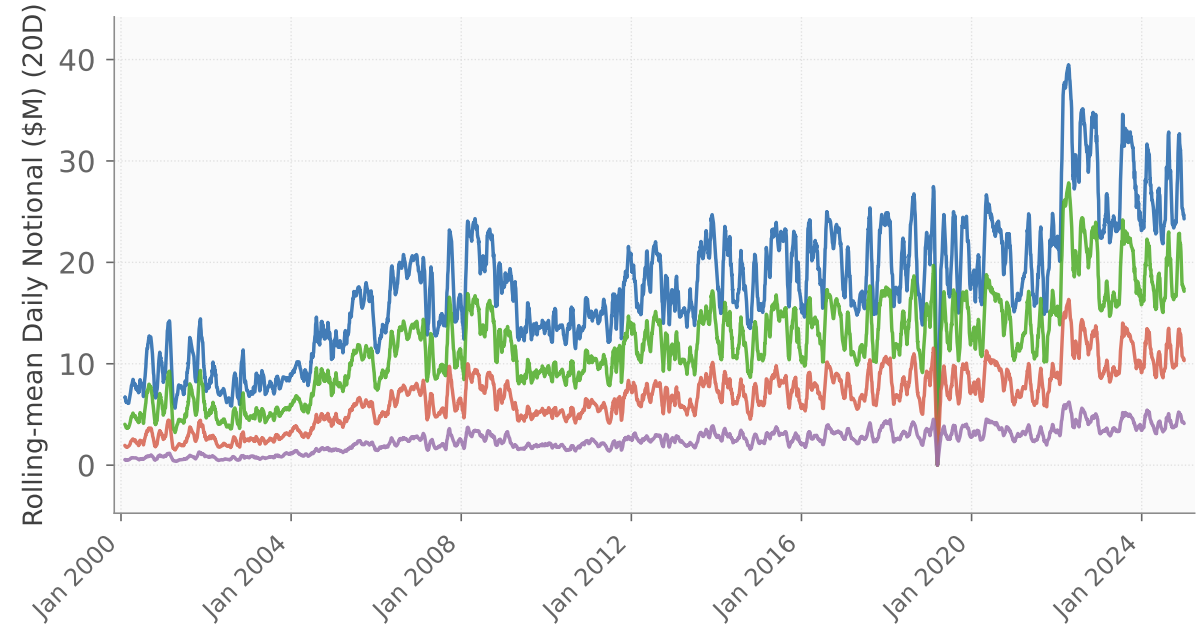
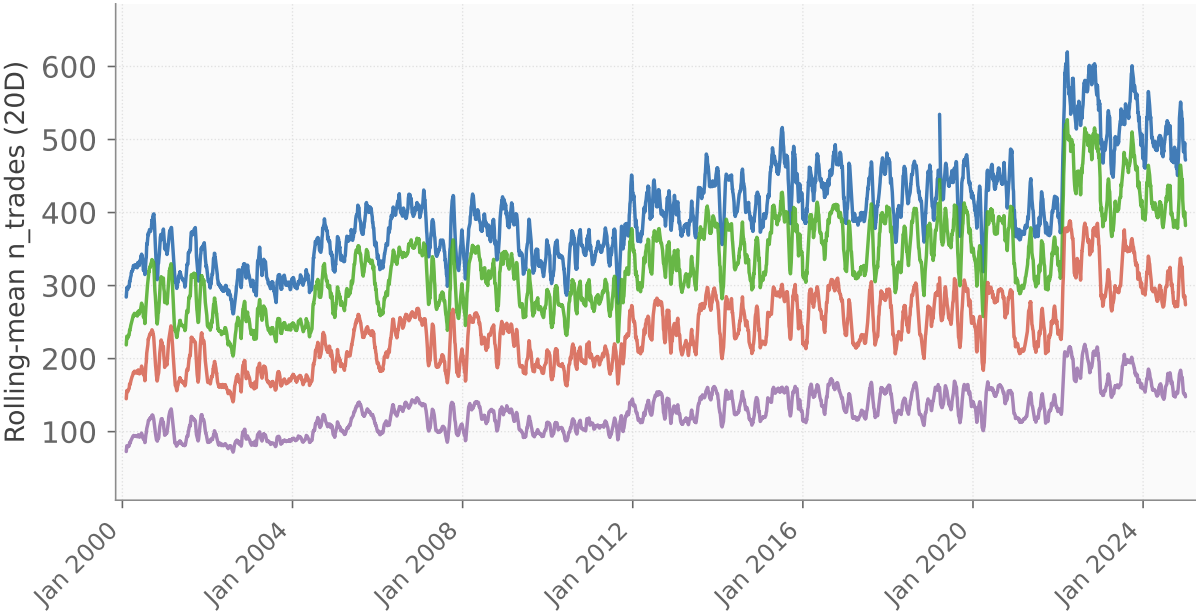
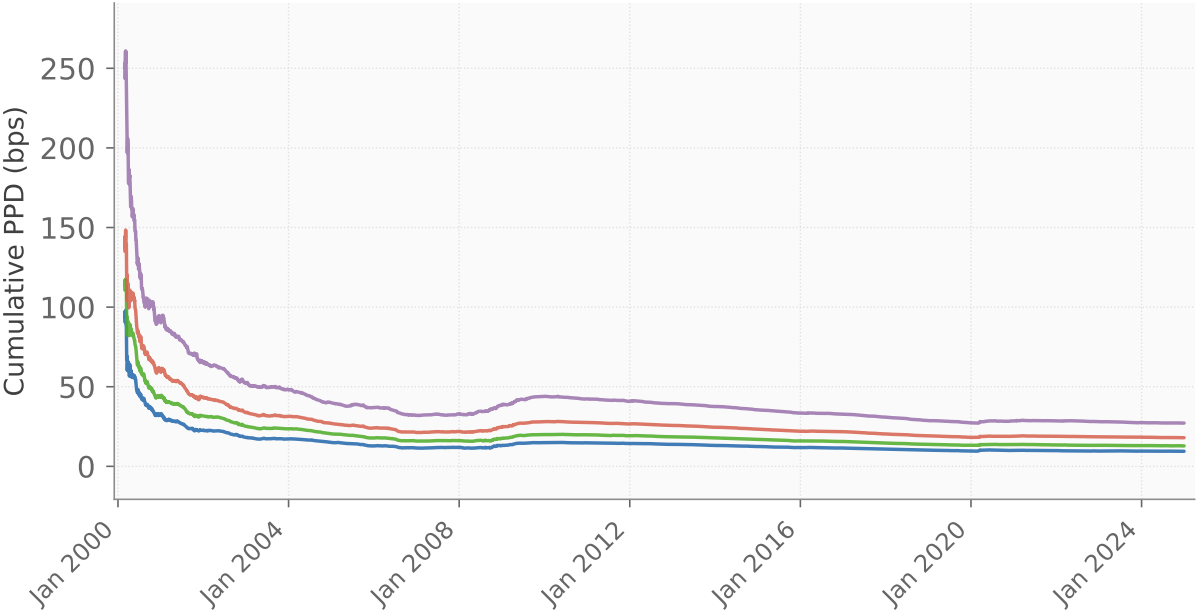
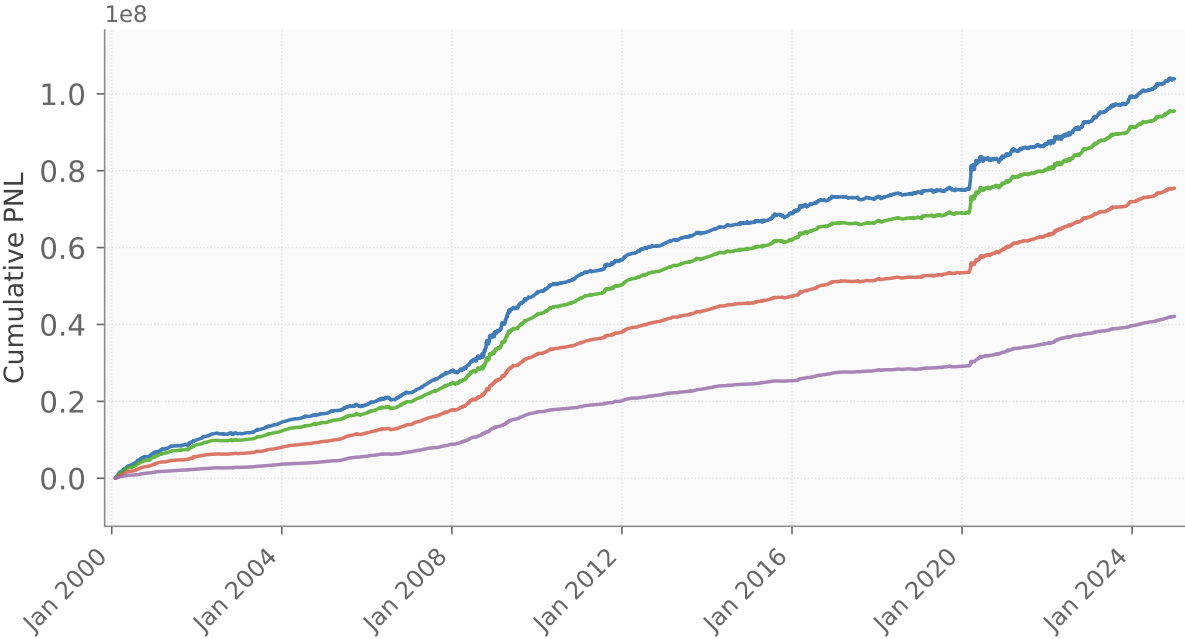
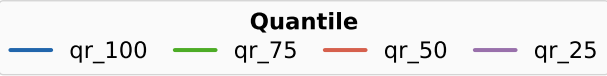
fret_OPCL_MR | pret_base | bs_m05_c250k — Temporal metrics



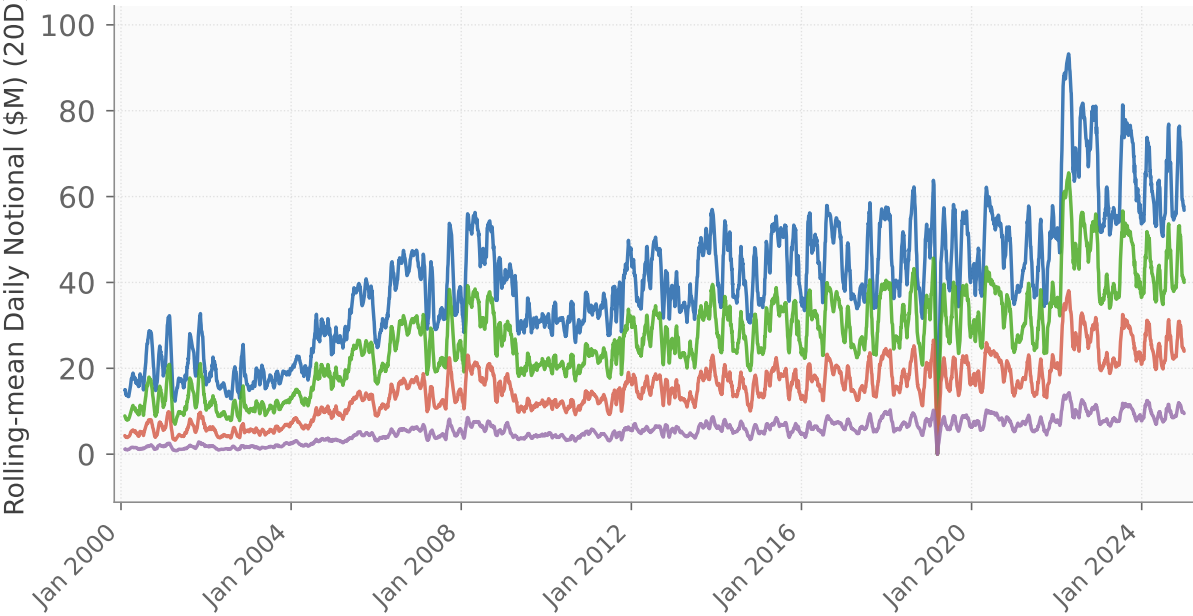
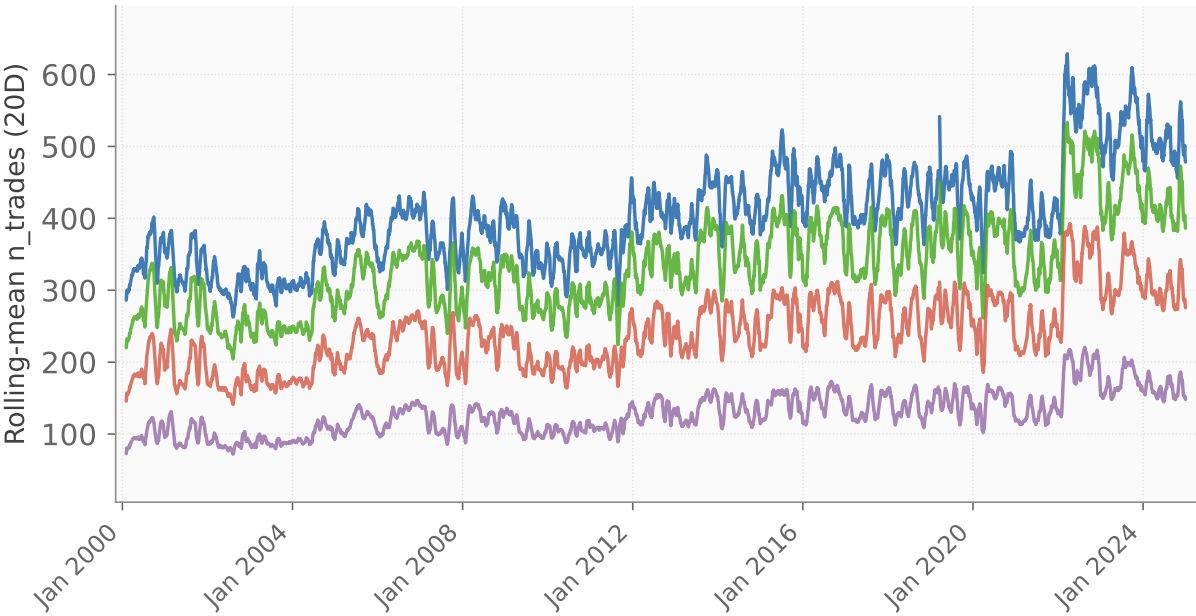
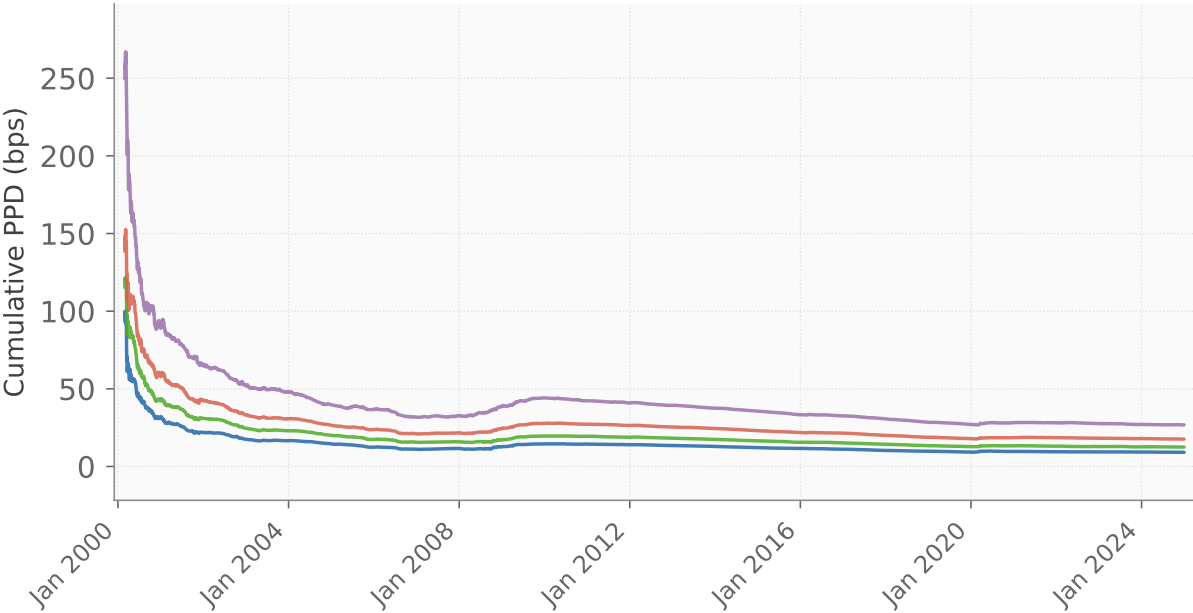
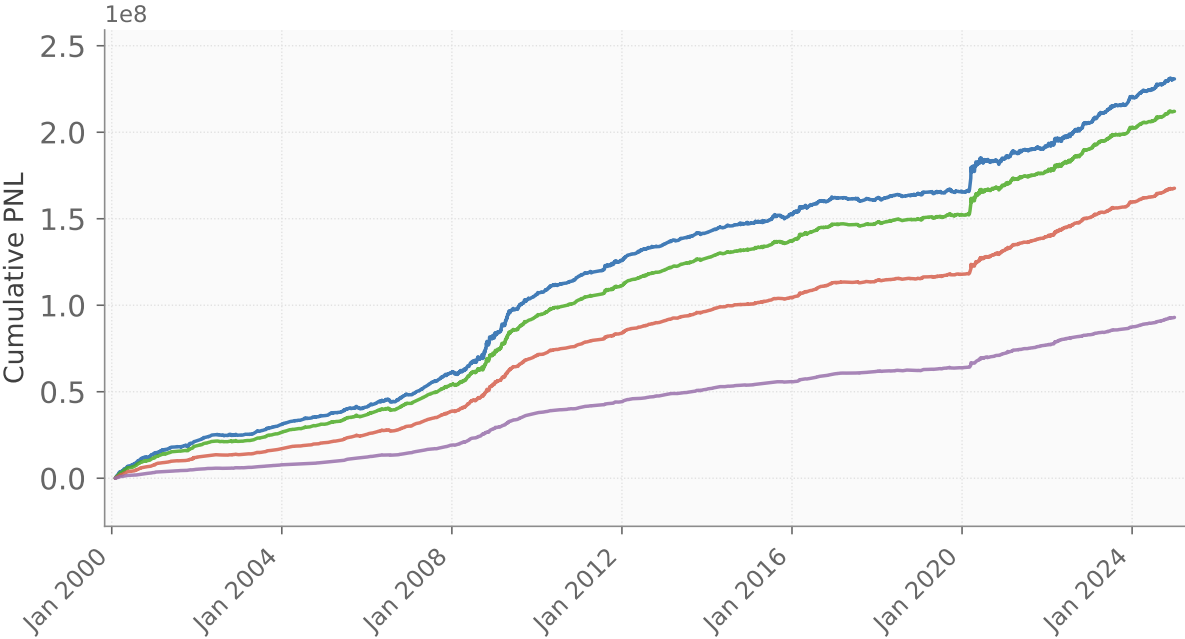
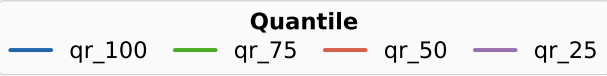
fret_OPCL_MR | pret_base | bs_m1_c500k — Temporal metrics



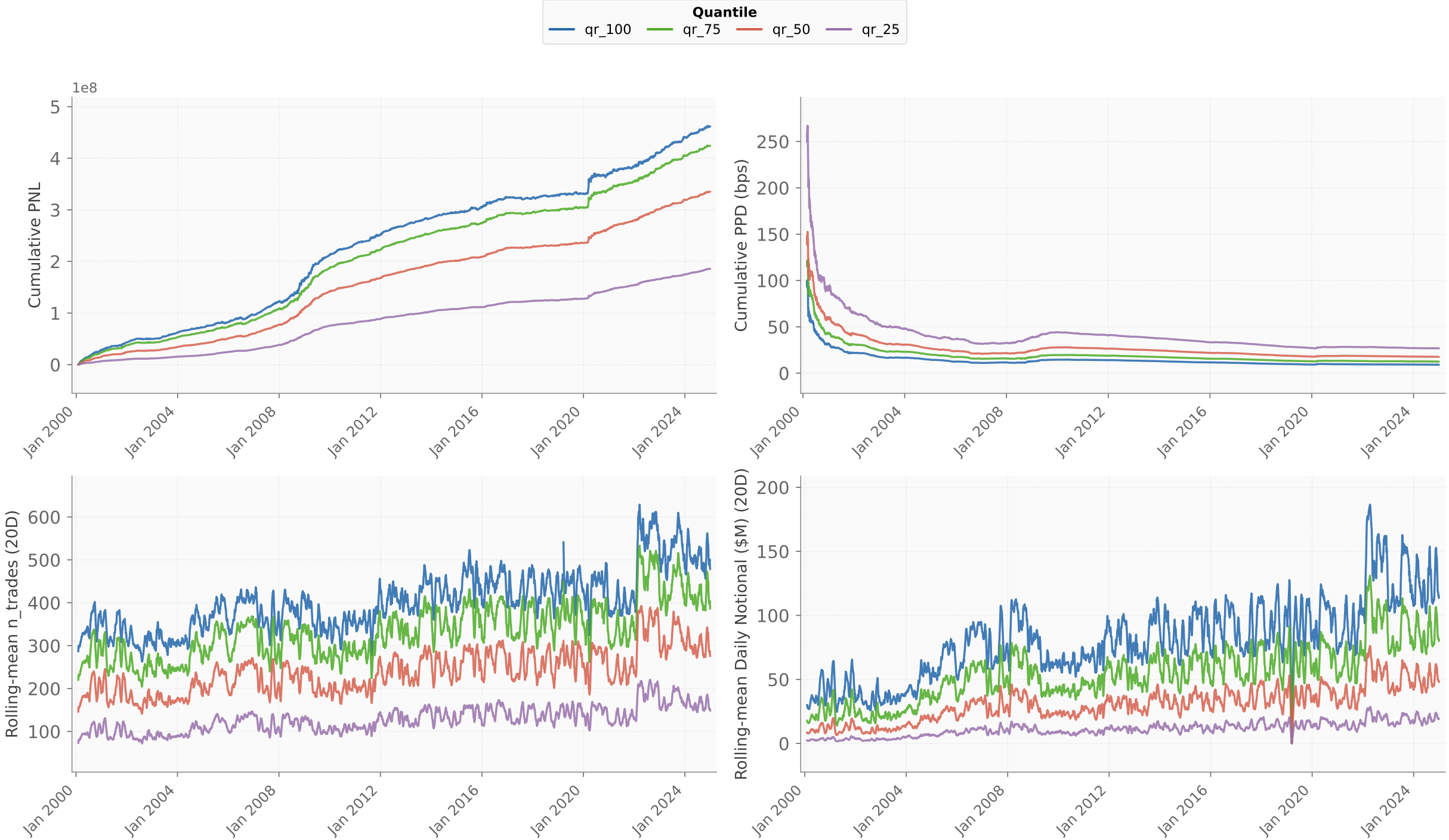
fret_OPCL_MR | pret_clus | bs_m025_c100k — Temporal metrics



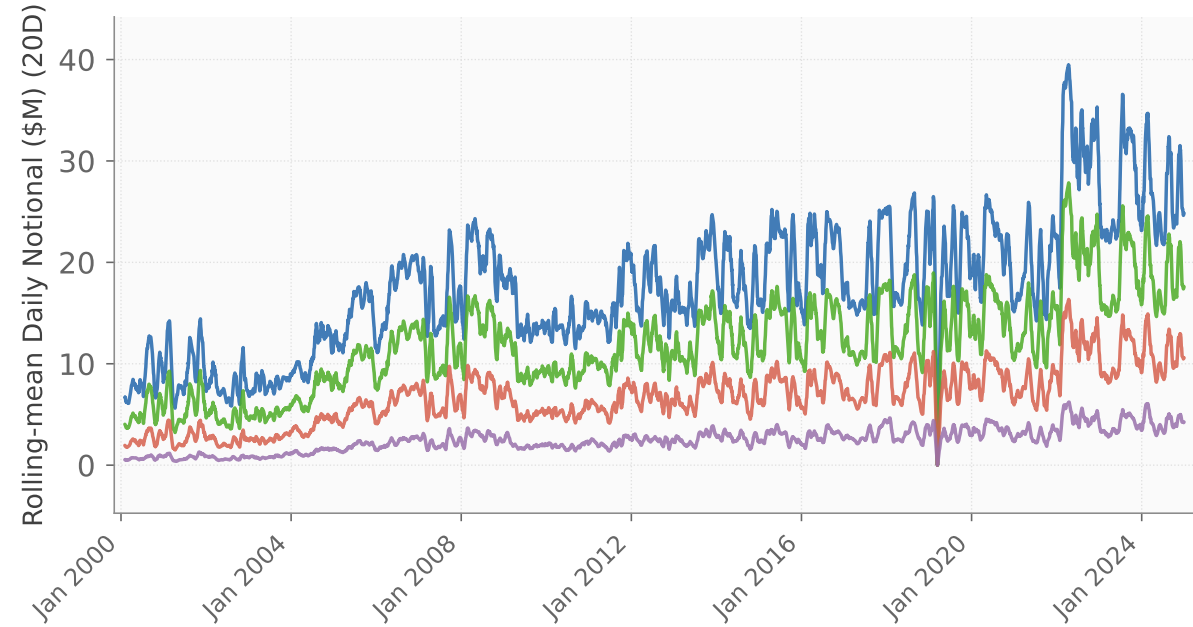
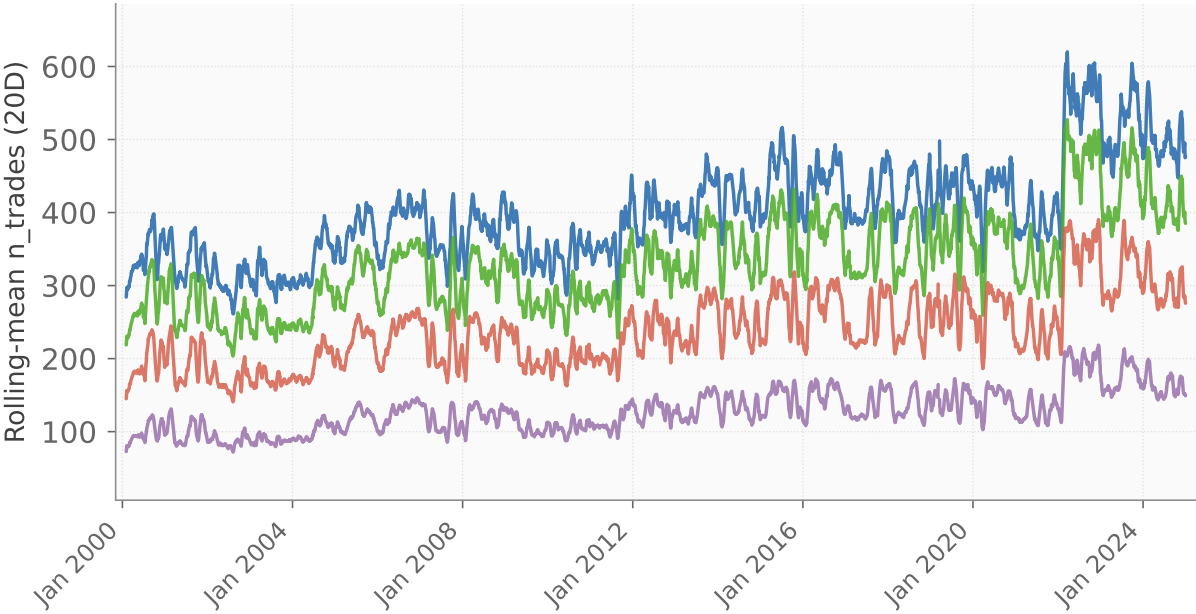
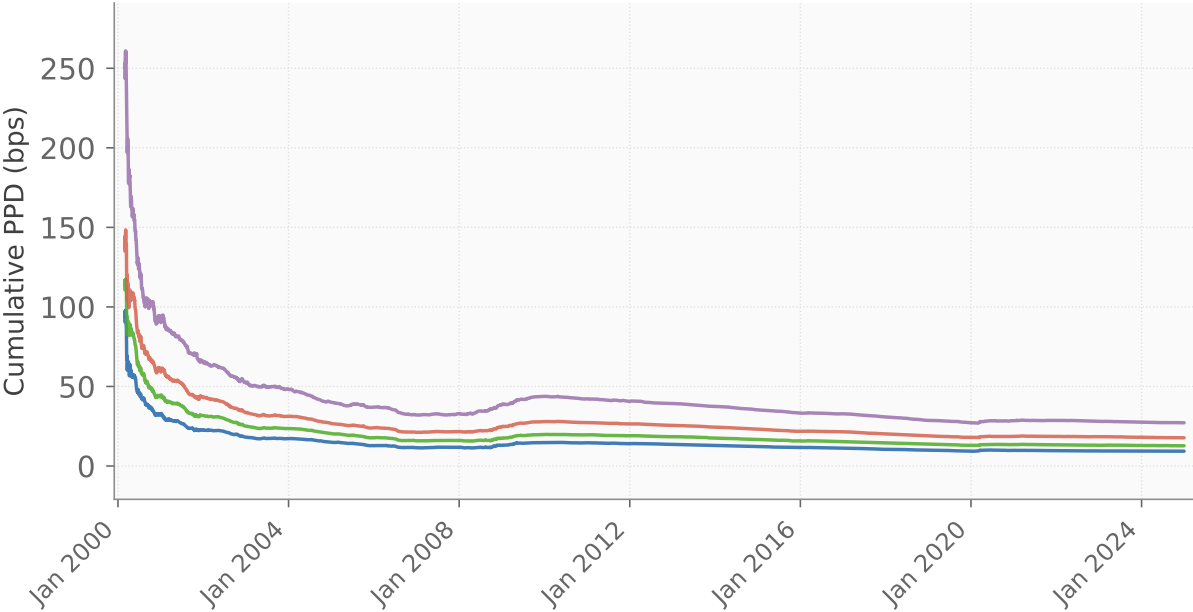
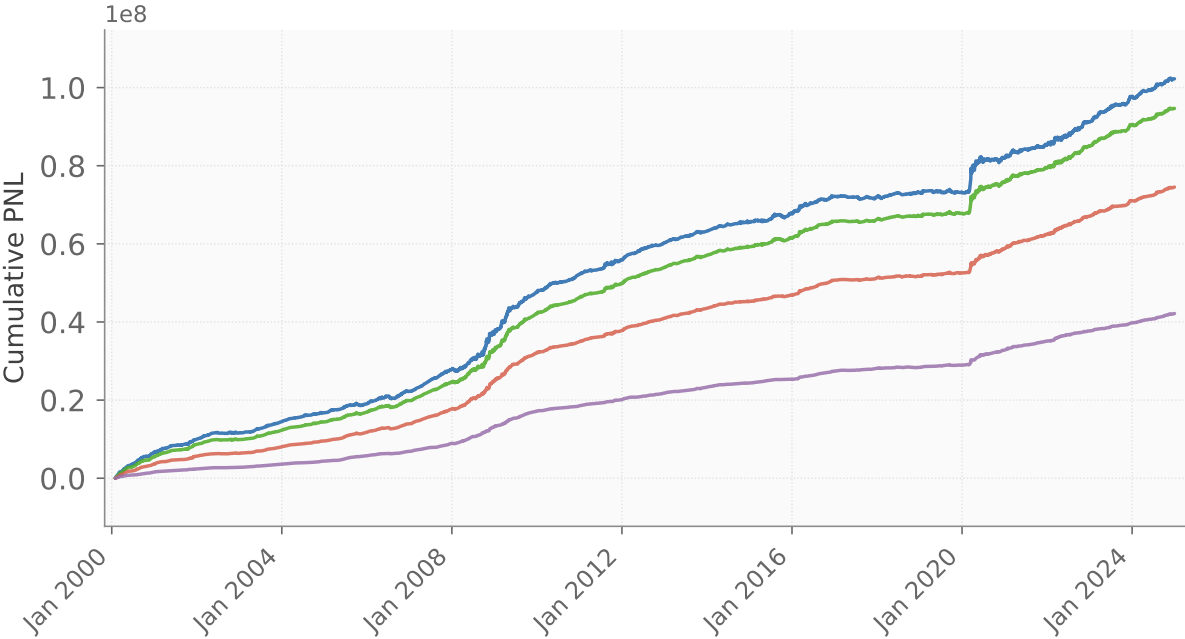
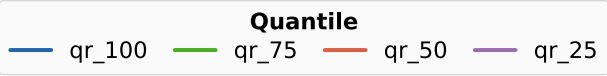
fret_OPCL_MR | pret_clus | bs_m05_c250k — Temporal metrics



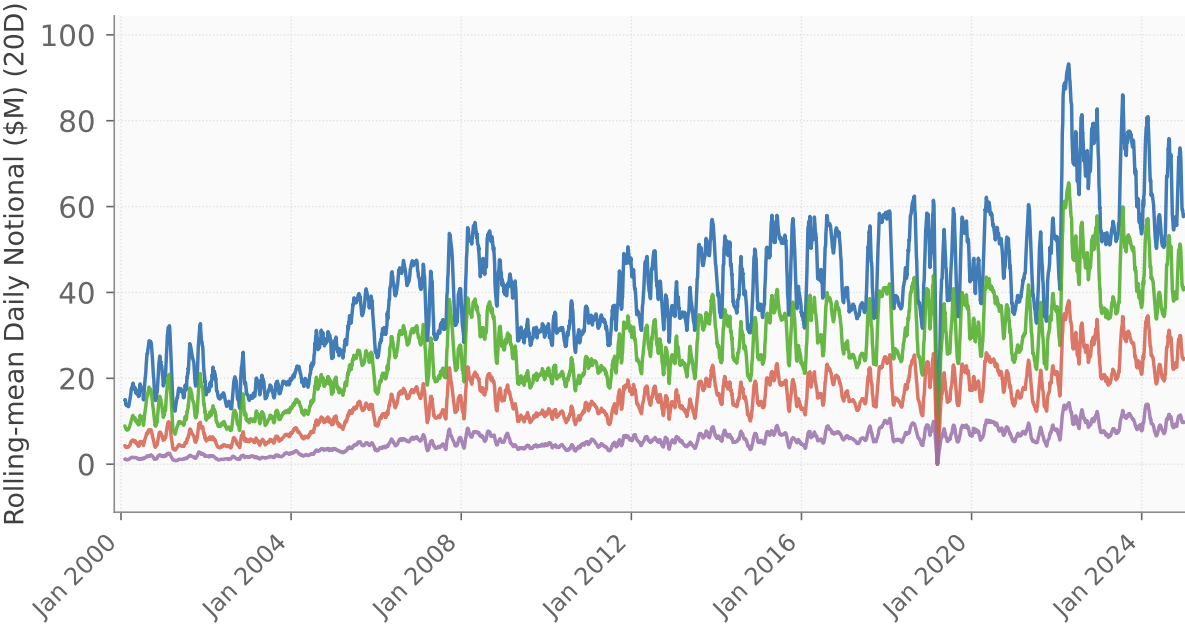
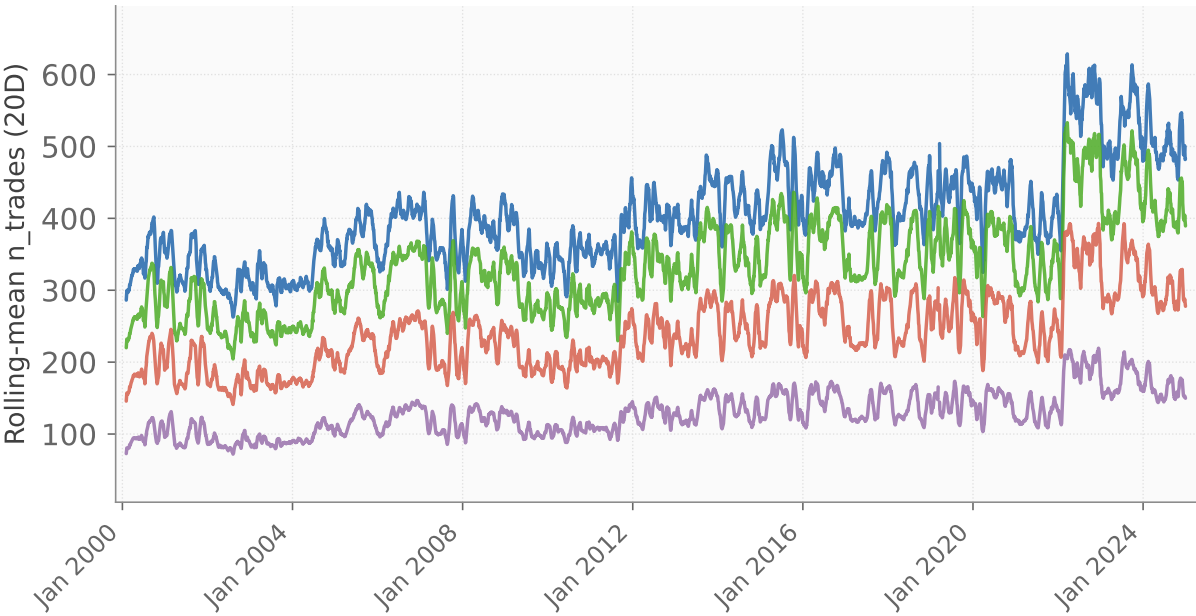
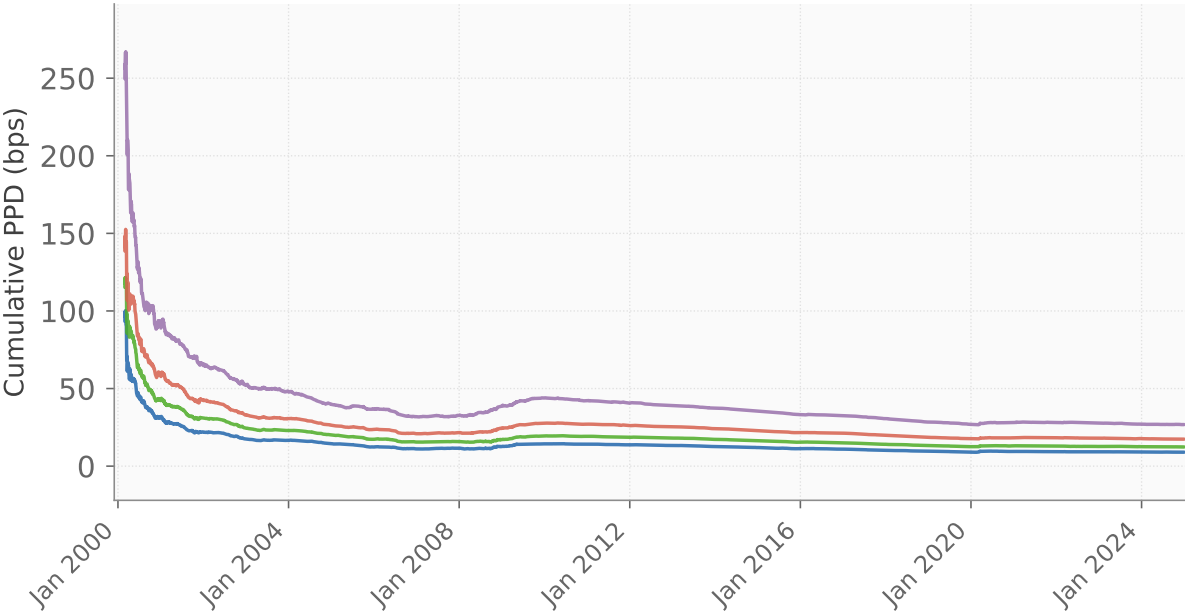
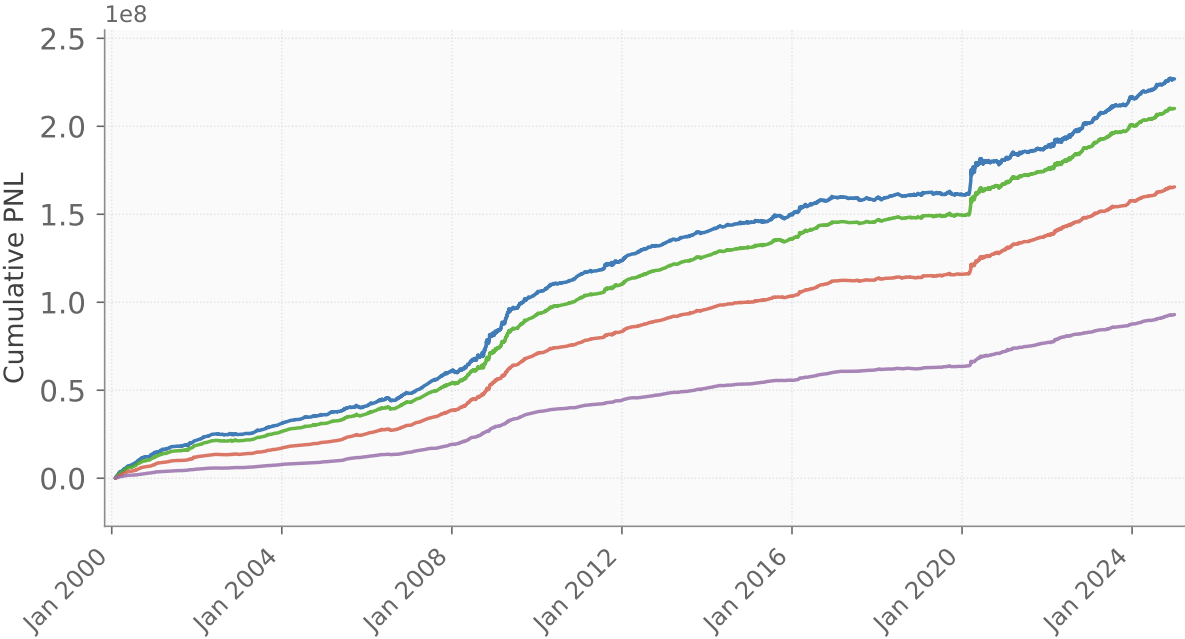
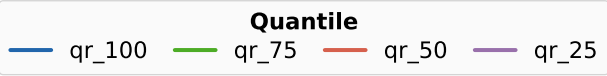
fret_OPCL_MR | pret_clus | bs_m1_c500k — Temporal metrics



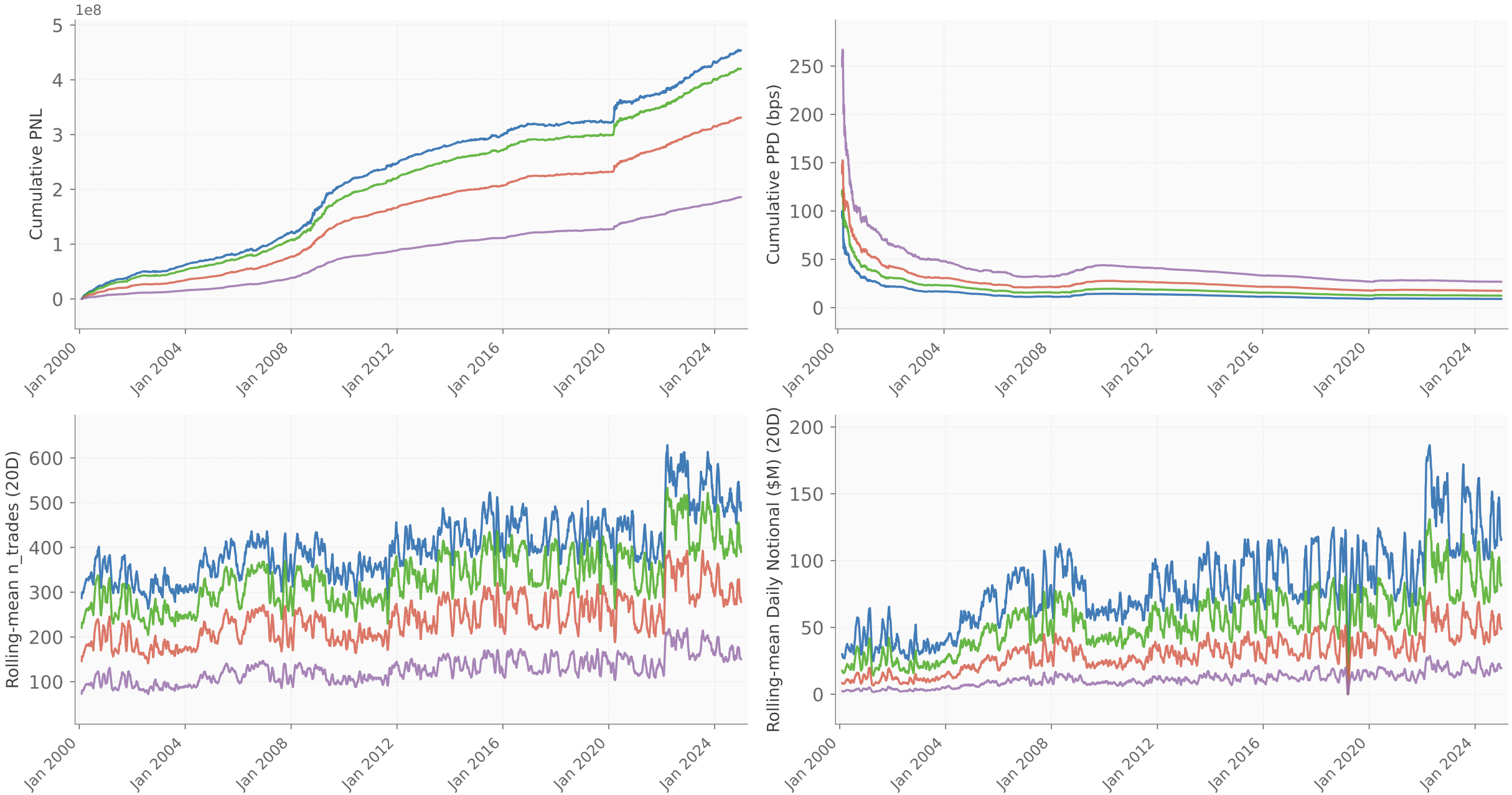
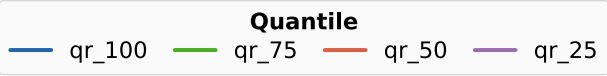
fret_OPCL_MR | pret_har | bs_m025_c100k — Temporal metrics



fret_OPCL_MR | pret_har | bs_m05_c250k — Temporal metrics



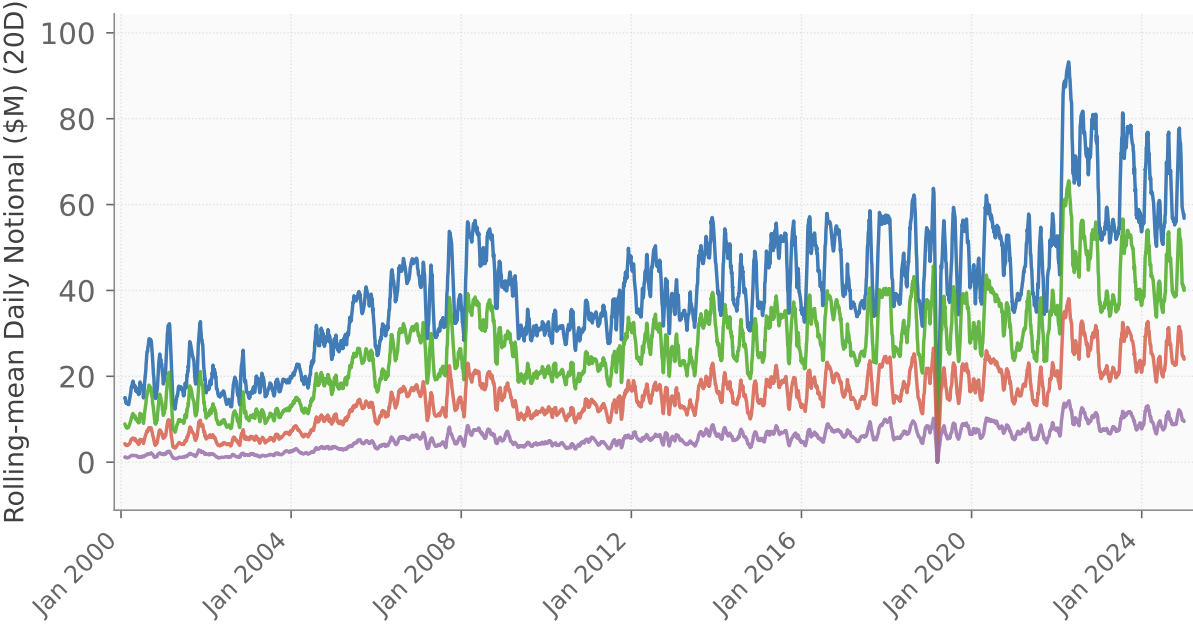
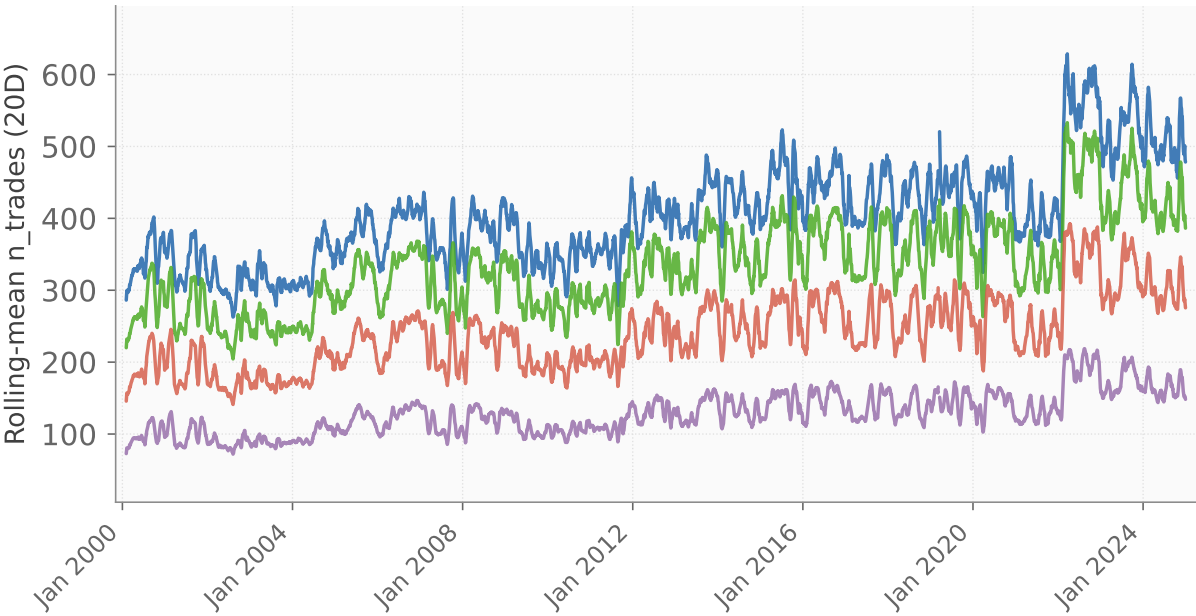
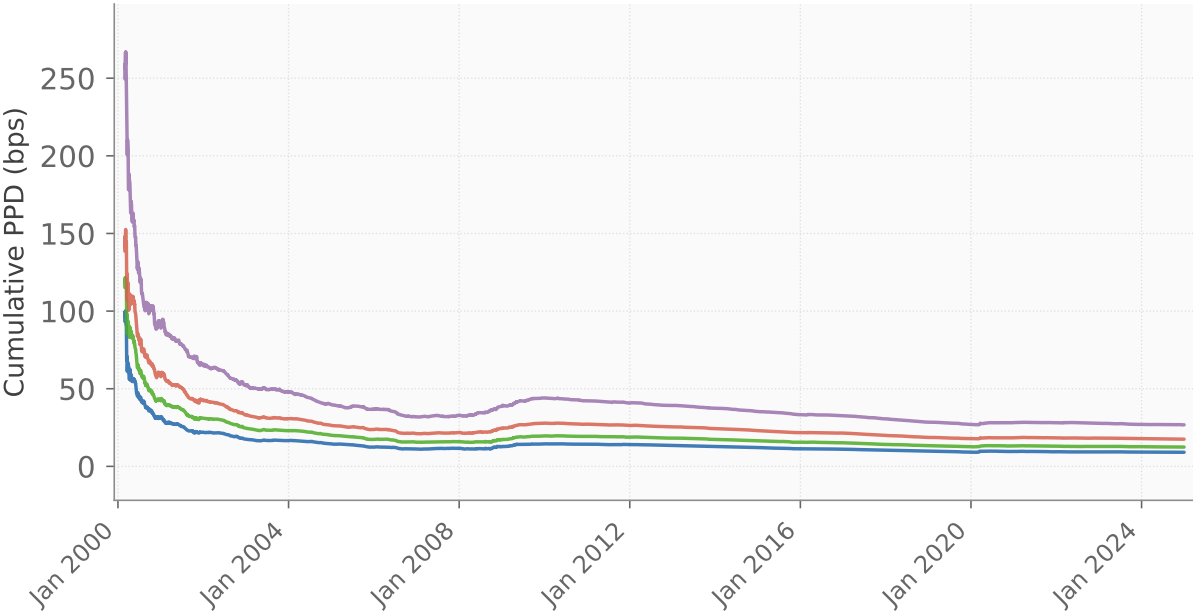
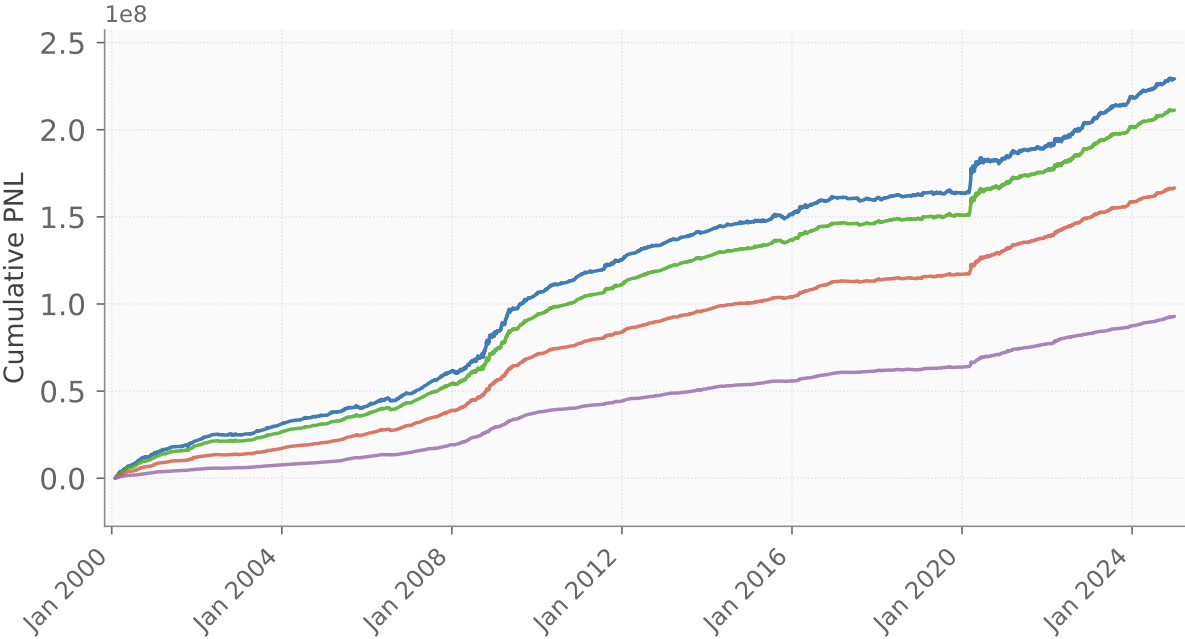
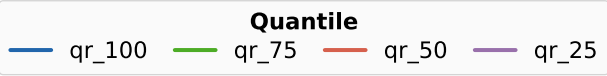
fret_OPCL_MR | pret_har | bs_m1_c500k — Temporal metrics



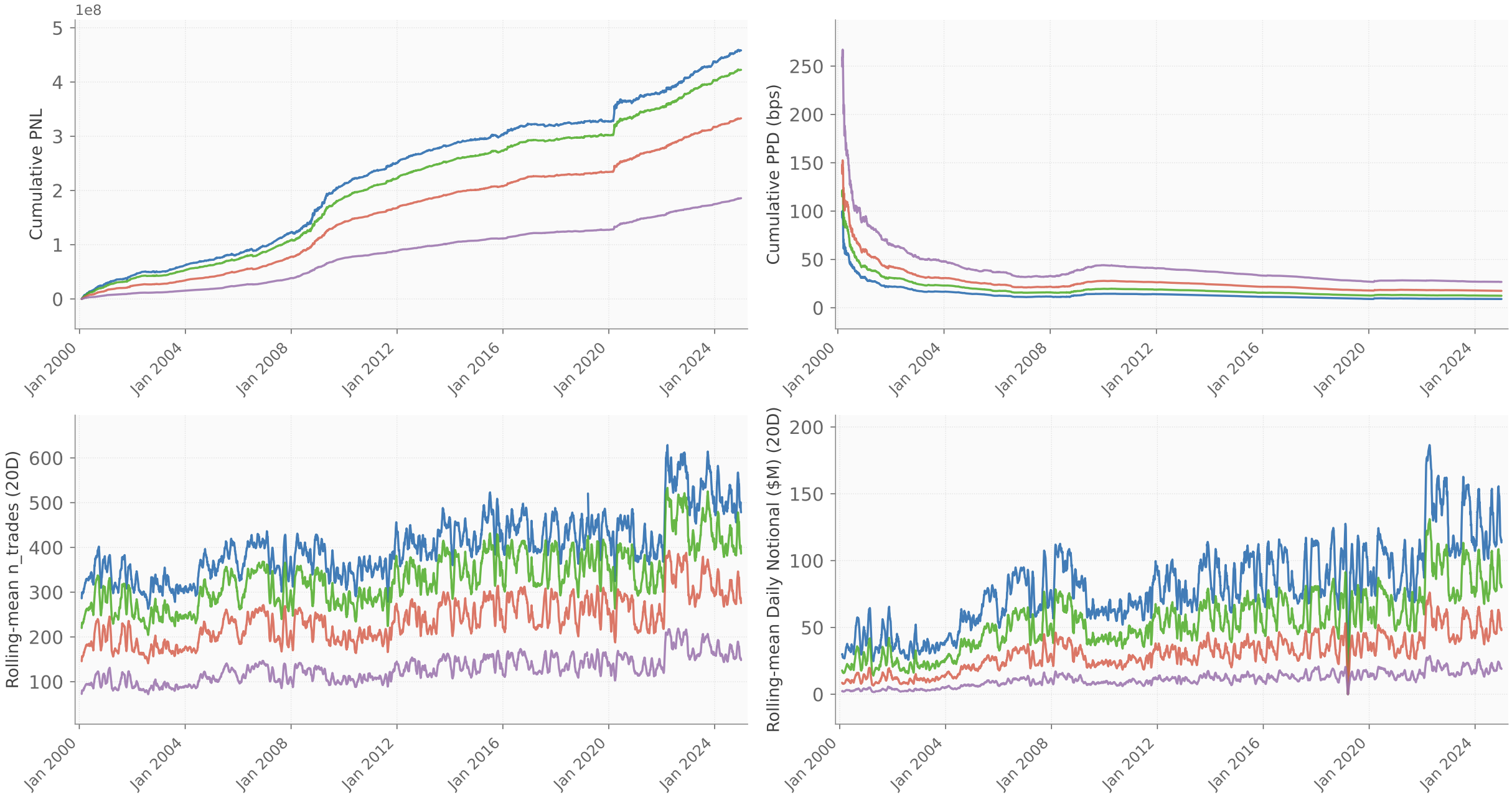
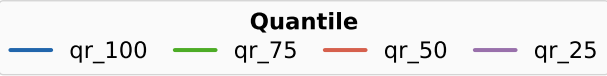
fret_OPCL_MR | pret_ols | bs_m025_c100k — Temporal metrics



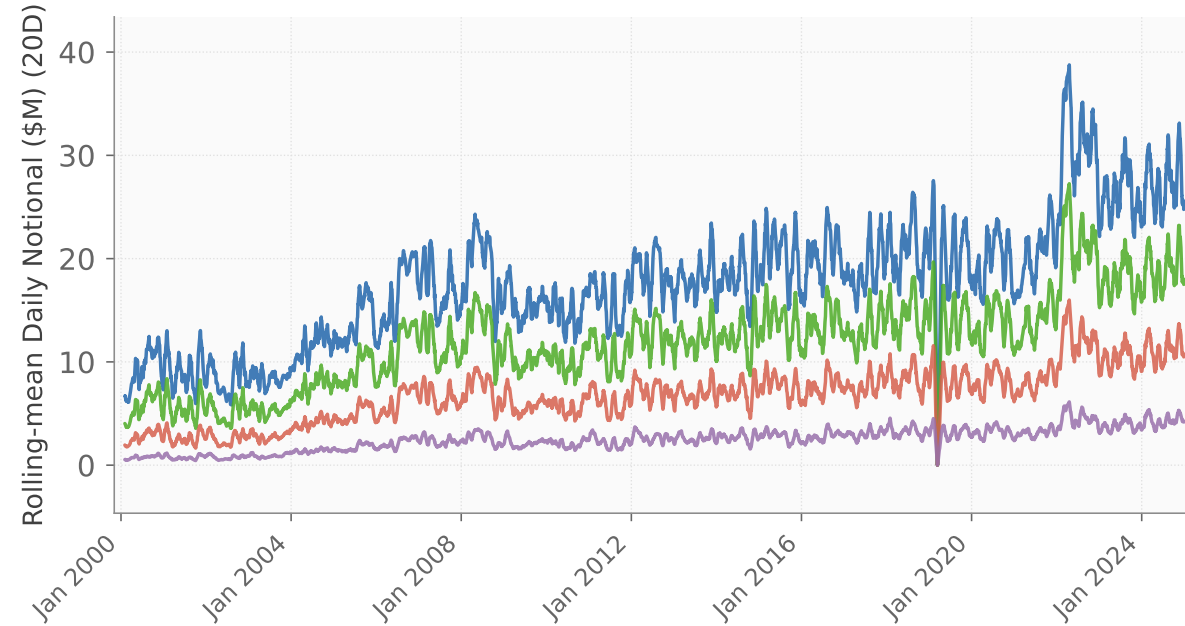
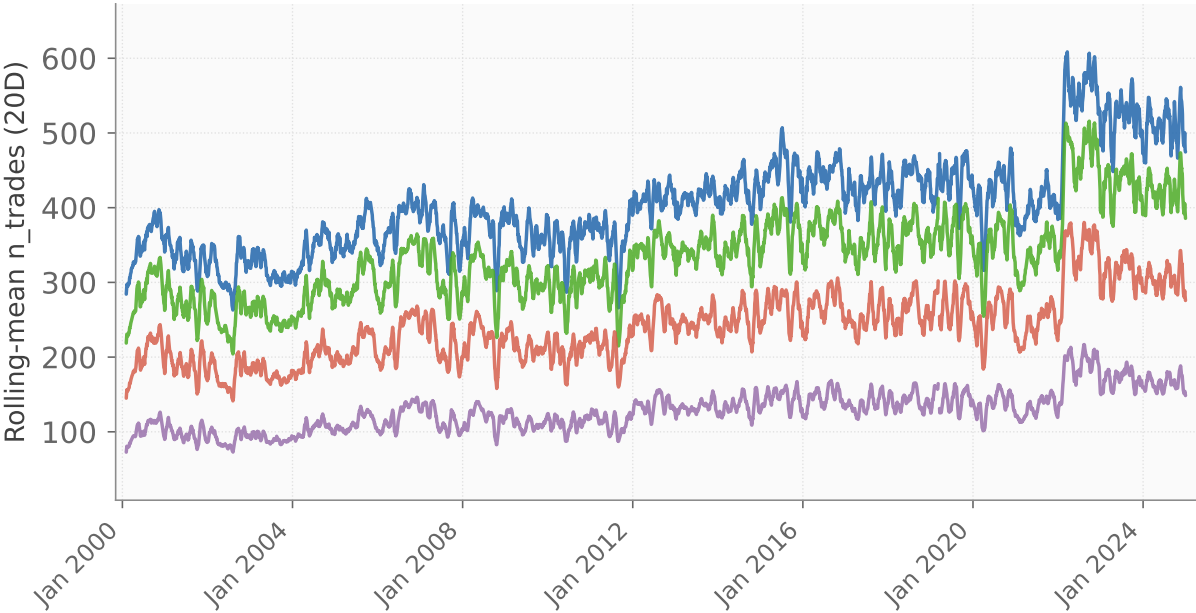
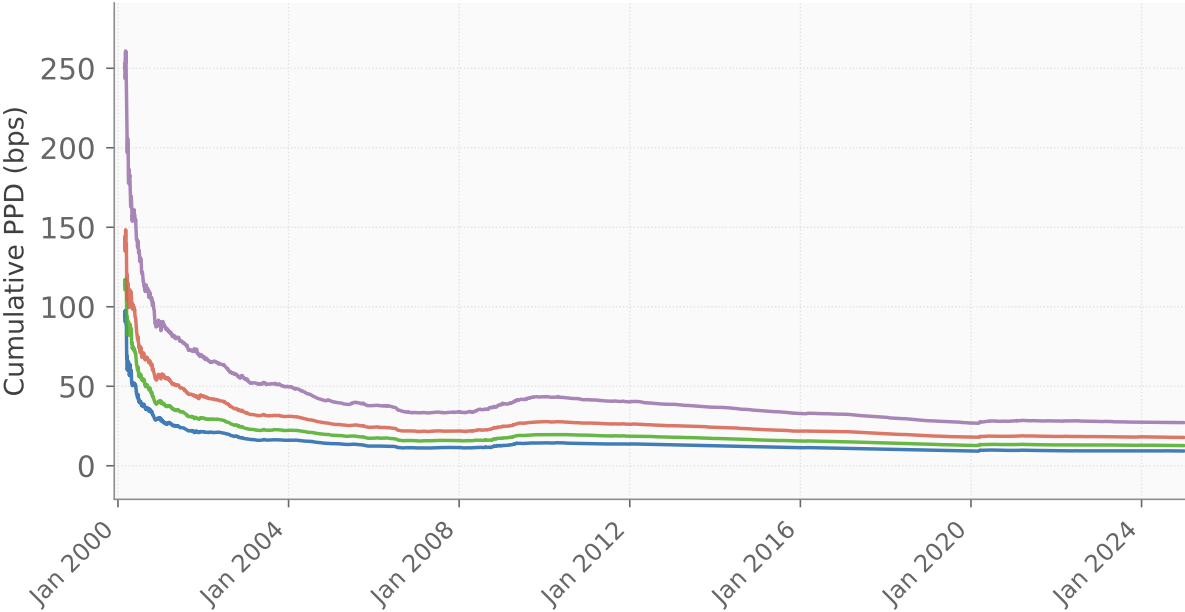
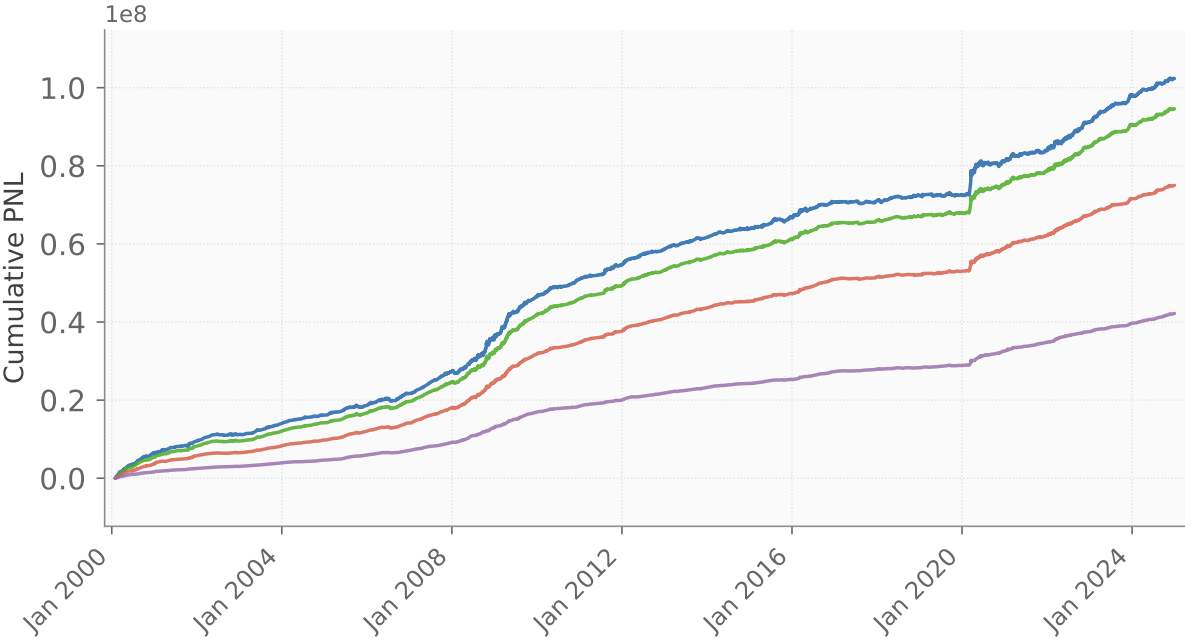
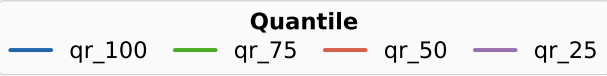
fret_OPCL_MR | pret_ols | bs_m05_c250k — Temporal metrics



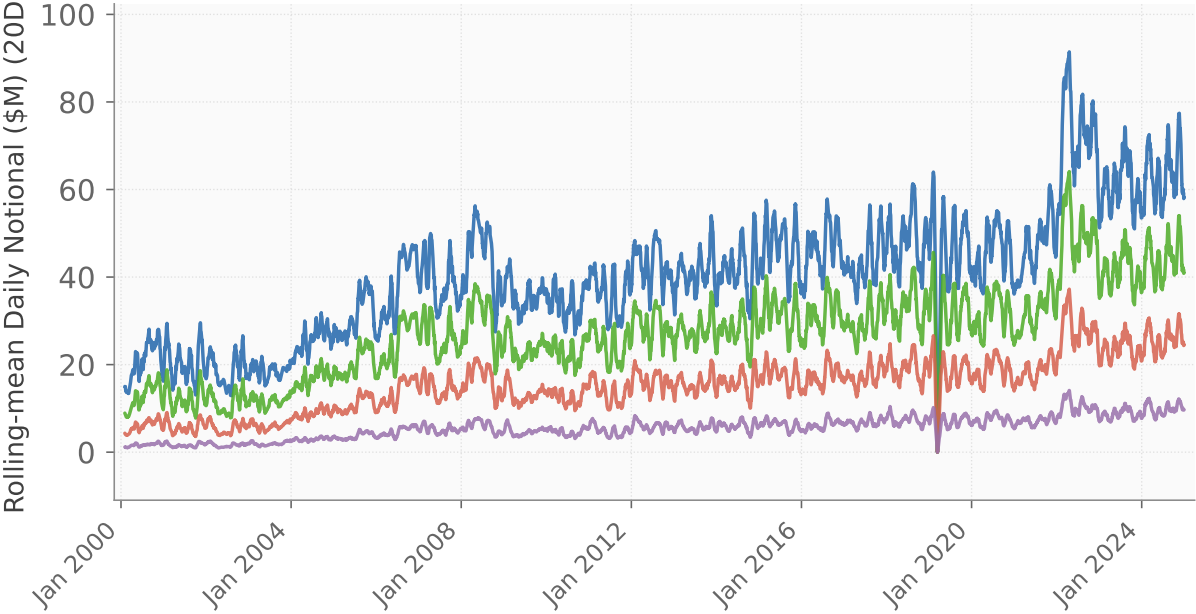
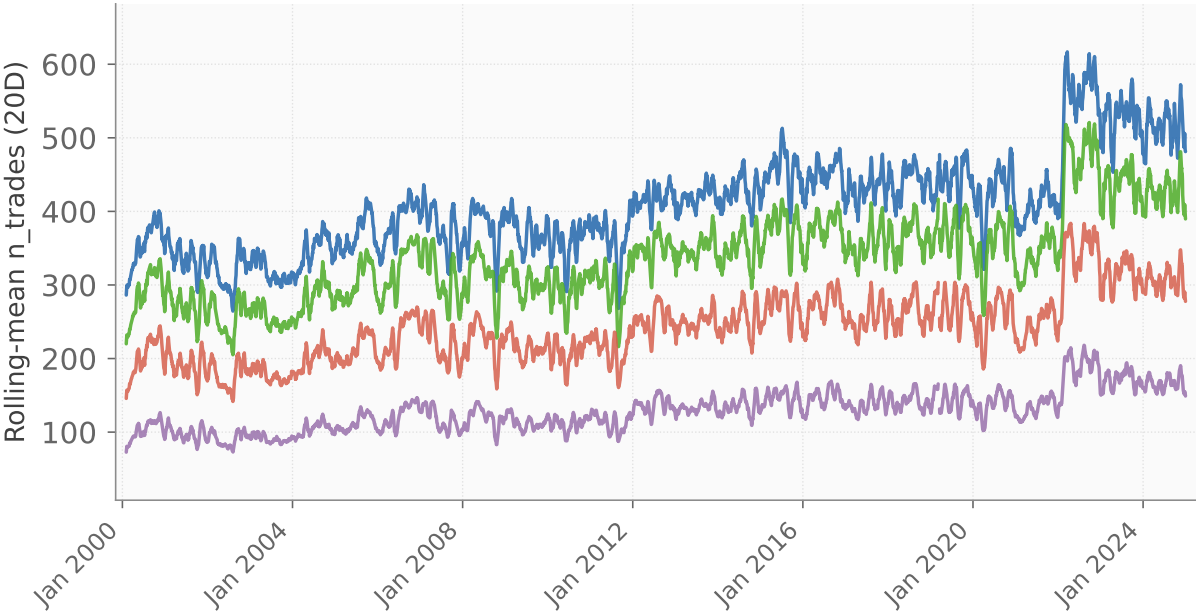
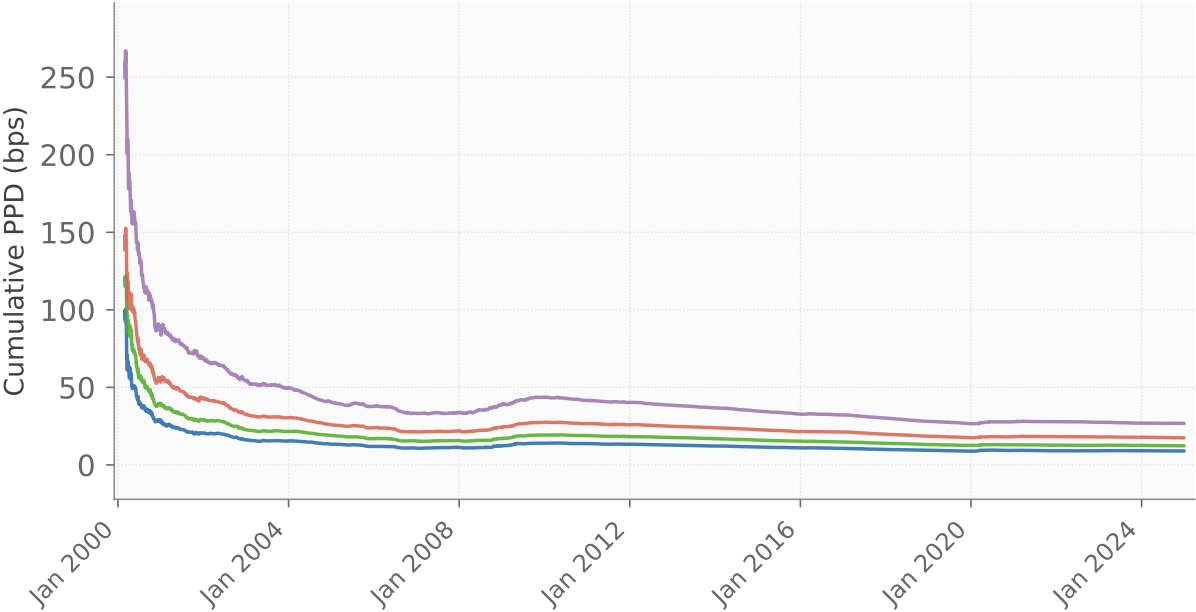
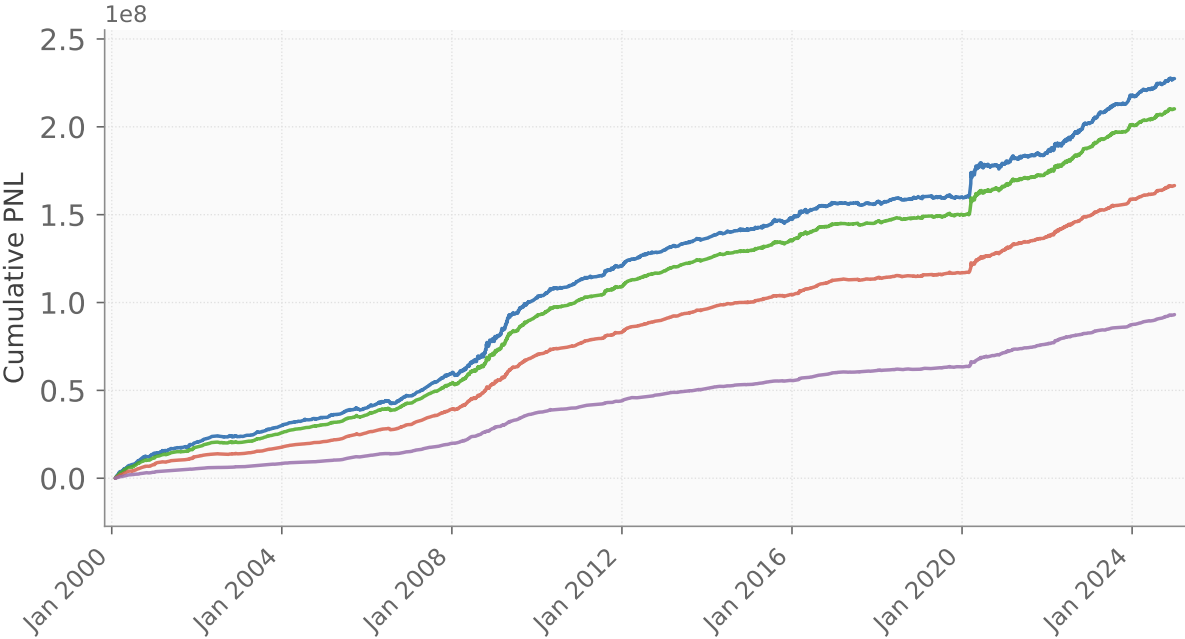
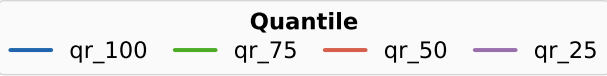
fret_OPCL_MR | pret_ols | bs_m1_c500k — Temporal metrics



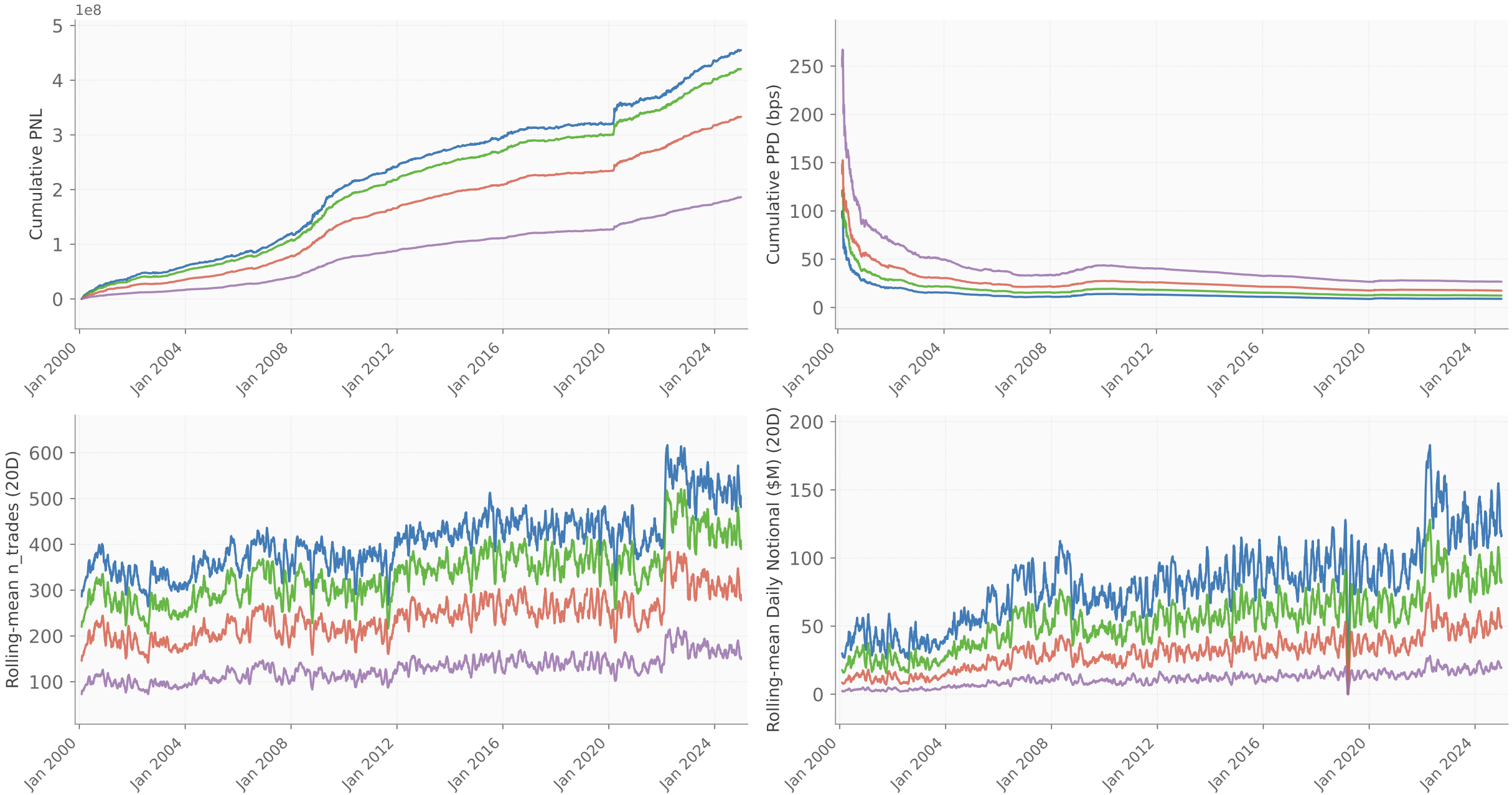
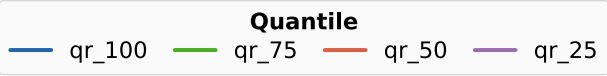
fret_OPCL_MR | pret_oracle | bs_m025_c100k — Temporal metrics



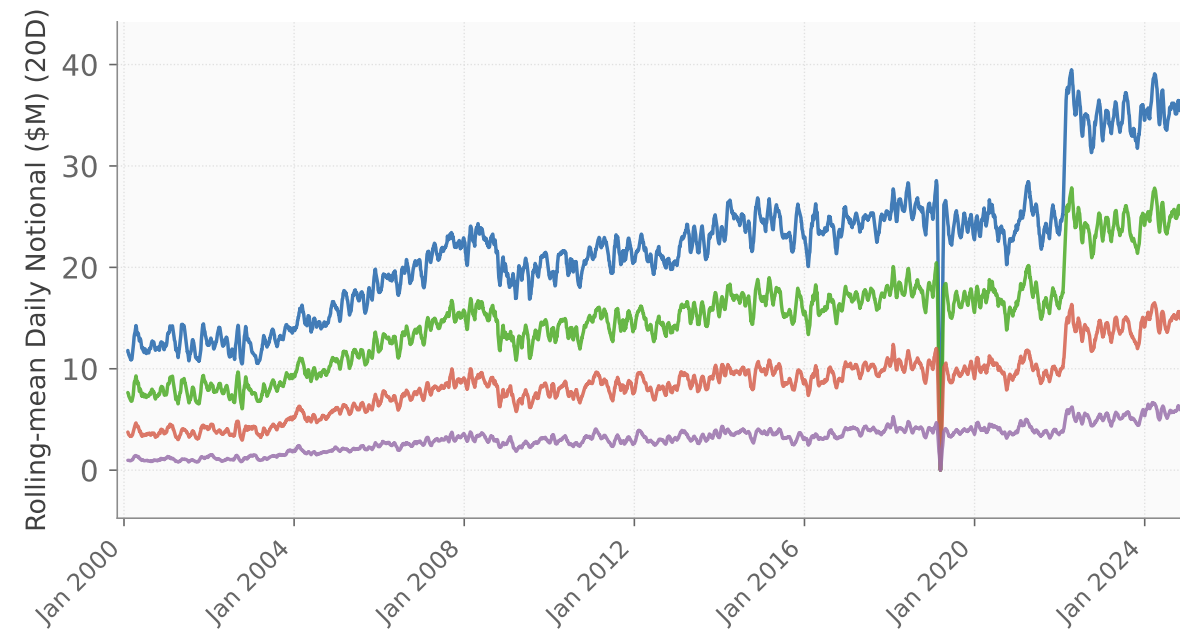
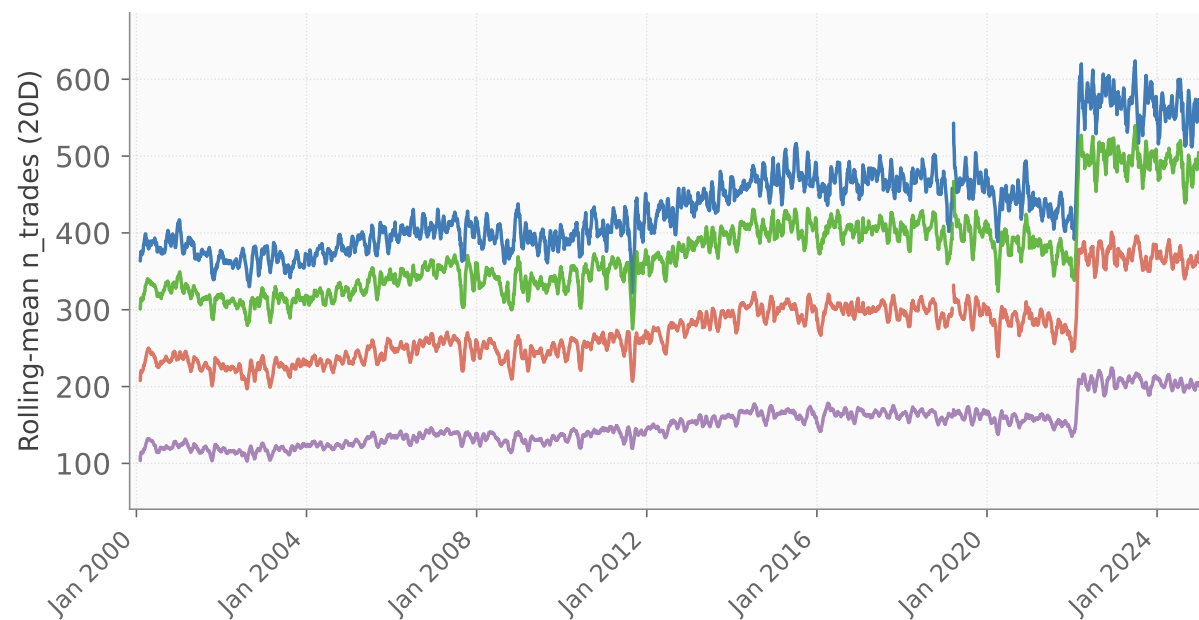
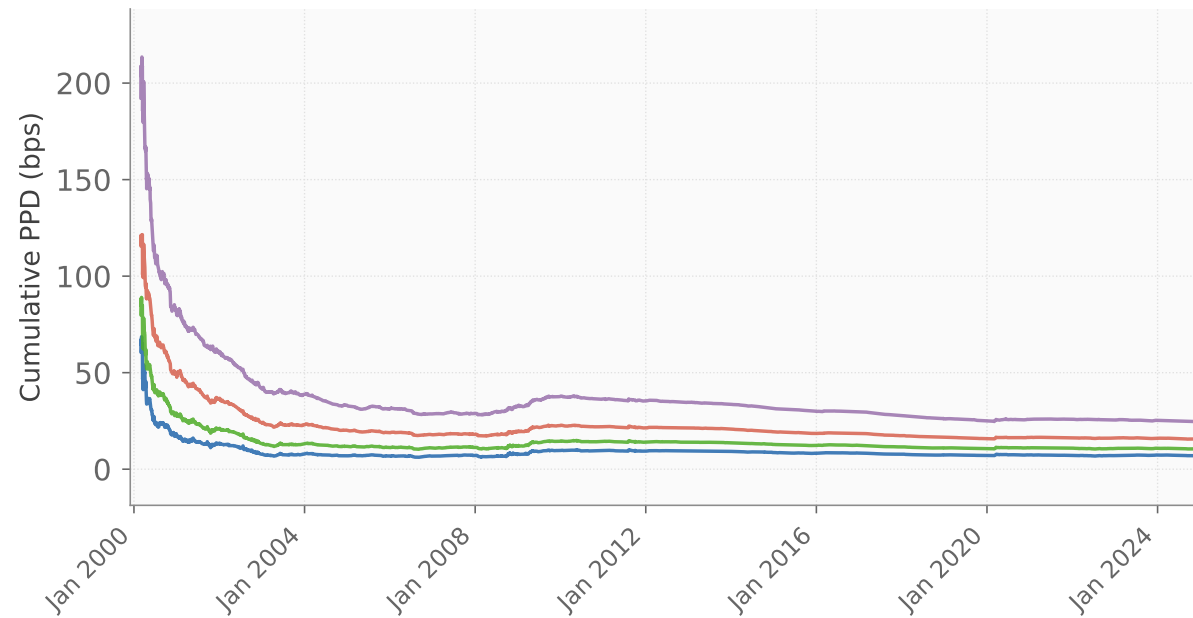
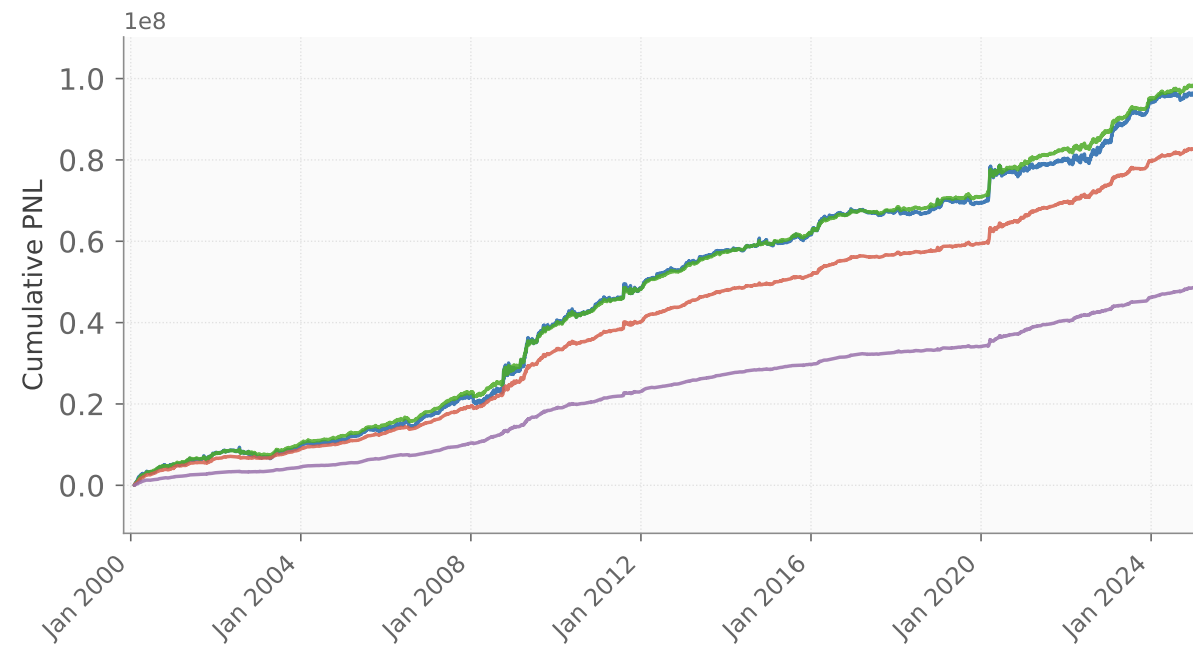
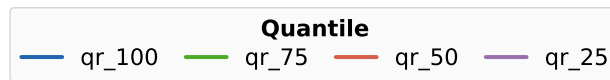
fret_OPCL_MR | pret_oracle | bs_m05_c250k — Temporal metrics



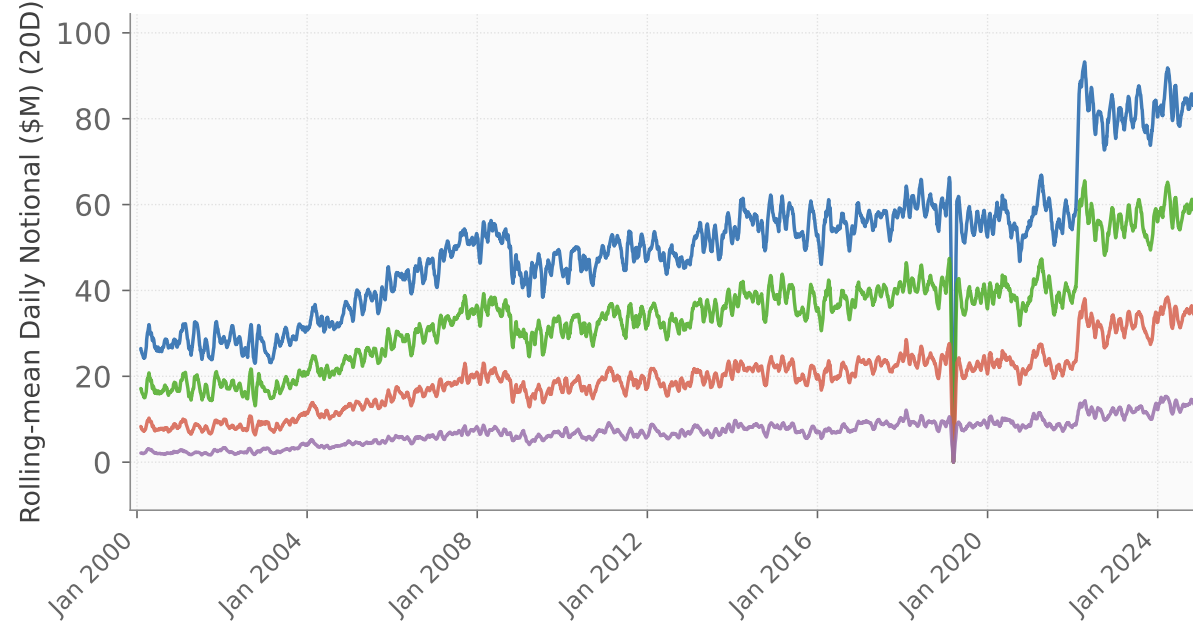
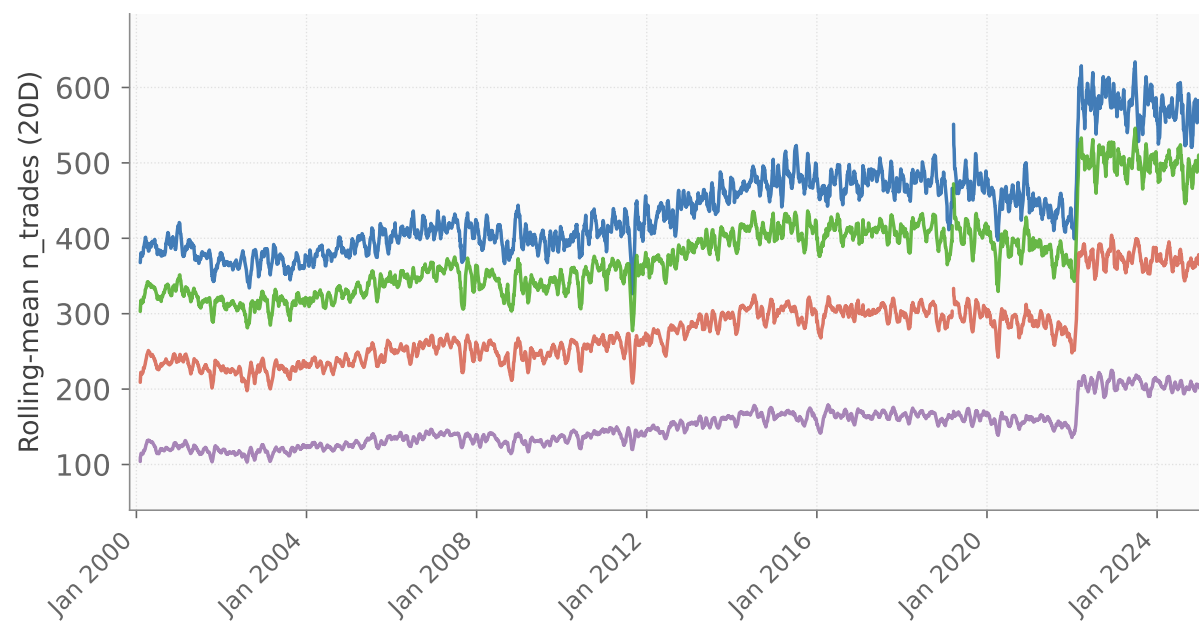
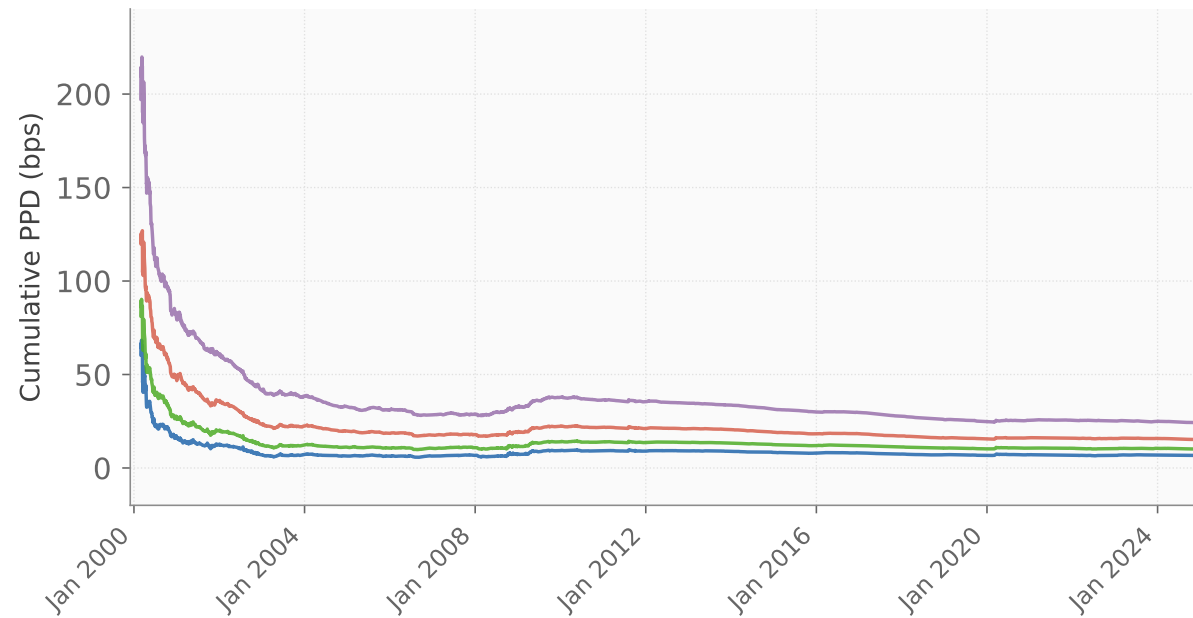
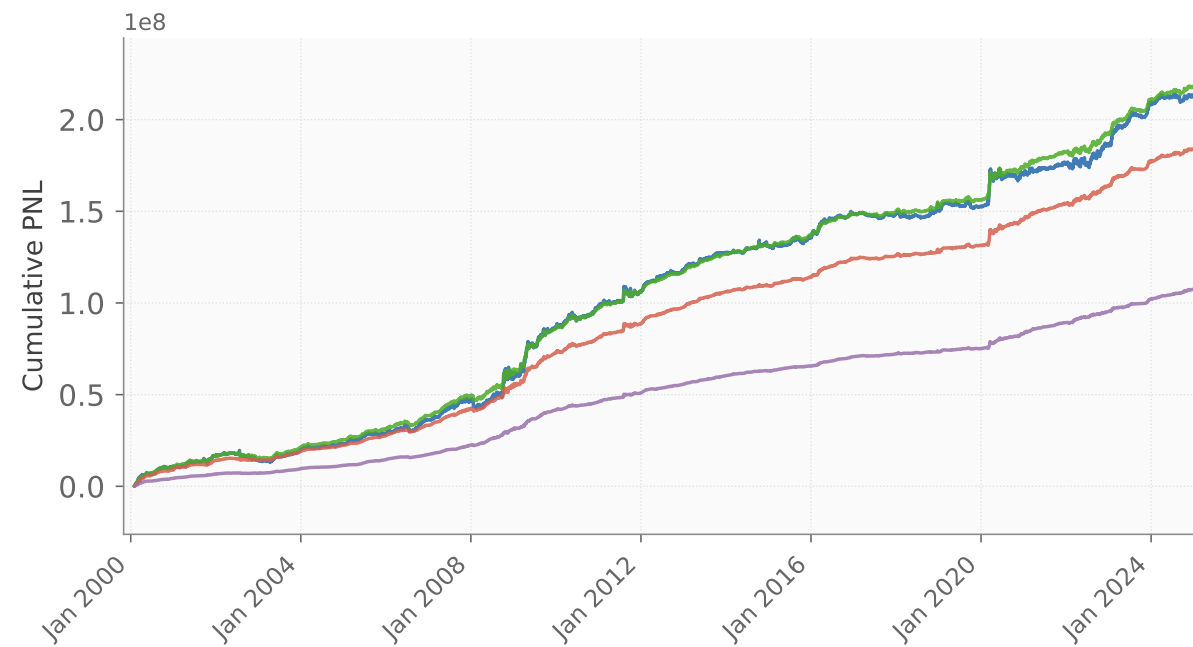
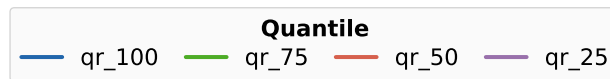
fret_OPCL_MR | pret_oracle | bs_m1_c500k — Temporal metrics



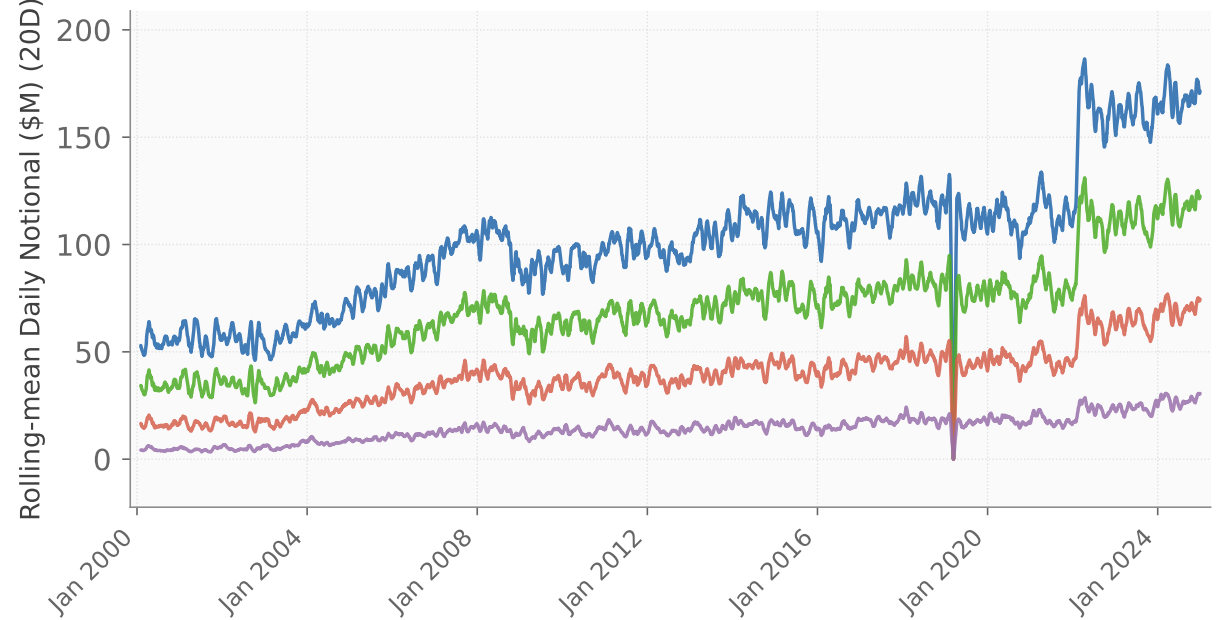
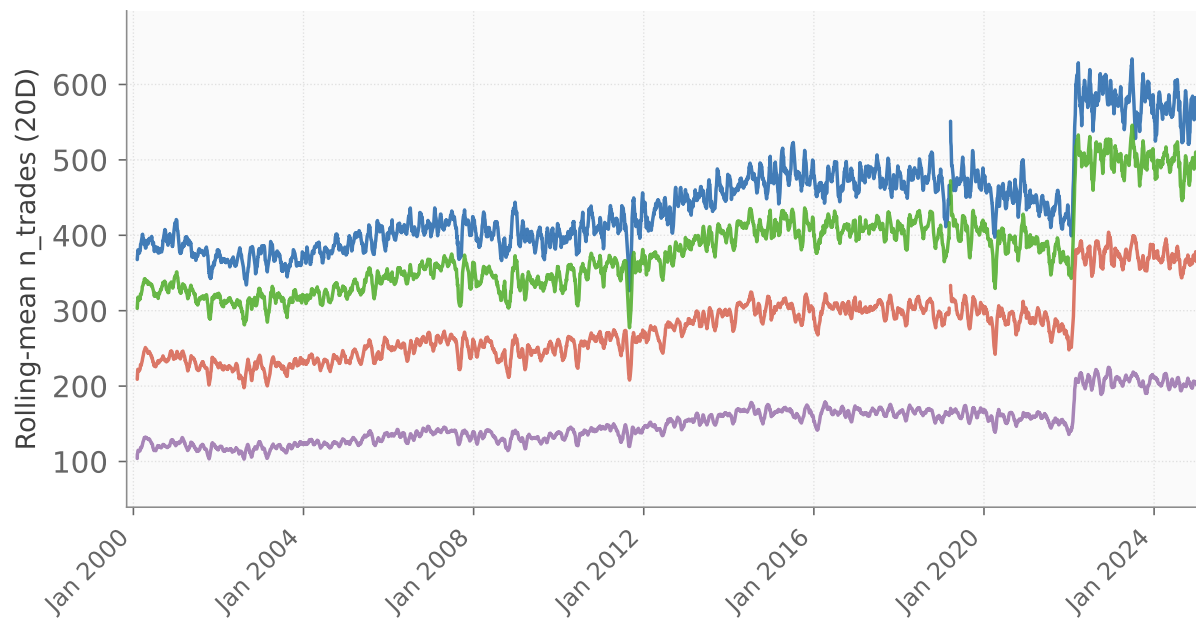
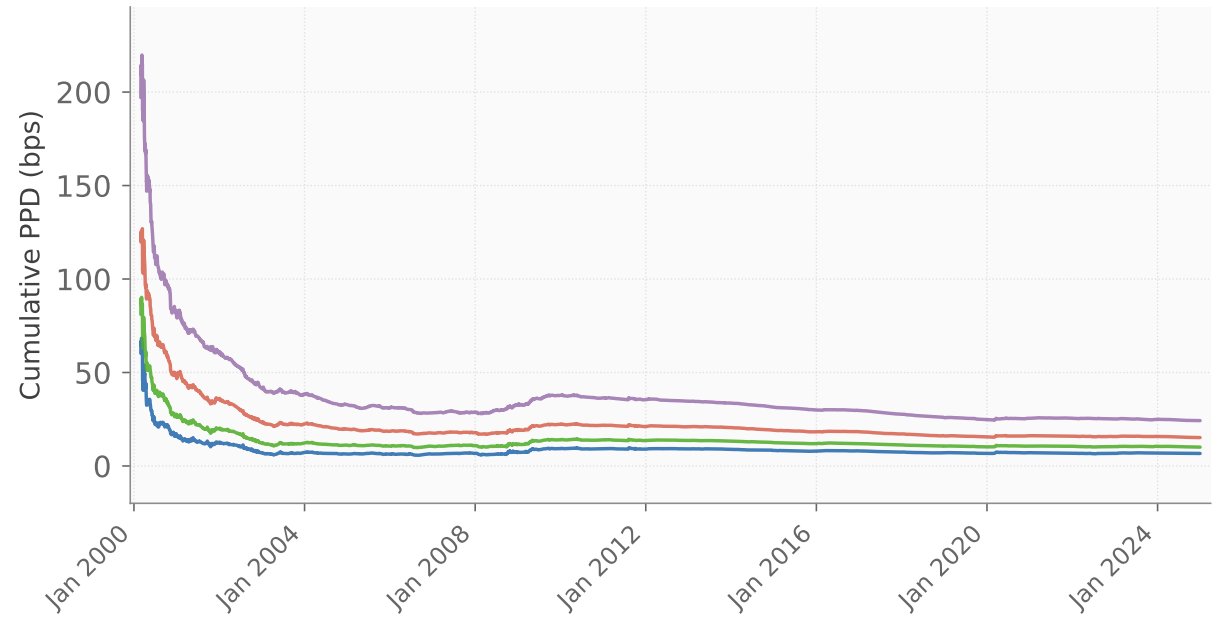
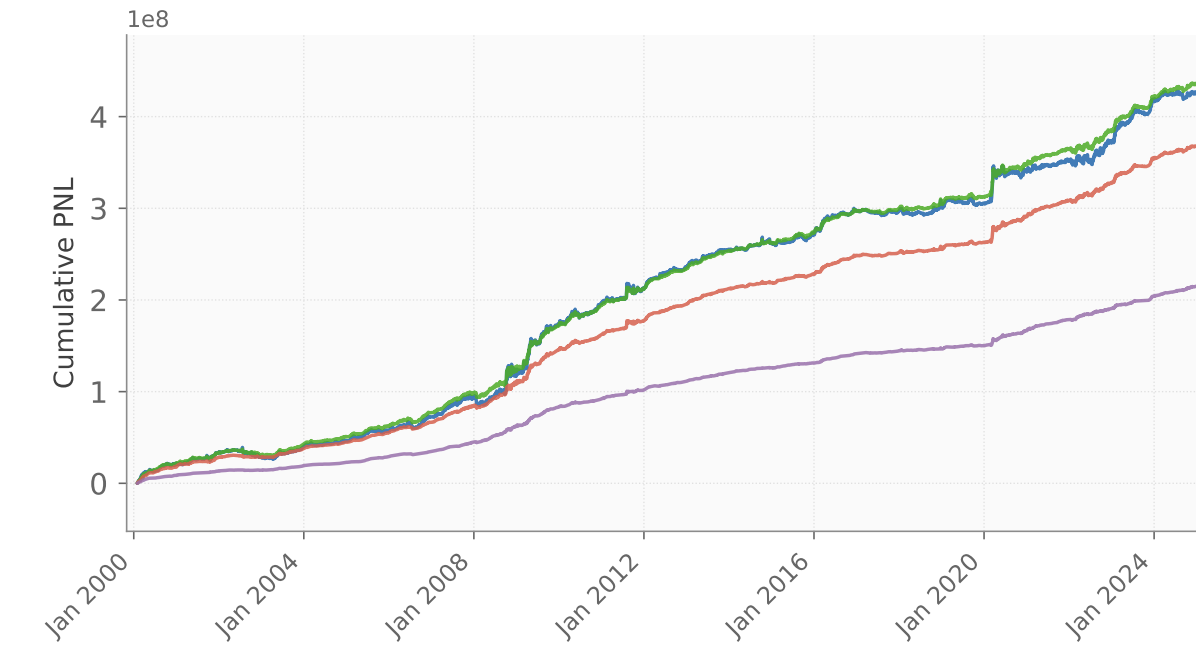
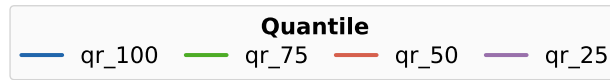
fret_OPCL_RR | pret_base | bs_m025_c100k — Temporal metrics



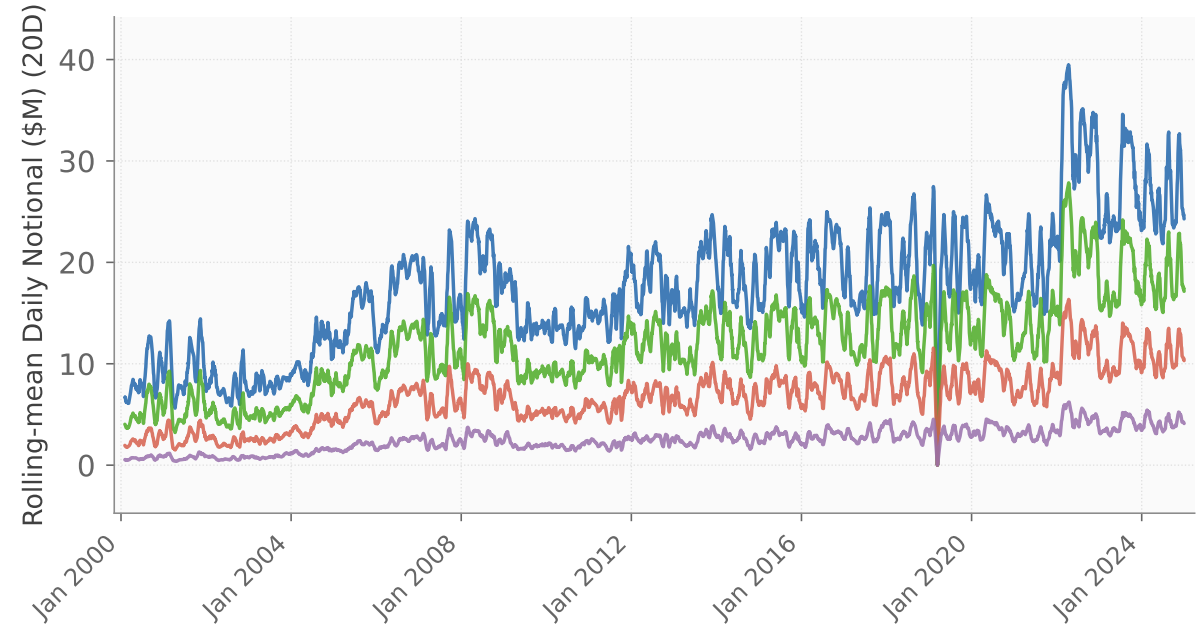
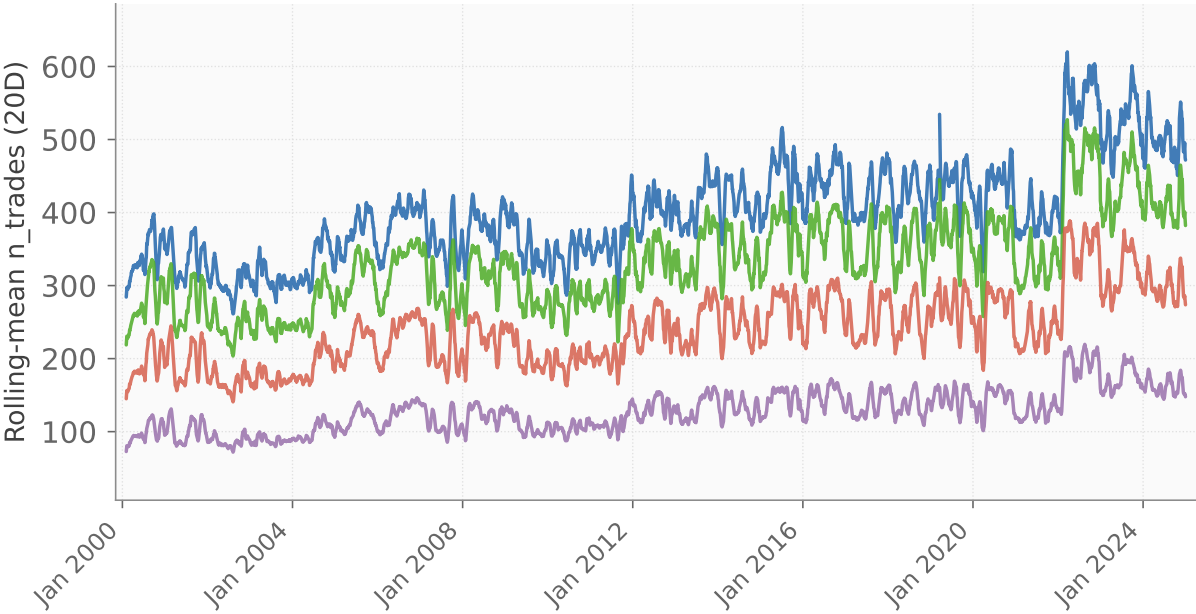
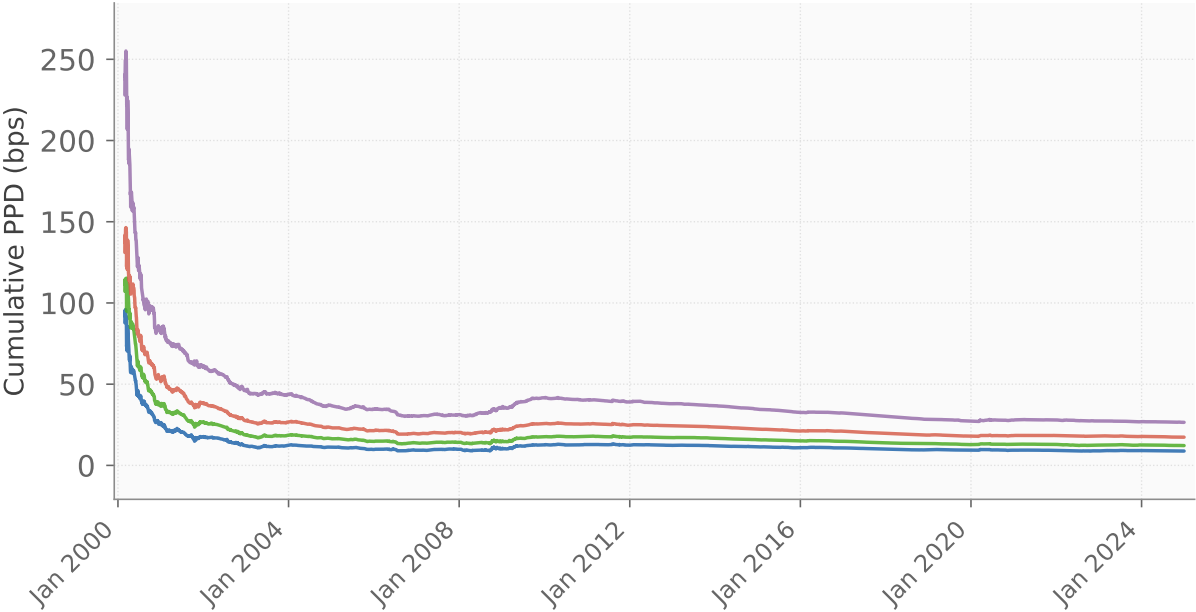
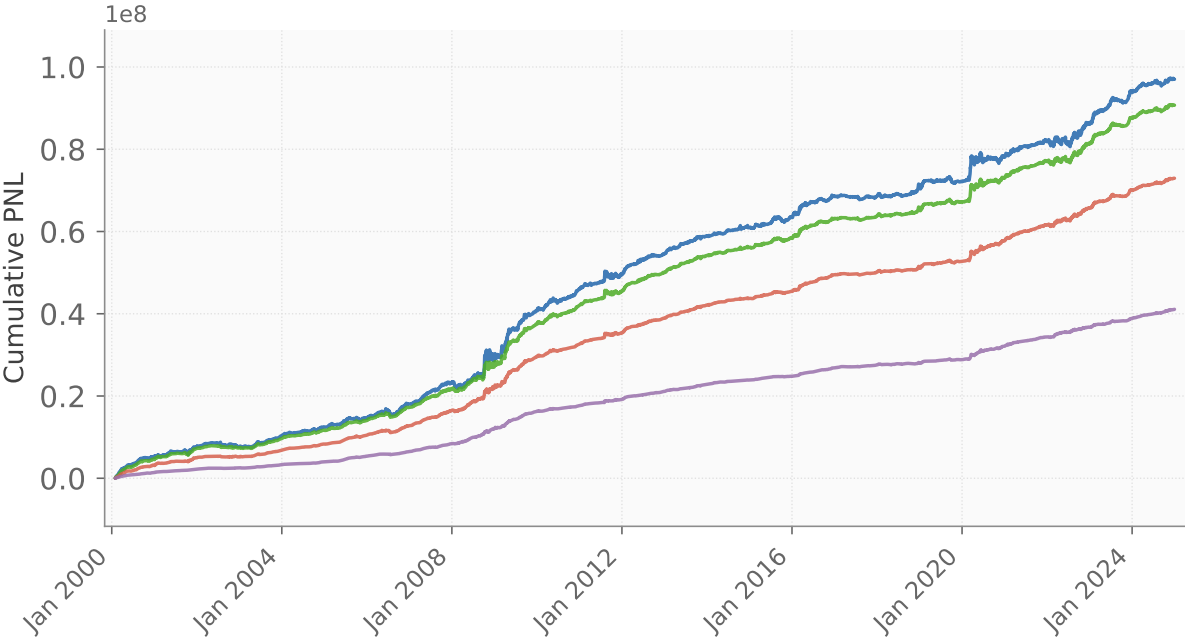
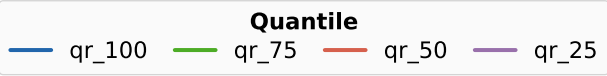
fret_OPCL_RR | pret_base | bs_m05_c250k — Temporal metrics



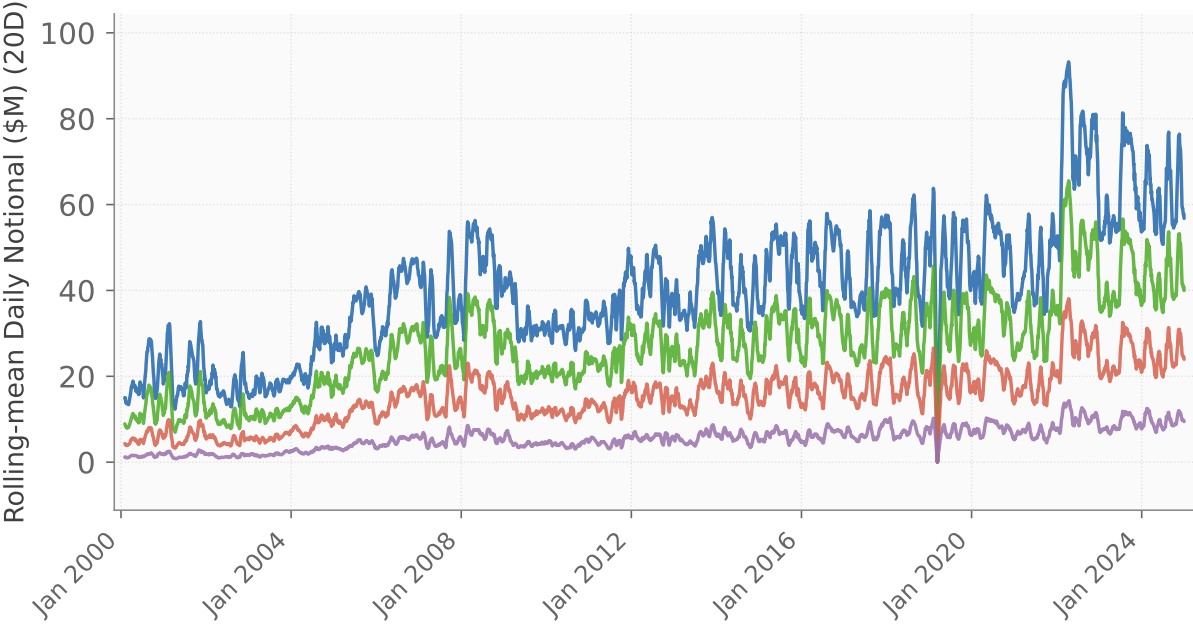
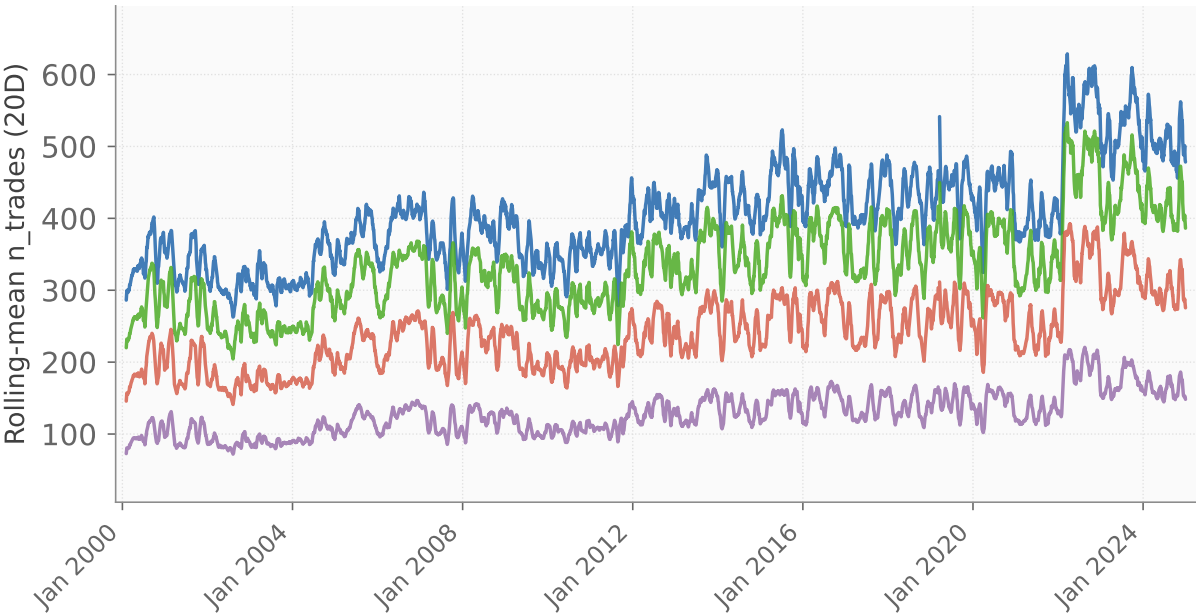
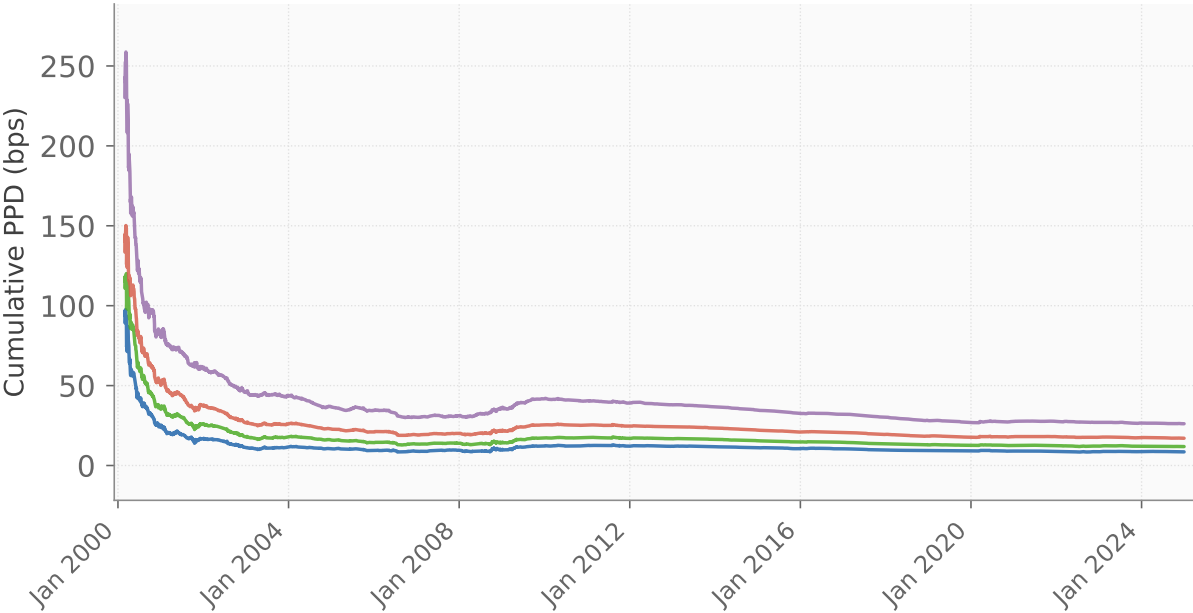
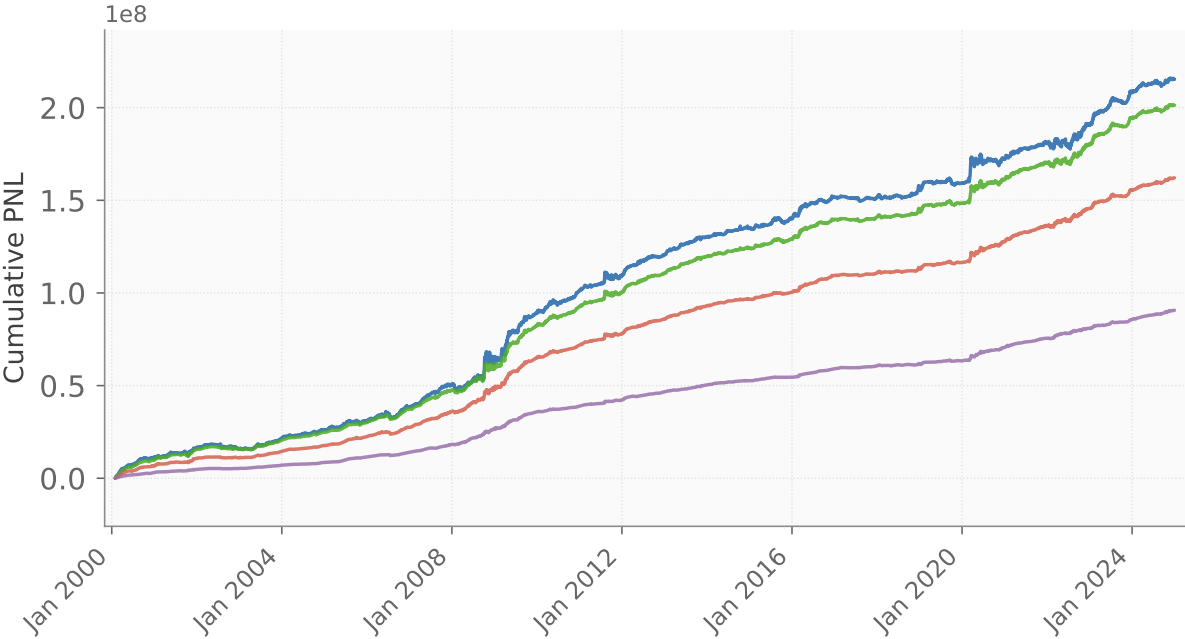
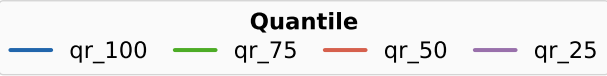
fret_OPCL_RR | pret_base | bs_m1_c500k — Temporal metrics



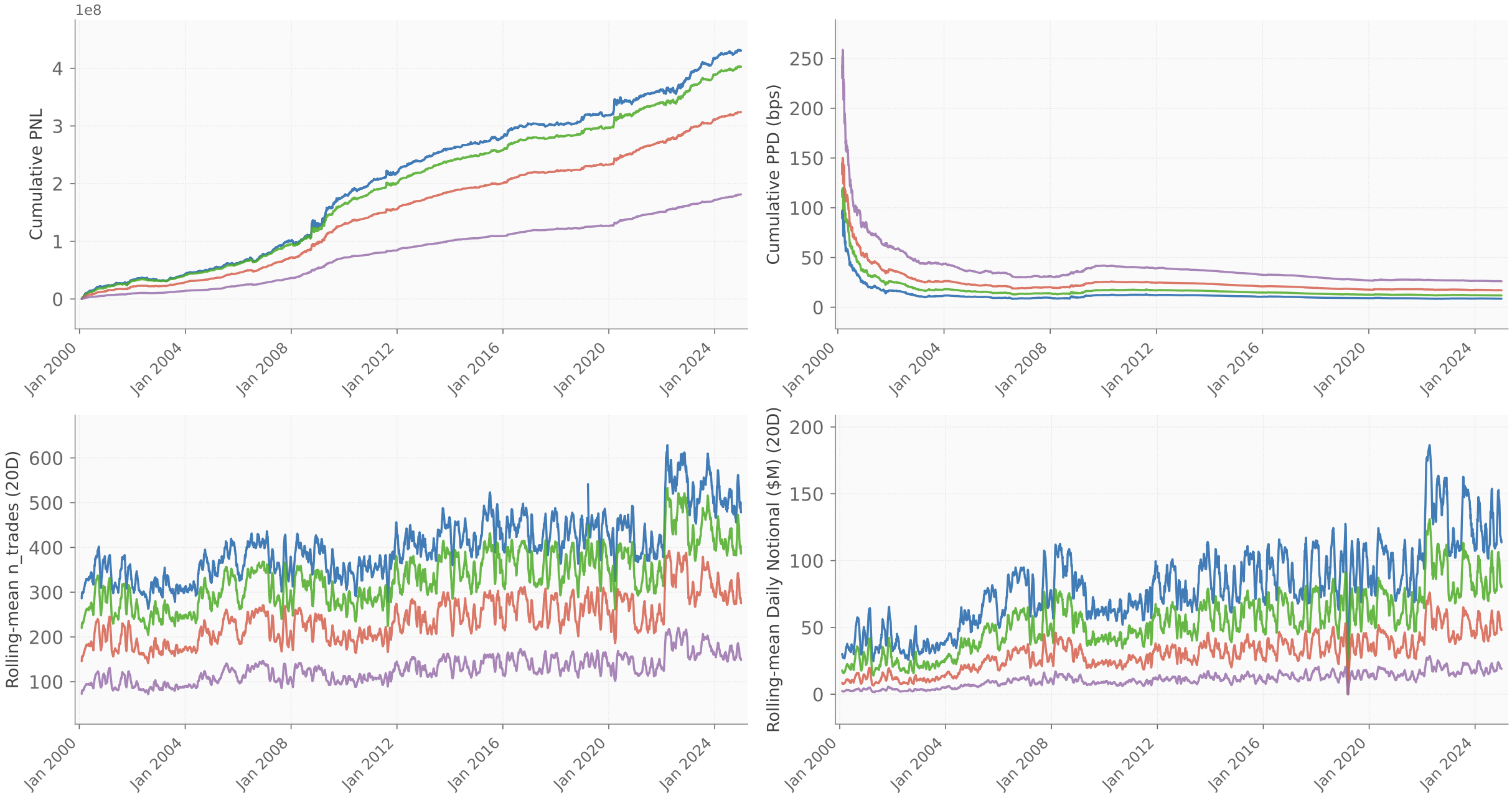
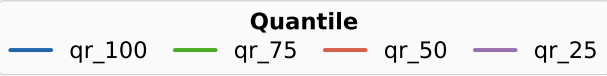
fret_OPCL_RR | pret_clus | bs_m025_c100k — Temporal metrics



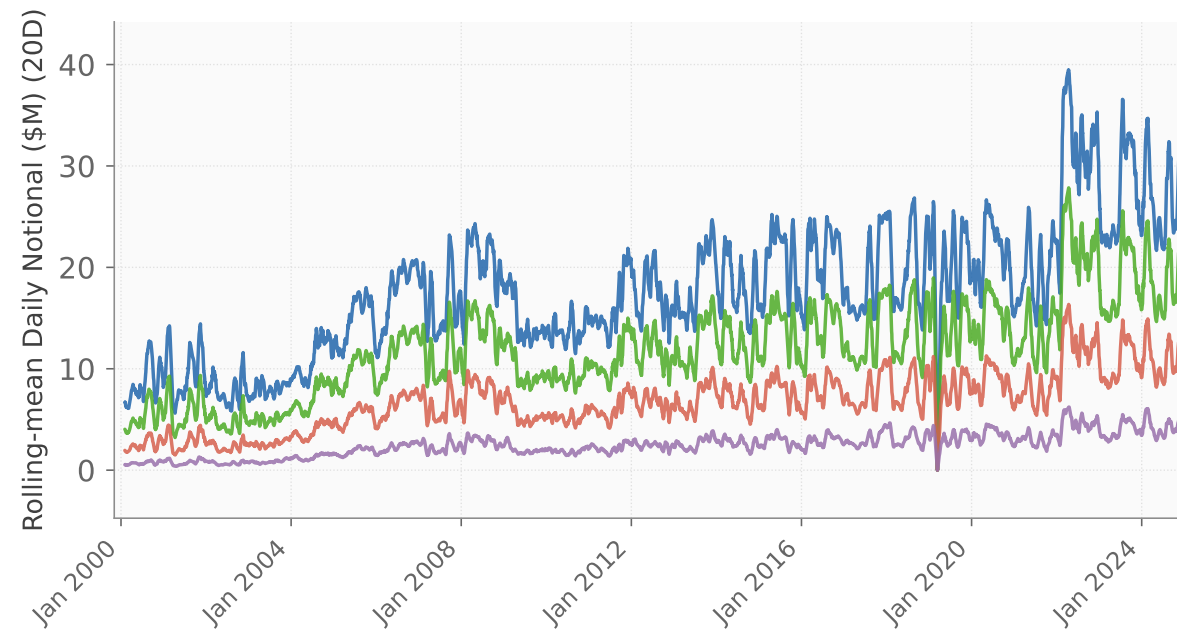
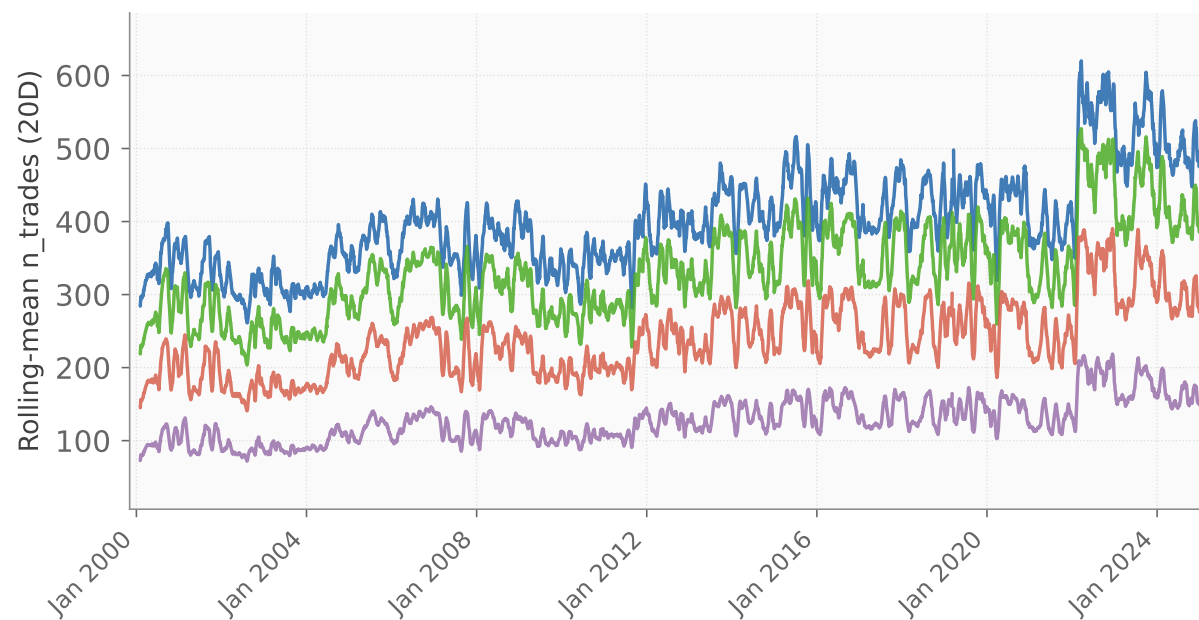
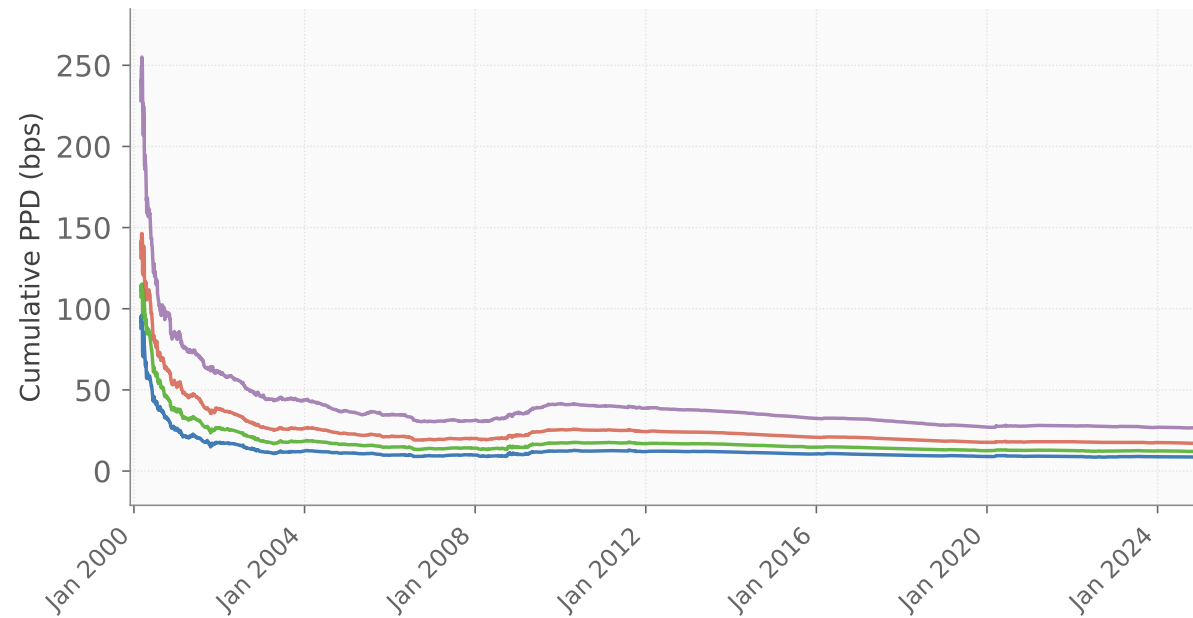
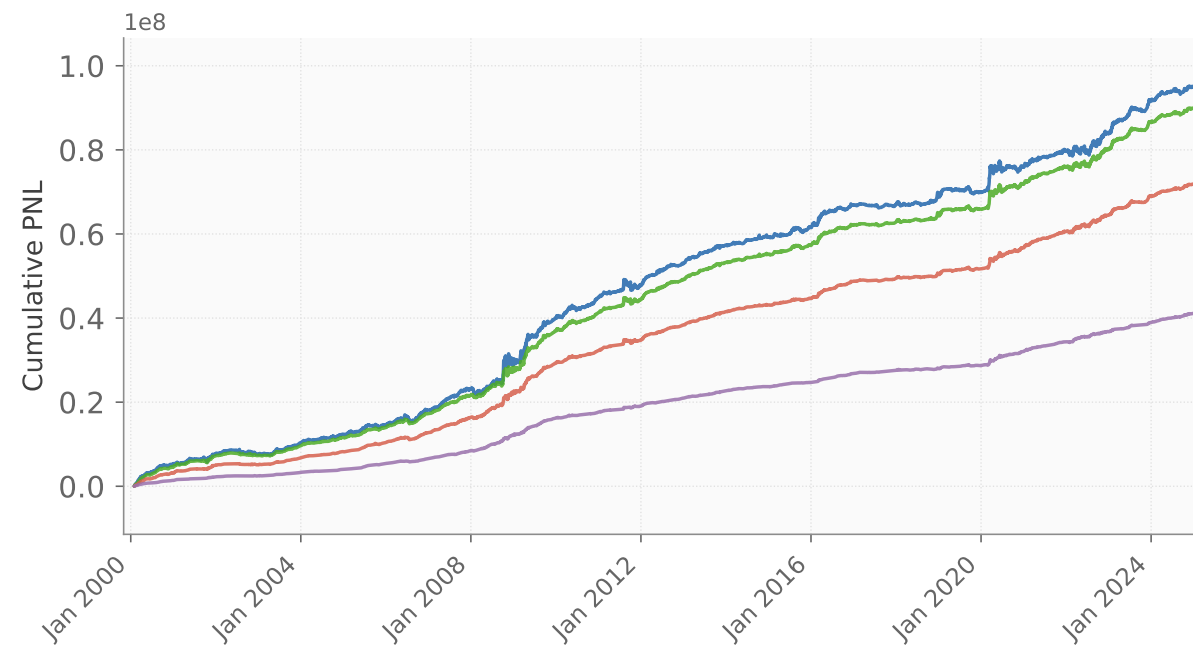
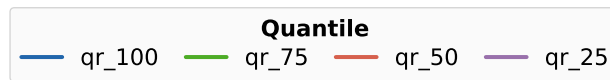
fret_OPCL_RR | pret_clus | bs_m05_c250k — Temporal metrics



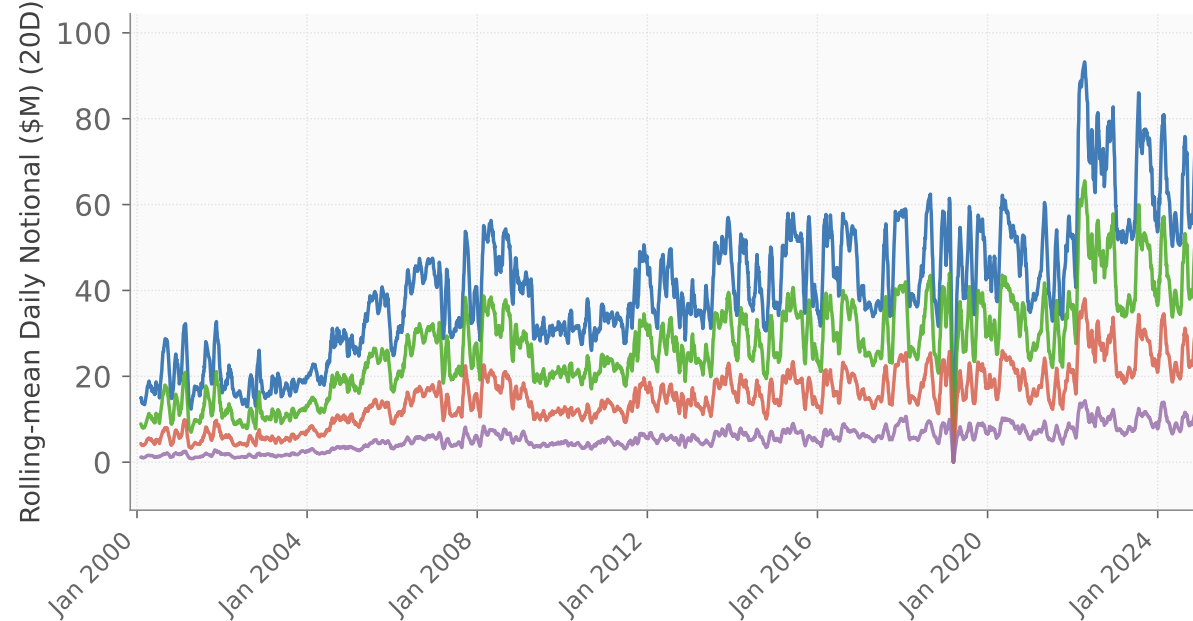
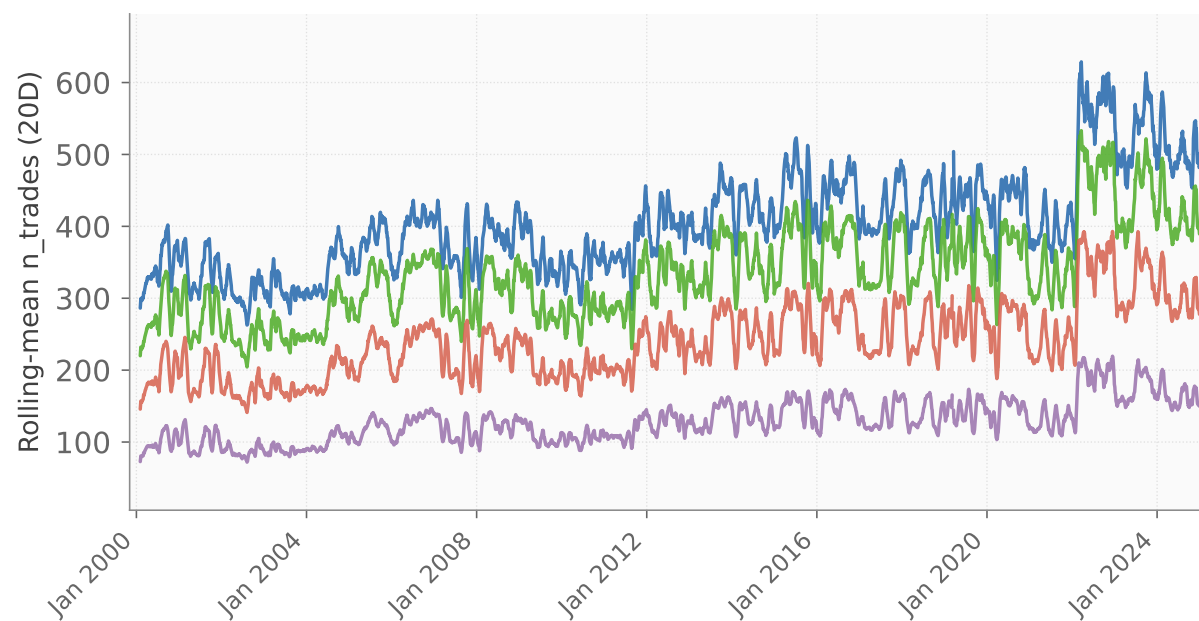
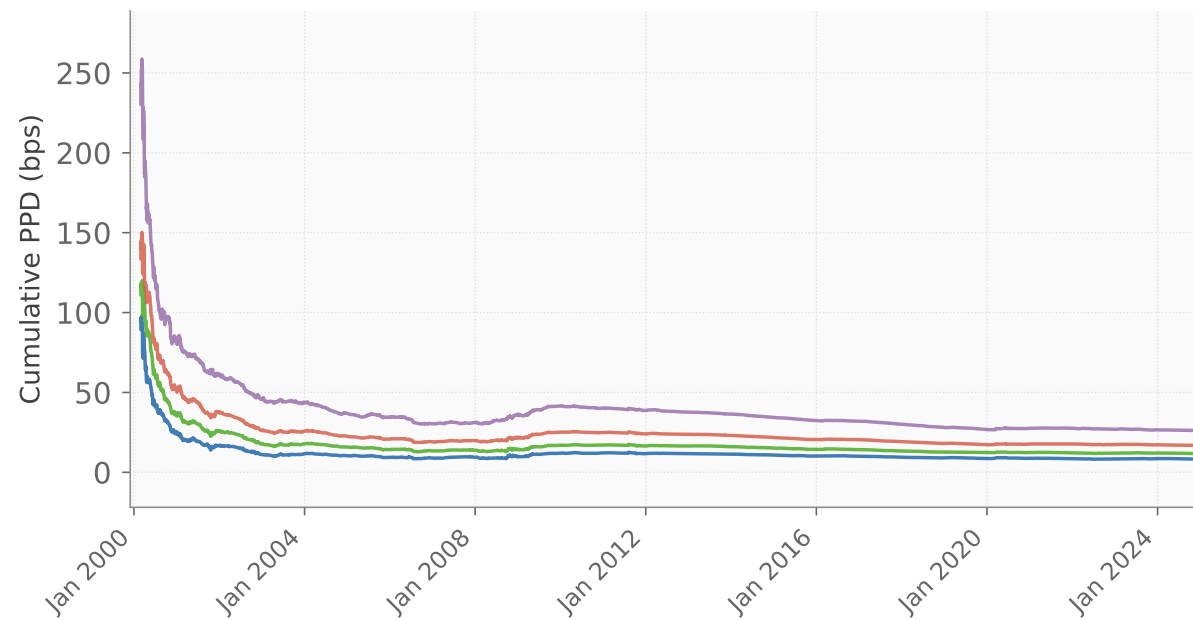
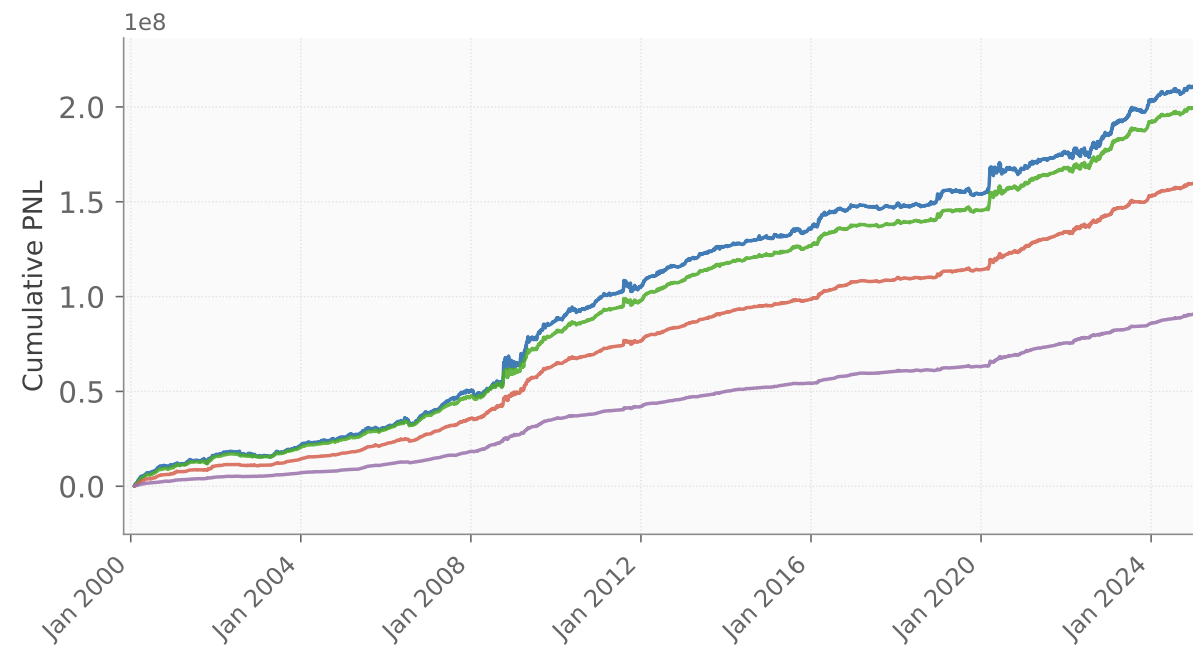
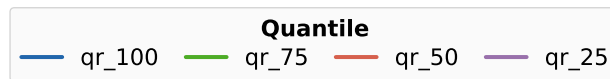
fret_OPCL_RR | pret_clus | bs_m1_c500k — Temporal metrics



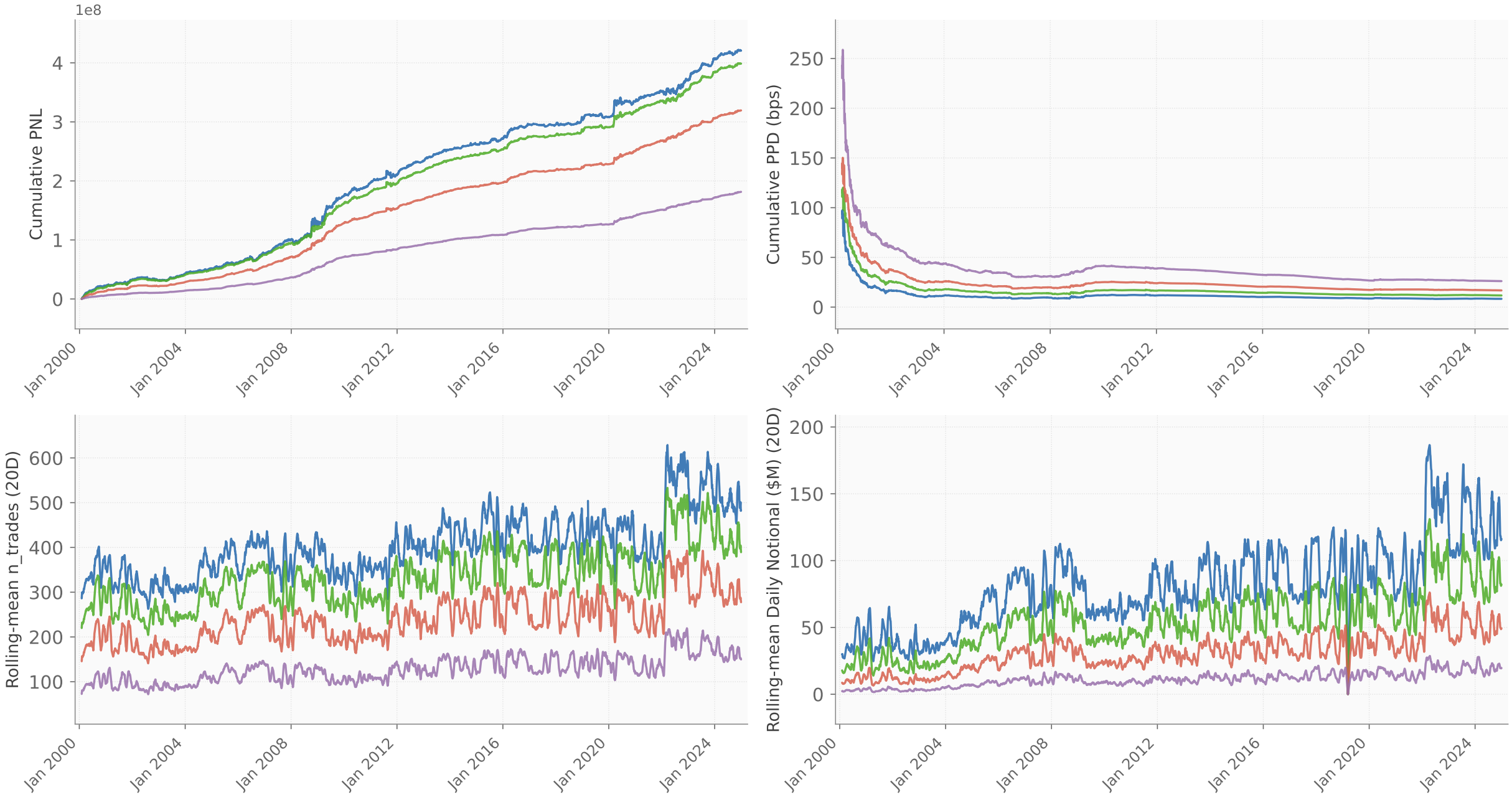
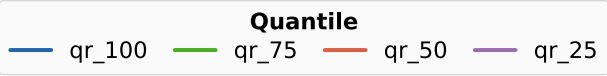
fret_OPCL_RR | pret_har | bs_m025_c100k — Temporal metrics



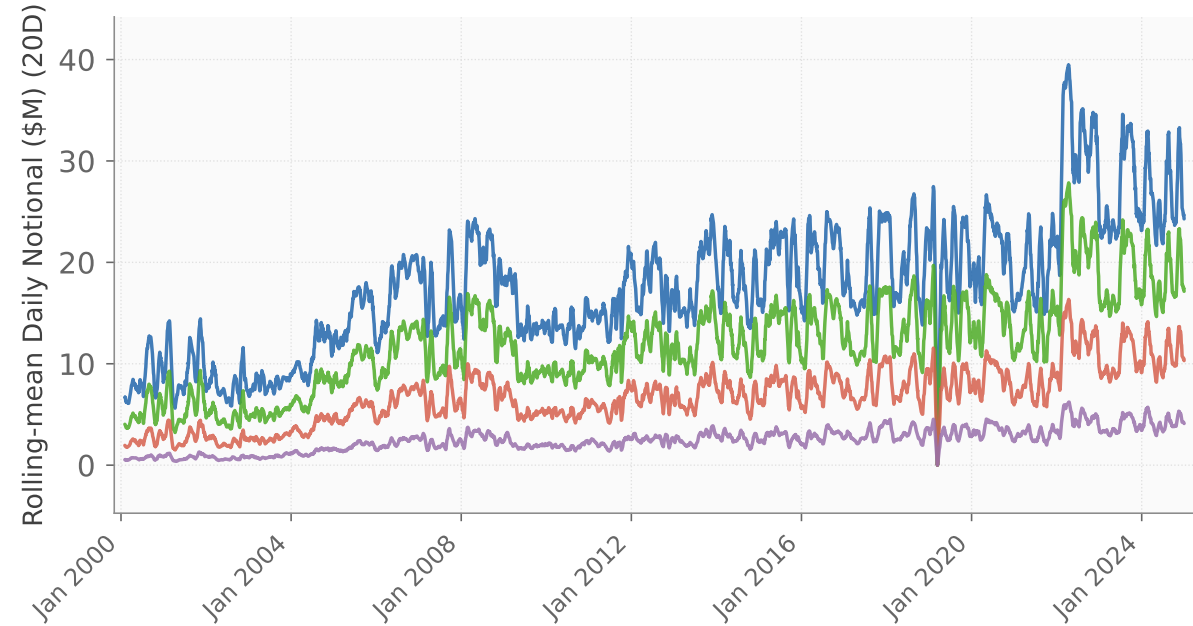
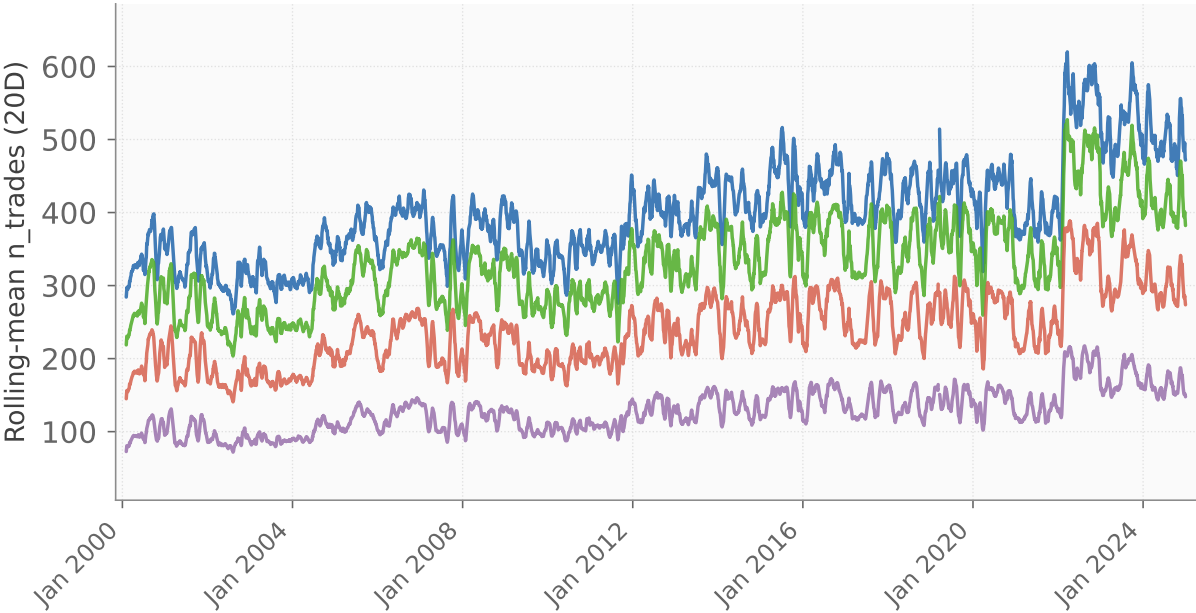
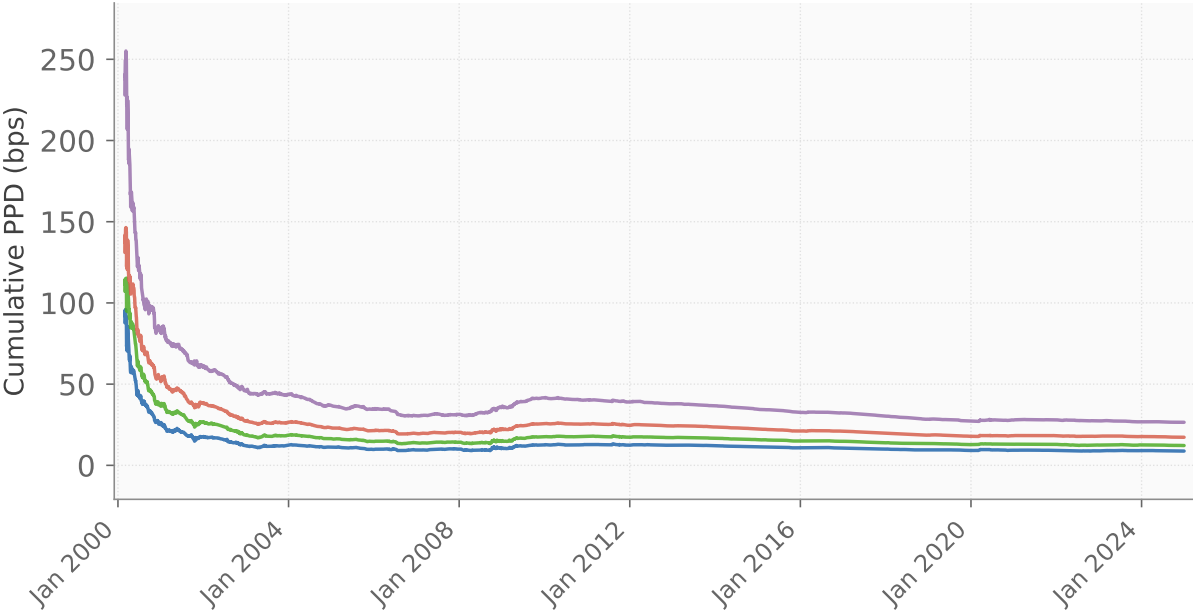
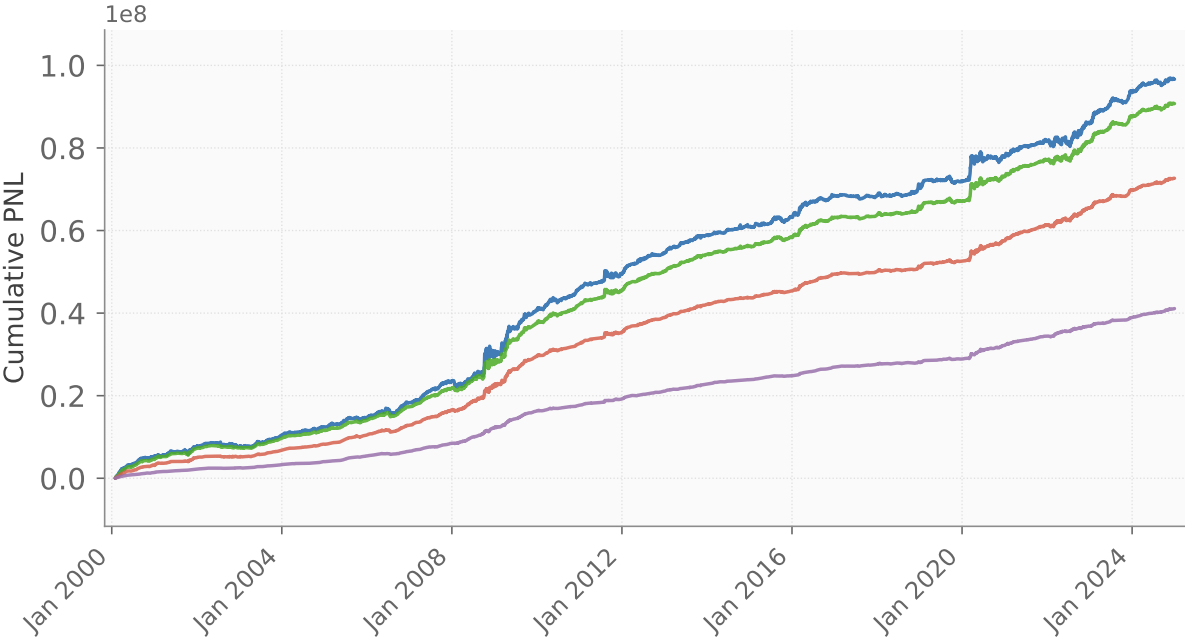
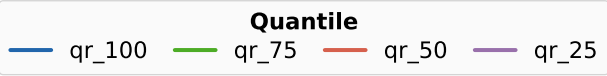
fret_OPCL_RR | pret_har | bs_m05_c250k — Temporal metrics



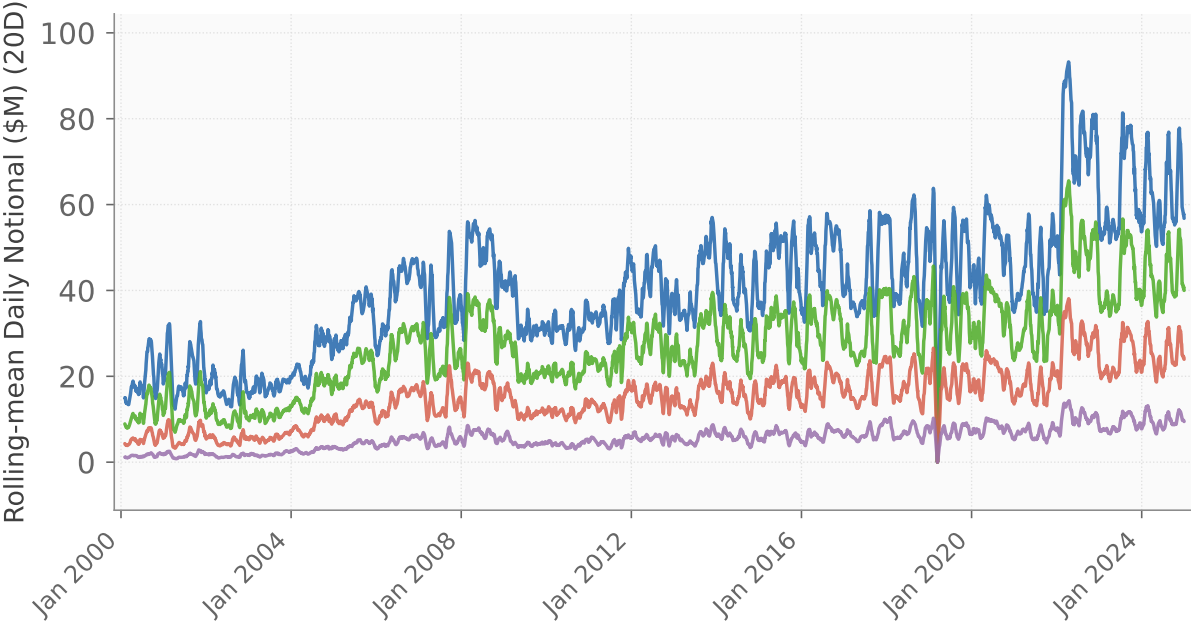
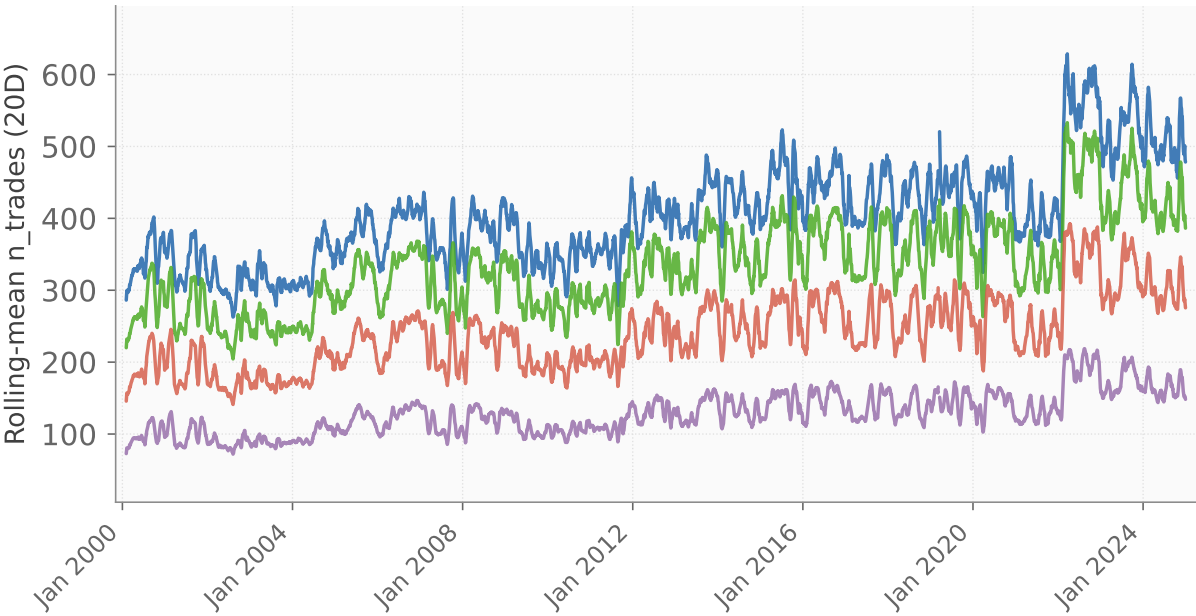
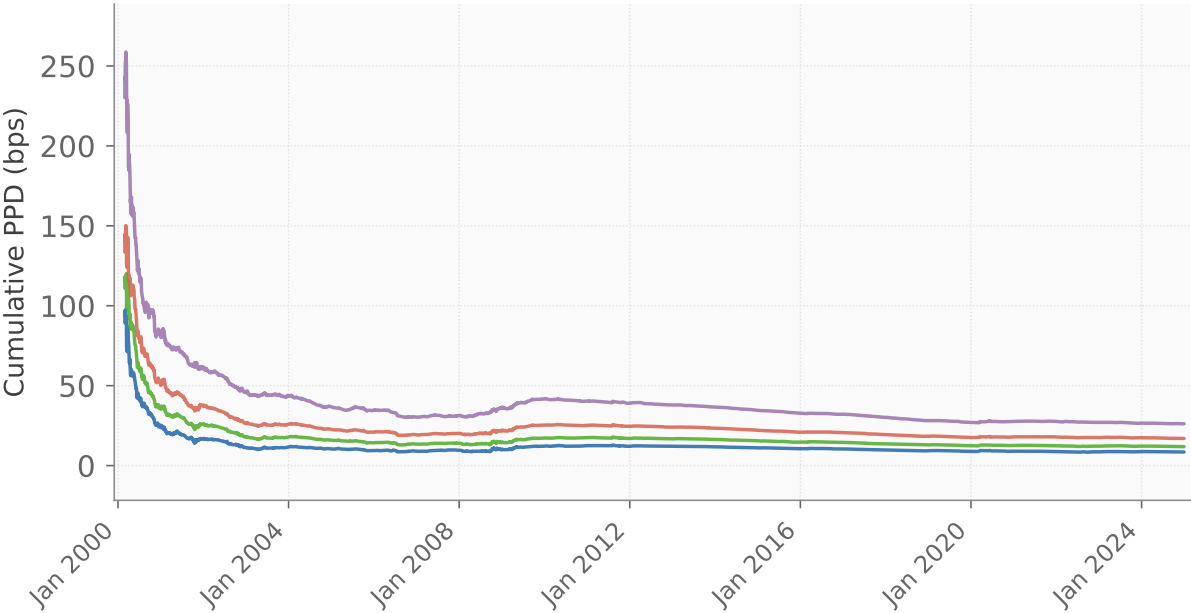
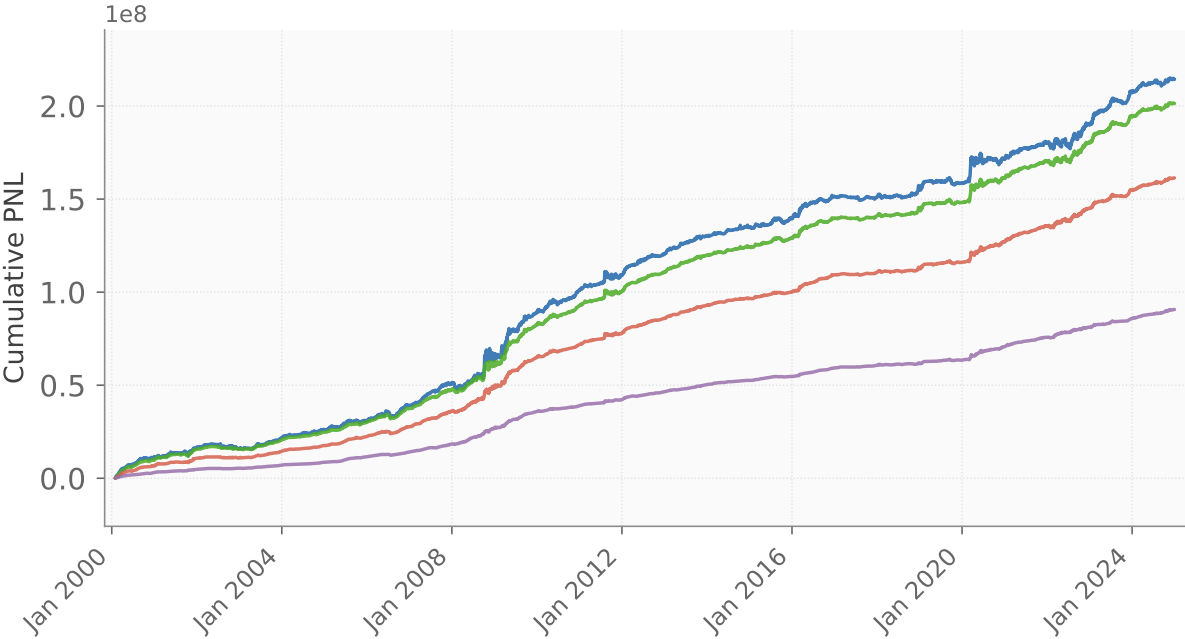
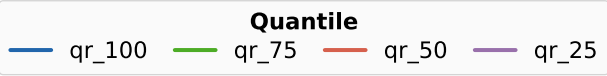
fret_OPCL_RR | pret_har | bs_m1_c500k — Temporal metrics



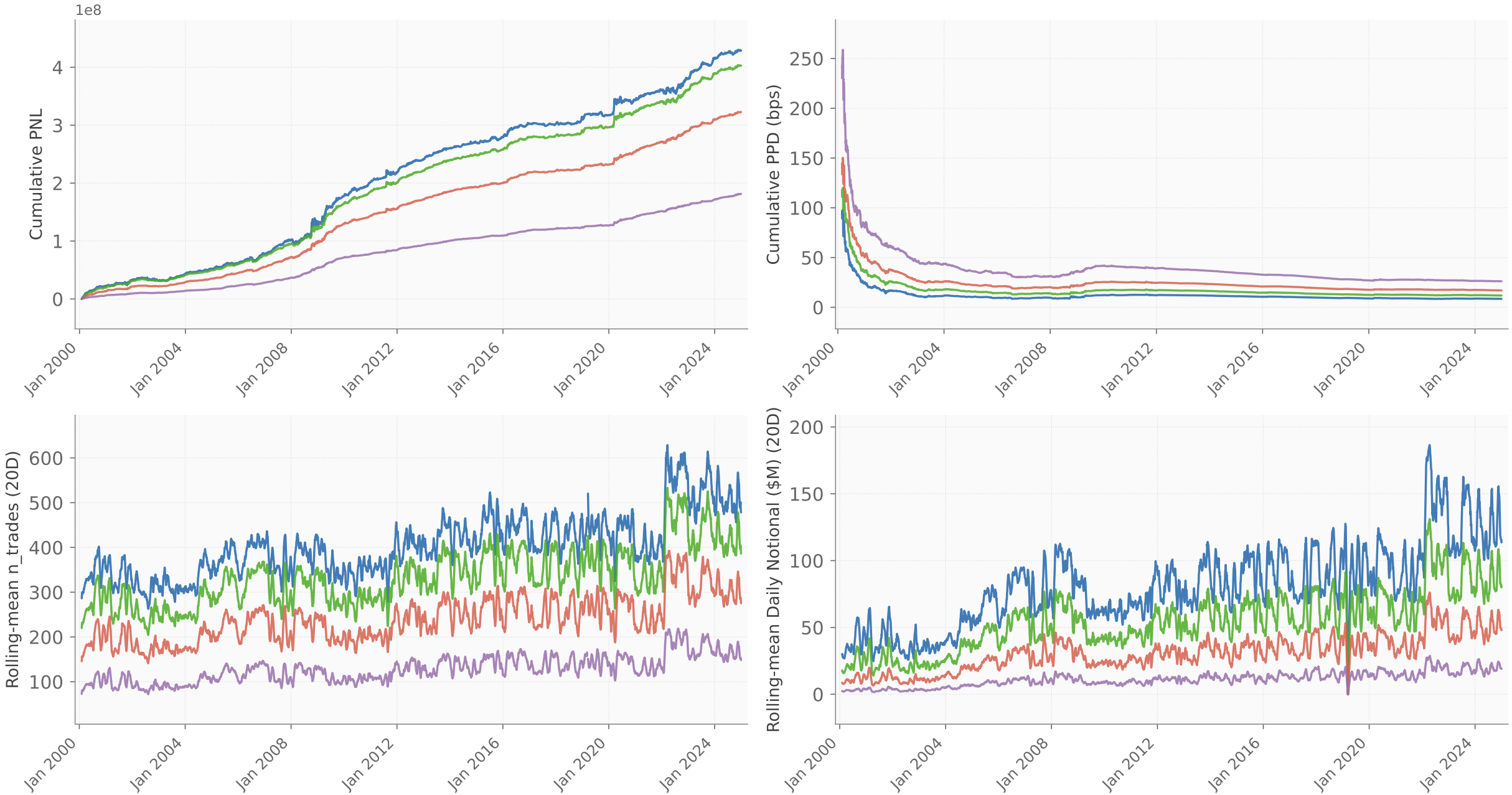
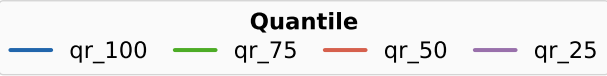
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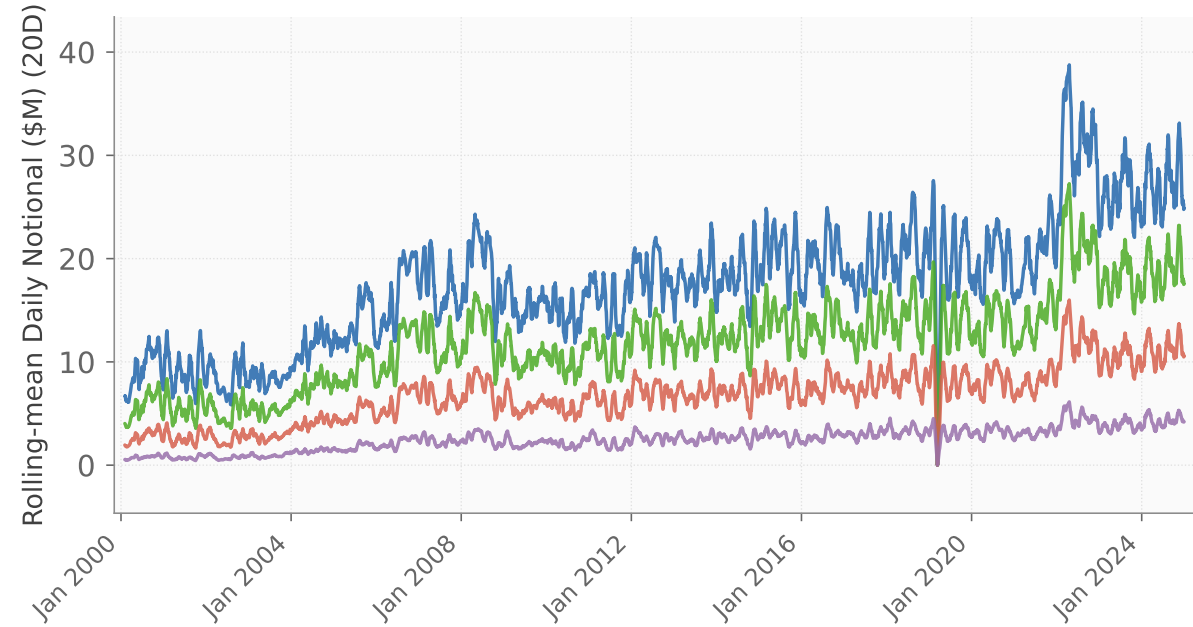
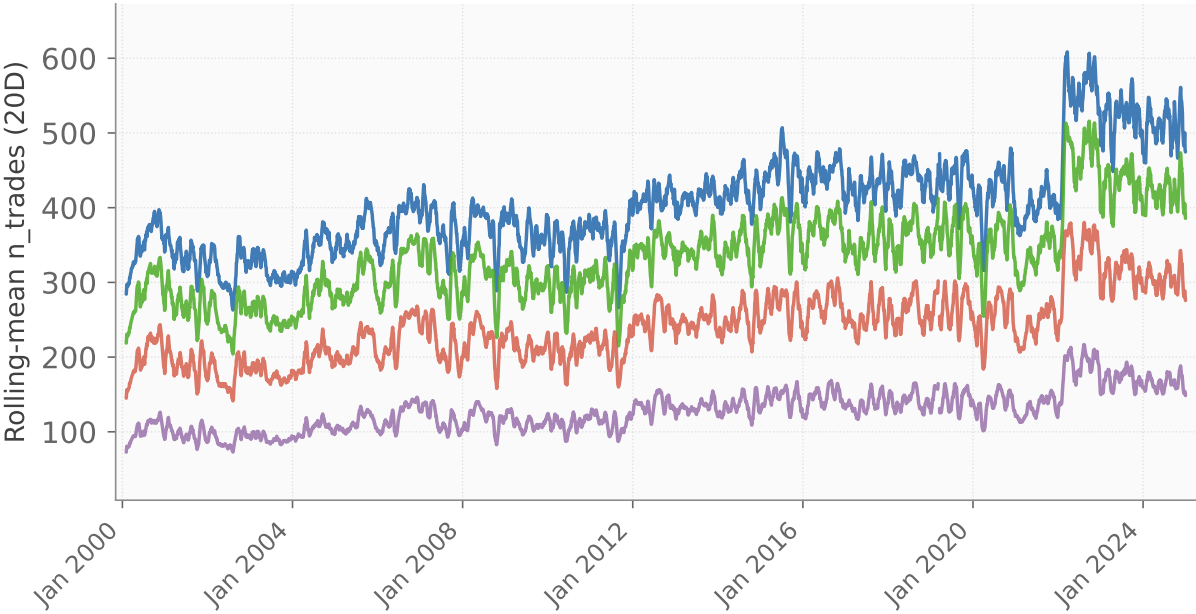
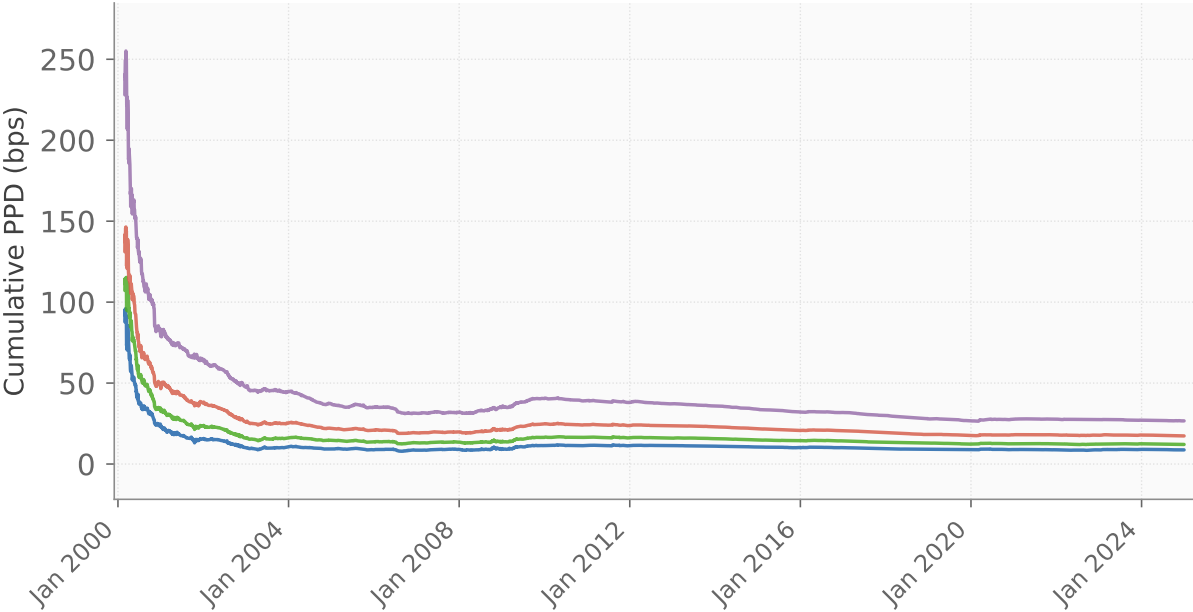
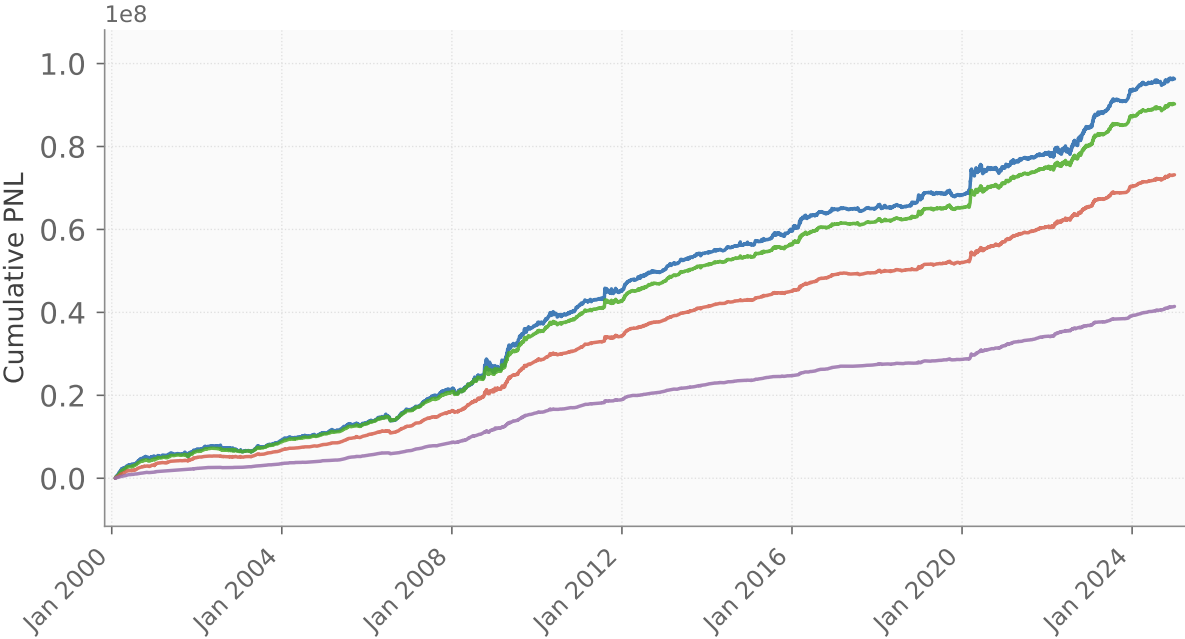
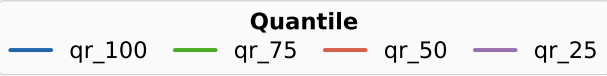
fret_OPCL_RR | pret_ols | bs_m05_c250k — Temporal metrics



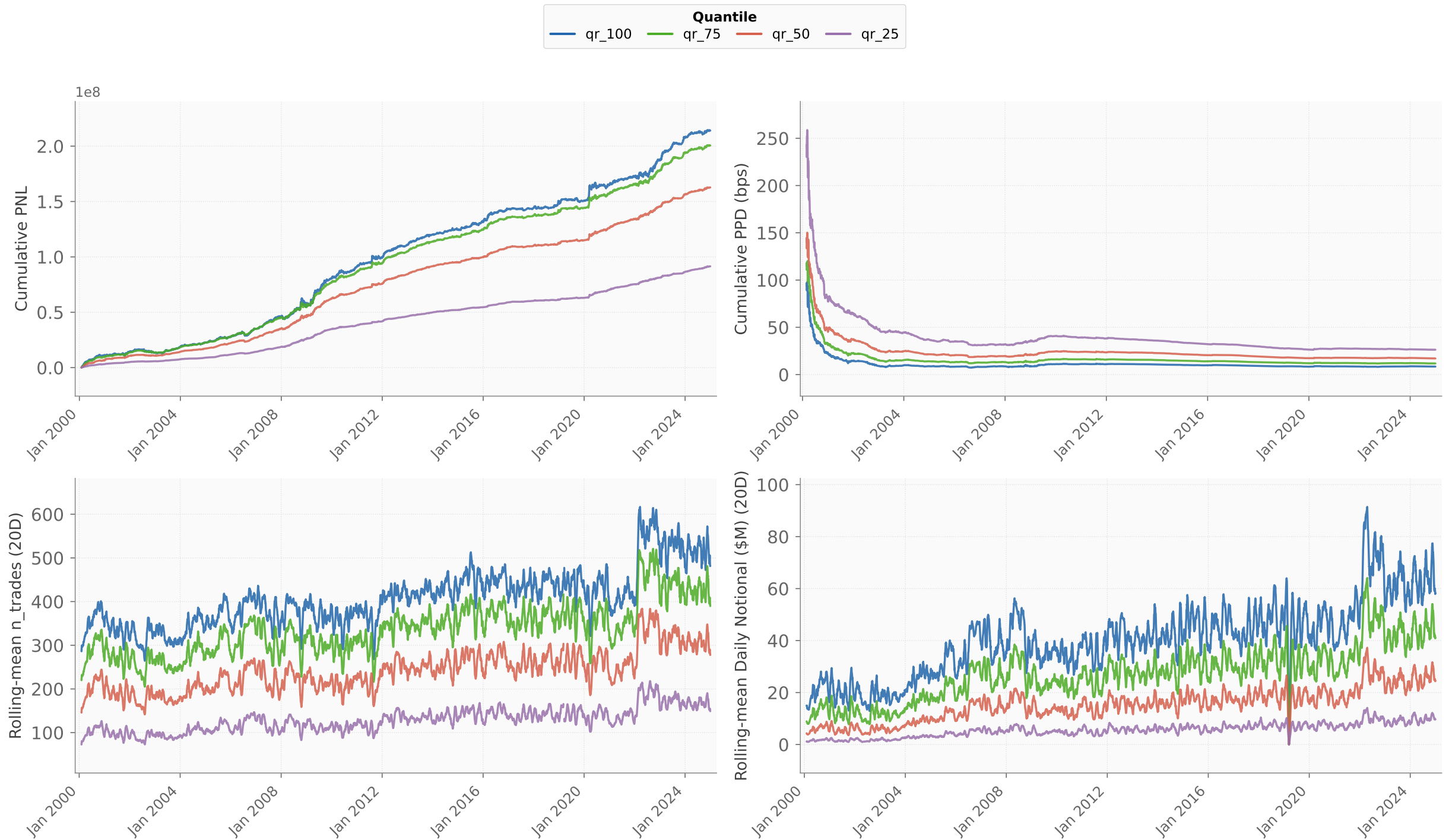
fret_OPCL_RR | pret_ols | bs_m1_c500k — Temporal metrics



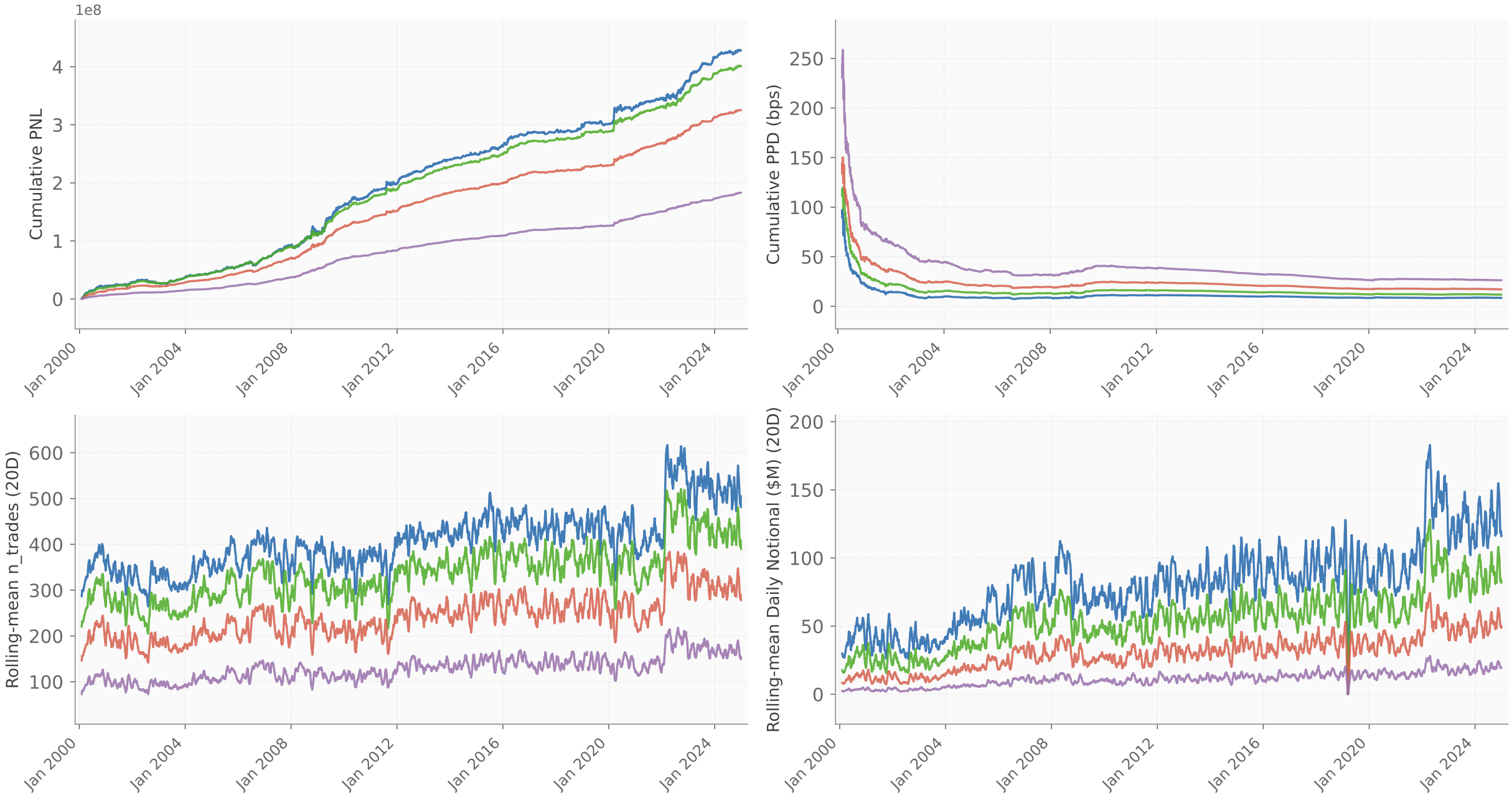
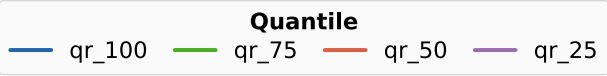
fret_OPCL_RR | pret_oracle | bs_m025_c100k — Temporal metrics



fret_OPCL_RR | pret_oracle | bs_m05_c250k — Temporal metrics



fret_OPCL_RR | pret_oracle | bs_m1_c500k — Temporal metrics



Outlier Tables

pnl — Top Highs & Lows

Type	Date	Signal	Bet	Target	Q	Value	z
High	2020-03-09	pret_base	bs_m1_c500k	fret_OP5CL_RR	qr_100	2.86016e+07	58.98
High	2020-03-10	pret_base	bs_m1_c500k	fret_OP5CL_RR	qr_100	2.3929e+07	49.34
High	2020-03-12	pret_base	bs_m1_c500k	fret_OP5CL_RR	qr_100	2.33873e+07	48.22
Low	2020-03-18	pret_base	bs_m1_c500k	fret_OP5CL_RR	qr_100	-3.16377e+07	-65.32
Low	2020-03-18	pret_base	bs_m1_c500k	fret_OP5CL_MR	qr_100	-2.71516e+07	-56.06
Low	2020-03-18	pret_base	bs_m1_c500k	fret_OP5CL_RR	qr_75	-2.31406e+07	-47.79

ppd — Top Highs & Lows

Type	Date	Signal	Bet	Target	Q	Value	z
High	2020-03-09	pret_oracle	bs_m05_c250k	fret_OP5CL_RR	qr_100	0.234899	16.11
High	2020-03-09	pret_ols	bs_m1_c500k	fret_OP5CL_RR	qr_100	0.234899	16.11
High	2020-03-09	pret_har	bs_m1_c500k	fret_OP5CL_RR	qr_100	0.234899	16.11
Low	2020-03-18	pret_base	bs_m025_c100k	fret_OP5CL_RR	qr_25	-0.31254	-21.59
Low	2020-03-18	pret_clus	bs_m025_c100k	fret_OP5CL_RR	qr_25	-0.312149	-21.57
Low	2020-03-18	pret_ols	bs_m025_c100k	fret_OP5CL_RR	qr_25	-0.312149	-21.57

Outlier Tables

sizeNotional — Top Highs & Lows

Type	Date	Signal	Bet	Target	Q	Value	z
High	2022-03-24	pret_base	bs_m1_c500k	fret_OP5CL_RR	qr_100	1.9774e+08	6.13
High	2022-03-24	pret_clus	bs_m1_c500k	fret_OPCL_MR	qr_100	1.9774e+08	6.13
High	2022-03-24	pret_oracle	bs_m1_c500k	fret_OPCL_MR	qr_100	1.9774e+08	6.13
Low	2019-03-14	pret_base	bs_m1_c500k	fret_OP5CL_RR	qr_50	0	-0.95
Low	2019-02-19	pret_base	bs_m025_c100k	fret_OPCL_RR	qr_25	0	-0.95
Low	2019-02-19	pret_base	bs_m025_c100k	fret_OPCL_RR	qr_100	0	-0.95

n_trades — Top Highs & Lows

Type	Date	Signal	Bet	Target	Q	Value	z
High	2024-12-23	pret_base	bs_m05_c250k	fret_OPCL_RR	qr_100	818	4.39
High	2024-12-23	pret_base	bs_m1_c500k	fret_OPCL_RR	qr_100	818	4.39
High	2024-12-23	pret_base	bs_m1_c500k	fret_OPCL_MR	qr_100	818	4.39
Low	2000-01-31	pret_oracle	bs_m1_c500k	fret_OP5CL_MR	qr_25	53	-1.89
Low	2000-01-31	pret_har	bs_m1_c500k	fret_OPCL_RR	qr_25	53	-1.89
Low	2000-01-31	pret_clus	bs_m1_c500k	fret_OP5CL_MR	qr_25	53	-1.89